

The background is a dark blue gradient. It features several large, overlapping circles in a medium blue color. Red arrows are scattered across the image: one at the top pointing up, one on the left pointing down, and two in the lower half forming a circular path around a central dark blue circle.

Principles of

Biological Autonomy

Francisco J. Varela

Edited and annotated by
Ezequiel A. Di Paolo and
Evan Thompson.

Foreword by
Amy Cohen Varela.

Principles of Biological Autonomy

Principles of Biological Autonomy

a new annotated edition

Francisco J. Varela

edited and annotated by Ezequiel A. Di Paolo and Evan Thompson

foreword by Amy Cohen Varela

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To the loving kindness of
Raul Varela and Corina Garcia,
sine qua non.

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Foreword

Francisco Varela's *Principles of Biological Autonomy* has the character of a treatise and the voice of a manifesto—two features that are found in his later and better known work, *The Embodied Mind* (Varela, Thompson, and Rosch, 1991).

It is striking to consider that a book of this philosophical depth and empirical breadth was published when its author was only thirty-three years old. It is a radical revision of our notions of the knower, the known, the nature of knowledge, and the idea of scientific research itself, buttressed by mathematical arguments, by what were at the time the latest developments in the theory of complex systems, and by Varela's investigations in biological systems. The *basso continuo* underlying the whole is a powerful and urgent appeal for us to follow this work of revision to its logical end and take up what he calls a *participatory epistemology*, leaving behind traditional views of objective understanding, ways of knowing that have long revealed their limits.

The view of knower as participant lowers the barrier between subject and object, between knower and known, and between self and other. It is an immersive view of human understanding that those who knew Varela, or know his work, will recognize as distinctive of his intellectual and personal style. Often unsettling, he is always arresting; engaging, his voice urges us, over and over, to take a step further, however uncertain the consequences may be. Setting out on the path that this work lays down means being able to entertain the possibility that certain mainstream views of reality and our place in it are just that: views or perspectives that can or must be changed if we hope to see concomitant changes in how we live and treat the world and our relations within it.

In the preface to this book, Varela writes:

A useful perspective does not require an objective, solid ground to which everything can be pinned down. The flavor of the epistemology espoused here . . . is that knowledge is indeed quite full of detail but hangs nowhere, apart from its tradition. . . . I believe that the importance of coming to grips with this realization, demanded by current research, and by logical rigor, is momentous to science, ethics, and personal life.

Those who have read *The Embodied Mind* will discover in *Principles of Biological Autonomy* the roots of the idea of groundlessness developed there, and see how it is, as the editors write, an “ur-text” for the enactive approach. This approach underscores the primacy of “personal life” or human experience in the epistemology of participation, transfiguring it into a full-fledged ethics or *caring epistemology*.

This new edition of *Principles of Biological Autonomy* does not just fill a gap in the textual traces left behind by Varela. Ezequiel Di Paolo and Evan Thompson weave their deep understanding into the fabric of Varela’s thought, illuminating its central concerns and clarifying the significance of parts which may be less accessible for nonspecialized readers. Their general introduction, the individual introductions to the three major parts of the book, and their addition of margin notes update Varela’s conceptualization in the light of current thinking in systems theory and the philosophy of cognitive science. While preserving the integrity of the original, they have entered into a truly enactive dialogue with the text. What emerges from this dialogue is a precious renewal, in which Varela’s voice is revitalized by theirs in counterpoint. I have no doubt that Francisco would have thoroughly enjoyed the dance with such minds. I salute these contributors for the care and brilliance with which they have prepared this volume. May its message, as crucial today as it was forty-five years ago, resonate far and wide.

Amy Cohen Varela

Introduction to the New Edition: A Message from the Margins

Francisco Varela wrote *Principles of Biological Autonomy* in his early thirties through the nomadic years that followed the coup d'état in his native Chile on September 11, 1973. He put the finishing touches on the manuscript in 1978 during his time as the scholar-in-residence at the Lindisfarne Association in New York City by invitation of William Irwin Thompson. He also gave a course at Lindisfarne based on the ideas of the book, as well as his research on the neurophysiology of visual perception and his growing interest in Buddhist philosophy. Lindisfarne proved to be a suitable atmosphere for nurturing his daring new ideas. The book brings together the deep contributions of a young scholar to a variety of fields, all in the service of shaping a new framework for understanding life and the mind that would continue to occupy Varela for the rest of his career.

Trained as a neurobiologist, Varela's scientific trajectory was itinerant, dwelling on territories as diverse as mathematics, systems theory, immunology, cognitive science, and philosophy, always informed by a keen awareness of the ongoing conversations in the social sciences, the humanities, and other epistemic traditions, including phenomenology, psychoanalysis, and Buddhist philosophy. Perhaps only someone able to regard mainstream science from these multiple perspectives, someone with an idiosyncratic history and assorted interests who enjoyed a view from the margins, would be able to see the recurring epistemic deadlocks and problems that reside at the blind spots of established disciplines. And perhaps only someone with the vitality of youth and the urgency of the nomad would find the courage to put his answers to these problems in the form of such an overarching treatise.

The publication of *Principles of Biological Autonomy* was welcomed by scholars and thinkers in general systems theory, cybernetics, mathematics, philosophy, and ecology. But partly because of its radical message, and partly due to the difficulties and heterogeneities in the text, the book did not reach a large audience of biologists, psychologists, or neuroscientists—those in disciplines that were (and still are) very much in need of a scientific framework for understanding autonomy. Having no further printings, *Principles of Biological Autonomy* soon acquired the status of an underground classic.

During the 1980s and 1990s, another demanding book coauthored by Varela also acquired cult status. It described his earlier work on autopoiesis with his colleague and mentor Humberto Maturana. Published by Robert S. Cohen and Marx W. Wartofsky in their series Boston Studies in the Philosophy of Science, *Autopoiesis and Cognition* became one of its bestselling volumes. The fresh reader would arrive at this book's radically novel approach to understanding living systems with some trepidation. One easily could be daunted by the mind-warping prose, the page-long paragraphs with scarce figures, and few, sometimes odd examples. Occasionally, they were punctuated by revelatory messages jumping off the page with illuminating clarity—but they ran contrary to everything that one had been taught about biology and cognition.

New printings and new translations deservedly brought *Autopoiesis and Cognition* out of obscurity, if not quite into the mainstream, in later years, and today this work is appreciated across a variety of disciplines. This has not been the fate of *Principles of Biological Autonomy*, which showcased Varela's further thinking after *Autopoiesis and Cognition*. *Principles of Biological Autonomy* soon went out of print, information about it became scarce, and copies became expensive and hard to find.

To the casual reader with an interest in cognitive science, Varela's work seemingly disappeared after *Autopoiesis and Cognition* and reemerged with the publication in 1991 of *The Embodied Mind*, written with Evan Thompson and Eleanor Rosch. The language of this book is very different, but the ideas are still radical and against established ways of thinking—still a view from the margins. But *The Embodied Mind* is an altogether different text from those early writings on autopoiesis. It is clearly argued and relies on empirical work and several philosophical traditions. The word “autopoiesis” hardly makes an appearance in it. Someone who has not followed Varela's work in the intervening years could be forgiven for thinking that there is little connection between this book and his earlier writings.

In fact, the connections are many and run deep. But to understand them, one must understand the development of Varela's thinking from the ideas of autopoiesis into broader concerns that emerged in part from unanswered questions in that early theory of living organization. *Principles of Biological Autonomy* is a key chapter in that story—a transition between periods and a precursor of Varela's later work, in particular on the enactive approach to cognition. References to autonomy and autonomous systems figure prominently in *The Embodied Mind* (some might regard *Principles of Biological Autonomy* as an enactive ur-text).

Varela put *Principles of Biological Autonomy* together out of his other writings with few but important changes, some textual glue, and little consideration for the fact that most readers are not simultaneously experts in biology, cybernetics, category theory, immunology, neuroscience, and the psychology of perception. In most cases, we would expect such an approach to yield a rather disjointed result. While the prose and technical discussions of the book are indeed heterogeneous and demand much from any reader who is not a polymath, a clear vision runs throughout the text. Central messages are expounded in one chapter and reappear in another in an apparently distant domain. They lure the reader with the potential payoffs of fully engaging this complex text to unravel the principles of autonomy across a range of scientific and philosophical questions.

The main theme in *Principles of Biological Autonomy* is the issue of autonomy. What does it mean to be an autonomous system? How is autonomy related to identity and interactions? Is it possible to approach this subtle and elastic idea scientifically? Varela answers the last question in the affirmative. But to arrive at his main proposal requires the careful examination of several preliminary issues, such as our roles as observers with multiple possibilities for making relevant distinctions, system-theoretic notions of structure and organization, and an exploration of the different kinds of explanations that we normally formulate in science and daily life. Perspectives on life, mind, and experience that have been deeply held for hundreds of years and have achieved the status of common sense beyond scientific and philosophical circles need to be revised or abandoned to broach the question of autonomy.

The whole exercise goes beyond the presentation of a theoretical proposal and invites readers to become aware of the insidious epistemology of control that frames our ways of thinking and being. *Principles of Biological Autonomy* can make the reader uneasy, primarily because of its consciousness-raising potential, its

invitation to reflexivity. For the same reason, reading it can be an exciting experience.

Varela's central message is summarized in his closure thesis: *Every autonomous system is organizationally closed*. This thesis is meant as a bid for thinking of examples that fit or may potentially not fit this definition. It is a *scientific* thesis, and at the same time a mind-teaser from which different thinking habits can form. Organizational or operational closure is a key concept not only in this book, but also in Varela's later work and enactive research. It describes a system that is organized as a network of processes that, through their ongoing operation, realize the enabling relations among themselves so that their network organization is sustained and distinguished as a unity in some domain. For the domain of biomolecular interactions, autonomy coincides with autopoiesis, the self-individuation of the unicellular organism. For other domains, such as the activity of immune networks or neurodynamic processes, autonomy is different from autopoiesis in the strict biomolecular sense. There is no restriction, in principle, to domains in which autonomous organization may be manifest. In each case, the closure thesis furnishes us with a test to settle the question of whether a system is autonomous.

Formulated in operational terms, the closure thesis is a powerful tool committed to fill in an important hiatus in the biological and cognitive sciences. The fact that these disciplines continue to debate the nature of life and mind, of agency and the self, circling around perennial questions with frustratingly slow progress, suggests that a fundamental gap remains in their conception of autonomy.

Nevertheless, the idea of autonomy as closure has inspired work in theoretical biology, systems biology, and origins of life studies. The closure thesis is a bold theoretical proposal that continues to be explored to this day. It paves the way for a naturalistic understanding of life and mind, without mysteries and without reduction of biology to physics or psychology to biology. Autonomy, to repeat, is a central and distinctive concept in the enactive approach, one that differentiates it from other contemporary perspectives on embodied cognition. In fact, the closure thesis is the ultimate antireductionist conceptual tool. It gives us a way of telling at what point we may say that a system and its domain of interaction have coemerged operationally at different scales and spheres, though not necessarily as neat hierarchical levels, with their corresponding phenomenology and regularities pertaining to that particular domain.

Varela explores several themes along and around the question of autonomy. These include the following:

- The links between autonomy, identity, and interactions.
- Autopoiesis as autonomy in the domain of molecular production. Models and manifestations of autopoiesis, implications for the phenomenology of living systems (behavior, ontogeny, evolution).
- The broader consequences of the autonomy perspective, cognitive domains, and the history of coupling.
- The formal relations between autonomy, self-reference, and recursivity.
- The descriptive complementarity between operational and symbolic discourse, and the kinds of explanations they serve.
- A call not to confuse the two kinds of discourse. Implications for the question of teleology and naturalizing the mind.
- The role of the active observer (community). An epistemology of participation.
- Formal approaches to distinctions, self-reference, and recursivity. Extensions to an indicational calculus to account for self-reference. Conceptual and formal approaches to complementarity, dialectics, adjointness, eigenbehaviors, and reflexive domains. Reflections on the power and limitation of mathematical models based on differential dynamics.
- Application of the closure thesis to immune networks as an alternative to clonal selection theory. Implications for autoimmunity and self-nonself distinction.
- Application of the closure thesis to the nervous system, dispensability of notions of information/representation, critique of input/output approaches to neural function, codependence between neurodynamics and sensorimotor couplings, and perceptual invariants.
- Epistemological implications of autonomy. The social dimension of the act of observing and autonomy manifested in social processes. Our own status as autonomous observers. A naturalized epistemology that includes ourselves within the world, and reflexive implications.

Today, we see a healthy exploration of ideas that not long ago could be found only at the fringes of the biological and cognitive sciences. An important participant in these conversations in the field

of embodied cognitive science is the enactive perspective initiated by Varela and colleagues. It has continued to develop with impact not only in cognitive science and philosophy of mind, but also in musicology, sports science, psychiatry, education science, health science, technology studies, literary studies, architecture, and the arts. In many cases, the term “enactive” is seen as a combination of embodied, situated, active, and participatory. The key message of this approach is that meaning is not something that we extract out of a pregiven world and process in our brains, but rather something that we are constantly engaged in creating together with the world and with others. What may not be immediately apparent is that in all these applications, the enactive ideas at play are framed and articulated by the concept of autonomy proposed in *Principles of Biological Autonomy*. Autonomy, even when not mentioned explicitly, underlies practically every work in the enactive approach. It may appear as a concern for the situated person and their own history and perspective, an examination of the disorders of the self in psychiatry, a hypothesis about habits and the learning of new skills, a discussion about minimal forms of mind, an exploration on the deeper links between affect and time, and so on. In all these cases, we are likely to witness the same kinds of thinking gestures that Varela developed in his first published book.

It is to bring not just this history, but also the very idea of autonomy to the center of current discussions that we have decided the time was ripe for a new edition of *Principles of Biological Autonomy*, some forty-five years after its publication. We hope to redress years of underserved inaccessibility. But even more, we hope to underscore the theoretical lineage of ideas that are very much alive and needed today. The autonomous perspective can provide tools for better framing our relations to technology. It can help us formulate simultaneously more powerful and caring outlooks on artificial intelligence, human communication, education, and health care. In contrast to the mindset of control, the epistemology of participation that emerges from the autonomy viewpoint is badly needed for reenacting our design strategies, our economic relations, our politics, and our participation in the rest of the natural world. These are some of the reasons that we think the messages expressed in *Principles of Biological Autonomy* should be brought to a new readership—from the margins into the main conversation.

About This Edition

The text for this edition of *Principles of Biological Autonomy* is largely faithful to the original, with its varying densities and technical vocabularies ranging from systems theory and mathematics to physiology and neuroscience, all underlaid by Varela's running epistemological commentary. All the edits are minor. We have fixed typographical errors, revised the bibliography correcting oversights, and introduced changes in some places to reflect more inclusive language. Also, a few issues with the mathematical notation have been dealt with and checked against the original publications, in a few cases introducing additional information from these sources.

We think that it is important for the text to convey an idea of Varela's vision during the first phase of his scientific career. Undoubtedly, had he written a similar book on the topic of autonomy in later years, the text would have been very different. Our aim is not to guess what such a book would look like, but rather to facilitate the reading of a rich and demanding work as it is. To achieve this, we have added a short introduction to each of the three parts of the book. These provide broad summaries as well as relevant context for Varela's ideas, making the key messages widely accessible. They also supply a brief overview of how these ideas continued to develop in follow-up work, by Varela and others, to this day.

Margin annotations to the original text complement these introductory essays and provide more detailed and specific information. These annotations are labeled "*Eds.*" to differentiate them from the original notes. They tend to fall into one or more of the following kinds:

1. Supplemental information when an original reference may not be recognizable to a wide readership.
2. Pointers to subsequent relevant work on a particular topic.
3. Clarification of meaning or context when a formulation may be ambiguous or open to contrasting interpretations.
4. Critical comments and cautionary remarks to avoid potential misreadings.

Abundant as they are, these annotations are by no means complete. Several passages without marginal comments probably deserve one, and most notes only scratch the surface of an important issue. We trust, however, that they serve the purpose of orienting the reader and suggesting pertinent additional material for further exploration.

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Preface: Information and Control Revisited

Systems are not found in nature, but only in the mind of man.

—C. Bernard, *Introduction to the Study of Experimental Medicine* (1949, 221)

What we are supplying are really remarks on the natural history of man; not curiosities however, but rather observations on facts which no one has doubted and which have only gone unremarked because they are always before our eyes.

—L. Wittgenstein, *Remarks on the Foundations of Mathematics* (1967, 141)

Two themes, in counterpoint, are the motif of this book. The first one is the autonomy exhibited by systems in nature. The second one is their cognitive, informational abilities.

These two themes stand in relation to one another as the inside and the outside of a circle drawn in a plane, inseparably distinct, yet bridged by the hand that draws them.

Autonomy means, literally, self-law. To see what this entails, it is easier to contrast it with its mirror image, allonomy or external law. This is, of course, what we call *control*. These two images, autonomy and control, do a continuous dance. One represents generation, internal regulation, assertion of one's own identity: definition from the inside. The other one represents consumption, input and output, assertion of the identity of the other: definition from outside. Their interplay spans a broad range, from genetics to psychotherapy.

We all know control well; it has been charted out and formalized. Hence the power of the computer and of consumer-oriented services. Its popular model is: something in/process/something out. We stand on both sides of in and out, whether an economic system, a compiler, or a person's mind. The fundamental paradigm

of our interaction with a control system is instruction, and the unsatisfactory results are errors.

Autonomy has been less fashionable. It is usually taken as a more vague and somewhat moralistic term, and waved off as a question of indeterminacy. There is little understanding of its generic import, let alone its representation in formal terms. The fundamental paradigm of our interaction with an autonomous system is a conversation, and its unsatisfactory results breaches of understanding.

One fundamental intention of this book is to bring the *interplay* of these two notions into the open, and to identify the underlying mechanisms that endow natural systems with autonomy. As it turns out, these mechanisms have to do with the pervasive circularities to be found in nature. We are led to consider in all seriousness the traditional image of the snake eating its own tail as the guiding image for autonomy as self-law and self-regulation. But what does this “self” mean, more precisely? A focus on this question is a guiding thread throughout this book, leading us to a characterization of self-referential, recursive processes and their properties as fundamental mechanisms of natural autonomy.

The way a system is identified and specified through our interactions with it is not separable from the way its *cognitive* performance is understood. The control characterization is intimately tied up with an understanding of information¹ as instruction and representation. Accordingly, to explore the way in which a system specifies its own identity is *also* to explore what its informational actions can possibly mean (Piaget, 1967). Thus, by discussing autonomy, we are led to a reexamination of the notion of information itself: away from instruction, to the way in which information is constructed; away from representation, to the way in which adequate behavior reflects viability in the system’s functioning rather than a correspondence with a given state of affairs.

It can be said that the central line of thinking in this book is to untie, explicitly, the knots of this inseparable trio: a system’s *identity*, its performance in its *interactions* with what it is not, and how we *relate* to these two distinct domains.² Different preferences and values attributed to this triad determine how we see a system, how we conceptualize information, and what role we attribute to ourselves in the whole process.

Stated in another way: Behind the predominant views on control and information-as-representation, we find a constellation of philosophical assumptions shaping the way we relate to the diversity of

¹I use the word information here in its most generic sense of *semeiosis*. Other connotations that the word has acquired in its Shannonian treatment are here strictly secondary. See for a discussion the excellent work of Nauta (1972), *The Meaning of Information*.

²*Eds.* This “inseparable trio” continues to be central to Varela’s later work. In attempting to define an organism in operational terms, we are obliged to recognize both the organism’s relation to its environment and our own relation as observers to both of them. Because of its focus on the organism, autopoietic theory is often seen as centered on the question of the system’s identity and the role of the observer. Less is said about the role of the environment in this entangled trio, even though enactive developments emphasize, as a corrective, the organism-environment history of interactions as a relation of mutual co-definition (Varela et al., 1991). Engaging with ecological psychology, enactive work has recently begun to turn more explicitly to understanding this relation in terms of development and place (Sepúlveda-Pedro, 2023).

sentient beings.³ By these I mean entities to which we are compelled to acknowledge an informational side, a mind of sorts, however opaque and simple. I am not talking about individual living beings only, but of many other aggregates such as ecological nets, managerial complexes, conversations, animal societies—in fact, wherever there is a sense of being distinct from a background, together with the capacity to deal with it via cognitive actions.

Since most of our lives is concerned with how we see other entities and how we comprehend what transpires between us, it is no wonder that the information sciences, understood in this broad sense, are loaded with philosophical and ethico-political connotations. It is my view that this area of science has been substantially modeled in the image of physics and its technological pathos. One essential difference here however is that *we* and the world that support us belong to the categories of sentient being and not of atoms and quasars. Consequently, the Promethean approach inherited from physics bounces back at us in a fast and dramatic way.

I am not being grandiose. The fact is that, after the Wars, scientific imagination turned from watts to bits, and in almost no time produced a dramatic change not only in the shape of what scientific research was about, but in the life of everybody as well. As if in a boiling pot, the images of information and information processes surround everybody and everything that has an interest in complex relationships, communication, and mind. Ideas, fields, and applications crisscross, and there is no sense of direction or unification such as the prewar science seemed to have. I am not seeking such unification here at all. I am instead identifying a *dominant assumption* that seems pervasive at every depth of this boiling pot, and I am proposing to explore an alternative.

Rosenberg (1974) has aptly characterized the dominant view of the information sciences as the “gestalt of the computer.” He is right, I believe, in a double sense. First, it is indeed like a perceptual gestalt in the sense of a favored perspective, making it very hard to step outside to contemplate where one is standing. Second, the computer indeed embodies the metaphor in terms of which everything else is measured. The fast pace in the field of design with its inherent manipulative ethos has overshadowed every other source of images and modes of understanding. Information, for the computer gestalt, becomes unequivocally what is *represented*, and what is represented is a *correspondence* between symbolic units in one structure and symbolic units in another structure. Representation is fundamentally

³A third and last member of this constellation of concepts is that of *subject*, as is understood currently in the image of a skull-bound individual. I will not enter in this book into the discussion of this central theme, although a few points are touched upon in the last chapter.

⁴Thus according to Newell and Simon (1976), this should be one of the basic building axioms of the information sciences.

a picture of the relevant surroundings of a system, although not necessarily a carbon copy.⁴

From the point of view of the natural (including the social) systems, the computer gestalt is, to say the least, questionable. There is nobody in the brain to whom we can refer to obtain an assignment of correspondences, and any attempt to view it as an input-output processing machine can be equally well interpreted as the machine's reducing *us* to an equally allonomous entity. With any of the variety of natural, autonomous systems, all we have is certain behavioral regularities, which are of interest to us as external observers having simultaneous access to the system's operation and to its interactions. Such regularities, when we choose to call them cognitive and informational, always refer us back to the unitary character of the system at hand, whether a cell, a brain, or a conversation. From this perspective, what we call a representation is not a correspondence given an external state of affairs, but rather a consistency with its own ongoing maintenance of identity. Such regularities, which we choose to call *symbolic*, are *not* operational for the system, for it is *we* who are establishing correspondence from a vantage point that is not in the system's operation. Thus, when we switch from a control to an autonomy perspective, what we call information differs from the computer gestalt in important ways. Every bit of information is relative to the maintenance of a system's identity, and can only be described in reference to it, for there is no designer. In this sense information is never picked up or transferred, nor is there any difference whatsoever between informational and non-informational entities in a system's ambient.⁵

Useful as it may be in the fields of design, the paradigm of cognitive processes as representations has been given a privileged status in our current thinking about cognition. It is well and good that we can sidestep these distinctions in the domain of design, or in some of our dealings with natural systems where they may be treated analogously. To take this approach as a *general* and universal strategy for all aspects of natural systems, including human transactions, seems incredibly limiting. In fact, it is not workable at all, as I shall argue in detail for the two richest cognitive systems in living beings: the immune and the nervous networks.

It is one of those interesting *corsi e ricorsi* of the history of ideas that the source of the computer gestalt was an understanding of living systems. From this initial inspiration, however, most of the emphasis seems to have shifted towards engineering and design, far more than into other areas. I am arguing, *again on the basis of*

⁵*Eds.* The criticism of representationalism is one of the central tenets of Varela's thought, autopoietic theory, and subsequent enactive work. It takes many forms, from pragmatic and epistemic to logical and ontological (Varela et al., 1991; Thompson, 2007; Noë, 2004; Hutto and Myin, 2012; Fuchs, 2017; Gallagher, 2017; Di Paolo et al., 2018, etc.). In this paragraph, Varela's point concerns the nonoperational character of representation-talk: while it may make sense to us observers in our symbolic domain of discourse to interpret some regularities in terms of semantic information, as such, these regularities play no operational role in the logic of autonomy (this point is elaborated in chapter 9). Varela's criticism resembles the homuncular infinite regression or mereological fallacy argument against representationalism (Kenny, 1971; Bennett and Hacker, 2003).

biological systems, that this predominant understanding is one-sided and incomplete.

I am claiming that information—together with all of its closely related notions—has to be reinterpreted as codependent or constructive, in contradistinction to representational or instructive. This means, in other words, a shift from questions about semantic correspondence to questions about *structural* patterns. A given structure determines what constitutes the system and how it can handle continuous perturbations from its surroundings, but needs no reference whatsoever to a mapping or representation for its operation. We don't ask what is the correspondence between an animal nervous system and "the" world in which it is, but rather what is the structure of the nervous system whereby it can effect a shaping of its domain of interactions. The notion of information as representation is ultimately independent of the system's structure; but it is for the external observer—better still, for the whole tradition describing the situation—that the externality of the supposed world to be mapped exists at all. By insisting on looking at cognitive processes as mapping activities, one systematically obscures the codependence, the intimate interlock between a system's structure and the domain of cognitive acts, the informative world which it *specifies through its operation*. Informational events have no substantial or out-there quality; we are talking literally about *in-formare*: that which is formed within. In-formation appears nowhere except in relative interlock between the describer, the unit, and its interactions.⁶

This idea is not really new; it has been familiar to many scientific and philosophical traditions. However, under the towering influence of positivism, it has been ignored in the language and empirical research of science by engineers, biologists, and educators alike. My argument takes root in science and attempts to redress this imbalance from the inside. This shift from a semantic to a structural point of view is, at this stage, a research program that is only beginning to unfold. In this book I shall explore and give substance to only a few items of this program. But unless we take into account that there is an autonomous side to many natural and social systems, we run into troubles, not only in the specifics of research and formalizations, but in the wider scale of our dealings with sentient beings, with life, with the environment, and in human communication. In this respect, the problems of biology are a microcosm of the global philosophical questions with which we grapple today.

One basic tenet in these pages is that I am not *against* what I have called the computer gestalt, nor am I rejecting it as useless.

⁶*Eds.* We see here an early statement of what will later become the explicit central claim of the enactive approach: cognitive systems enact a world of significance. This happens because of the codependent relations that are specified between agents and the world by the need to remain alive and to continue to be autonomous. Later work elaborates this relation of codependence using the term *sense-making* (Varela, 1984b; Weber and Varela, 2002; Di Paolo, 2005; Thompson, 2007, 2011).

I am saying it is *limited* and workable only in situations of restricted autonomy and fixed in-formation, which are actually the only situations where control and representational views can emerge at all. The perspective I am sketching here can accommodate this standard view, and set it more in balance. Accordingly, one very important concept throughout this book is that of complementarity, and the constructive interplay between two interdependent visions that raises one's level of understanding to a new level. I am not an "autonomist" at war with control engineers; but I do want to state clearly that the control view, if taken alone, leads to inadequacies in our understanding of natural systems, and to important epistemological and political difficulties. I am not saying what is better; I am stating alternatives.

The fact that I take here the point of view of a natural historian (and a biologist at that) means also that there is no discontinuity with the way in which these concepts can be seen to apply to human cognitive capacities, and, for that matter, to societal dynamics.⁷ What we see operating in greater detail in natural autonomy—the actual subject of this book—is a reflection of what we ourselves are immersed in.

In strict accordance with this view of in-formation, we shall see that the presence of the observer (of the observer-community, of the tradition) becomes more and more tangible, to the extent that we have to build upon a style of thinking where the description reveals the properties of the observer rather than obscuring them (as von Foerster has aptly remarked). It is a view of participatory knowledge and reality, which we see rooted in the cognitive, informational processes of nature from its most elementary cellular forms. There are, in fact, two distinct ways in which the irruption of the observer becomes apparent in this presentation.

On the one hand, we see the necessity of acknowledging the role of the process through which we distinguish the unities or entities we talk about: the way the world is split into distinct compartments, and the way such discriminations and distinctions are related by levels and relationships between levels. Thus, the maintenance of a system's identity—its autonomy—is a distinct and irreducible domain with respect to the functioning of that system in its interactions. These two phenomenal domains are related only through *our* descriptions, and these relationships do not enter into the operation of the system we are concerned with. Each of these views is complementary to the other, and we need to make them explicit.

⁷Eds. This is an early statement of a central enactive idea: the *continuity between life and mind*, in this case explicitly reaching from biology to human society. This concept is understood in a nonreductive sense in the enactive literature. It entails a conceptual as well as a phenomenal continuity without denying qualitative differences in forms of life and mind. Such differences, however, demand to be explained naturalistically, excluding conceptual breaks or gaps (see Thompson, 2007; Di Paolo et al., 2017, 2018). For a similar notion of continuity, see Dewey (2008, 30).

A second way in which the observer enters into this view of unities and their information is that we ourselves fall into the same class—there is continuity in the biological sense, and in the cognitive mechanisms that operate elsewhere in nature. Thus what is basically valid for the understanding of the autonomy of living systems, for cells and frogs, carries over to *our* nervous system and social autonomy, and hence to a naturalized epistemology, which is not without its consequences. It forces us to a renewed understanding of what physical nature can be that is inseparable from our biological integrity, and what we ourselves can be that are inseparable from a tradition.⁸

In point of fact, these two modes of bootstrapping the observer-community close the circle for this presentation and make it hang together cohesively. It embodies, once again, the same basic notion that a useful perspective does not require an objective, solid ground to which everything can be finally pinned down. The flavor of the epistemology espoused here, and assumed by the structure of the book, is that knowledge is indeed quite full of detail but hangs nowhere, apart from its tradition, and leads nowhere except to a new interpretation within that tradition. I believe that the importance of coming to grips with this realization, demanded by current research and by logical rigor, is momentous in science, ethics, and personal life. If these pages contribute an inch to that awareness, they will have succeeded amply in my eyes. Yet I immediately reiterate the caveat that I shall *not* dwell in these matters in great detail. My subject is autonomy in natural systems. All the connections and implications with epistemology and human affairs will be pointed at, but not explored at any length.

There are certain persons who have influenced this book so pervasively that it seems fair to mention them at the outset. They are Humberto Maturana, Heinz von Foerster, Gregory Bateson, and Jean Piaget. Of all of them, only Maturana is truly represented in the content of these pages.⁹ What has always inspired and maintained my interest in these thinkers is that they refuse to set lines of demarcation, and yet keep a fixed gaze along a line of thinking. Thought comes alive, and it enlivens the stillness of disciplines. Further, all of them make natural history a permanent source of observation; there is concern with contemplation rather than design. A third trait common to these men, and of great importance to me, is the presence of an experimental epistemology as an explicit background to consider information and mind in its fullest sense, be it in Balinese dance or frog's vision. This breadth of concerns, embodied in delicate

⁸*Eds.* A corollary of the continuity between life and the mind is that in researching the mind, we must eventually seek to understand our own activities as researchers. Reflexivity, already explicit in autopoietic theory (Maturana and Varela, 1980, 1987), is central to the enactive approach (e.g., Stewart, 2010). It takes the form of a call for a circulation between science and experience (Varela et al., 1991) and a perspective “of *participation* and *interpretation* where subject and object are inseparably meshed” (Varela, 1984a, 322). In rejecting the God’s-eye view for science, our situatedness as a community of researchers is seen both as an advantage—Varela admired the claim by Hans Jonas (1966, 91) that “life can be known only by life” (Varela, 2000, 15)—and as an explanatory goal that underlines the ethical aspects of investigating the mind.

⁹*Eds.* Varela called himself “a student of Gregory Bateson” (Varela, 1979b, 14). The two interacted closely toward the end of Bateson’s life (see Thompson, 1987). Indeed, Varela finished writing this book when he was a scholar-in-residence at the Lindisfarne Association in 1978, following Bateson’s time as scholar-in-residence in 1977. Both had been invited by William Irwin Thompson. For more on the relationship between Varela and Thompson, see Thompson (1989). For Varela’s reflections on Heinz von Foerster, see Varela (1996). Jean Piaget’s work is discussed in chapter 15 and in Varela et al. (1991). Piaget’s work on equilibration continues to be influential in enactive theory (Di Paolo et al., 2017).

interplay between a view of the general and the texture of the specific, is what seems to be called for.

I have tried not to be idiosyncratic in my use of language, and sparse in introducing new nomenclature. Yet it seems inevitable to introduce a certain number of new concepts and definitions if one is to point to an uncharted ground at all. Thus, for example, I could have couched the discussion of autonomy in terms of whole systems or totalities, but these notions have acquired many connotations which might have obscured what I wanted to say; this pitfall, in fact, lurks throughout the book. Pointers and cross-references are provided in every case to related literature.

The text unfolds in three parts. These cover, respectively: autonomy of living systems as a source of characterization of autonomy in general; forms of representing complementarity and circularities; and the cognitive capacities of autonomous systems and their codependent information. Each one of these parts can be read somewhat independently of the rest, according to the reader's inclination.

I cannot say too strongly that this book is offered in the spirit of synthesis and exploration, not of treatise, dogma, or set opinion. Nothing presented here can be regarded as fixed, and I am well aware of it. In fact, the very nature of the subject, of sketching a gestalt switch about natural information and control, is predictably intricate and likely to yield mixed results. It involves partly a reinterpretation of what seems already available to us, conceptually and experimentally, and partly a rather difficult process of conceiving new designs and adjusting to new perspectives. The whole thing is actually quite shifty. My rationale for publishing it is that there is *enough* here to start and liven a discussion, enough of a body of work to create debate. I simply look at all of this as signposts in a vast shapeless landscape, where many routes may be taken, and perhaps, where we may choose to return to where we started with the feeling of seeing it again, more sharply.

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equally in debt. Perhaps more than to anybody in particular I owe to the collective mind of the people of Chile, where I found my inspiration.

The initial stages of this work were carried out at and with the support of the University of Chile. It continued at the Universidad Nacional in Costa Rica, which generously housed me in the difficult period after I left my country torn by civil war. A major portion of this book was produced while I was with the School of Medicine at the University of Colorado. I am indebted to David Moran and David Whitlock for giving me ample freedom there. The final stages of writing were carried out at the Institute of Behavioral Science, University of Colorado at Boulder.

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This book is based on work carried over a period of eight years of concern with the central questions of autonomy and information. A basic motive for writing it was the feeling that it was time to pull the disparate threads produced during this time a bit closer, and to offer a panoramic view of where I am trying to look. Except for a few chapters, all of the material included has been published before. I have reworked and rewritten all of these source papers extensively, and added whatever connecting links seemed missing at this stage. In spite of this, the reader will have to bear with a certain amount of repetition and differences in style. Many of the source papers were written in collaboration with other authors: Joseph Goguen, Louis Kauffman, Humberto Maturana, Nelson Vaz, and Ernst von Glasersfeld. In reworking the papers for this book, I have surely done violence to their initial style and intention. I am deeply grateful to all these collaborators for letting me do so; whatever success I have had in conveying an interesting idea should be shared by them in full. In all cases, at the end of the chapter I have listed the sources explicitly.

To my wife Leonor I owe more than acknowledgment; I owe all that comes from vast, nourishing love.

Francisco J. Varela, 1979

I Autonomy of the Living and Organizational Closure

So long as ideas of the nature of living things remain vague and ill-defined, it is clearly impossible, as a rule, to distinguish between an adaptation of the organism to the environment and a case of fitness of the environment for life, in the very most general sense. Evidently to answer such questions we must possess clear and precise ideas and definitions of living things. Life must by arbitrary process of logic be changed from the varying thing which it is into an independent variable or an invariant, shorn of many of its most interesting qualities to be sure, but no longer inviting fallacy through our inability to perceive clearly the questions involved.

—L. Henderson, *The Fitness of the Environment* (1913)

It may be said that the metaphysical hypothesis of a dialectic of Nature becomes more interesting when it is used to explain the passage from inorganic matter to organic bodies, and the evolution of life on earth. This is true. But it should be noted that this *formal* interpretation of life and evolution will never be more than a pious dream as long as scientists have no way of using the notions of totality and totalisation as a guiding hypothesis. Nothing is gained by proclaiming that the evolution of the species or the appearance of life are moments of the “dialectic of Nature” as long as we are ignorant of *how* life appeared and *how* species are transformed. For the present, biology, in its actual research, remains positivistic and analytical. It is possible that a deeper knowledge of its object, through its contradictions, will force biology to consider the organism in its totality, that is to say, dialectically, and to consider all biological facts in their relation of interiority. This is *possible*, but it is *not certain*.

—J. P. Sartre, *Critique of Dialectical Reason* (2004, 34)

Introduction to Part I

Part I introduces the reader to the key ideas of the theory of autopoiesis, the organizational criterion for describing living systems. Most of the content in this part derives from Varela's collaborative work with Humberto Maturana and is based on their joint writings, particularly a technical report produced for the Biological Computer Lab (BCL) at the Department of Electrical Engineering of the University of Illinois (Maturana and Varela, 1975).

The BCL was a research institution founded in 1958 by Heinz von Foerster, and its mission was to continue to elaborate and apply the more visionary ideas of the first wave of cybernetics. Cybernetics research had been gradually channeled into disciplines such as control theory and the nascent artificial intelligence (often acknowledged to have its founding event in 1956 at an eight-week summer research workshop at Dartmouth College). The more speculative and philosophical branches of cybernetics, including work by W. Ross Ashby, Gordon Pask, Stafford Beer, and von Foerster himself, found a home at the BCL until its closure in 1974.

Maturana made regular, extended visits to the BCL starting in the late 1960s, at a time when Francisco Varela, who was already a member of his lab in Chile, went to Harvard to complete his PhD thesis (in a record time of two years) in the laboratory of Torsten Wiesel. Both researchers entered into a fruitful period of collaborative writing between 1970 and 1973, when the core ideas of the theory of autopoiesis took shape. In 1973, they published a book in Spanish, *De Máquinas y Seres Vivos*, which covered the main body of this work (Maturana and Varela, 1973). Its English translation became the influential book *Autopoiesis and Cognition* (Maturana and Varela, 1980). That book was published in the influential series Boston Studies in the Philosophy and History of Science thanks to Robert S. Cohen and Max Wartofsky, the series editors, who took a strong interest in the work. Some of these core ideas also had appeared in Maturana's 1970 BCL report 9, entitled "Biology of Cognition," and their joint BCL report 9.4 from 1975.

This collaborative work in the early 1970s marked the first key period of Varela's scientific career, with conceptual contributions to biology and cognitive science that were so significant (even when their radical character resulted in a very slow percolation into these wider communities) that most academics would consider them sufficient achievements for a lifetime. Varela was compelled to abandon his assistant professor position at the University of Chile in

1973 following the military coup against the democratically elected government of Salvador Allende (Varela, 1979b, 1996). Like many artists, intellectuals, and activists, he was forced into exile, effectively interrupting his collaboration with Maturana until the 1980s.

Even though they are largely based on previously published work, the chapters in part I show signs of Varela's increasingly maturing vision and his own interpretation of the meaning of "autopoiesis" as a scientific concept and as a perspective, which was beginning to branch from Maturana's own interpretation (Maturana and Varela, 1981; Varela, 2011).

In chapter 1, Varela situates the need for the concept of autopoiesis both historically and in the approaches to biology of the time. Rejecting vitalism, biology largely embraces two complementary kinds of mechanicism: the reduction of biological phenomena to physicochemical processes and the shift of biological phenomena from the individual organism to the species effected by evolutionary explanations. Both of these miss the phenomenon of the organism itself, and with notable exceptions, the question of what is common to all forms of life is hardly ever asked.

Providing an answer to this question, while staying within the realm of scientific explanations, is the goal of the theory of autopoiesis introduced in chapter 2. Following cybernetic work, a distinction is introduced between the structure and the organization of a system. The organization describes the set of relations between the components that make up a class of systems. Its structure is the actual instantiation of these relations. "Organization" is the more abstract term, while "structure" is more concrete. There is complementarity between these concepts, and Varela highlights the role of a community of observers in navigating these dualities. For the most part, the distinction between these ideas remains watertight in the rest of the book.

"Autopoiesis" refers to a class of physical systems instantiating a *particular organization*—namely, a self-maintaining organization wherein component processes continuously regenerate the set of relations that realize this organization and constitute the system as a concrete unity in physical space. The proposal is that all living systems are autopoietic systems (i.e., they are organized in this fashion), and vice versa. Studying the phenomenon of life requires accounting for both the organizational and the structural aspects of an organism.

The autopoietic perspective asserts that the organism plays a central role in biology. It gives us conceptual resources to define life

not as a contingent list of properties (e.g., reproduction, growth, and sensitivity), but rather as physical systems organized in a particular way. It decouples questions of what life is from other phenomena that also concern biologists, such as explaining the diversity of organisms and their adaptive relations to the environment—phenomena that, lacking a scientific definition of life, tend to be borrowed in order to fulfil its role.

After elaborating the epistemological implications of this proposal, Varela presents a computational model of autopoiesis in chapter 3. This classical model prefigures much of the work in the field of artificial life that took place in the 1990s and adds much-needed concreteness to a rather abstract and dense set of ideas.

Chapter 4 digs deeper into the organizational implications of autopoiesis, distinguishing relations of topological constitution (defining spaces and boundaries), specificity (what produces what), and order (how everything is regulated). These types of relations are mutually recursive, highlighting the inherent circularity of the idea of autopoiesis. These relations, it is claimed, obtain in the living cell, and when considering the origin of life, we consider it at the level of the instantiations of these relations, and not at the level of what specific molecules or chemical reactions are involved. An example of such an organizational approach is Eigen and Schuster's hypercycle model of molecular evolution, which focuses on relations of specification but not on topological relations; hence, it lacks an account of compartmentalization.

Chapter 5 develops the implications of autopoiesis for evolution. It distinguishes autopoiesis from reproduction and claims that the latter is subordinated to the former. This was a bold claim at the time, as it is now, when evolutionary thinking remains at the core of theoretical biology. The chapter also introduces the notion of "ontogeny," defining it as the plastic structural changes undergone by an organism while conserving its autopoiesis. The emphasis in both cases is on the claim that notions of coding, information, and fitness belong to the domain of symbolic discourse, not to the concrete operation of autopoiesis. This view does not deny natural selection, but it does not emphasize it either, speaking of general patterns of structural and organizational change during evolution and setting the stage for the specifically autopoietic conception of evolution as *natural drift*, presented later in Maturana and Varela (1987).

Chapter 6 develops additional biological and epistemological implications of autopoiesis. It reiterates the importance of the

organization/structure duality and the need for a formal theory of biological phenomena based on not losing sight of this complementarity, as opposed to focusing only on structural phenomena (e.g., biochemical reactions), without placing them within the organizational framework of autopoiesis. This leads to a series of complications of the epistemic task of observing and explaining biological phenomena, placing special importance on the acts of observing and making distinctions, determining what counts as a unity, what counts as a component, and what phenomenological domains are entailed by them. In performing these epistemic acts, the observer may capture the self-affirming properties of autopoietic dynamics or may fail to do so. The chapter briefly illustrates some of the phenomena that may be thus distinguished, in terms of behavior, viability, and cognitive and interactional domains.

Part I culminates with the formulation in chapter 7 of the central thesis of the book, the closure thesis. Here, we see Varela following his own interpretive path. By examining the questions of multicellularity and social systems as systems composed (among other things) of autopoietic units, Varela cautions against the quick interpretation of any complex self-maintaining system as an instantiation of autopoiesis. In some cases, such as multicellular organisms, it makes sense to describe them as autopoietic systems of higher order. But in other cases (say, relations in the family and other social systems) the term is at best a metaphor, at worst a crude distortion of the phenomenology of such systems. And yet this phenomenology often resembles the logic of self-production and self-distinction of autopoiesis, up to a point.

For Varela, to make sense of these resemblances, the way forward was to introduce the concept of *autonomy* as a generalization of autopoiesis. Autonomy, according to him, can account for the phenomena of wider self-sustaining systems. The key technical innovation developed by Varela is to equate this loose idea of autonomy with the technical concept of organizational closure, which restates the criteria of recursive organization of processes forming a unity, but this time in any space where they may exist, not only in the space of biochemical transformations.

The ideas in part I continued to be very influential in subsequent work. They motivated the critical elaborations of organizational approaches in biology, such as the work of Alvaro Moreno and colleagues (e.g., Moreno and Mossio, 2015). They were highly influential in Lynn Margulis's vision of life and evolution, as well as

on current work on origins of life studies and synthetic biology (e.g., Margulis and Sagan, 1995). The computational autopoiesis model, as mentioned, was inspirational for work in artificial life, particularly the subbranch concerned with artificial chemistries and protocell modeling. And the ideas of autonomy and organizational/operational closure have resurfaced in the twenty-first century as a keystone of enactive theory.

1 Autonomy and Biological Thinking

1.1 Evolution and the Individual

1.1.1

The description, invention, and manipulation of unities generated through distinctions is at the basis of all scientific—and rational—enquiry.

This is no less true of living unities.¹ What is peculiar to them, however, is that in our common experience, living unities assert their individuality, that is, living things appear as *autonomous* unities of bewildering diversity endowed with the capacity to reproduce. In these encounters, autonomy appears so obviously an essential feature of living systems that whenever something is observed that seems to have it, the naive approach is to deem it alive. Yet, autonomy, although continuously revealed in the self-asserting capacity of living systems to maintain their identity through the active compensation of deformations, seems so far to be the most elusive of their properties.

Autonomy and diversity, the maintenance of identity and the origin of variation in the mode in which this identity is maintained, are the basic challenges presented by the phenomenology of living systems to which people have for centuries addressed their curiosity about life.

In the search for an understanding of autonomy, classical thought, dominated by Aristotle, created vitalism by endowing living systems with a nonmaterial purposeful driving component that attained expression through the realization of their forms. After Aristotle, and as variations of his fundamental notions, the history of biology records many theories that attempt in one way or another to encompass all the phenomenology of living systems under some

¹*Eds.* In their early writings, Varela and Maturana often used the term *living unity* to describe a biological individual. The term underlines a notion of individuality and identity as a topological aspect of autopoietic organization. Organisms are produced by the activity of processes that generate and maintain their component parts *and* they self-distinguish as coherent entities that define a domain of interactions with the environment (hence their dual character as composite and unitary).

peculiar organizing force (Hall, 1969). However, the more biologists looked for the explicit formulation of one or another of these special organizing forces, the more they were disappointed by finding only what they could find anywhere else in the physical world: molecules, potentials, and blind material interactions governed by aimless physical laws. Thence, under the pressure of unavoidable experience and the definite thrust of Cartesian thought, a different outlook emerged, and *mechanicism* gradually gained precedence in the biological world by insisting that the only factors operating in the organization of living systems were physical factors, and that no nonmaterial vital organizing force was necessary. In fact, it seems now apparent that any biological phenomenon, once properly defined, can be described as arising from the interplay of physico-chemical processes whose relations are specified by the context of its definition.²

²Eds. According to Georges Canguilhem (2008, 59–74), the history of biology has been a pendular oscillation between vitalism and mechanicism, to the point that one suspects that it is the result of an “undiscovered dialectic” attributable to the phenomenon of life itself. For him, vitalism conserves a stubborn vitality, not as an empirical claim but as a philosophical question animating styles of thinking to the present day, with or without the explicit use of the label. See Donohue and Wolfe (2023).

Diversity has been removed as a source of bewilderment in the understanding of the phenomenology of living systems by Darwinian thought and particulate genetics, which have succeeded in providing an explanation for it without resorting to any peculiar directing force. Yet the influence of these notions, through their explanation of evolutionary change, has gone beyond the mere accounting for diversity: It has shifted completely the emphasis in the evaluation of the biological phenomenology from the individual to the species, from the unity to the origin of its parts, from the present organization of living systems to their ancestral determination.

Today the two streams of thought represented by the physico-chemical and the evolutionary explanations are braided together. The molecular analysis seems to allow for the understanding of reproduction and variation; the evolutionary analysis seems to account for how these processes might have come into being. Apparently we are at a point in the history of biology where the basic difficulties have been removed.

1.1.2

Biologists, however, are uncomfortable when they look at the phenomenology of living systems as a whole. Many manifest this discomfort by refusing to say what a living system is.³ Others attempt to encompass present ideas under comprehensive theories governed by organizing notions, such as information-theoretic principles (e.g., Miller, 1965), that require of biologists the very understanding that they want to provide.

³Some interesting examples of this discomfort can be found in the discussions transcribed in the series edited by Waddington (1972), where a number of prominent biologists voiced their opinions on the subject.

The ever present question is: *What is common to all living systems that allows us to qualify them as living?* If not a vital force, if not an organizing principle of some kind, then what?

In other words, notwithstanding their diversity, all living systems share a common organization which we implicitly recognize by calling them “living.” There is no clear understanding or formulation of such an organization. In fact, the very question is odd to most biologists—with some notable exceptions⁴—because the great developments in molecular biology have led to an overemphasis on isolated components, and to a disregard of questions pertaining to what makes the living system a whole, autonomous unity that is alive regardless of whether it reproduces or not. As a result, processes that are history dependent (evolution, ontogenesis) and history independent (individual organizations) have been confused. But these two kinds of process must be kept separate and accounted for in related, but distinct terms. A very good recent example is Monod’s idea of the teleonomic apparatus as a characterization of the living organization (Monod, 1971). The capacity and tendency of the genetic material to reproduce and preserve itself generation after generation through the encoding of molecular species in the DNA is pointed at by Monod as the key to life. This, however, pushes all the properties pertaining to the individual unit as a coherent cooperative whole (the functioning cell, for example) into a single molecular species, the DNA, which now contains some abstract description of the teleogenic project of the cell. Ironically, by pushing this kind of mechanicism to such an extreme, Monod finds himself philosophically close to the vitalists, who insisted on a similar reduction of the “life” characteristics to some component other than the cooperative relations of the cellular unity⁵ (Maturana, 1978; Barthélemy-Madaule, 1972). Indeed, teleonomic and evolutionary considerations leave the question of the nature of the organization of the living unity untouched.⁶

1.1.3

Our endeavor is to understand the nature of living organization. However, in this approach we make a starting point of the unitary character of a living system. I maintain that evolutionary thought, through its emphasis on diversity, reproduction, and the species in order to explain the dynamics of change, has obscured the necessity of looking at the autonomous nature of living units for the understanding of biological phenomenology. Also I think that the maintenance of identity and the invariance of defining relations

⁴The notable exceptions that come to mind are Paul Weiss (1969), and Conrad Waddington (1972).

⁵A remarkable passage in the book says: “The *ultima ratio* of all the teleonomic structures and performances of living beings is thus enclosed in the sequences of residues making up polypeptide fibers, ‘embryos’ of the globular proteins which in biology play the role Maxwell assigned to his demons a hundred years ago. In a sense, a very real sense, it is at this level of chemical organization that the secret of life lies, if indeed there is any one such secret” (Monod, 1971, 95).

⁶*Eds.* Monod’s reduction of living phenomena to the activity of special molecular mechanisms finds its evolutionary counterpart in the work of George C. Williams (1966). Williams explains traits of apparent benefit to a group or population as forms of phenotypic mediation that facilitate the spread of genes presumed responsible for such traits. This idea was later radicalized by the concept of the *selfish gene* explored by Richard Dawkins (1976). Varela’s point, that Monod’s reduction of teleonomy to DNA sequences is ironically close to vitalism, resembles more recent criticisms of attributing agential powers to genes (see Walsh, 2015; Thompson, 2007, 160). In both cases, on the biomolecular side and the evolutionary side, a homuncular fallacy is at play that misses, as Varela says, the “uncomfortable” phenomenon of the organism as a whole and the nature of living organization.

⁷Eds. Varela may not have been aware of Helmuth Plessner's work, which was similarly directed at understanding organisms as unities. Plessner's 1928 magnum opus, *Levels of Organic Life and the Human*, has been translated only recently into English (Plessner, 2019). For more on Plessner's work, including its relation to autopoiesis, see Michelini (2021).

⁸Eds. As he notes in the next section (1.2.2), Varela uses *machine*, *mechanism*, and *system* as approximately equivalent terms. The connotations, despite the caveats, sound unfortunate to those who are critical of the suitability of machine metaphors to describe organisms (e.g., Nicholson, 2013; Dupré, 2012, 71). Varela's stance should be understood along the lines proposed by organizational approaches in biology (e.g., Moreno and Mossio, 2015). Modern mechanistic approaches (Craver, 2007) may not be incompatible with this view, insofar as it also stresses the possibility of multiple nonreducible levels of explanation (e.g., Bechtel, 2011). Bich and Bechtel (2021) make the case for an integration of organizational and mechanistic traditions; see also Lee (2023). *Mechanism*, however, can have unclearly demarcated meanings (Nicholson, 2012). The use of these terms wanes in Varela's later work (*autopoietic system* is the preferred option in Maturana and Varela, 1987). His approach is best described as *operational*, which he contrasts with *symbolic* explanations at length in chapter 9. This contrast is largely maintained in his later work and in contemporary enactive perspectives.

in the living unities are at the base of all possible ontogenic and evolutionary transformation in biological systems, and this I intend to explore. Thus, our purpose in this first part of the book is to understand the organization of living systems in relation to their unitary character.⁷

1.2 Molecules and Life

1.2.1

Our approach will be mechanistic: No forces or principles will be adduced which are not found in the physical universe. Yet our problem is the living organization, and therefore our interest will not be in properties of components, but in processes and relations between processes realized through components.⁸

This is to be clearly understood. An explanation is always a reformulation of a phenomenon in such a way that its elements appear operationally connected in its generation. Furthermore, an explanation is always given by us as observers, and it is central to distinguish in it what pertains to the system as constitutive of its phenomenology from what pertains to the needs of our domain of description, and hence to our interactions with it, its components, and the context in which it is observed. Since our descriptive domain arises because we simultaneously behold the unity and its interactions in the domain of observation, notions arising from cognitive and expositional needs in the domain of description do not pertain to the explanatory notions for a constitutive organization of the unity (phenomenon). We shall return to this important issue very often in this book.

Furthermore, an explanation may take different forms according to the nature of the phenomenon explained. Thus, to explain the movement of a falling body one resorts to properties of matter, and to laws that describe the conduct of material bodies according to these properties (kinetic and gravitational laws), while to explain the organization of a control plant one resorts to relations and laws that describe the conduct of relations. In the first case, the materials of the causal paradigm are bodies and their properties; in the second case, they are relations and their relations, independently of the nature of the bodies that satisfy them. In this latter case, in our explanations of the organization of living systems, we shall be dealing with the relations that the actual physical components must satisfy to constitute such a system, not with the identification of these components. It is our *assumption* that there is an organization that is

common to all living systems, whichever the nature of their components. Since our subject is this organization, not the particular ways in which it may be realized, we shall not make distinctions between classes or types of living systems.

1.2.2

By adopting this philosophy, we are in fact just adopting the basic philosophy that animates cybernetics and systems theory, with the qualifications to these names that were discussed in the Preface. This is, I believe, nothing more and nothing less than the essence of a modern mechanicism. In saying that living systems are “machines” we are pointing to several notions that should be made explicit. First, we imply a nonanimistic view, which it should be unnecessary to discuss any further. Second, we are emphasizing that a living system is defined by its organization, and hence that it can be explained as any organization is explained, that is, in terms of relations, not of component properties. Finally, we are pointing out from the start the dynamism apparent in living systems and which the word “machine” or “system” connotes.⁹

We are asking, then, a fundamental question: Which is the organization of living systems, what kind of machines are they, and how is their phenomenology, including reproduction and evolution, determined by their unitary organization?

⁹In this book “machines” and “systems” are used interchangeably. They obviously carry different connotations, but the differences are inessential, for my purpose, except in seeing the relation between the history of biological mechanism and the modern tendency for systemic analysis. Machines *and* systems point to the characterization of a class of unities in terms of their organization.

Sources

(Maturana and Varela, 1975), (Varela et al., 1974).

2 Autopoiesis and the Organization of Living Systems

2.1 The Duality between Organization and Structure

2.1.1

Machines and biology have been, since antiquity, closely related. From the zoological figures present in astronomical simulacra, through renaissance mechanical imitations of animals, through Descartes's airpipe nerves, to present-day discussions on the computer and the brain, runs a continuous thread. In fact, the very name *mechanism* for an attitude of inquiry throughout the history of biology reveals this at a philosophical level (Hall, 1969; de Solla Price, 1964). More often than not, mechanism is mentioned in opposition to vitalism, as an assertion of the validity of the objectivity principle in biology: there are no purposes in animal nature. Its apparent purposefulness is similar to the purposefulness of machines.¹ Yet, the fact that one picks machines as a set of objects comparable to living systems deserves a closer look. What in machines makes it possible to establish such a connection?

If one is to have an understanding of a given class of machines, it is obviously insufficient to give a list of its parts or to define its purpose as a human artifact. The way to avoid both insufficiencies is to describe the permitted interrelations of the machine components, which define the possible transitions that the machine can undergo. This, on the one hand, goes beyond the mere listing, and on the other, implies the nature of the output that determines the purpose of the machine. Notably, when looking at the components, one sees that not all of their properties have equal importance. If one is to instantiate (construct or implement) a certain machine, then, in choosing the components, one will take into account only those component properties that satisfy the desired interrelations leading

¹*Eds.* The notion of organisms as purposeless must be understood as a call to bracket teleological thinking. It arises from the separation between operational and symbolic discourse. In chapter 9, Varela argues that both kinds of discourse are necessary. In later work, notions such as *codefinition* rather than separation between domains become more prevalent (Varela, 1991b, 2011, 614), culminating in a conception of organisms as *intrinsically purposeful* based on autopoietic principles (Weber and Varela, 2002); see also Thompson (2007, 140–149, 453–454). By not assuming purposes, teleology may be genuinely naturalized. *Machine*, moreover, carries the suggestion of an *external* rather than *intrinsic* purpose, which is the point of the critique of cybernetics in Jonas (1966). Canguilhem (2008, 76) advises against seeing machines as "theorems solidified" and ignoring the relation between mechanism and finalism.

to the expected sequence of transitions that constitutes the machine description. This is tantamount to saying that the components may be any components at all *as long as their possible interrelations satisfy a given set of desired conditions*. Alternatively, one can say that what specifies a machine is the set of component's interrelations, regardless of the components themselves.

The *relations* that define a machine as a unity, and determine the dynamics of interactions and transformations it may undergo as such a unity, we call the *organization* of the machine. The actual relations that hold between the components that integrate a concrete machine in a given space constitute its *structure*.² The organization of a machine (or system) does not specify the properties of the components that realize the machine as a concrete system; it only specifies the relations that these must generate to constitute the machine or system as a unity. Therefore, the organization of a machine is independent of the properties of its components, which are arbitrary, and a given machine can be realized in many different ways by many different kinds of components. In other words, although a given machine can be realized by many different structures, for it to constitute a concrete entity in a given space its actual components must be defined in that space, and have properties that allow them to generate the relations that define it.³

We are thus saying that what defines a machine organization is relations, and hence that the organization of a machine has no connection with materiality, that is, with the properties of the components that define them as physical entities. In the organization of a machine, materiality is implied but does not enter *per se*. A Turing machine, for example, is a certain organization; there seems to be a hopeless gap between the way in which a Turing machine is defined and any possible instance (electrical, mechanical, etc.) of it. This has been pointed out by workers in the field of cybernetics. As Ashby puts it:

Cybernetics started by being closely associated in many ways with physics, but it depends in no essential way on the laws of physics or on the properties of matter. . . . The materiality is irrelevant, and so is the holding or not of the ordinary laws of physics. *The truths of cybernetics are not conditional on their being derived from some other branch of science.* (Ashby, 1956, 1).

Wiener was pointing to this when he emphasized the primacy of “ ‘information,’ not matter or energy. No materialism which does not admit this [distinction], can survive at the present day” (Wiener, 1961, 132).

²It is very unfortunate that in the cybernetics and systems literature, these two terms are used in very many different ways. For example, in Klir's terminology, structure is closer to what I call here organization (Klir, 1969). The present usage, however, does not seem to depart very radically from that of most authors. See Maturana (1975).

³*Eds.* The distinction between structure and organization is inspired in cybernetics, which sought to investigate the logic of abstract relations between processes. Cybernetics influenced computationalism, and it is unsurprising that Varela's remarks on the irrelevance of materiality resemble the functionalist thesis of multiple realizability (Villalobos and Razeto-Barry, 2020). Distinguishing between structure and organization enables a series of strong formal proposals, but it amounts to an impregnable separation between matter and form (DiFrisco, 2014; Moreno and Mossio, 2015). This calls for a synthetic countermove where materiality is brought back into the story, as in the emphasis on material constraints in organizational approaches in biology (Moreno and Ruiz-Mirazo, 1999) and on precariousness in the enactive literature (Beer and Di Paolo, 2023; Di Paolo, 2009; Di Paolo and Thompson, 2014). Varela's focus on the re-enchantment of the concrete and on lived experience in his later work can also be seen in terms of the relevance of materiality (Varela, 1991a).

There are several other situations where a similar disjunction between materiality and organization appears. Take, for instance, symmetry. One clearly has empirical examples of symmetry. Yet, one can formulate a theory of it in which materiality concepts do not enter at all. Still, it is possible to transport this theory with no modification to a different context where materiality does appear, as in particle physics. Certainly several other examples exist.

2.1.2

The objection might arise that the notion of organization belongs to a more inclusive field, that of mathematics. This objection, however, carries no weight, because the explanatory value of the notions under discussion correlate with empirical circumstances, artificial or natural, that embody them. Thus, there is the symmetry of natural objects and there is the mathematics of symmetry. Similarly, there is the experience of magnetism and there is the mathematics of magnetism. They do not superimpose, but one embodies the other. From this point of view there is no difference between physics and, say, cybernetics. What makes physics peculiar is the fact that the materiality *per se* is implied; thus, the structures described embody concepts that are derived from materiality itself, and do not make sense without it. Despite any advances, in physics one is looking at the structure of materiality. Whether these basic structures are subsumed in such constructs as self-fields is of no import to our argument.

Furthermore, there are no differences in the explanatory paradigm used in the formulation of, say, atomic theory or control theory. In both cases we are dealing with an attempt to reformulate a given phenomenology in such terms that its components are causally connected. Yet in one case the notions are directly related with materiality, while in the other case materiality does not enter at all.

We thus believe that the classical distinction between synthetic and analytic *should be refined*. Within the synthetic one should distinguish two levels: the *materially* synthetic (i.e., where materiality enters *per se* into consideration), and the *nonmaterially* synthetic (i.e., where materiality is implied but is, as such, irrelevant).⁴

In this light, one should look closely at the consequences of the basic assertion for biological mechanism: Living systems are machines of one or several well-defined classes. This is to say: The definitional element of living unities is a certain organization (the set of interrelations leading to a given form of transitions) independent of the structure, the materiality that embodies it; not

⁴*Eds.* The cybernetic aim to say something general about large classes of systems does not entail that materiality is irrelevant to those systems. A series of constraints must be in place for a material system to instantiate an abstract machine (e.g., sufficient heat dissipation to guarantee the operation of semiconductor components in a computer). Moving from design to scientific goals, these material constraints must be made explicit by organizational approaches in biology. Inspired by Varela's work, Alvaro Moreno and colleagues have carefully looked at the material constraints entailed by the ideas of autopoiesis and autonomy, such as constraints on the energy flows necessary for autonomy to be physically realized (e.g., Moreno and Ruiz-Mirazo, 1999; Moreno and Mossio, 2015).

⁵*Eds.* This does not mean that the components are fully independent of their interrelations, especially if we give materiality a more central role than in early autopoiesis theory. We notice that it is possible to privilege interrelations, thus avoiding reductionism and accepting Varela's point that it is through these relations that we can specify the organization of living systems, while still considering that the materiality of components constrains organizational descriptions (e.g., the limits at which material processes operate, thermodynamic considerations, temporal constraints, etc.).

⁶*Eds.* This is an *epistemic* sense of irreducibility: the observer has to consider the complementary organizational and structural aspects of an organism. Varela's later thought is closer to Hans Jonas (1966, 81), for whom life introduces an *ontological* revolution by conserving an identity of form *through* material flows. This is the kind of entity that physics excludes to formulate its general laws (Wigner, 1970). Physics seeks to explain symmetries and invariance, but biological systems are characterized by regimes of extended criticality and symmetry breaking, making them inherently *historical* (Longo and Montévil, 2013; Montévil, 2022). The observer remains central in this stronger notion of irreducibility. Life requires an understanding of interiority that necessitates, according to Jonas, the observer's own resources as an organism.

the nature of the components, but their interrelations.⁵ There are three main consequences of this assertion:

1. Any explanation of a biological system must contain at least *two complementary aspects*, one referring to it as an organization, and the other referring to it as a structure, as an instance. The first must account for the specific (dynamic) configuration of components that define it; the second must account for how its particular components enter into the given interrelations that constitute it.
2. Any biological system *can* be treated in terms of the properties of its actual components as a physical system. There is no limitation whatsoever on doing so, except for the number of variables that one might have to consider. But this is only a problem in computation. Eventually, one should be able to have a physical description as accurate as needed of any biological system. Although such an analysis is insufficient, it is necessary in order to point to the specific structure(s) of biological systems, so that it will be possible to make sense of a given form of interrelations.
3. Insofar as the physical analysis of biological systems is still physics, what is *specific* to biology is precisely the analysis of the class of machines that living systems are, and the changes that these undergo in time. Thus, the specific aspects of any biological explanation belong to the second level outlined above, and are necessarily not deducible from physics. In this sense, biology is not reducible to physics (although the explanatory paradigm is the same, as noted above).⁶ Reduction is used here to mean a program which would make it eventually possible to derive biology from physical chemistry, in order to produce a unified science (Schaffner, 1967; Roll-Hansen, 1969).

Thus, in this complementary organization/structure we find the first important dimension in which the descriptions of a system reflect back our own descriptive maneuvers. It is clear that the need to include both the organization and the structure of a machine for a complete explanation depends entirely on what we, as a community of observers, consider adequate. Such dualities in descriptions are a running theme throughout this book (cf. part II).

2.1.3

The *use* to which a machine can be put by a user is not a feature of the organization of the machine, but of the domain in which the machine operates, and belongs to our description of the machine in

a context wider than the machine itself. This is a significant notion. Human-made machines are all made with some purpose, practical or not—some aim (even if it is only to amuse) that is specified. This aim usually appears expressed in the product of the operation of the machine, but not necessarily so. However, we use the notion of purpose when talking of machines because it calls into play the imagination of the listener and reduces the explanatory task in the effort of conveying the organization of a particular machine. In other words, with the notion of purpose we induce the listener to invent the machine we are talking about. This, however, should not lead us to believe that purposes, or aims, or functions, are to be used as constitutive properties of the machine that we describe with them; such notions belong to the domain of observation, and cannot be used to characterize any particular type of machine organization. The product of the operations of a machine, however, can be used to this end in a nontrivial manner in the domain of descriptions generated by the observer.

This is a very essential instance of the distinction, made before, between notions that are involved in the explanatory paradigm for a system's phenomenology, and notions that enter because of needs of the observer's domain of communication. To maintain a clear record of what pertains to each domain is an important methodological tool, which we use extensively. It seems an almost trivial kind of logical bookkeeping, yet it is too often violated by usage.⁷

2.2 Autopoietic Machines

2.2.1

That living systems are machines cannot be shown by pointing to their components. Rather, one must show their organization in a manner such that the way in which all their peculiar properties arise becomes obvious. In order to do this, we shall first characterize the kind of systems that living systems are, and then show how the peculiar properties of the living may arise as consequences of the organization of this kind of machines.

2.2.2

There are systems that maintain some of their variables constant, or within a limited range of values. This is, in fact, the basic notion of *stability or coherence*, which stands at the very foundation of our understanding of systems (e.g., Wiener, 1961). The way this is expressed in the organization of these machines must be one that

⁷Eds. Purposes as such are not found, according to Varela, as processes or as components that constitute the operation of a machine. Logical bookkeeping avoids the dangers of mixing operational and symbolic discourses (see chapters 8 and 9). We can draw a parallel with the distinction between personal and subpersonal explanations (Dennett, 1969) and with the dangers of mereological fallacies (Bennett and Hacker, 2003). Observing this distinction does not by itself resolve the problem of how the two domains of discourse relate.

⁸Eds. According to Maturana's introduction to Maturana and Varela (1980), he coined the term *autopoiesis* to capture the circular organization of living systems. For Varela's account of the origins of the term, see Varela (1996). The term had been used in this same sense before by Georges Canguilhem, who in an essay from 1951 refers to the "autopoietic character of organic activity" (Canguilhem, 2008, 9). See Etzeberria and Wolfe (2023) for reflections on Canguilhem and autopoiesis. The idea of organisms as self-organized, circular relations of production goes back to Kant (1978). See also Juarrero Roqué (1985) and Thompson (2007, 129–140).

⁹Eds. Conditions (1) and (2) may be described as *self-production* and *self-distinction*, respectively (Di Paolo, 2018).

¹⁰Eds. The comparison with homeostasis provides intuitive support for a dynamical interpretation of autopoiesis (see Maturana and Varela, 1980, 78). In strict terms, however, it is problematic. Autopoiesis entails an *organization* that remains invariant under the operation of its structural realization. Being organizational rather than structural, the condition of autopoiesis admits no deviations other than total loss of autopoiesis (see, e.g., section 6.2.1). Nor does autopoiesis describe any mechanism that would be sensitive to potential loss of its invariant organization and be able to prevent it. *Homeostasis* refers precisely to these two capabilities that autopoiesis lacks: the ability to return to acceptable bounds once a variable has escaped them and the ability to avoid crossing these bounds in the first place. This is not a small problem. It motivates the introduction of the concept of *adaptivity* in the enactive approach (Di Paolo, 2005), precisely because something like the homeostatic interpretation is used to buttress an attempt to naturalize teleology based on autopoiesis (Weber and Varela, 2002). See also Razeto-Barry (2012).

defines the process as occurring completely within the boundaries of the machine that the very same organization specifies. Such machines are homeostatic machines, and all feedback is internal to them. If one says that there is a machine M in which there is a feedback loop through the environment, so that the effects of its output affect its input, one is in fact talking about a larger machine M' which includes the environment and the feedback loop in its defining organization.

The idea of autopoiesis capitalizes on the idea of homeostasis, and extends it in two significant directions: first, by making every reference for homeostasis internal to the system itself through mutual interconnection of processes; and secondly, by positing this interdependence as the very source of the system's identity as a concrete unity which we can distinguish. These are systems that, in a loose sense, produce their own identity: they distinguish themselves from their background. Hence the name autopoietic, from the Greek αὐτός = self, and ποιεῖν = to produce.⁸

*An autopoietic system is organized (defined as a unity) as a network of processes of production (transformation and destruction) of components that produces the components that: (1) through their interactions and transformations continuously regenerate and realize the network of processes (relations) that produced them; and (2) constitute it (the machine) as a concrete unity in the space in which they exist by specifying the topological domain of its realization as such a network.*⁹

It follows that an autopoietic machine continuously generates and specifies its own organization through its operation as a system of production of its own components, and does this in an endless turnover of components under conditions of continuous perturbations and compensation of perturbations. Therefore, an autopoietic machine is a homeostatic (or rather a relational-static) system that has its own organization (defining network of relations) as the fundamental invariant.¹⁰ This is to be clearly understood. Every unity has an organization specifiable in terms of static or dynamic relations between elements, processes, or both. Among these possible cases, autopoietic machines are unities whose organization is defined by a particular network of processes (relations) of production of components, the autopoietic network, not by the components themselves or their static relations. Since the relations of production of components are given only as processes, if the processes stop the relations of production vanish; as a result, for a machine to be autopoietic, its defining relations of production must be continuously regenerated by the components

which they produce. Furthermore, the network of processes which constitute an autopoietic machine is a unitary system in the space of the components that it produces and that generate the network through their interactions.

It is important to realize that we are not using the term organization in the definition of an autopoietic machine in a transcendental sense, pretending that it has an explanatory value of its own. We are using it only to refer to the specific relations that define an autopoietic system. Thus, autopoietic organization simply means processes concatenated in a specific form: a form such that the concatenated processes produce the components that constitute and specify the system as a unity. It is for this reason that we can say that if at any time this organization is actually realized as a concrete system in a given space, then the domain of the deformations that the system can withstand without loss of identity (that is, maintain its organization) is the domain of changes in which it exists as a unity.

2.2.3

The autopoietic network of processes defines a class of systems. The boundaries of this class are, of course, not sharp, and this comes about because of the nature of the approach we have taken. First, we have taken as a starting point the fact that systems arise as a result of our processes of distinction through some favored criteria. Thus, there will be many different ways in which both the system and its components can be classified, and in which its boundary can be specified. A similar statement is true about the notion of production of components. Depending on the domain of discourse we choose, this notion will vary in connotations. In order to remove such ambiguities, we would have to give rather precise definitions of these words, probably through some mathematical formalism. This we shall not do. It would defeat the very purpose of conveying an intuition about the living organization in a clear form. A second reason for eschewing excessive qualifications is that we characterized autopoietic machines in the context of certain specific objects called living systems, and more concretely, living *cells*. Thus we have in mind, and will keep in mind, such systems as our reference point in order to give the appropriate connotations to notions such as production and boundary. This particular frame of reference does make autopoietic systems into a recognizable class. For example, in a human-made machine in the physical space, say a car, there is an organization given in terms of a concatenation of processes, yet these processes are in no sense processes of production of the

components which specify the car as a unity, since the components of a car are all produced by other processes, which are independent of the organization of the car and its operation. Machines of this kind are non-autopoietic dynamic systems. In a natural physical unity like a crystal, the spatial relations among the components specify a lattice organization that defines it as a member of a class (a crystal of a particular kind), while the kinds of components that constitute it specify it as a particular case in that class. Thus, the organization of a crystal is specified by the spatial relations that define the relative positions of its components, while these specify its unity in the space in which they exist—the physical space. This is not so with an autopoietic machine. In fact, although we find spatial relations among its components whenever we actually or conceptually freeze it for an observation, the observed spatial relations do not (and cannot) define it as autopoietic. This is so because the spatial relations between the components of an autopoietic machine are specified by the network of processes of production of components that constitute its organization, and they are therefore necessarily in continuous change. A crystal organization, then, lies in a different domain than the autopoietic organization: a domain of relations between components, not of relations between processes of production of components; a domain of processes, not of concatenations of processes. We normally acknowledge this by saying that crystals are static.¹¹

¹¹ *Eds.* This paragraph highlights the intended interpretation of the formal definition of autopoiesis by pointing to living cells as the case that helps pin down the otherwise loose meanings of terms such as *process*, *transformation*, *production of components*, appearing in the definition. The contrasting cases of a car and a crystal underscore the intended meanings. Note, however, that a better comparison would not have been with an already formed crystal, but rather with the process of crystalization (Simondon, 2020). In this case, the differences are more subtle. Using this example to help fix the unspecified terms, we may argue that the structure of a crystal at the boundary with a saturated milieu constitutes a pattern that continuously regenerates and realizes itself. This would at least suffice to claim that crystalization fulfills condition (1) of autopoiesis: self-production. It would be harder to adjust terminological meanings to claim that it also fulfills condition (2), or: self-distinction. In Simondon's terms, the process of crystalization, though individuating, is not self-individuating, which is to say that it is not fully autopoietic (Thompson, 2011).

Whether one can classify as autopoietic other systems (such as social or physical) is, of course, dependent on whether one can give a precise connotation to the idea of component production processes and the generation of a boundary in some appropriate space where the components exist, and yet not violate the usage of words, such as production, so as to render them meaningless. I will return to this point later in the book, but it is fair to anticipate my view in saying that I see autopoiesis as *one* possible form of autonomy (or organizational closure, as defined later), and that this term should be restricted to systems, whether natural or artificial, that are characterized by a network that is, or resembles very closely, a chemical network.

2.2.4

The consequences of the autopoietic organization are paramount:

1. Autopoietic machines are autonomous; that is, they subordinate all changes to the maintenance of their own organization,

independently of how profoundly they may otherwise be transformed in the process. Other machines, henceforth called *allopoietic* machines, have as the product of their functioning something different from themselves (as in the car example). Since the changes that allopoietic machines may suffer without losing their definitory organization are necessarily subordinated to the production of something different from themselves, they are not autonomous.

2. Autopoietic machines have individuality; that is, by keeping their organization as an invariant through its continuous production, they actively maintain an identity that is independent and yet makes possible their interactions with an observer. Allopoietic machines have an identity that depends on the observer and is not determined through their operation, because its product is different from themselves; allopoietic machines do have an externally defined individuality.¹²
3. Autopoietic machines are unities because, and only because, of their specific autopoietic organization: Their operations specify their own boundaries in the processes of self-production. This is not the case with an allopoietic machine, whose boundaries are defined completely by the observer, who, by specifying its input and output surfaces, specifies what pertains to it in its operations.¹³
4. Autopoietic machines do not have inputs or outputs. They can be perturbed by independent events and undergo internal structural changes which compensate for these perturbations. If the perturbations are repeated, the machine may undergo repeated series of internal changes, which may or may not be identical. Whichever series of internal changes takes place, however, they are always subordinated to the maintenance of the machine organization, a condition which is definitive for the autopoietic machines. Thus if there is a relation between these changes and the course of perturbations to which we may point, it pertains to the domain in which the machine is observed, but not to its organization. Although an autopoietic machine can be treated as an allopoietic machine, this treatment does not reveal its organization as an autopoietic machine. In fact, autopoietic and allopoietic descriptions of a system are complementary pairs, depending on the observer's needs. They are a particular instance of what, later on, we characterize as the universal duality between autonomous and control descriptions (cf. part II).¹⁴

¹² *Eds.* Individuality is understood here not just as type-identity (i.e., being a member of a particular class, which is ascribable to any entity given its organization, whether autopoietic or not). The key difference is that autopoietic identity is maintained as a result of its operation following condition (1). Because of this self-referentiality, autopoietic individuality is also a token-identity, a self-affirmation of the continuity of *this* particular organism. This identity is not based on external ascriptions by an observer.

¹³ *Eds.* Point 3 is not a repetition of point 2; it implicitly refers to condition (2) in the definition; that is, the specification of a topological domain in which the system is realized as a unity. Varela emphasizes the cell boundary as an illustration of this unity character (with its inside/outside topology). Other processes, apart from the cell membrane, also subserve self-distinction in living cells; for example, its cytoskeleton (Fletcher and Mullins, 2010) and CRISPR-Cas immunity (Rath et al., 2015).

¹⁴ *Eds.* Another way to express this point is to say that autopoietic systems do not operate by following external instructions. They respond to "inputs" as perturbations to their ongoing dynamic processes. If this situation changes, they cease to be autopoietic because they cease to be structure-determined. This does not mean that interactions with the environment are unstructured or present no regularities, but simply that they cannot be treated as lawful cause/effect relations in an autopoietic description. This switch from an input/output view to one based on perturbations without loss of autopoiesis is what is known as *structural coupling* (chapter 5).

2.2.5

The actual way in which an organization such as the autopoietic organization may in fact be implemented in the physical space—that is, the physical structure of the machine—varies according to the nature (properties) of the physical materials which embody it. Therefore there may be many different kinds of autopoietic machines in the physical space (physical autopoietic machines); all of them, however, will be organized in such a manner that any physical interference with their operation outside their domain of compensations will result in their disintegration, that is, in the loss of autopoiesis. It also follows that the actual way in which the autopoietic organization is realized in one of these machines (its structure) determines the particular perturbations it can suffer without disintegration, and hence the domain of interactions in which it can be observed.¹⁵ These features of the actual concreteness of autopoietic machines embodied in physical systems allow us to talk about particular cases, to put them in our domain of manipulation and description, and hence to observe them in the context of a domain of interactions that is external to their organization. This has two fundamental consequences:

¹⁵ *Eds.* This is how the domain of compensations and the domain of interactions are defined (i.e., as the compensations and interactions that the system may undergo without losing autopoiesis). These define what Varela describes as *cognitive domains*, an idea that he further develops in part III.

1. We can describe physical autopoietic machines, and also manipulate them, as parts of a larger system that defines the independent events which perturb them. Thus, as noted above, we can view these perturbing independent events as inputs, and the changes of the machine that compensate for these perturbations as outputs. To do this, however, amounts to treating an autopoietic machine as an allopoietic one, and we thereby recognize that if the independent perturbing events are regular in their nature and occurrence, an autopoietic machine can in fact be integrated into a larger system as a component allopoietic machine, without any alteration in its autopoietic organization.
2. We can analyze a physical autopoietic machine in its physical parts, and treat all its partial homeostatic and regulatory mechanisms as allopoietic machines (submachines) by defining their input and output surfaces. Accordingly, then, these submachines are necessarily components of an autopoietic machine and are defined by relations which they satisfy in determining its organization. The fact that we can divide physical autopoietic machines into parts does not reveal the nature of the domain of interactions that they define as concrete entities operating in the physical universe.

2.3 Living Systems

2.3.1

If living systems are machines, that they are physical autopoietic machines is trivially obvious; they transform matter into themselves in a manner such that the product of their operation is their own organization. However, we deem the converse is also true: A physical system, if autopoietic, is living. In other words, we claim that the notion of *autopoiesis is necessary and sufficient to characterize the organization of living systems*.¹⁶ This proposed equivalence raises, of course, quite a number of philosophical arguments with a long-standing history in the philosophy of biology. It seems useful to comment very briefly on three of them:

1. Machines and systems are generally viewed as natural or human-made artifacts with completely known properties that make them, at least conceptually, perfectly predictable. Contrariwise, living systems are *a priori* frequently viewed as ultimately unpredictable systems, with purposeful behavior similar to ours. If living systems were machines, they could be made by humans and, according to the view mentioned above, it seems unbelievable that humans could manufacture a living system. This view can be easily disqualified, because it either implies the belief that living systems cannot be understood because they are too complex for our meager intellect and will remain so, or that the principles that generate them are intrinsically unknowable; either implication would have to be accepted *a priori* without proper demonstration. There seems to be an intimate fear that the awe of life and the living would disappear if a living system could be not only reproduced, but designed by humans. This is nonsense; the beauty of life is not a consequence of its inaccessibility to our understanding.¹⁷
2. To the extent that the nature of the living organization is unknown, it is not possible to recognize when one has at hand, either as a concrete synthetic system or as a description, a system that exhibits it. Unless one knows which is the living organization, one cannot know which organizations are alive. In practice, it is accepted that plants and animals are living, but they are characterized as living through the enumeration of certain properties. Among these, reproduction and evolution appear as determinants, and for many observers the condition of living appears subordinated to the possession of these properties. However,

¹⁶ *Eds.* This is a proposal rather than a claim that can be logically established, since it depends on what we are willing to accept as inescapable aspects of life. Some researchers consider autopoiesis as necessary but not sufficient for life (Bitbol and Luisi, 2004; Bourguin and Stewart, 2004); see the discussion in Thompson (2007). Self-assembled micelles and vesicles are self-producing and self-distinguishing, and thus autopoietic, but they do not possess chemical networks capable of metabolizing external elements. If this property is deemed crucial for life, then such adaptive capacities should be part of its definition. Di Paolo (2018) shows that an internal tension in the definition of autopoiesis can be resolved by including such adaptive properties, suggesting a tighter link between life and mind than in autopoietic theory.

¹⁷ *Eds.* The study of artificial protocells is an active branch of synthetic biology and origins of life studies (e.g., Luisi et al., 2006; Stano and Mavelli, 2015). Protocells form as self-assembled, self-catalyzing containers (typically lipid vesicles) that may be capable of replication and homeostatic behavior.

when these properties are incorporated in a concrete or conceptual human-made system, those who do not accept emotionally that the nature of life can be understood immediately apprehend other properties as relevant, and manage to refrain from accepting any synthetic system as living by continually specifying new requirements.

3. It is very often assumed that observation and experimentation are alone sufficient to reveal the nature of living systems, and no theoretical analysis is expected to be necessary, still less sufficient, for a characterization of the living organization. It would take too long to state why we depart from this radical empiricism. Epistemological and historical arguments more than justify the contrary view: Every experimentation and observation implies a theoretical perspective, and no experimentation or observation has significance or can be interpreted outside the theoretical framework in which it took place.

2.3.2

Our endeavor has been to put forth a characterization of living systems, such that all their phenomenology could be understood through it. We have tried to do this by pointing at autopoiesis in the physical space as a necessary and sufficient condition for a system to be a living one.

To know that a given aim has been attained is not always easy. In the case at hand, the only possible indication that we have attained our aim is the reader's agreement that all the phenomenology of living systems is illuminated by this view, and that reproduction and evolution indeed require and depend on autopoiesis. The following pages are devoted to this thesis.

Sources

(Maturana and Varela, 1975), (Varela and Maturana, 1972).

3 A Tessellation Example of Autopoiesis

3.1 The Model

3.1.1

To make the foregoing a bit less abstract, I wish to present at this point a simple model that displays the autopoietic organization. This model is presented in an imaginary, two-dimensional space of components, in the manner of tessellation automata (Burks, 1970). As will be obvious, the model is inspired by the kind of chemical productions existing in a living cell; in fact, the model can be taken as a simplification of such productions.

This model is significant in two respects: On the one hand, it permits the observation of the autopoietic organization at work in a system simpler than any known living system, as well as its spontaneous generation from components; on the other hand, it may permit the development of formal tools for the analysis and synthesis of autopoietic systems.¹

3.1.2

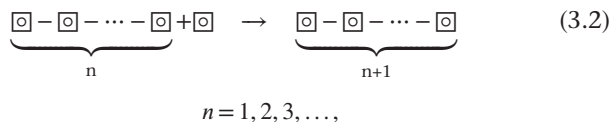
The model consists of a two-dimensional universe where numerous elements $\circ$ (“substrates”) and a few $*$ (“catalysts”) move randomly in the spaces of a quadratic grid. These elements are endowed with specific properties that determine interactions that may result in the production of other elements $\square$ (“links”) having properties of their own, including the capability of interactions (“bonding”). Let the interactions and transformations be as follows:

Composition:

$$* + 2\circ \rightarrow * + \square \quad (3.1)$$

¹*Eds.* This computational model of autopoiesis prefigures work in the field of artificial life. Varela was interested in using minimal models to clarify complex ideas (e.g., Varela, 1988). He was a prominent figure in the establishment of the artificial life scene, coorganizing with Paul Bourguine the First European Conference on Artificial Life (Varela and Bourguine, 1992). The model was reimplemented by Milan Zeleny (1977), and later by Barry McMullin (1997; 2004) (see appendix A). Minimal abstract models of autopoiesis continue to be studied (e.g., Agmon et al., 2016; Beer, 2015; Breyer et al., 1998; Egbert and Di Paolo, 2009).

Concatenation:



Disintegration:



²Eds. Barry McMullin studied and replicated this model in the 1990s and found that link particles tend to bond prematurely, becoming immobile and unavailable for membrane repair. He and Varela uncovered an unreported interaction in the original algorithm (see notes in appendix A) that prevented this problem (McMullin and Varela, 1997; McMullin, 1997, 2004). For an overview, see Thompson (2007, 107–112).

³Details of the computation are given in appendix A. To facilitate appreciation of the development of the model, figures 3.1 and 3.2 are drawn from a computer printout, changing the symbols actually used in the computations.

The interaction (3.1) between the catalyst $*$ and two substrate elements $2\bigcirc$ is responsible for the composition of an unbonded link $\square$. These links may be bonded through the interaction (3.2), which concatenates these bonded links to unbranched chains of $\square$'s. A chain so produced may close upon itself, forming an enclosure which we assume to be penetrable by the $\square$'s but not by $*$. Disintegration, (3.3), is assumed to be independent of the state of links $\square$, i.e., whether they are free or bound, and can be viewed either as a spontaneous decay or as a result of a collision with a substrate element.²

In order to visualize the dynamics of the system, we show two sequences (figures 3.1 and 3.2) of successive stages of transformation as they were obtained from the printout of a computer simulation of this system.³

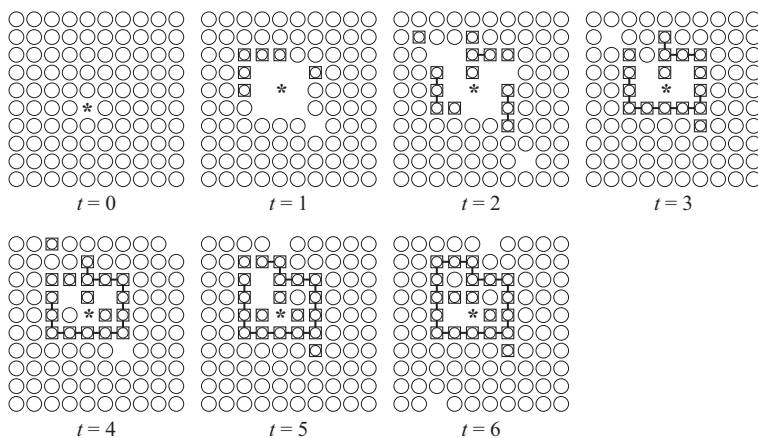
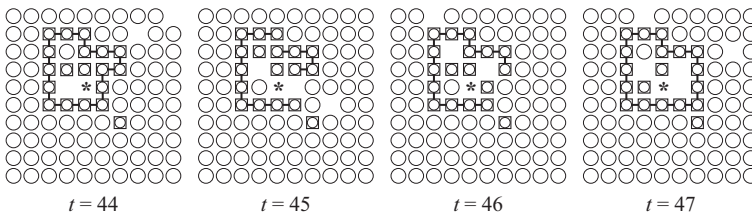


Figure 3.1

The first seven instants ($0 \rightarrow 6$) of one computer run, showing the spontaneous emergence of a unit in this two-dimensional domain. Interactions between substrate $\bigcirc$ and catalyst $*$ produce chains of bonded links $\square$ which eventually enclose the catalyst, thus closing a network of interactions, which constitutes an autopoietic unity in this domain. Redrawn from (Varela et al., 1974).

**Figure 3.2**

Four successive instants ($44 \rightarrow 47$) in the same computer run of figure 3.1, showing regeneration of the boundary broken by spontaneous decay of links. Ongoing production of links reestablishes the unity's border under changes of form and turnover of components (Varela et al., 1974).

If an $\square$ -chain closes on itself enclosing an element $*$ (figure 3.1), the $\square$'s produced within the enclosure by the interaction (3.1) can replace in the chain, via (3.2), the elements $\square$ that decay as a result of (3.3) (figure 3.2). In this manner, a unity is produced, which constitutes a network of productions of components that generate and participate in the network of productions that produced these components by effectively realizing the network as a distinguishable entity in the universe where the elements exist. Within this universe these systems satisfy the autopoietic organization.⁴ In fact, the element $*$ and elements $\circ$ produce elements $\square$ in an enclosure formed by a two-dimensional chain of $\square$'s; as a result the $\square$'s produced in the enclosure replace the decaying $\square$'s of the boundary, so that the enclosure remains closed for $*$ under continuous turnover of elements, and under recursive generation of the network of productions, which thus remains invariant (figures 3.1 and 3.2). This unity cannot be described in geometric terms, because it is not defined by the spatial relations of its components. If one stops all the processes of the system at a moment at which $*$ is enclosed by the $\square$ -chain, so that spatial relations between the components become fixed, one indeed has a system definable in terms of spatial relations, that is, a crystal, but not an autopoietic unity.

⁴*Eds.* Strictly speaking, the catalyst $*$ is not itself produced by the system; rather it is externally given (see McMullin 2004). Nevertheless, the model shows how a self-sustaining organization emerges in a model without being explicitly coded in it, demonstrating what Beer and Di Paolo (2023) call "systemic precariousness."

3.2 Interpretations

3.2.1

It should be apparent from this model that the processes generated by the properties of the components [(3.1)–(3.3)] can be concatenated in a number of ways. The autopoietic organization is but one

of them, yet it is the one that, by definition, implies the realization of a dynamic unity. The same components can generate other, allopoietic organizations; for example, a chain which is defined as a sequence of $\square$'s is clearly allopoietic, since the production of the components that realize it as a unity does not enter into its definition as a unity. Thus, the autopoietic organization is neither represented nor embodied in (3.1)–(3.3), as in general no organization is represented or embodied in the properties that realize it.

In the case described, as in a broad spectrum of other studies that can generically be called tessellation automata (von Neumann, 1966; Burks, 1970), the starting point is a generalization of the physical situation.⁵ In fact, one defines a space where spatially distinguishable components interact, thus embodying the concatenation of processes which lead to events among the components. This is of course what happens in the molecular domain, where autopoiesis as we know it takes place. For the purpose of explaining and studying the notion of autopoiesis, however, one may take a more general view, as we have done here, and revert to the tessellation domain, where physical space is replaced by any space (a two-dimensional one in the model), and molecules by entities endowed with some properties. The phenomenology is unchanged in all cases: the autonomous self-maintenance of a unity while its organization remains invariant in time.

It is apparent that in order to have autopoietic systems, the components cannot be simple in their properties. In the present case we required that the components have specificity of interactions, forms of linkage, mobility, and decay. None of these properties are dispensable for the formation of this autopoietic system. The necessary feature is the presence of a boundary that is produced by a dynamics such that the boundary creates the conditions required for this dynamics.

3.2.2

It is interesting to note that, though inspired by the idea of autopoiesis, this tessellation automaton is of some independent interest (see also Zeleny and Pierre, 1976; Zeleny, 1977). It is fundamentally distinct from other tessellation models, such as Conway's well-known game of "life" (Gardner, 1971) and the other lucid games proposed by Eigen and Winkler 1975, because in these models the essential property studied is that of reproduction and evolution, and not that of individual self-maintenance.⁶ In other words, the process by which a unity maintains itself is fundamentally different

⁵Eds. *Cellular automata*, as these models are known, are an active branch of study in computational mathematics and physical modeling (Berto and Tagliabue, 2022). They do not represent a generalization of a physical situation, but rather an abstraction of it, where processes have been simplified, typically by assuming discrete spatiality and temporality, as well as a discrete set of states and fixed local rules of change. These abstractions sometimes afford useful conceptual clarity. John Conway's *Game of Life* has been explored precisely for its relevance in modeling self-maintenance of dynamic patterns (e.g., "gliders") in the context of clarifying autopoiesis and related concepts. See Randall Beer (2004; 2015; 2020b).

⁶Eds. Several cellular automata models have been used to study self-reproducing patterns (von Neumann, 1966; Langton, 1984). Since the 1990s, interest has shifted to the study of emergent and self-maintaining rather than self-replicating structures, particularly in the area of artificial chemistry, of which this model may be said to be a precursor (e.g., Banzhaf and Yamamoto, 2015; Fontana and Buss, 1994; Dittrich et al., 2001).

from the process by which it can duplicate itself in some form or another. Production does not entail reproduction, but reproduction does entail some form of self-maintenance or identity. In the case of von Neumann, Conway, and Eigen, the question of the identity or self-maintenance of the unities they observe in the process of reproducing and evolving is left aside and taken for granted; it is not the question these authors are asking at all.

How easily an autopoietic model like the one presented above could be generalized to three dimensions is an open question. There seems to be no conceptual difficulty involved. However, it is my suspicion that one further dimension will require a significant increase in complexity of the properties in the components, and the simplicity of the rules embodied in (3.1)–(3.3) will be lost. One further study that would be worth some attention is how much more complexity in the rules of interactions would be required for this simple two-dimensional model to reproduce itself, thus bringing it closer to the studies of von Neumann and Eigen mentioned above. This study seems particularly important in the light of the difficult problem posed by the combination between self-maintenance and reproduction, which requires some combination of purely dynamic processes of production and (discrete) processes of specification of components (e.g., Eigen, 1971; Pattee, 1977).

Another interesting point about this tessellation model is that the properties of the components that are minimally required to produce the autopoietic dynamic could be illuminating for the kind of properties required in the molecular domain.⁷ Thus, these kinds of studies potentially hold a key both for the synthesis of molecular autopoietic units, and for the understanding of neobiogenesis. We believe that the synthesis of molecular autopoiesis can be attempted at present, as suggested by studies like those on microspheres and liposomes (Fox, 1965; Bangham, 1968) when analyzed in the present framework. Consider, for example, a liposome whose membrane lipidic components are produced and/or modified by reactions among its components that take place only under the conditions of concentration produced within the liposome membrane. Such a liposome would constitute an autopoietic system. Experiments along these lines are only beginning (Guiloff, 1981).

Source

(Varela et al., 1974).

⁷Eds. Varela collaborated with Pier Luigi Luisi in efforts to produce minimal chemical autopoietic systems. They studied the synthesis of self-replicating bounded chemical structures (micelles); see Luisi and Varela (1989). Luisi continued to develop the study of minimal chemical autopoiesis and has investigated the emergence of life from a systems biology point of view (Luisi, 2016). Synthetic protocells following similar principles, such as the formation of enclosed lipid vesicles, are studied in the field of synthetic biology using a combination of chemical and computational models (Solé et al., 2007).

4 Embodiments of Autopoiesis

4.1 Autopoietic Dynamics

4.1.1

That a cell is an autopoietic system is apparent in its life cycle. What is not obvious is how the cell is a molecular embodiment of autopoiesis, as should be apparent from its analysis in terms of what we may call the “dimensions” of its autopoietic dynamics.¹

1. *Production of Constitutive Relations.* Constitutive relations are relations that determine the topology of the autopoietic organization, and hence its physical boundaries. The production of constitutive relations through the production of the components that hold these relations is one of the defining dimensions of an autopoietic system. In the cell such constitutive relations are established through the production of molecules (proteins, lipids, carbohydrates, and nucleic acids) that determine the topology of the relations of production in general, that is, molecules that determine the relations of physical neighborhood necessary for the components to hold the relations that define them. The cell defines its physical boundaries through the production of constitutive relations that specify its topology. There is no specification within the cell of what it is not.
2. *Production of Relations of Specifications.* Relations of specifications are relations that determine the identity (properties) of the component of the autopoietic organization, and hence, in the case of the cells, its physical feasibility. The establishment of relations of specification, through the production of components that can hold these relations, is another of the defining dimensions of an autopoietic system. In the cell such relations of specification are produced mainly through the production of nucleic acids and

¹*Eds.* With minor edits, chapter 4 reproduces several passages of chapter 3 in Maturana and Varela (1980). It inherits much of the dense language of that work. Its main goal is to show how a living cell actually instantiates an autopoietic organization. To do this, the cell is described along three dimensions of production of relations: topology, specificity, and order. These dimensions are coordinated into an organization that is kept invariant while structural aspects along each dimension may change. What defines a cell is this invariance and not, say, the activity involving particular components, such as DNA, that participate in the network of relations just as other components do.

proteins that determine the identity of the relations of production in general. In the cell this is obviously obtained, on the one hand, by relations of specificity between DNA, RNA, and proteins, and on the other hand, by relations of specificity between enzymes and substrates. Such production of relations of specificity holds only within the topological substrate defined by the production of relations of constitution. There is no production in the cell as an autopoietic system of relations of specification that do not pertain to it.

3. *Production of Relation of Order.* Relations of order are those that determine the dynamics of the autopoietic organization by determining the concatenation of the production of relations of constitution, specification, and order, and hence its actual realization. The establishment of relations or order through the production of components that control the production of relations (of constitution, specification, and order) constitute the third dimension of the autopoietic dynamics. In the cell, relations or order are established mainly by the production of components (metabolites, nucleic acids, and proteins) that control the speed of production of relations of constitution, specification, and order. Relations of order thus make up a network of parallel and sequential relations of constitution, specification, and order that constitute the cell as an invariant dynamic topological unity. There is no ordering through the autopoietic organization of the cell of processes that do not belong to it.

If one examines a cell, it is apparent that DNA participates in the specification of polypeptides, and hence of proteins, enzymatic and structural, which specifically participate in the production of proteins, nucleic acids, lipids, glucides, and metabolites. Metabolites (which include all small molecules, monomers or not, produced in the cell) participate in the regulation of the speed of the various processes and reactions that constitute the cell, establishing a network of interrelated speeds in parallel and sequentially interconnected processes, both by gating and by constitutive participation, in such a way that all reactions are functions of the state of the transforming network that they integrate. All processes occur bound to a topology determined by their participation in the processes of production of constitutive relations.

4.1.2

In current usage, cellular processes are simplified by supposing that specification is mostly effected by nucleic acids, constitution by

proteins, and order (regulation) by metabolites. The autopoietic process, however, is closed in the sense that it is entirely specified by itself, and such simplification represents our cognitive relation with it, but does not operationally reproduce it. In the actual system, specification takes place at all points where its organization determines a specific process (protein synthesis, enzymatic action, selective permeability); ordering takes place at all points where two or more processes meet (changes of speed or sequence, allosteric effects, competitive and noncompetitive inhibition, facilitation, inactivation) determined by the structure of the participating components; constitution occurs at all places where the structure of the components determines physical neighborhood relations (membranes, particles, active sites in enzymes). What makes this system a unity with identity and individuality is that all the relations of production are coordinated in a system describable as having an invariant organization. In such a system any deformation at any place is compensated for, not by bringing the system back to an identical state in its components such as might be described by considering its structure at a given moment, but rather by keeping its *organization* constant as defined by the relation of the productions that constitute autopoiesis. The only thing that defines the cell as a unity (as an individual) is its autopoiesis, and thus, the only restriction put on the existence of the cell is the maintenance of autopoiesis. All the rest (that is, its structure) can vary: Relations of topology, specificity, and order can vary as long as they constitute a network in an autopoietic space.

4.2 Questions of Origin

4.2.1

The production of relations of constitution, specification, and order are not characteristic of autopoietic systems. They are inherent in unitary interactions in general, and in molecular interactions in particular; they depend on the properties of the units or molecules as expressed in the geometric and energetic relationships that they may adopt. Thus, the geometric properties of the molecules determine the relations of constitution—that is, the topology, the physical neighborhoods, or the spatial relations into which they may enter. The chemical properties of the molecules determine their possible interactions, and hence the relations of specificity, which are a dimension independent of the relations of constitution. Together they determine the sequence and concatenation of molecular interactions, that is, relations of order.

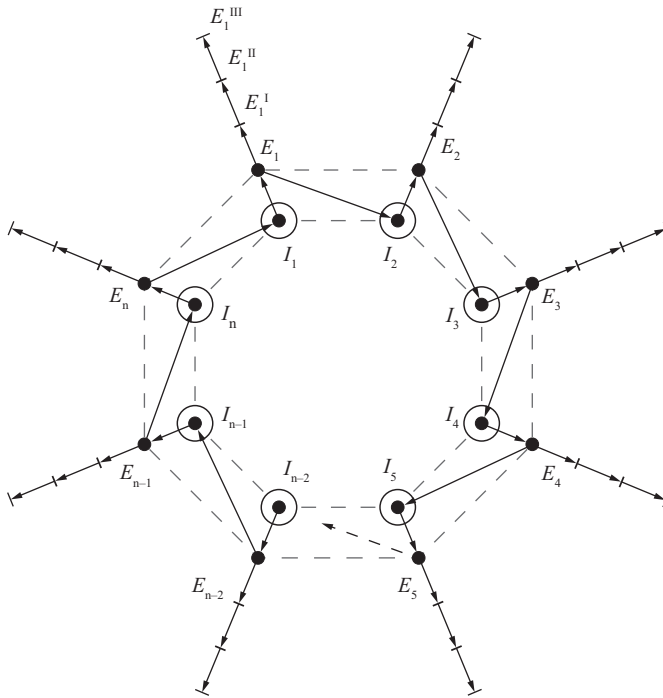
Accordingly, autopoiesis may arise in a molecular system if the relations of production are concatenated in such a way that they produce components specifying the system as a unity that exists only while it is actively produced by such concatenation of processes. This is to say that autopoiesis arises in a molecular system only when the relation that concatenates these relations is produced and maintained constant through the production of the molecular components that constitute the system through this concatenation. Thus, in general, the question of the origin of an autopoietic system is a question about the conditions that must be satisfied for the establishment of an autopoietic dynamics. This problem, then, is not a chemical one, in terms of what molecules took or can take part in the process, but a general one of what relations the molecules or any constitutive units should satisfy.

A clear example of this situation is Eigen's studies on the origin of life (Eigen, 1971, 1973; Eigen and Schuster, 1977), with the successive steps of stability in chemical reactions that could have led to a cell-like system, and in particular, to something like a genetic code. By analytic methods derived from nonequilibrium thermodynamics, combined with computer simulations, Eigen shows how selective pressures could have been brought to bear in the process of molecular evolution. Interestingly enough, he concludes that of central importance to this process is a circular concatenation of processes, such as the hypercycle of figure 4.1. In this generalized situation, the processes of specification, constitution, and order are related in a typically autopoietic fashion, although Eigen has not put emphasis on boundary generation, since his interest lies in the processes of specification.²

²Eds. Eigen and Schuster's hypercycle is described as an example of organizational closure (i.e., an autonomous system) in chapter 7. John Maynard Smith (1979) provides an early discussion of the relevance of the hypercycle for origins of life studies because it can help explain the accurate replication of long RNA molecules. He also notices that the lack of compartmentalization in the model is an issue, albeit not from an organizational perspective, as Varela does, but from an evolutionary one concerned with solving the paradoxical codependence of enzymes and large genomes in prebiotic evolution; see (Maynard Smith and Szathmáry, 1997).

4.2.2

The establishment of an autopoietic system cannot be a gradual process: Either a system is an autopoietic system or it is not. In fact, its establishment cannot be gradual because an autopoietic system is defined as a system, that is, it is defined as a topological unity by its organization. Thus, either a topological unity is formed through its autopoietic organization, and the autopoietic system is there and remains, or there is no topological unity, or else a topological unity is formed in a different manner and there is no autopoietic system but there is something else. Accordingly, there are not and cannot be intermediate systems. We can describe a system and talk about it as if it were a system that would, with little transformation, become an autopoietic system, because we can imagine different systems with

**Figure 4.1**

Eigen's self-producing hypercycle. An RNA-like molecule I_i serves as the specification for a catalytic molecule E_i . Each branch from E_i , may include several other processes (e.g., polymerization, regulation), but one of these branches provides a coupling to the carrier I_{i+1} . These linkages close, so that E_n enhances the formation of I_1 . The hypercycle, as studied through a system of nonlinear differential equations, is postulated as a unit of selection in the early evolution of life. Redrawn from (Eigen, 1974).

which we can compare it, but such a system would be intermediate only in our description, and in no organizational sense would it be a transitional system.

In general the problem of the origin of autopoietic systems has two aspects: One refers to their feasibility, and the other to the possibility of their spontaneous occurrence. The first aspect can be stated in the following manner: The establishment of any system depends on the presence of the components that constitute it, and on the kinds of interactions into which they may enter: thus, given the proper components and the proper concatenation of their interactions, the system is realized. The concrete question of the feasibility of a molecular autopoietic system is, then, the question of the conditions in which different chemical processes can be concatenated to form topological unities that constitute relational networks in

the autopoietic space. The second aspect can be stated in the following manner: Given the feasibility of autopoietic systems, and given the existence of terrestrial autopoietic systems, there are natural conditions under which these may be spontaneously generated. Concretely the question would be: Which were or are the natural conditions under which the components of the autopoietic systems arose or arise spontaneously on Earth? This question cannot be answered independently of the manner in which the feasibility question is answered, particularly with regard to the feasibility of one or several different kinds of molecular autopoietic systems. The presence today of one mode of autopoietic organization on Earth (the nucleic-acid-protein system) cannot be taken to imply that the feasibility question has only one answer.

Source

(Maturana and Varela, 1975).

5 The Individual in Development and Evolution

5.1 Introduction

Living systems embody the living organization.¹ Living systems are autopoietic systems in the physical space. The diversity of living systems is apparent; it is also apparent that this diversity depends on reproduction and evolution. Yet, reproduction and evolution do not enter into the characterization of the living organization as autopoiesis, and living systems are defined as unities by their autopoiesis. This is significant because it makes it evident that the *phenomenology* of living systems depends on their being autopoietic unities. In fact, reproduction requires the existence of a unity to be reproduced, and it is necessarily secondary to the establishment of such a unity; evolution requires reproduction and the possibility of change, through reproduction of that which evolves, and it is necessarily secondary to the establishment of reproduction.² It follows that the proper evaluation of the phenomenology of living systems, including reproduction and evolution, requires their proper evaluation as autopoietic unities.

5.2 Subordination to the Condition of Unity

Unity (distinguishability from a background, and hence from other unities) is the sole necessary condition for existence in any given domain. In fact, the nature of a unity and the domain in which it exists are specified by the process of its distinction and determination; this is so regardless of whether that process is conceptual (as when a unity is defined by an observer through an operation of distinction in her domain of discourse and description), or physical (as when an autonomous unity comes to be established through the

¹*Eds.* With minor edits, chapter 5 reproduces several passages of chapter 4 in Maturana and Varela (1980).

²*Eds.* This was a bold statement at the time (as it is now), bearing in mind Dobzhansky's (1973) dictum: "*Nothing in biology makes sense except in the light of evolution.*" The central message of autopoietic theory is not a negation of the importance of evolution but a foregrounding of the neglected autopoietic organization of living systems. Still, the case may have been overstated, as we point out in section 5.4.1.

actual working of its defining properties that assert its distinction from a background through their actual operation in the physical space). Accordingly, different kinds of unity necessarily differ in the domain in which they are established, and having different domains of existence, they may or may not interact as these domains do or do not interact.

Unity distinction, then, is not an abstract notion of purely conceptual validity for descriptive or analytical purposes, but is an operative notion referring to the process through which a unity becomes asserted or defined: *the conditions that specify a unity determine its phenomenology*. In living systems, these conditions can be traced to their autopoietic organization. In fact, autopoiesis implies the subordination of all change in the autopoietic system to the maintenance of its autopoietic organization, and since this organization defines the system as a unity, it implies total subordination of the phenomenology of the system to the maintenance of its unity. This subordination has the following consequences:

1. The establishment of a unity defines the domain of its phenomenology, but the way the unity is constituted—its structure—defines the kind of phenomenology that it generates in that domain. It follows that the particular form adopted by the phenomenology of each autopoietic (biological) unity depends on the particular way in which its individual autopoiesis is realized. It also follows that the domain of ontogenic transformations (including conduct) of each individual is the domain of the homeorhetic³ trajectories through which it can maintain its autopoiesis.
2. All the biological phenomenology is necessarily determined and realized through individuals (that is, through autopoietic unities in the physical space), and consists in all the paths of transformations that they undergo, singly or in groups, in the process of maintaining invariant their individual defining relations. Whether or not in the process of their interactions the autopoietic unities constitute additional unities is irrelevant for the subordination of the biological phenomenology to the maintenance of the identity of the individuals. In fact, if a new unity is produced that is not autopoietic, its phenomenology, which will necessarily depend on its organization, will be biological or not according to its dependence on the autopoiesis of its components, and will accordingly depend or not depend on the maintenance of these as autopoietic units. If the new unity is autopoietic, then its phenomenology is biological and obviously depends on the

³Eds. The term *homeorhesis* was introduced into biology by Conrad H. Waddington (1957, 32) to describe a system's robust tendency to return to a particular *trajectory* after a perturbation, in contrast to *homeostasis*, which typically implies a tendency to return to a particular *state*.

maintenance of its autopoiesis, which in turn may or may not depend on the autopoiesis of its components.

3. The identity of an autopoietic unity is maintained as long as it remains autopoietic, that is, as long as it, as a unity in the physical space, remains a unity in the autopoietic space, regardless of how much it may otherwise be transformed in the process of maintaining its autopoiesis.

5.3 Plasticity of Ontogeny: Structural Coupling

Ontogeny is the history of the structural transformation of a unity. Accordingly, the ontogeny of a living system is the history of maintenance of its identity through continuous autopoiesis in the physical space. From the mere fact that a physical autopoietic system is a dynamic system, realized through relations of production of components that imply concrete physical interactions and transformations, it is a necessary consequence of the autopoietic organization of a living system that its ontogeny should take place in the physical space.

Since the way an autopoietic system maintains its identity depends on its particular way of being autopoietic (that is, on its particular structure), different classes of autopoietic systems have different classes of ontogenies. Moreover, since an autopoietic system does not have inputs or outputs, all the changes that it may undergo without loss of identity, and hence with maintenance of its defining relations, are necessarily determined by its invariant organization. Consequently, the phenomenology of an autopoietic system is necessarily always commensurate with the deformations that it suffers without loss of identity, and with the deforming environment in which it lies; otherwise it would disintegrate.

As a consequence of the invariance of the autopoietic organization, the way the autopoiesis is realized in any given unity may change during its ontogeny, with the sole restriction that this should take place without loss of identity, that is, through uninterrupted autopoiesis. Although the changes that an autopoietic system may undergo without loss of identity while compensating its deformations under interactions are determined by its organization, the sequence of such changes is determined by the sequence of these deformations. There are two sources of deformations for an autopoietic system as they appear to be to an observer: one is constituted by the environment as a source of independent events in the sense that these are not determined by the organization of the system;

the other is constituted by the system itself as a source of states that arise from compensations of deformations, but that themselves can constitute deformations that generate further compensatory changes. In the phenomenology of the autopoietic organization these two sources of perturbations are indistinguishable, and in each autopoietic system they braid together to form a single ontogeny. Thus, although in an autopoietic system all changes are internally determined, for an observer its ontogeny partly reflects its history of interactions with an independent environment. Accordingly, two otherwise equivalent autopoietic systems may have different ontogenies.

In summary: *the continued interactions of a structurally plastic system in an environment with recurrent perturbations will produce a continual selection of the system's structure.* This structure will determine, on the one hand, the state of the system and its domain of allowable perturbations, and on the other hand will allow the system to operate in an environment without disintegration. We refer to this process as structural coupling (Maturana, 1978). If we can consider the system's environment also as a structurally plastic system, then the system and the environment will have an interlocked history of structural transformations, selecting each other's trajectories.

Thus, we again find the relevance of the position taken by the observer and his cognitive needs. An observer beholding an autopoietic system as a unity in a context that he also observes, and that he describes as its environment, may distinguish in it internally and externally generated perturbations, even though these are intrinsically indistinguishable for the dynamic autopoietic system itself. The observer can use these distinctions to make statements about the structural coupling of the system which he observes, and he can use this history to describe an environment (which he infers) as the domain in which the system exists. He cannot, however, infer from the observed correspondence between the ontogeny of the system and the environment that this ontogeny describes, or from the environment in which he sees it, a constitutive *representation* of these in the organization of the autopoietic systems, for this would only mean a confusion of observational perspectives across a logical type.⁴ The continuous correspondence between conduct and environment revealed during ontogeny is the result of the invariant nature of the autopoietic organization, and not of the existence of any representation of the environment in it; nor is it at all necessary that the autopoietic system should "obtain" or develop such a representation to persist in a changing environment. To talk

⁴*Eds.* This key point is the basis for Maturana and Varela's criticism of representationalist thinking in biology, particularly neurobiology, and it was later developed by Varela and colleagues into a critique of the representational theory of mind in cognitive science. See Varela et al. (1991).

in a meaningful and sophisticated way about a representation of the environment in the organization of a living system may be essential in our explanatory discourse (see Chapter 9), but it is inessential to define what makes a certain system a unit. Informational notions such as representations only become necessary to explain phenomena that unities can exhibit over long spans of time and with a certain degree of reliability. But this is another matter than defining the organization of a unit. We shall return later to this important duality of the observer's perspective.

5.4 Reproduction and the Complications of the Unity

5.4.1

Reproduction requires a unity to be reproduced; this is why reproduction is operationally secondary to the establishment of the unity, and it cannot enter as a defining feature of the organization of living systems.⁵ Furthermore, since living systems are characterized by their autopoietic organization, reproduction must necessarily have arisen as a complication of autopoiesis during autopoiesis, and its origin must be viewed and understood as secondary to, and independent of, the origin of the living organization. The dependence of reproduction upon the existence of the unity to be reproduced is not a trivial problem of precedence, but is an operational problem in the origin of the reproduced system and its relations with the reproducing mechanism. In order to understand reproduction and its consequences in autopoietic systems we must briefly analyze the operational nature of this process in relation to autopoiesis.

There are three phenomena that must be distinguished in relation to the notion of reproduction: replication, copying, and self-reproduction.

1. *Replication.* A system that successively generates unities different from itself, but in principle identical to each other, and with an organization that the system determines in that process, is a replicating system. Replication, then, is not different from repetitive production. Any distinction between these processes arises as a matter of description in the emphasis that the observer puts on the origin of the equivalent organization of the successively produced unities, and on the relevance that this equivalence has in a domain different from that in which the repetitive production takes place. Thus, although all molecules are produced by specific molecular and atomic processes that can at least in principle be repeated, only when certain specific kinds of molecules are

⁵*Eds.* This statement seems logically flawless, but it relies on having assumed the separation of processes of autopoiesis and processes of reproduction. It is possible to distinguish autopoiesis and reproduction in *organizational* terms at the level of a single unity. But this does not rule out, for instance, reproductive processes subserving autopoiesis, as with cells that avoid bursting by dividing in two, or the participation of autopoietic processes in larger autonomous cycles of chemical recycling involving several unities, including transgenerational cycles. These *structural* complications (see section 5.4.2) do not erase the distinction between autopoiesis and reproduction, but it does cast doubt on the obviousness of the statement that reproduction requires the existence of a unity to be reproduced. There is a slippage from a logical relation to a temporal one in the corollary that reproduction must necessarily have arisen as a complication of autopoiesis. Ideas on the origin of life seem to be converging on a perspective in which both types of processes *co-arise* in ecological communities combining individual and collective activity (Moreno, 2019; Ruiz-Mirazo et al., 2020). This perspective, described as *ecopoiesis*, a term coined by Robert H. Haynes (see, e.g., Lovelock, 1987), resembles what Beth Dempster and Donna Haraway call *sympoiesis* (Dempster, 2000; Haraway, 2016) and interestingly extends the autopoietic view.

⁶*Eds.* An example of replication is the production of cell components, such as proteins, by the cellular machinery. Notice that replicated units are independent of each other.

⁷*Eds.* Copying focuses on transferred patterns between a model and a copy. If the new copy replaces the model in the process, new copies will form a historical chain.

⁸*Eds.* Mitosis is a paradigmatic case of self-reproduction. A cell divides into two cells that are not structurally identical but share the same organization. The processes of reproduction overlap with the processes of production. Successive reproductive events form a historical chain.

produced in relation to the cellular activities (proteins and nucleic acids) by certain repeatable molecular concatenations, is their production called replication. Such a denomination, then, makes reference only to the context in which the identity of the successively produced molecules is deemed necessary, not to a unique feature of that particular molecular synthesis.⁶

2. *Copying.* Copying takes place whenever a given object or phenomenon is mapped by means of some procedure upon a different system, so that an isomorphic object or phenomenon is realized in it. In the notion of copying the emphasis is put on the mapping process, regardless of how this is realized, even if the mapping operation is performed by the model unit itself.⁷
3. *Self-reproduction.* Self-reproduction takes place when a unity produces another one with a similar organization to its own, through a process that is coupled to the process of its own production. It is apparent that only autopoietic systems can have molecular self-reproduction, because only they are realized through a process of self-production (autopoiesis) in the physical space.⁸

For an observer there is reproduction in all these three processes, because she can recognize in each of them a unitary pattern of organization which is embodied in successively generated systems through the three well-defined mechanisms. The three processes, however, are intrinsically different because their dynamics gives rise to different phenomenologies, which appear particularly distinct if one considers the network of systems generated under conditions in which change is allowed in the process of reproduction of the successively embodied pattern of organization. Thus, in replication and copying the mechanism of reproduction is necessarily external to the pattern reproduced, while in self-reproduction it is necessarily identical to it. Furthermore, only in self-copying and self-reproduction can the reproducing mechanism be affected by changes in the unities produced that embody the pattern reproduced. It should be clear that the historical interconnections established between independent unities through reproduction vary with the mechanism through which reproduction is achieved.

5.4.2

In living systems presently known on Earth, autopoiesis and reproduction are directly coupled, and hence these systems are truly self-reproducing systems. In fact, in them, reproduction is a moment in autopoiesis, and the same mechanism that constitutes

one constitutes the other. The consequences of such a coupling are paramount: (1) Self-reproduction must take place during autopoiesis; accordingly the network of individuals thus produced is necessarily self-contained in the sense that it does not require for its establishment a mechanism independent of the autopoietic determination of the self-reproducing unities. Such would not be the case if reproduction were attained through external copying or replication. (2) Self-reproduction is a form of autopoiesis; therefore, variation and constancy in each reproductive step are not independent, and both must occur as expressions of autopoiesis. (3) Variation through self-reproduction of the way the autopoiesis is realized can only arise as a modification during autopoiesis of a preexisting functioning autopoietic structure; consequently, variation through self-reproduction can only arise from perturbations that require further complications to maintain autopoiesis invariant. The history of self-reproductively connected autopoietic systems can only be one of continuous complication of the structures of autopoiesis.⁹

Again, let us note that notions such as coding, message, or information are not, strictly speaking, applicable to the phenomenon of self-reproduction; their use in the description of its mechanism constitutes an attempt to represent it on another descriptive level.

5.5 Evolution, a Historical Network

5.5.1

A historical phenomenon is a process of change in which each of the successive states of a changing system arises as a modification of a previous state in a causal transformation, and not *de novo* as an independent occurrence.¹⁰ The notion of history may either be used to refer to the antecedents of a given phenomenon as the succession of events that gave rise to it, or be used to characterize the given phenomenon as a process. Therefore, since a causal explanation is always given in the present as a reformulation of the phenomenon to be explained in the domain of interactions of its components (or of isomorphic elements), the history of a phenomenon as a description of its antecedents cannot contribute to its explanation, because the antecedents are not components of the phenomenon which they precede or generate. Conversely, if history as a phenomenon is to be explained in the present as a causal network of sequentially concatenated events in which each event is a state of the network that arises as a transformation of the previous state, then it follows that

⁹*Eds.* This paragraph recognizes the structural links between autopoiesis and reproduction, but the idea requires clarification. It is written with single-cell organisms in mind (some statements do not apply to multicellulars). Self-reproduction overlaps with autopoiesis, but it is not clear whether it must take place *entirely* as a part of autopoiesis or whether it is a *form* of autopoiesis. The ambiguity here is: whose autopoiesis? Variations may thus arise not as complications during autopoiesis either before or after reproduction, but rather during the reproductive event (e.g., statistical noise in chemical concentrations can trigger different cellular dynamics and lead to cell differentiation). If “larger” (i.e., ecological, intergenerational) self-preserving patterns of reproduction co-arise with autopoiesis, these patterns may realize their own organization (e.g., influencing reproductive rates). The historical chain may or may not lead to complications of the structures of autopoiesis.

¹⁰*Eds.* This corresponds to a *historical chain*. Historicity is more strongly defined in terms of the historical chains in which symmetries are broken and require novel descriptions, as in the appearance of biological novelty (e.g., Montévil, 2022; Di Paolo et al., 2022).

¹¹ *Eds.* See chapters 8 and 9 on operational and symbolic explanations.

although history cannot contribute to explaining any phenomenon causally, it can permit an observer to account for the origin of a phenomenon as a state in a causal (historical) network. Observers can do this because they have independent observational (or descriptive) access to the different states of the historical process.¹¹

It is in this context that the phenomenology of autopoietic systems must be considered when viewed in reference to evolution. Biological evolution is a historical phenomenon, and as such it must be explained in the present context by its reformulation as a historical network constituted through the causal interactions of coupled or independent biological events. Furthermore, biological events depend on the autopoiesis of living systems; accordingly, our aim here is to understand how evolution is defined as a historical process by the autopoiesis of the biological unities.

5.5.2

If by evolution we refer to what has taken place in the history of transformation of terrestrial living systems, then evolution as a process is the history of change of a pattern of organization embodied in independent unities sequentially generated through reproductive steps, in which the particular defining organization of each unity arises as a modification of the preceding one (or ones), which thus constitutes both its sequential and its historical antecedent. Consequently, evolution requires sequential reproduction and change in each reproductive step. Without sequential reproduction as a reproductive process in which the defining organization of each unity in the sequence constitutes the antecedent for the defining organization of the next one, there is no history; without change in each sequential reproductive step, there is no evolution. In fact, sequential transformations in a unity without change of identity constitute its ontogeny, that is, its individual history if it is an autopoietic unit.

Reproduction by replication or copying of a single unchanging model implies an intrinsic decoupling between the organization of the unities produced and their producing mechanism. As a consequence, any change in the reproduced pattern of organization embodied in the unities successively produced by replication or copying from a single model, can only reflect the ontogenies of the reproducing systems or the independent ontogenies of the units themselves. The result is that under no circumstances in these non-sequential reproductive cases does a change in the organization of a unity affect the organization of the others yet to be produced, and, independently of whether they are autopoietic or not, they do

not constitute a historical network, and no evolution takes place. The collection of unities thus produced constitutes a collection of independent ontogenies.

In sequential reproduction, as it occurs in self-reproducing systems that attain reproduction through autopoiesis, or as it occurs in those copying systems in which each new unity produced constitutes the model for the one, the converse is true. In these cases, there are aspects of the defining organization of each unity that determine the organization of the next one by their direct coupling with the reproductive process, which is thus subordinated to the organization of the reproduced unities.¹² Consequently, changes in these aspects of the organization of the unities sequentially generated that occur either during their own ontogeny, or in the process of their generation, necessarily result in the production of a historical network. The unities successively produced unavoidably embody a changing pattern of organization in which each state arises as a modification of the previous one. In general, then, sequential reproduction with the possibility of change in each reproductive step necessarily leads to evolution, and in particular, in autopoietic systems evolution is a consequence of self-reproduction.

¹² *Eds.* The use of “organization” instead of “structure” in this passage suggests that reproduction is considered an organizational aspect. As defined, reproduction assumes conservation of the *autopoietic* organization. Therefore, something more specific than autopoietic organization is implied here, since these organizations seem to be susceptible of alteration during reproduction. But this is not clarified.

5.5.3

Ontogeny and evolution are completely different phenomena, both in their appearance and in their consequences. In ontogeny—the history of transformation of a unity—the identity of the unity, in whatever space it may exist, is never interrupted. In evolution—a process of historical change—there is a succession of identities, generated through sequential reproduction, which constitute a historical network, and that which changes (evolves), namely the pattern of organization of the successively generated units, exists in a different domain than the units that embody it. A collection of successive ontogenies in whose organization an observer can see relations of maintained change, but that have not been generated through sequential reproduction, do not constitute an evolving system, not even if they reflect the continuous transformation (ontogeny) of the system that produced them. It is inadequate to talk about evolution in the history of change of a single unity in whatever space it may exist; unities only have ontogenies. Thus, it is inadequate to talk about the evolution of the universe, or the chemical evolution of Earth; one should only talk about the ontogeny of the universe or the chemical history of Earth.¹³ Also, there is a biological evolution only in that there is sequential reproduction of living systems; if

¹³ *Eds.* That is, provided that one is willing to use the term *evolution* with the technical meaning given here. Since studying the history of Earth or the universe is likely to demand a different operational vocabulary, nothing prevents the term *evolution* from being used with a correspondingly specific meaning in those disciplines.

there were non-self-reproducing autopoietic systems before that, their different patterns of organization did not evolve, and there was only the history of their independent ontogenies.

5.5.4

Selection, as a process in a population of unities, is a process of differential realization in a context that specifies the unitary organizations that can be realized. In a population of autopoietic unities selection is a process of differential realization of autopoiesis, and hence, if these are self-reproducing autopoietic unities, of differential self-reproduction. Consequently, if there is sequential reproduction, and the possibility of change in each reproductive step, then selection can make the transformation of the reproducible pattern of organization embodied in each successive unity a recursive function of the domain of interactions which that very same autopoietic unity specifies. If any system that is realized is necessarily adapted in the domain in which it is realized, and adaptation is the condition of possible realization for any system, then evolution takes place only as a process of continued adaptation of the unities that embody the evolving pattern of organization. Accordingly, different evolving systems will differ only in the domain in which they are realized, and hence in which selection takes place, not in whether they are adaptive or not. Thus, evolution in self-reproducing living systems that maintain their identity in the physical space (as long as their invariant autopoietic organization is commensurate with the restrictions of the ambient in which they exist) is necessarily a process of continued adaptation, because only those of them whose autopoiesis can be realized reproduce, regardless of how much the way they are autopoietic may otherwise change in each reproductive step.¹⁴

¹⁴ *Eds.* In this discussion, adaptation is considered a relation to the environment that conserves autopoiesis. Therefore, it is not a graded notion, nor is it attributable to parts or traits of the organism. An organism is either adapted and alive or not adapted and dead. This contrasts with uses of the term in evolutionary biology, where organisms are sometimes described as bearers of differentially adapted traits, referring to differences in the number of viable offspring or other factors that define fitness attributable to those traits. Rejecting these uses does not invalidate a sensible meaning of selection as resulting from differential self-reproduction that recursively influences the domain of interaction and stabilizes the difference in reproductive patterns. There may be a suggestion of something akin to ideas of niche construction here, but the point is not developed.

5.5.5

A species is the result of the selection process in a population or collection of populations of reproductively interconnected individuals, which are thus nodes in a historical network. These individuals share a genetic pool, that is, a fundamentally equivalent pattern of autopoietic organization under historical transformations. Historically, a species arises when a reproductive network of this kind develops an independent reproductive network as a branch, which, by being an independent historical network (reproductively separated) has an independent history. It is said that what evolves is the species and that the individuals in their historical existence are subordinated to this evolution. In a superficial descriptive sense this

is meaningful, because a particular species as an existing collection of individuals represents continuously the state of a particular historical network in its process of becoming a species; and if described as a state of a historical network, a species necessarily appears in a process of transformation. Yet the species exists as a unit only in the historical domain, while the individuals that constitute the nodes of this historical network exist in the physical space.¹⁵ Strictly, a historical network is defined by each and every one of the individuals that constitute its nodes, but it is at any moment represented historically by the species as the collection of all the simultaneously existing nodes of the network; in fact, then, a species does not evolve, because as a unity in the historical domain it only has a history of change. What evolves is a pattern of autopoietic organization embodied in many particular variations in a collection of transitory individuals that together define a reproductive historical network. Thus, the individuals, though transitory, are essential, not dispensable, because they constitute a necessary condition for the existence of this historical network that they define. The species is a descriptive notion that represents a historical phenomenon; it does not constitute a *causal* component in the phenomenology of evolution.

¹⁵ *Eds.* The distinction is clear, but it is more applicable to the idea of lineage or phylogeny as the history of species rather than to species themselves. There are ways to define species as a collective concept in non-historical terms, such as typologically and/or in terms of reproductive viability, although these definitions are not unproblematic (Wheeler and Meier, 2000). A species (and other taxonomic ranks) could in principle be conceived as further organizational specifications of autopoiesis.

5.5.6

It cannot be too strongly emphasized that for evolution to take place as an actual history of change of a pattern of organization through its embodiment in successively generated unities, reproduction must allow for change in the sequentially reproduced organization. In present living systems reproduction takes place as a modification of autopoiesis and is bound to it. This is to be expected. Originally many kinds of autopoietic unities were probably formed, which would mutually compete for the precursors. If any class of them had any possibility of self-reproduction, it is evident that it would immediately displace through selection the other, non-reproducing forms. The onset of the history of self-reproduction need not have been complex; for example, in a system with distributed autopoiesis mechanical fragmentation is a form of self-reproduction. Evolution through selection would appear, with the enhancement of those features of the autopoietic unities that facilitated their fragmentation (and hence the regularity and frequency of self-reproduction) to the extent of making it independent of external accidental forces.

It is at this point that we can see the difference between borderline cases of autopoietic units (such as the model structures discussed in chapter 3) and the chemical networks operative in cellular

systems. Simple chemical structures, as we know them, have no form of reliable but flexible reproduction, and thus are evolutionarily uninteresting, even if they qualify as autopoietic systems. In contrast, the phenomenology that cellular systems can generate is immense. One outstanding question in this respect is whether there is actually any way of realizing an autopoietic system with an interesting evolutionary phenomenology except through the components which constitute present living cells. But this need not concern us here.

In brief then, once the simplest self-reproducing process takes place in an autopoietic system, evolution is on its course and self-reproduction can enter a history of change, with the ensuing total displacement of any coexisting non-self-reproducing autopoietic unities. *Hence the linkage between autopoiesis and self-reproduction in terrestrial living systems.* Of course it is not possible to say now what actually took place in the origin of biological evolution. The fact is that in present-day living systems self-reproduction is crucially associated with nucleic acids and their role in protein specification. It seems that this could not have been so if the nucleic-acid-protein association were not a condition virtually constitutive of the original autopoietic process, which was secondarily associated with reproduction and variation, as suggested by the studies of Eigen (1971) and Eigen and Schuster (1977) (figure 4.1).¹⁶

Source

(Maturana and Varela, 1975).

¹⁶Eds. Varela was probably unaware of the work by Tibor Gánti on the Chemoton, published in Hungarian in 1971, the same year as Eigen's influential paper, and translated into English in 2003. This model combines three chemically coupled cycles corresponding to metabolism, topological closure, and a template polymerization cycle that regulates growth and potentially reproduction. Models such as this one may retain the conceptual minimalism of computational autopoiesis but allow the study of phenomena involving metabolism-reproduction coupling; for instance variations on the Chemoton suggest that it may provide a plausible basis for explaining the origin of genomes (Fernando and Di Paolo, 2004; Szathmáry et al., 2005).

6 On the Consequences of Autopoiesis

6.1 Introduction

Autopoiesis in the physical space is necessary and sufficient to characterize a system as a living system.¹ Reproduction and evolution as they occur in the known living systems, and all the phenomena derived from them, arise as secondary processes subordinated to their existence and operation as autopoietic unities. Hence, the biological phenomenology is founded in the phenomenology of autopoietic systems in the physical space. For a phenomenon to be a biological phenomenon it is necessary that it depend in one way or another on the autopoiesis of one or more physical autopoietic unities. This has been the argument so far. Let us now follow some of its implications.

6.2 Biological Implications

6.2.1

We first consider autopoiesis in the physical space. A living system is a living system because it is an autopoietic system in the physical space, and it is a unity in the physical space because of its autopoiesis as a mechanism of identity. Accordingly, any structural transformation that a living system may undergo in maintaining its identity must take place in a manner determined by, and subordinated to, its defining autopoiesis; hence, in a living system loss of autopoiesis is disintegration as a unity and loss of identity—that is, death.²

The physical space is defined by components that can be determined by operations that characterize them in terms of properties such as masses, forces, accelerations, distances, fields, etc. Furthermore, such properties themselves are defined by the interactions of

¹*Eds.* With minor edits, chapter 6 reproduces several passages of chapter 5 in Maturana and Varela (1980). The qualifier “in physical space,” clarified in section 6.2.1, is meant to avoid confusion. This becomes clearer in chapter 7, where autopoiesis is interpreted as being definitionally realized in the domain of molecular processes, in contrast to broader kinds of autonomous organizations. The implications of the materiality of autopoiesis have been more explicitly explored by other authors (e.g., Fleischaker, 1988; Moreno and Ruiz-Mirazo, 1999).

²*Eds.* This and other references to disintegration and loss of identity are examples of what Beer and Di Paolo (2023) describe as “systemic fragility,” a basic form of precariousness. These references also underscore the all-or-nothing character of autopoiesis.

the components that they characterize. In the physical space thus understood, two essential kinds of phenomenology can take place according to the way the components participate in their generation, namely, statical and mechanistic (dynamic, machinelike). The statical phenomenology is a phenomenology of relations between *properties* of components; the mechanistic phenomenology is a phenomenology of relations between processes realized through the properties of components.

What about the biological phenomenology of individual living systems? That is, what about the phenomenology of autopoietic systems that takes place in the physical space? Since a living system is defined as a system by the concatenation of processes of production of components that generate the processes that produce them and constitute the system as a unity in the physical space, biological phenomena are necessarily phenomena of relations between processes that satisfy the autopoiesis of the participant living systems. Accordingly, under no circumstances is a biological phenomenon defined by the properties of its component elements; it is always defined and constituted by a concatenation of processes in relations subordinated to the autopoiesis of at least one living system. For example, the accidental collision of two running animals, as a bodily encounter of living systems, is not a biological phenomenon (even though it may have biological consequences), but the bodily contact of two animals in courtship is.³

³*Eds.* This is a useful example amid the dense prose of this chapter. The point is to draw the contours of what makes a phenomenon specifically biological. The answer given here is a direct dependence on autopoiesis.

Strictly, then, although biological and statical phenomena are physical phenomena because they are realized through the properties of their physical components, they differ because statical phenomena are phenomena of relations between properties of components (as previously defined), while biological phenomena are phenomena of relations between processes. Therefore, biological phenomena, as phenomena of relations between processes, are a subclass of the mechanistic phenomena that constitute them, and they are defined through the participation of these processes in the realization of at least one autopoietic system. The phenomenology of living systems, then, is the mechanistic phenomenology of physical autopoietic machines.

6.2.2

We now arrive at the duality of organization and structure. As the mechanistic phenomenology of physical autopoietic machines, the biological phenomenology is perfectly well defined and, in

principle, amenable to theoretical treatment through the theory of autopoiesis. It follows that such a theory, as a formal theory, will be a theory of the concatenation of processes of production that constitute autopoietic systems, and not a theory of properties of components of living systems. This says nothing,⁴ however, of the difficulties of such a formal theory. In fact, it is apparent that we are at a stage where analytical tools for the understanding of cooperative, parallel processes are meager, as is dramatically shown in the work of Goodwin (1970; 1976), based on dynamical system modeling. These difficulties, however, may be more theoretical than practical, in view of the possibility of complementary modes of description for systems, and in particular for autopoietic systems, which we discuss later on in part II.

It also follows that a theoretical biology would be possible only as a theory of biological phenomenology, and not as the application of physical or chemical notions, which pertain to a different phenomenological domain, to the analysis of biological phenomena. In fact, it should be apparent now that any attempt to explain a biological phenomenon in statical or non-autopoietic mechanistic terms would be an attempt to reformulate it in terms of relations between properties of components, or relations between processes that do not produce a unity in the physical space, and hence would necessarily fail. Since a biological phenomenon takes place through the operation of components, it is always possible to abstract from it component processes that can be adequately described in statical or non-autopoietic mechanistic terms, because as abstracted processes they in fact correspond to statical or allopoietic mechanistic phenomena. In such a case, any connection between the statical or non-autopoietic mechanistic processes and the biological phenomenon from which the observer abstracts them is provided by the observer who considers both simultaneously, as we often need to do.

This is, in other words, the *duality* between *organizational* and *structural descriptions*. We seem to be unable to characterize a class or organization unless there is some way of relating such relations in some particular structure. Conversely, no specific structure can serve to account for the phenomenology it generates, unless characterized in terms of the class of organization to which it belongs. Thus we need to preserve the relation between organization and structure of a system, but at the same time not to confuse the two kinds of description, as, apparently, it is easy to do.⁵

⁴Eds. This clarification (from this point to the end of the paragraph) has been added by Varela for the current text. It indicates that autopoietic theory places a broad, systemic constraint on any formal theory of biological phenomena, which Varela seemed to envision in the light of modeling and analytical work in theoretical biology.

⁵Eds. The added gloss on the original (Maturana and Varela, 1980, 114) in this paragraph serves to clarify Varela's epistemology. It is not simply that we may be confused into thinking that the study of biological components without reference to autopoiesis is sufficient for understanding biological phenomena. It is rather that the duality between components and the whole is inescapable and demands that we be attentive to both domains without confusing them.

The biological phenomenon proper, however, is not and cannot be captured by purely structural explanations, which necessarily remain a reformulation of a phenomenon in a non-autopoietic phenomenological domain. A biological explanation must be a reformulation in terms of processes subordinated to autopoiesis.

6.2.3

An adequate theory of biological phenomena should permit the analysis of the dynamics of the concrete components of a system in order to determine whether or not they participate in processes that integrate a biological phenomenon. In fact, no matter how much we think we understand biological problems today, it is apparent that without an adequate theory of autopoiesis it will not be possible to answer questions such as: given a dynamic system, what relations should I observe between its concrete components to determine whether or not they participate in processes that make it a living system? Or: given a set of components with well-defined properties, in what processes of production can they participate so that the components can be concatenated to form an autopoietic system? The answer to these questions is essential if one wants to solve the problem of the origin of living systems on Earth.⁶ The same question must be answered if one wants to design a living system. In particular, it should be possible to determine from theoretical biological considerations which relations should be satisfied by any set of components if these are to participate in processes that constitute an autopoietic unity. Whether or not one may want to make an autopoietic system is, of course, an ethical problem. However, if our characterization of living systems is adequate, it is apparent that they could be made at will. What remains to be seen is whether such a system has already been made by humans although unwittingly, and if so, with what consequences.⁷

⁶*Eds.* See Fleischaker (1988) for an early attempt at breaking autopoiesis into a series of minimal criteria.

⁷*Eds.* As noted in chapter 2, an important branch of current work in synthetic biology is concerned with the modeling and synthesis of protocells (Luisi et al., 2006; Stano and Mavelli, 2015; Solé et al., 2007).

Finally, the characterization of living systems as physical autopoietic systems must be understood as having universal value, that is, autopoiesis in the physical space must be viewed as defining living systems anywhere in the universe, however different they may otherwise be from terrestrial ones. This is not to be considered as a limitation of our imagination, nor as a denial that there might exist still unimagined complex systems. It is a statement about the nature of the biological phenomenology: The biological phenomenology is neither more nor less than the phenomenology of autopoietic systems in the physical space.

6.3 Epistemological Consequences

6.3.1

The basic epistemological question in the domain of the biological problems is that which refers to the validity of the statements made about biological systems. It is now obvious that scientific statements made about the universe acquire their validity through their operative effectiveness in their application in their purported domain. Yet any observation, even one that permits us to recognize the operational validity of a scientific statement, implies an epistemology: a body of explicit or implicit conceptual notions that determines the perspective of the observations and, hence, what can and what cannot be observed, what can and what cannot be validated by its operative effectiveness, and what can and what cannot be explained by a given body of theoretical concepts.

This has been a fundamental problem in the conceptual and experimental handling of the biological phenomena, as is apparent in the history of biology, which reveals a continuous search for the definition of the biological phenomenology in a manner such that would permit its complete explanation through well-defined notions and, accordingly, its complete validation in the observational domain. In this respect, evolutionary and genetic notions have been so far the most successful.

Yet these notions alone are insufficient because, although they provide a mechanism for historical change, they do not adequately define the basis of biological phenomenology. In fact, evolutionary and genetic notions (by emphasizing generational change) treat the species as the source of all biological order, showing that the species evolves while the individuals are transient components whose organization is subordinated to its historical phenomenology. However, since the species is, concretely at any moment, a collection of individuals capable in principle of interbreeding, it turns out that what would define the organization of individuals is either an abstraction, or something that requires the existence of well-defined individuals to begin with. Where does the organization of the individual come from? What is the mechanism for its determination?

This difficulty cannot be solved on purely evolutionary and genetic arguments, since it is apparent to everyone (including evolutionists and geneticists) that any attempt to overcome it by resorting to other, comprehensive notions is doomed to failure if they do not provide us with a mechanism to account for the phenomenology

⁸*Eds.* Biological individuality, here equated with the concept of organism, cannot be explained in terms of genetic or evolutionary processes that assume it. This is one of the main drives behind the theory of autopoiesis. Biological individuality remains a complex and controversial topic, in part because it challenges well-established patterns of hylomorphic thinking, and in part also because of the sometimes surprising complexity of biological phenomena. For two philosophical treatments of individuality that are relevant to Varela's work, see Jonas (1968) and Simondon (2020). For a contemporary account of concepts of biological individuality, see Lidgard and Nyhart (2017).

⁹*Eds.* At this point, Varela omits the claim that the biological domain is fully defined and self-contained, not requiring additional notions (Maturana and Varela, 1980, 116). This assertion was probably too strong for Varela, as can be seen from his more nuanced epistemology later in this book (chapters 8, 9, and 16), as well as in practically all his subsequent work.

of the individual. Such is the case when some sort of preformism is introduced by applying informational notions at the molecular level (nucleic acids or proteins); or when organismic notions are used that emphasize the unitary character of living systems but do not provide a mechanism for the definition of the individual. These notions fail because they imply the validity of the same notion that they are supposed to explain.⁸

As is apparent from all that has been said, the key to the understanding of biological phenomenology is the understanding of the organization of the individual. We have claimed that this organization is the autopoietic organization. Furthermore, we have shown that this organization and its origin are fully explainable with purely mechanistic notions that are valid for any mechanistic phenomenon in any space, and that once the autopoietic organization is established, it determines an independent phenomenological subdomain of the mechanistic phenomenology: the domain of the biological phenomena.⁹

The development of the Darwinian notion of evolution, with its emphasis on the species, natural selection, and fitness, had an impact in human affairs that went beyond the explanation of diversity and its origin in living systems. It had sociological significance because it seemed to offer an explanation of the social phenomenology in a competitive society, as well as a scientific justification for the subordination of the destiny of the individuals to the transcendental values supposedly embodied in notions such as humankind, the state, and society. In fact, the social history of human beings shows a continuous search for values that explain or justify human existence, as well as a continuous use of transcendental notions to justify social discrimination, slavery, economic subordination, and political submission of persons, individually or collectively, to the design or whim of those who pretend to represent the values contained in those notions. For a society based on economic discrimination, competitive ideas of power, and subordination of the citizen to the state, the notions of evolution, natural selection, and fitness (with their emphasis on the species as the perduring historical entity maintained through the dispensability of transient individuals) seemed to provide a biological (scientific) justification for its economic and social structure. It is known on biological grounds that what evolves is humanity as the species *Homo sapiens*. It is also known on biological grounds that competition participates in the specification of evolutionary change even in humans. It is true that under the laws of natural selection the individuals most apt in the features which

are favorably selected survive, or have reproductive advantages over the others, and that the others do not contribute or contribute less to the historical destiny of the species. Thus, from the Darwinian perspective it seemed that the role of the individual was to contribute to the perpetuation of the species, and that all that one had to do for the well-being of humankind was to let the natural phenomena follow their course. Science, biology, appeared as justifying the notion “anything for the benefit of humanity.”

We have shown, however, that these arguments are not valid in justifying the subordination of the individual to the species, because biological phenomenology is based on the autonomy of the individuals, and without individuals there is no biological phenomenology whatsoever.¹⁰ The organization of the individual is autopoietic, and upon this fact rests all its significance: it becomes defined through its existing, and its existing is autopoietic.

Thus in the realm of biology we see reflected the ethical and, ultimately, political choice of leaving out the view of the autonomy of things, whether animals or humans. The understanding of life becomes a mirror of our epistemological choices, which carry over to human actions.

6.3.2

A *phenomenological domain* is defined by the properties of the unity or unities that constitute it, either singly or collectively through their transformations or interactions. Thus, *whenever a unity is defined, or a class or classes of unities are established that can undergo transformations or interactions, a phenomenological domain is defined.*¹¹

Two phenomenological domains intersect only to the extent that they have common generative unities, that is, only to the extent that the unities that specify them interact; otherwise they are completely independent, and obviously they cannot generate each other without transgressing the domains of relations of their respective specifications. Conversely, one phenomenological domain can generate unities that define a different phenomenological domain, but the new domain is specified by the properties of the new unities, not by the phenomenology that generates them. If this were not the case, the new unities would not in fact be different unities, but would be unities of the same class that generated the parental phenomenological domain; and they would generate a phenomenological domain identical to it.

Autopoietic systems do generate different phenomenological domains by generating unities whose properties are different from

¹⁰ *Eds.* There are many other reasons why such ideological extrapolations of evolutionary theory are invalid, from suspect use of metaphors to plain poor science; see, for example, Levins and Lewontin (1987) and Lewontin and Levins (2007).

¹¹ *Eds.* A phenomenological domain describes the phenomena (events, processes, changes, relations) that pertain to a particular system as a whole. Thus, a car may move fast down a road, may be stuck in traffic, may be washed or painted, etc. These are phenomena that belong to the car's domain. The workings of a car's components, however, belong to the domain of interactions among parts, which is a different domain. For instance, an oil leak causes a breakdown that leads to engine failure. But it does not determine what happens to the car (e.g., whether it may be brought safely to a stop). These are phenomena in the car's domain and will depend on relations within this domain (its speed, the road and weather conditions, etc.). The car's speed is not found in its engine.

those of the unities that generate them. These new phenomenological domains are subordinated to the phenomenology of the autopoietic unities because they depend on them for their actual realization; but they are not determined by them: they are only determined by the properties of their originating unities, regardless of how these were originated. One phenomenological domain cannot be explained by relations that are valid for another domain; this is a general statement, which applies also to the different phenomenological domains generated through the operation of autopoietic systems. Accordingly, as an autopoietic system cannot be explained through statical or non-autopoietic mechanistic relations in the space in which it exists, but must be explained through autopoietic mechanistic relations in the mechanistic domain, so the phenomena generated through interactions of autopoietic unities must be explained in the domain of interactions of the autopoietic unities through the relations that define that domain.

6.3.3

The domain of *interactions* of an autopoietic unity is the domain of all the deformations that it may undergo without loss of autopoiesis. Such a domain is determined for each unity by the particular mode through which its autopoiesis is realized in the space of its components, that is, by its structural coupling. It follows that the domain of interactions of an autopoietic unity is necessarily bounded, and that autopoietic unities with different structures have different domains of interactions.¹² Furthermore, an *observer* can consider the way in which an autopoietic system compensates its deformations as a *description* of the deforming agent that he sees acting upon it, and the deformation suffered by the system as a representation of the deforming agent. However, since the domain of interactions of an autopoietic system is bounded, an observer of an autopoietic system can describe entities external to it (by interacting with them) that the system cannot describe, either because it cannot interact with them or because it cannot compensate the deformations which these cause.

The domain of all the interactions an autopoietic system can enter into without loss of identity is its *cognitive domain*; or, in other words, the cognitive domain of an autopoietic system is the domain of all the descriptions that it can possibly make.¹³ Accordingly, for any autopoietic system its particular mode of autopoiesis determines its cognitive domain and hence its behavioral diversity,

¹²Eds. The domain of interactions is bounded by the viability and structure of the autopoietic system. If an interaction provokes a fatal breakdown, this defines a point on the boundary of the domain of interactions. If, given its current structure, an interaction is out of the reach of the autopoietic system, say, because of its limited sensory capacity, it also lies outside the bounds of the domain of interaction. However, none of this means that there may not be countless different interactions within this domain.

¹³Eds. The term *cognitive* may sound dissonant, as with the use of the term *behavior* in section 15.1.2. Both cases may seem too broadly applied. Later work on enaction attempts to specify behavior and cognition as activities involving the bringing forth of significance or relevance, often described as sense-making. What is conserved in this view is the dependence of sense-making on the structure of the sense-maker, as well as on the environment.

and it follows that the cognitive domain of an autopoietic system changes along with its ontogeny and structural coupling.

We shall explore later in this book (part III) the implications that the proper characterization of autonomy has within the domain of cognition. However, we anticipate here a few of these implications, in the light of the dependence of the cognitive domain upon the autopoietic organization of the individual.

The cognitive domain of any autopoietic system is necessarily relative to the particular way in which its autopoiesis is realized. Also, if knowledge is, in some suitable sense, *descriptive conduct*, then knowledge is relative to the cognitive domain of the knower. Therefore, if the way in which the autopoiesis of an organism is realized changes during its ontogeny, the actual knowledge of the organism (its descriptive repertoire) also changes: knowledge, then, is necessarily always a reflection of ontogeny of the knower, because ontogeny as a process of continuous structural change without loss of autopoiesis is a process of continuous specification of the behavioral capacity of the organism,¹⁴ and hence of its actual domain of interactions. Intrinsically, then, no “absolute” knowledge is possible, and the validation of all possible relative knowledge is attained through successful autopoiesis or *viability*.

¹⁴*Eds.* Or, rather, continuous change of the behavioral capacity, since this may lose specificity when skills degrade due to illness, injury, or aging, or gain specificity when new skills are acquired.

6.3.4

Autopoietic systems may interact with each other under conditions that result in structural (behavioral) coupling. In this coupling, the autopoietic conduct of an organism *A* becomes a source of deformation for an organism *B*, and the compensatory behavior of organism *B* acts, in turn, as a source of deformation of organism *A*, whose compensatory behavior acts again as a source of deformation of *B*, and so on recursively until the coupling is interrupted. In this manner, a chain of interlocked interactions develops. In each interaction the conduct of each organism is constitutively independent in its generation of the conduct of the other, because it is internally determined by the structure of the behaving organism only; but it is for the other organism, while the chain lasts, a source of compensable deformations that can be described as meaningful in the context of the coupled behavior. *These are communicative interactions*. If the coupled organisms are capable of plastic behavior that results in their respective structures becoming permanently modified through the communicative interactions, then their corresponding series of structural changes (which would arise in the context of their

¹⁵ *Eds.* As previously noted, these descriptors (*descriptive*, *consensual*, *communicative*, and certainly *linguistic*) may seem too broadly applied here. We need further specifications to bring these terms closer to their more widespread meanings. This is actually made possible by the concept of *autonomy* that Varela introduces and explores in the rest of the book. In this way, social interaction and other forms of social agency, such as languaging, may be defined beyond mere couplings or recursive history of couplings between autonomous agents (De Jaegher and Di Paolo, 2007; Di Paolo et al., 2018).

¹⁶ *Eds.* The terminology of *non-intersecting domains* is abandoned by Varela from this point on. The dependence of a domain on the unity that generates it entails that what happens in one domain cannot fully *determine* what happens in another one. It does not mean that it can have no influence (e.g., see section 10.1.1). Models of constitutive and interactive aspects of autonomy illuminate this relation of dependence without full determination (Beer, 2020a). There are relevant notions of *intersection* that we may wish to conserve in order to account for regularities among different phenomena (e.g., material overlap; the engine of this car belongs to the domain of interaction among components, not to the domain of the car as a unity, but it is still the engine of *this* car). The problem of not doing this is the risk of a kind of extreme internalism verging on solipsism (Varela, 2011, 614); see also Di Paolo (2009).

coupled deformations without loss of autopoiesis) will constitute two historically interlocked ontogenies that generate an interlocked consensual domain of behavior, which becomes specified during its process of generation. Such a *consensual* domain of communicative interactions, in which the behaviorally coupled organisms orient each other with modes of behavior whose internal determination has become specified during their coupled ontogenies, is a *linguistic* domain.¹⁵

In such a consensual domain of interactions the conduct of each organism may be treated by an observer as constituting a connotative description of the conduct of the other, or, in her domain of description as an observer, as a consensual denotation of it. Thus, communicative and linguistic interactions are intrinsically not informative; organism *A* does not and cannot determine the conduct of organism *B*, because due to the nature of the autopoietic organization itself, every change that an organism undergoes is necessarily and unavoidably determined by its own organization. A linguistic domain, then, as a consensual domain that arises from the coupling of the ontogenies of otherwise independent autopoietic systems, is intrinsically noninformative, even though an observer, by neglecting the internal determination of the autopoietic systems that generate it, may describe it *as if it were so*. Phenomenologically, the linguistic domain and the domain of autopoiesis are different, and although one generates the elements of the other, they do not intersect.¹⁶

Sources

(Maturana and Varela, 1975), (Maturana, 1975).

7

The Idea of Organizational Closure

7.1 Higher-Order Autopoietic Systems

7.1.1

Whenever the conduct of two or more unities is such that there is a domain in which the conduct of each one is a function of the conduct of the others, it is said that they are coupled in that domain. Coupling arises as a result of the mutual modifications that interacting unities undergo in the course of their interactions without loss of identity. If the identity of the interacting unities is lost in the course of their interactions, a new unity may be generated as a result of it, but no coupling takes place. In general, however, coupling leads also to the generation of a new unity that may exist in a *different domain* in which the component coupled unities retain their identity. The way in which this takes place, as well as the domain in which the new unity is realized, depends on the properties of the component unities.

7.1.2

Coupling in living systems is a frequent occurrence, and the nature of the coupling of living systems is determined by their autopoietic organization. This is so because autopoietic systems can interact with each other without loss of identity as long as their respective paths of autopoiesis constitute reciprocal sources of compensable perturbation. Furthermore, due to their organization, autopoietic systems can couple and constitute a new unity while their individual paths of autopoiesis become reciprocal sources of specification of each other's environment, if their reciprocal perturbations do not overstep their corresponding ranges of tolerance for variation without loss of autopoiesis. As a consequence, the coupling remains

invariant, while the coupled systems undergo structural changes that are generated through the coupling and hence commensurate with it. These considerations also apply to the coupling of autopoietic and non-autopoietic unities, with obvious modifications in relation to the retention of identity of the latter. In general, then, the coupling of autopoietic systems with other unities, autopoietic or not, is realized through their autopoiesis. That coupling may facilitate autopoiesis requires no further discussion, and that this facilitation may take place through the particular way in which the autopoiesis of the coupled unities is realized has already been said. It follows that selection for coupling is possible, and that through evolution under a selective pressure for coupling a composite system can be developed (evolved) in which the individual autopoiesis of every one of its autopoietic components is subordinated to an environment defined through the autopoiesis of all the other autopoietic components of the composite unity. Such a composite system will necessarily be defined as a unity by the coupling relations of its component autopoietic systems in a space that the nature of the coupling specifies, and will remain a unity as long as the component systems retain their autopoiesis, which allows them to enter into those coupling relations.

7.1.3

A system generated through the coupling of autopoietic unities may, on a first approximation, be seen by an observer as autopoietic to the extent that its realization depends on the autopoiesis of the unities that integrate it. Yet, if such a system is not defined by relations of production of components that generate these relations and define it as a unity in a given space, but by other relations (either between components or between processes), then it is *not* an autopoietic system, and the observer is mistaken. The apparent autopoiesis of such a system is incidental to the autopoiesis of the coupled unities that constitute it, and not intrinsic to its organization; the mistake of the observer, therefore, lies in the fact that he sees the system of coupled autopoietic unities as a unity in his perceptual domain in other terms than those defined by its organization.¹

Contrariwise, if a system is realized through the coupling of autopoietic unities and is defined by relations of production of components that generate these relations and constitute it as a unity in some space, then it is an autopoietic system in that space, regardless of whether the components produced coincide with the unities that generate it through their coupled autopoiesis. If the autopoietic

¹*Eds.* An example would be a *biofilm*. Microbial cells adhere to a surface and produce an extracellular matrix composed of polymers, proteins, and lipids. This becomes a complex habitat for further growth and colonization, allowing communication via quorum sensing and horizontal gene transfer and resulting in added resistance to antibiotics. Biofilms are described as an emergent form of communal bacterial life (Flemming et al., 2016). Illustrating the point about the observer's perspective, some argue that it is possible to look at biofilm in different ways, including as nascent multicellular organisms (Penesyan et al., 2021).

system thus generated is a unity in the physical space, it is a living system. If the autopoiesis of an autopoietic system entails the autopoiesis of the coupled autopoietic unities that realize it, then it is called an autopoietic system of *higher order*.²

7.1.4

An autopoietic system can become a component of another system if some aspects of its path of autopoietic change can participate in the realization of this other system. As has been said, this can take place in the present through a coupling that makes use of the homeorhetic resorts of the interacting systems, or through evolution by the recursive effect of a maintained selective pressure on the course of transformation of a reproductive historical network that results in a subordination of the individual component autopoiesis (through historical change in the way these are realized) to the environment of reciprocal perturbations that they specify. Whichever is the case, an observer can describe an autopoietic component of a composite system as playing an *allopoietic role* in the realization of the larger system that it contributes to realizing through its autopoiesis. In other words, the autopoietic unity functions in the context of the composite system in a manner that the observer would describe as allopoietic.

Thus this allopoietic function is a feature of an alternative description by the observer, who changes the domain of description (from internal causal relations to external constraints) and the level of the system under consideration (from the autopoietic system as a unit, to the system plus its environment as a unit). To confuse these two forms of description would obscure both the mode in which an autopoietic unity becomes one, and the mode in which it can constitute a higher-order unity. The proper presentation of this feature of observation is through the duality of autonomy and control in the observer's cognition.

7.1.5

If the autopoiesis of the component unities of a composite autopoietic system conform to allopoietic roles that through the production of relations of constitution, specification, and order define an autopoietic unit, then the composite system becomes in its own right an autopoietic unity of second order. This has actually happened on Earth with the evolution of the multicellular pattern of organizations.³ When this occurs, the component (living) autopoietic systems necessarily become subordinated, in the way

²*Eds.* A relevant case concerns the idea of *symbiogenesis* to explain the evolutionary origins of eukaryotes. This was first suggested by Konstantin Mereschkowski in 1905 (Martin and Kowallik, 1999), who described plastids (cell organelles) in plant cells as originally endosymbiotic cyanobacteria. The evolutionary implications of this idea were developed by Boris Kozo-Polyansky (2010) in the 1920s and further advanced in the 1960s by Lynn Margulis (1981). Drawing on autopoietic theory, Margulis proposed the term *holobiont* to refer to a unified system comprising symbiont and host (see O'Malley, 2017). This would be a candidate for a higher-order autopoietic system.

³*Eds.* Maturana and Varela (1987) describe multicellular organisms as second-order autopoietic systems. Varela's later works (1991b, 88–89; Weber and Varela 2002, 115) suggest that multicellulars are constituted as *autonomous systems* whose identity depends on the establishment of a somatic ecology that regulates interlevel conflict (Buss, 1987). Immune networks and the nervous system provide such systemwide regulation and contribute to the organism's individuality. So do local-global interactions between cells and the extracellular matrix (Varela and Frenk, 1987; Bich et al., 2019). Multicellularity remains challenging because various (potentially) autonomous levels are involved in regulating the mechanisms of cell division, signaling, differentiation, and aggregation. Not all multicellulars are the same. Highly integrated ones, such as metazoans, regulate development through systemwide constraints, such as intercellular signals that modulate cell differentiation. Others do not (See Moreno and Mossio, 2015).

they realize their autopoiesis, to the constraints (maintenance) of the autopoiesis of the higher-order autopoietic unity which they, through their coupling, define topologically in the physical space. If the higher-order autopoietic system undergoes self-reproduction (through the self-reproduction of one of its component autopoietic unities or otherwise), an evolutionary process begins in which the evolution of the pattern of organization of the component autopoietic systems is necessarily subordinated to the evolution of the pattern of organization of the composite unity.

Furthermore, it is to be expected that if the proper contingencies are given, higher-order autopoietic unities will be formed through selection. In fact, if coupling arises as a way of satisfying autopoiesis, then the more stable that coupling is, the more stable will be any second-order unity formed from previous autopoietic systems. However, in an intuitive sense, a very stable condition for coupling appears if the unity organization is precisely geared to maintaining this organization—that is, if the unity becomes autopoietic. It seems, then, that there is an ever-present selective pressure for the constitution of higher-order autopoietic systems from the coupling of lower-order autopoietic unities of higher order—a pressure imposed by the circumstances under which a unity can be specified in a given space.⁴

⁴*Eds.* This is possibly a teleological slippage. If a system composed of autopoietic units also realizes an autopoietic organization, it will tend to add stability to its autopoietic components. *Once realized*, higher-order autopoietic systems will likely be evolutionarily more stable than otherwise similar configurations that are not autopoietic. But this does not mean there is a selective pressure *for* the constitution of such higher-order units—they simply may not have been realized in the first place.

7.2 Varieties of Autonomous Systems

7.2.1

Biological phenomena depend upon the autopoiesis of the individuals involved; thus, there are biological systems that arise from the coupling of autopoietic unities, some of which may even constitute autopoietic systems of higher order. What about human social systems; are they, as systems of coupled human beings, also biological systems? Or, in other words, to what extent are the relations that characterize a human society isomorphic to the autopoiesis of the individuals that integrate it?

The answer to this question is not trivial and requires considerations that, in addition to their biological significance, have ethical and political implications. This is obviously the case, because such an answer requires the *characterization of the relations that define a society as a unity* (a system), and whatever we may say biologically will apply in the domain of human interactions directly, either by use or abuse, as we saw with evolutionary notions. In fact, no position or view that has any relevance in the domain of human

relations can be deemed free from ethical and political implications, nor can a scientist consider himself alien to these implications.⁵

The difficulties of characterizing the defining relations and the extent of the implications of such characterizations extend to many kinds of unities that are part of, or close to, human life, such as families, ecosystems, economies, managerial complexes, nations, clubs—in brief, every natural system. As in the case of living systems, what is apparent is a *degree of autonomy* in the way such unities are present in our experience. They have defined a domain or space in which they exist (usually not the physical space), and they have components that integrate them and relations among these components such that the unity attains coherence and can be distinguished through the interdependence of components. How are we to deal with this variety of autonomous systems?

⁵*Eds.* Some of these themes are further discussed in chapter 16. See also Varela (1999a), Protevi (2009) and Di Paolo and De Jaegher (2022).

7.2.2

In general, the actual recognition of an autopoietic system poses a cognitive problem that has to do both with the capacity of the observer to recognize the *relations* that define the system as a unity, and with his capacity to distinguish the *boundaries* that delimit this unity in the space in which it is realized (his criteria of distinction). Since it is a defining feature of an autopoietic system that it should specify its own boundaries, a proper recognition of an autopoietic system as a unity requires that the observer perform an operation of distinction that defines the limits of the system in the same domain in which it specifies them through its autopoiesis. If this is not the case, he does not *observe* the autopoietic system as a unity, even though he may *conceive* it. Thus, in the present case, the recognition of a cell as a molecular autopoietic unity offers no serious difficulty, because we can identify the autopoietic nature of its organization, and can interact visually, mechanically, and chemically with one of the boundaries (membrane) that its autopoiesis generates as an interface to delimit it as a three-dimensional physical unity.

7.2.3

What other autonomous systems have in common with living systems is that in them too, the proper recognition of the unity is intimately tied to, and occurs in the *same* space specified by, the unity's organization and operation. This is precisely what autonomy connotes: assertion of the system's identity through its functioning in such a way that observation proceeds through the coupling

between the observer and the unit in the domain in which the unity's operation occurs.

What is unsatisfactory about autopoiesis for the characterization of other unities mentioned above is also apparent from this very description. The relations that characterize autopoiesis are relations of *productions* of components. Further, this idea of component production has, as its fundamental referent, chemical production.⁶ Given this notion of production of components, it follows that the cases of autopoiesis we can actually exhibit, such as living systems or model cases like the one described in chapter 3, have as a criterion of distinction a topological boundary, and the processes that define them occur in a physical-like space, actual or simulated in a computer.

Thus the idea of autopoiesis is, by definition, restricted to relations of productions of some kind, and refers to topological boundaries. These two conditions are clearly unsatisfactory for other systems exhibiting autonomy. Consider for example an animal society: certainly the unity's boundaries are not topological, and it seems very farfetched to describe social interactions in terms of "production" of components. Certainly these are not the kinds of dimensions used by, say, the entomologist studying insect societies. Similarly, there have been some proposals suggesting that certain human systems, such as an institution, should be understood as autopoietic (Beer, 1975b; Zeleny and Pierre, 1976; Zeleny, 1977). From what I said above, I believe that these characterizations are category mistakes: they confuse autopoiesis with autonomy. I am saying, in other words, that we can take the lessons offered by the autonomy of living systems and convert them into an operational characterization of *autonomy in general*, living and otherwise.

7.2.4

Autonomous systems are mechanistic (dynamic) systems defined as a unity by their organization. *We shall say that autonomous systems are organizationally closed.*⁷ *That is, their organization is characterized by processes such that (1) the processes are related as a network, so that they recursively depend on each other in the generation and realization of the processes themselves, and (2) they constitute the system as a unity recognizable in the space (domain) in which the processes exist.*

Several comments are in order:

1. The processes that specify a closed organization may be of any kind and occur in any space defined by the properties of the

⁶Eds. Here, Varela is setting the stage for introducing the idea of *autonomy* as a generalization of *autopoiesis*. One aspect of this generalization is that autonomy will not only be concerned with biomolecular processes of production of components, but also with more general kinds of processes as well, such as patterns of electrochemical neural activity (chapter 15). We should note that the term *production* is flexible, and it is to some degree justifiable to conceive of autonomous systems as generating and sustaining their own identity, and in this sense, *producing* themselves in the space in which they exist. Autonomous neural patterns exist in a "space" of relational electrochemical activity, which is not identical to, nor less physical than, the three-dimensional space where neural tissue is biochemically produced. Hence the further clarification: autopoiesis concerns *biomolecular* processes behind the production of components.

⁷Eds. Throughout this book, Varela uses *organizational* closure. In later writings, he switched to *operational* closure (Maturana and Varela, 1987; Bourguine and Varela, 1992), which is more commonly found in recent literature. See Fleischaker (1988, 38). The two terms are generally deemed synonymous, although Thompson suggests a nuanced distinction: "*Organizational closure* refers to the self-referential (circular and recursive) network of relations that defines the system as a unity, and *operational* closure to the reentrant and recurrent dynamics of such a system", (2007, 45).

components that constitute the processes. Instances of such processes are production of components, descriptions of events, rearrangements of elements, and in general, computations of any kind, whether natural or artificial. In this sense, whenever the processes are defined and their specificity is introduced in the characterization of organizational closure, a particular class of unities is defined. Specifically, if we consider processes of production of components, which occur in the physical space, organizational closure is identical with autopoiesis.

2. The processes that participate in systems may combine and relate in many possible forms. Organizational closure is but *one* form, which arises through the circular concatenation of processes to constitute an interdependent network. Once this circularity arises, the processes constitute a self-computing organization, which attains coherence through its own operation, and not through the intervention of contingencies from the environment. Thus the unity's boundaries, in whichever space the processes exist, is indissolubly linked to the operation of the system. If the organization's closure is disrupted, the unity disappears. This is characteristic of autonomous systems.
3. We can interact with and recognize an autonomous system because there is a criterion for distinguishing it in some space. However, if such a distinction is, at closer inspection, not associated with the system's operation, then either the unity is not organizationally closed, or else the observer is describing it in a dimension that is not the one in which the organizational processes occur. Only when organization and distinction are linked do we have an autonomous system, and this can only occur through organizational closure.⁸
4. In a sense, the idea of organizational closure generalizes the classical notion of stability of a system that cybernetics inherited from classical mechanics, proposed by Andronov and Pontryagin in the 1930s.⁹ This is so to the extent that one can, in this formalism, represent a system as a network of interdependent variables, whose pattern of coherence (in the stable trajectories of the phase space) affords a criterion of distinction (the variables are assumed to be observables). Many models of this sort exist in the literature, among them the hypercycle studied by Eigen and Schuster (1978).

Thus, in some instances, the stability of a dynamical system can be taken as a representation of the organizational closure of

⁸*Eds.* We may identify an apparently autonomous system by convention or as a hypothesis (e.g., an insect colony or a conversation). We refine these distinctions by asking whether they are associated with how the system operates and whether the condition of organizational closure obtains in them.

⁹*Eds.* In 1937, Aleksandr A. Andronov and Lev S. Pontryagin developed the notion of "roughness" of a dynamical system, a precursor of the idea of structural stability. See Aubin and Dalmedico (2002).

an autonomous system. But these two ideas, dynamical stability and organizational closure, are not to be confused, the former being a specific case of the latter since stability is a particular rendering of invariance. In fact, the framework of differentiable dynamics that gives rise to the notion of stability cannot accommodate a number of mechanistic systems that are of interest to us in general (such as nervous systems, conversations, and the like), because they are some levels removed from their physico-chemical underpinnings. Further, in this classical representation, the interdependence of the processes is not made explicit but remains implicit in the formalism, so that the very mechanism of autonomy is obscured. These limitations are reflected very dramatically in previous attempts to use the differentiable approach for a general treatment of autonomous, viable natural systems (e.g., Iberall, 1972). We shall return to this question of formalization of autonomy later on, in chapters 10 and 13 (see especially section 13.11.1).¹⁰

¹⁰*Eds.* There are different notions of stability not clearly distinguished here. One is dynamical stability, in the sense of a system's states evolving toward an attracting set; another is structural stability, in the sense of a system's behavior being robust to parametrical change. A priori, neither sense is equivalent to the idea of organizational closure, although autonomous systems may exhibit either kind of stability. When Varela states that stability is a particular rendering of invariance, this should *not* be read as saying that dynamically or structurally stable systems are a *subset* of organizationally closed systems. The relation between these classes is not obvious and merits further investigation.

In a very similar vein, organizational closure is close to, but distinct from, feedback, to the extent that the latter requires and implies an external source of reference, which is completely absent in organizational closure. A network of feedback loops mutually interconnected is organizationally closed, and in fact, this sort of analysis can be useful in some cases. But what we should never forget is that one of the central intentions of the study of autopoiesis and organizational closure is to describe a system with no input or outputs (which embody their control or constraints) and to emphasize their autonomous constitutions; this point of view is alien to the Wienerian idea of feedback *simpliciter* (cf. Bateson, 1977).

In the present approach, the notion of stability is generalized to that of *coherence* or *viability understood as the capacity to be distinguished in some domain*, and the representation of such coherence is generalized to any form of indefinite recursion of defining processes such that they generate the unitary character of the system.

5. In the characterization of organizational closure, nothing prevents the observer himself from being part of the process of specifying the system, not only by describing it, but by being one link in the network of processes that defines the system. This situation is peculiar in that the describer cannot step outside of the unity to consider its boundaries and environment simultaneously, but it is associated with the unit's functioning

always as a determining component. Such situations, to which most of the autonomous social systems belong, are characterized by a dynamics in which the very description of the system makes the system different. At each stage, the observer relates to the system through an understanding, which modifies her relationship to the system. This is, properly speaking, the hermeneutic circle of interpretation-action, on which all human affairs are based.

6. As in the case of autopoiesis, the organizational closure generates a unity, which in turn specifies a phenomenological domain. Thus with each organizational closed class of unities a unique phenomenology is associated. Whenever such phenomenology is extensive, in diversity and importance, a proper name is given both to the phenomenology and the kind of organizational closure, as in the case of autopoiesis and biological phenomenology. Another example is closure through linguistic interactions and the phenomenology of communication.

Furthermore, it is clear that once a unit is established through closure, it will specify a domain with which it can interact without loss of its closure or loss of identity. Such a domain is a domain of descriptive interaction relative to the environment as beheld by the observer, that is, is a *cognitive domain* for the unity. Mechanisms of identity, the generation of phenomenology, and a cognitive domain are all related notions that are grouped around the specification of an organization through closure in a given domain.¹¹

7.2.5

The role that living systems play in the characterization of organizational closure in one of *paradigmatic case*. Autopoiesis is a case of, and not synonymous with, organizational closure, and the autonomy of living systems is a case of, and not synonymous with, autonomy in general. However, because of the kind of detail we have in our knowledge of living systems, and because there are some particularly minimal cases such as the cell, the basis of autonomy is clearer in living systems, whence their exemplary character. There is a mass of experience and tradition in biology that suggests and confirms the autopoietic nature of the living organization.

Furthermore, it would seem that in all natural systems so far studied in any detail, the recursive interdependence of their processes has been revealed. To substantiate this claim, it is not possible to simply go through empirical evidence in different fields. This is so because the way in which empirical evidence is organized is,

¹¹ *Eds.* Other approaches have gravitated toward related notions of closure to define living systems. Jean Piaget (1967) described organisms as organizationally closed by being made of elements that maintain a cyclic organization by regenerating themselves through chains of mutual dependence. Robert Rosen elaborated a metabolism-repair formalism that allowed him to propose that organisms are “closed to efficient causation,” (Rosen, 1991, 244). Drawing on related ideas (e.g., Eigen, 1971; Dyson, 1982), Stuart Kauffman (1986) proposed a notion of catalytic closure in which catalyzed chemical reactions produce their own catalysts as a step toward proto-organisms. This is similar to Howard Pattee’s (1973) notion of statistical closure. Tibor Gánti’s minimal model of a living system, the Chemoton, includes a chemical “motor” whose reactions must be closed to provide dynamic stability, (Gánti, 2003, 86). Pattee (1982) also used the term *semantic closure* to describe how molecular strings in the cell act as semi-otic tokens in hierarchies of the translation and control of biochemical processes that produce the very strings and their relations in a self-referential manner (see Etzeberria and Moreno, 2001; Rocha, 2001). For critical discussions and elaborations of these ideas, see Cárdenas et al. (2010), Chemero and Turvey (2008), Juarrero (2023), and Mossio et al. (2009a). Moreno and Mossio (2015) propose that biological organization operates under a regime of closure of constraints. For discussions of varieties of closure, see the extensive special issue edited by Chandler and Van de Vijver (2000).

in itself, a function of the basic theoretical perspective one adopts. Thus our approach proceeds in the opposite direction: we will make this background of knowledge into a theoretical assumption, and then proceed to apply it to several domains and prove its validity by means of its fertility.

This basic theoretical assumption, I now make explicit in the following:

Closure Thesis

Every autonomous system is organizationally closed.

By a “Thesis” I mean here a heuristic guide, based on empirical evidence, that gives some precise meaning to an intuitive notion. In this sense is similar to Church’s Thesis in the theory of computation, where the vague notion of computability is made equivalent to that of a recursive function, because nothing that, in our culture, is consensually accepted as an effective procedure has ever been found not to be reducible to a recursive function. Similarly here, the vague notion of autonomy is made equivalent to that of organizational closure, because of our previous knowledge of autonomy of natural systems. The task is, then, to use the idea of organizational closure and its consequences to explore the phenomena of autonomy.¹²

¹²Eds. The closure thesis is Varela’s central theoretical contribution in this book. It provides a bridging principle between the theory of autopoiesis and the wider phenomena of autonomy in biological, cognitive, and social systems, which before this point was only vaguely conceived. The thesis should be read as a proposal and an invitation to find counterexamples (i.e., cases of autonomy in which we would not find organizational closure). This thesis has played a fundamental role in Varela’s later work (1991b; 1997) and in theoretical developments in the enactive approach (Di Paolo, 2005, 2009; De Jaegher and Di Paolo, 2007; Di Paolo et al., 2017, 2018; Thompson, 2007). However, to help emphasize autonomy as a constitutively active concept, a further qualification has been added: every autonomous systems is operationally closed *under precarious conditions* (Di Paolo, 2009; Di Paolo and Thompson, 2014; Beer and Di Paolo, 2023).

7.2.6

There are paramount consequences if a system exhibits organizational closure. This is so because closure and the system’s identity are interlocked, in such a way that it is a necessary consequence for an organizationally closed system to subordinate all changes to the maintenance of its identity. This we discussed extensively in relation to living systems, and again, their behavior can be taken as paradigmatic. What is seldom realized is that if we can legitimately say that, for example, a corporate structure has organizational closure, the same kind of self-maintenance of identity will carry over unchanged to this phenomenological domain. This is *not* to say that some social systems are living systems and behave as such, as has been so often stated: it means that organizational closure generates a domain of *autonomous behavior* in this unity that is livinglike, but of quite different characteristics. The practical consequences of this view of social situations are, I believe, quite dramatic, for it forces us to distinguish very clearly between the organization of, say a corporation, and the purpose that is ascribed to it. If the corporation exhibits closure, no matter what our description of the system’s purpose is, its behavior will be such that all perturbations and changes

will be subordinated to the maintenance of the system's identity. This is so even when we may treat perturbations from the environment as controlling inputs. Such controlling inputs belong to an alternative description of the system (cf. chapter 10), revealing a phenomenology that is complementary, but not reducible, to its autonomous behavior. For such systems, all apparent informational exchanges with its environment will be, and can only be, treated as perturbations within the processes that define its closure, and thus no "instructions" or "programming" can possibly exist. The observer may change her descriptions and consider the regularities between ambient perturbations and the system's regularities in compensations, but all interpretations of such regularities as information flow are relative to the system's closure and can only be understood in reference to its functioning.

It is just as well to realize, with these considerations, that this revision of control and information has ethical and political implications that are very concrete and cannot be avoided. I will *not* discuss them in this book at any length. I do want to make it clear that the idea of autonomy and its consequences are *not* restricted to biological, natural systems, but can encompass human and social systems *as well*. Here, I can only phrase the arguments for biological cases and draw the epistemological inferences. This represents, not a limitation of the applicability of the ideas, but a limitation of my ability to cover the subtleties of the extension to the social realm. Others have been more articulate about some of these implications. For a discussion on the specific ideas of autonomy and closure for socio-political systems see Bråten (1978), Alker (1976), Beer (1972; 1975a; 1975b), Schwember (1976), Burns (1976), and most especially the work of Dupuy and Robert (Dupuy and Robert, 1976; Robert, 1980), which studies the way in which control notions shape the delivery of social services.¹³ Other, more general discussions in consonance with the questions discussed here are Goffman (1974), Berger and Luckman (1966), Morin (1973; 1977), Castoriadis (1975), Flores and Winograd (1978), and Moscovici (1968; 1972).

7.2.7

The detailed discussion of autonomy of living systems, their characterization as autopoietic systems, and the generalization of the autonomy of living systems to the Closure Thesis, has set a clear agenda for the remainder of our investigation. There are two distinct themes that interpenetrate. On the one hand, there is the role and presence of the observer, who sets criteria for distinctions in different

¹³ *Eds.* Varela was cautious about the unqualified application of the idea of autopoiesis to social systems (e.g., Varela, 1989). See also Protevi (2009). This is one motivation for introducing the more encompassing idea of autonomy. Autopoiesis, as a concept, continued to be an inspiration for social scientists, notably in work on social systems by Luhmann (1996), on law by Teubner (1993), and on computer systems design and discourse by Winograd and Flores (1986); see also Bednarz (1988) and Mingers (1995) and the lively debate on the subject edited by Fleischaker (1992). The references to Beer and Schwember show Varela's interest in Project Cybersyn (Medina, 2011), an attempt at a planned socialist economy based on cybernetic principles in Chile (1971–1973). That Varela's thinking on these matters was subtly dialectical at this time is substantiated by references to Sartre (1960), used as an epigraph for part I; and Kosík (1967), indicated in the bibliography as a significant influence for this book. Varela continued to have conversations on these topics with social thinkers such as Cornelius Castoriadis and Jean-Pierre Dupuy (e.g., Castoriadis, 2011; Dupuy and Varela, 1992).

domains and is capable of alternative descriptions or different views of a system. On the other hand, there is the role of recursive, self-referential phenomena in determining a system's identity, which generates, for each class of unities, a cognitive domain. These two main themes converge and become operationally one in the cases where the describer and system's processes are the same. These topics we will consider successively in the chapters that follow.

Sources

(Maturana and Varela, 1975), (Varela and Goguen, 1978), (Varela, 1978), (Varela, 1981).

II Descriptions, Distinctions, and Circularities

Die Fehler der Beobachter entspringen aus den Eigenschaften des menschlichen Geistes. Der Mensch kann und soll seine Eigenschaften weder ablegen noch, verleugnen. Aber er kann sie bilden und ihnen eine Richtung geben. Der Mensch will immer tätig sein.

—J. W. von Goethe, *Beobachtung und Denken* (circa 1794), (von Goethe, 1893, 42).

A universe comes into being when a space is severed or taken apart ... By tracing the way we represent such a severance, we can begin to reconstruct, with an accuracy and coverage that appears almost uncanny, the basic forms underlying linguistic, mathematical, physical, and biological science, and can begin to see how the familiar laws of our own experience follow inexorably from the original act of severance.

—G. Spencer-Brown, *Laws of Form* (1969, v)

Introduction to Part II

Varela introduces the closure thesis, according to which every autonomous system is organizationally closed, in chapter 7. Autonomy generalizes the concept of autopoiesis to cases other than organic individuation. From here, we would expect a detailed analysis of examples of autonomy to follow. But this will not take place until part III, where Varela makes the case for the autonomous activity of immune networks and neural/sensorimotor dynamics, and suggests other forms of autonomy in the social domain. Before this, Varela explores epistemological issues and formal frameworks that serve to better ground the ideas surrounding the closure thesis. The themes of circularity, complementarity, self-reference, and the role of the observer are the main threads in part II and connect this part to the rest of the book.

Chapters 8 and 9 elaborate the distinction between operational and symbolic discourse already presented in part I. They reiterate the dangers of assigning operational efficacy (in the language of mechanisms, processes, dynamics, etc.) to phenomena that the observer distinguishes by relating the system as it is with its larger context, including other systems, past events, desired outcomes, etc. Examples of such symbolic phenomena include notions of semantic information, goals, functions, coding, purposes, motivations, and so forth. For Varela, it is a fallacy to ignore the distinction between operational and symbolic discourse, a message reminiscent of similar warnings such as Alfred N. Whitehead's fallacy of misplaced concreteness, Daniel C. Dennett's distinction between personal and subpersonal discourse, and other mereological fallacies particularly common in neuroscience (Bennett and Hacker, 2003).

Falling into these traps is relatively harmless when dealing with nonautonomous systems, as it is possible to assign symbolic value to inputs and outputs and to the system's responses, as well as using similar language to describe the interaction between its parts. A valid way of understanding a malfunctioning car engine may express relations between components in intentional and informational language (e.g., "the temperature sensor is sending a wrong signal"). Even though this is not how the mechanisms actually operate (sensors are not agents), the symbolic language does not do violence to the kind of system in question and is particularly useful in the context of design and engineering. (Although it is possible to miss operational aspects that are hard to describe symbolically, in the context of design, they tend to be intentionally minimized.)

In the case of autonomous systems, by contrast, the misplacement of symbolic notions within the operational domain is a sure way of contradicting their autonomy, and hence an invalid way of conceiving, modeling, or formalizing this idea. The reason for this is that autonomy is defined as the formal (though not material) closure of an organization of processes. This condition describes a network of self-referring operational relations. When we project symbolic notions onto the operation of this network, we are effectively opening it by bringing into play relations that are external to it or not contemporaneous, thus effectively undoing the postulate of autonomy.

This more or less repeats the autopoietic perspective developed in part I. But for Varela, the story doesn't end here. To assume that a proper understanding of autonomy requires only operational explanations, and that symbolic discourse is merely a matter of communicational convenience (which is what autopoietic theory seems to suggest), is to privilege one kind of cognitive engagement over a different one that has historically proved very useful. For Varela, both kinds of discourse need to engage in complementary acts of explanation and understanding (borrowing the *Erklärung-Verstehung* distinction from the social sciences). He notes that some symbolic explanations involve the abbreviation of operational processes or their linkage across scales and systems. In other words, some symbolic descriptions are deemed to be compatible with operational explanations, even when they are too complex to formulate explicitly. These are *admissible* symbolic descriptions (as a contrast, think of symbolic explanations that are difficult to reconcile with existing scientific knowledge, such as extrasensory perception). The term "admissible" still indicates a privileging of operational discourse. A fuller complementarity requires looking at how symbolic discourse drives and orients the development of scientific/mechanistic ideas according to the needs and interests of a community—an idea that Varela doesn't elaborate on much but remains latent.

Complementarities are common and reappear often and at different levels. In chapter 10, Varela explores the relation between the acts of distinction and indication (drawing a boundary and placing a mark on one side of it) and the dualities that emerge in those acts, dualities between parts and whole, system and environment, structure and organization, networks and trees, etc. Here we are all the time talking about both the observed phenomena and the way that the observer approaches their study, a duality in its own right epitomizing the participatory epistemology characteristic of

Varela's work and the later enactive approach. This is unusual in scientific discourse because the epistemic assumptions, the distinctions and indications, are often implicit (but never absent). But for Varela, these are part and parcel of the story of autonomy because the phenomena in question are not "ontologically flat," and their dualities not only admit but require perspectival shifts. Performing these shifts without enough skill, let alone without awareness, will not help us move forward.

Looking at the particular case of the network/tree duality allows Varela to express the idea of complementarity in the mathematical language of adjointness; that is, as mutually defined formal operations between structured domains (trees and nets categories). Think of adjoint operators as dual perspectives on the same question (e.g., the most uncrackable code/the most efficient code-breaking method); to think of one side, one must take the other side into consideration. This naturally leads Varela to explore a dialectical outlook on dualities, not so much conceiving of dialectics as a method but as an invitation to hold together both parts of a distinction and, more specifically, different "levels" involved in giving rise to a distinction such that complementarities may be encompassed in the very dialectical movement between vantage points. This is the reason why the study of autonomy is neither reductionist nor holistic, but rather requires a constant back-and-forth between the attitudes on both sides of this duality.

We notice that both the formal notion of adjointness and the excursus into dialectics attempt to bring into focus the relations of mutual definition between both sides of a distinction. In a sense, they are a way of bracketing a static act of *indication* or focusing on one part of a distinction. It is this act, however, and its dynamic reinterpretation that are at the center of the formalizations developed in chapters 11, 12, and 13. These chapters can be dense even for readers with mathematical training. They include large sections dedicated to theorems and proofs that serve to establish the coherence and validity of the formalism and move towards its main implications, but readers who are not particularly interested in them may safely skip them.

Chapter 11 introduces the calculus of indications developed by George Spencer-Brown in his book *Laws of Form*, a mathematical work particularly appreciated in cybernetic circles in the 1970s. Varela considered this calculus a natural way of formalizing distinctions as a basic cognitive act, as well as a tool that can be employed to arrive at fundamental insights.

The formalism becomes more directly relevant for the question of autonomy by means of an extension introduced by Varela and developed in collaborative work with Joseph Goguen and Louis Kauffman. Chapter 12 presents this extended calculus by looking at the reentry of expressions (patterns of indications) that in some way refer to themselves. (This is notably the case of the definition of autonomy as the organization of a network of processes that realizes its own organization.) Reentrant or self-referring expressions enrich the laws of form by adding a third state to the indicated (marked) or not indicated (unmarked) static states. This third state is an oscillation or waveform that does not settle into either of the two static states, very much in a spirit similar to the perspectival back-and-forth mentioned earlier. Reentry introduces dynamics in Spencer-Brown's calculus. Mathematically, extending a formalism by admitting its paradoxical or undefined states is often a way of opening up its applicability (think of complex numbers). The chapter explores the duality of patterns of indication as formal structures (e.g., a self-referential statement) and as dynamic forms (e.g., perspective shifts). Reentrant expressions often lead to infinite waveforms, which can nevertheless be tractable since the limit of finite expressions exhibiting relations of partial order.

Chapter 13 elaborates some formal ramifications of the extended calculus. It starts by introducing the notion of eigenbehavior, the state or value that results after recursive applications of an operator and that may be a fixed point or a dynamic pattern and may also be history dependent. Varela then elaborates a formalism that will lead to the possibility of equations expressing invariants or eigenbehavior at multiple levels of distinction. He focuses on operator domains, a way of describing algebras involving operations ranked according to the number of arguments involved. Any expression in an operator domain corresponds to a tree structure, which may be infinite but again approachable through limiting cases of partially ordered finite trees (and reminiscent of the infinite tree representations in recurrent networks). This eventually leads to the possibility of equations and solutions in algebras that naturally admit indicational reentry. The ideas are exemplified in an examination of situations of double-bind in social interactions that result in dynamic eigenbehaviors. They also serve to show the limitations of differential equation approaches (the most common tool in physical sciences) for capturing self-reference.

The chapters in part II showcase the range of Varela's interests, even when nominally focusing on one central thread: the

circularities in the idea of autonomy. Varela did not continue to pursue the more formal aspects of his work. We may surmise one reason for this. The extended calculus and the algebras for self-referential processes seem to be subject to a trade-off between encompassing the expressibility of complementarities and practical applications. The more suitably they capture the nuances and complexities of self-reference, the less immediately useful they seem when attempting to model concrete biological systems. This is a trade-off that Varela seems aware of, and it has been the similar fate of other formal approaches in biology, such as Robert Rosen's (M, R) systems. Some of the limitations of dynamical formalisms are these days partially overcome with the use of computational models that are able to express conditional transitions between modes of operation or levels of description. But they demand a designer's prior knowledge of exactly how these transitions should work.

In contrast, the epistemological insights gleaned from these formalisms have remained central to Varela's thought and continued to develop in his enactive work and his concern for a circulation between science and lived experience. These insights acquire tangible concreteness in the two case studies of autonomy in part III.

8

Operational Explanations and the Dispensability of Information

8.1 Introduction

The study of autopoiesis makes it very clear that we cannot avoid putting at the center of our attention the ways in which *our* choices and cognitive properties are reflected, time and again. It would seem that the farther we move from the idealized billiard-ball world of nineteenth-century physics, the more difficult it is to contemplate one's explanations of a phenomenal domain without putting in, at the same time and at the center, the observing agent.¹

In this part we show how the study of autonomy and system's descriptions in general cannot be distinguished from a study of the describer's properties, and that the system and observer appear as an inseparable pair. Further, we develop a dualistic-complementarity approach to the descriptive properties of the observer. This we do, first, by a detailed study of a central issue that was raised in the study of living systems, namely, the question of *purpose* and *information* in the characterization of the living organization.

¹*Eds.* The role of the observer as central to any inquiry about life, mind, and experience is a constant in all of Varela's work. It takes the shape of an explicit circulation between theorizing and disciplined approaches to experience in the enactive approach (Varela et al., 1991; Varela, 1999b; Varela and Depraz, 2005; Stewart, 2010; Thompson, 2007).

8.2 Purposelessness

8.2.1

Teleology, teleonomy, and information are notions employed in discourse, pedagogical and explanatory, about living systems, and it is sometimes asserted that they are essential definitory features of their organization.

Our present aim is to show that in the light of the preceding discussion, these and other notions are unnecessary for the *definition* of the living organization, and that they belong to a descriptive domain

²*Eds.* As noted in section 2.1.1, without fully abandoning the distinction between operational and symbolic discourse introduced in this chapter, Varela's attitude toward teleology in organisms developed in his later work. Following Kant's conception of the organism and the philosophy of Hans Jonas, he sought deeper connections between the purposes and the organization of living beings than merely consigning teleology to the domain of descriptions. The goal was to do this without re-inserting purposes into the organism's organization but understanding the historical reciprocity and codefinition between organism and environment as a knotted dialectic (Varela, 1991b; Weber and Varela, 2002). See Di Paolo (2005) and Thompson (2007, 140–149, 453–454).

³*Eds.* We may wonder at what point we get to enact this capacity for descriptions without purposes ever entering into the story. In autopoietic theory, the suggestion is that we reach a realm of purposes and norms, only with language (Maturana and Varela, 1987). Nonlinguistic animals would not have this capacity. The enactive perspective rejects this hard break in life-mind continuity even when language is still considered a "game-changer" (Thompson, 2007; Di Paolo et al., 2017, 2018).

⁴*Eds.* For an account of biological function that attempts to ground the teleology and normativity of this notion on the organization of the living system, see Mossio et al. (2009b) and Moreno and Mossio (2015).

distinct from and independent of the domain in which the living system's *operations* are described.²

8.2.2

It is usually maintained that the most remarkable feature of living systems is a purposeful organization, or what is the same, the possession of an internal project or program represented and realized in and through their structural organization. Thus, ontogeny is generally considered as an integrated process of development towards an adult state, through which certain structures are attained that allow the organism to perform certain functions according to the innate project that defines it in relation to the environment. Also, phylogeny is viewed as the history of adaptive transformations through reproductive processes aimed at satisfying the project of the species, with complete subordination of the individual to this end. Furthermore, it is apparent that there are organisms that may even appear capable of specifying some purpose in advance (as in the writing of this book) and conduct all their activities towards this attainment. This element of apparent purpose (the possession of a project or program in the organization of living systems), which has been called teleonomy without implying any vitalistic connotations, is frequently considered as a necessary, if not as a sufficient, definitional feature for their characterization (e.g., Monod, 1971; Ayala, 1970).

Purpose or aims, however, are not features of the organization of any machine (allo- or autopoietic): these notions belong to the domain of our discourse *about our actions*, that is, they belong to the domain of *communicative descriptions*, and when applied to a machine, or any system independent of us, they reflect our considering the machine or system in some encompassing context.³ In general, the observer puts the machine either conceptually or concretely to some use, and thus defines a set of circumstances that cause the machine to change, following a certain path of variations in its output. The connection between these outputs, the corresponding inputs, and their relation with the context in which the observer includes them, determines what we call the aim or purpose of the machine; this aim necessarily lies in that domain of the observer that defines the context and establishes the nexuses.

Similarly, the notion of function arises in the description made by the observer of the components of a machine or system in reference to an encompassing entity, which may be the whole machine or part of it, and whose states constitute the goal that the changes in the components are to bring about.⁴

In saying that a function of P is ϕ , we must pay closer attention to the character of ϕ . It must be something like “circulation,” “support,” etc. All these notions suppose a larger, more embracing conceptual scheme: circulation *in* something, support *of* something. A functional description necessarily includes a larger context to which ϕ makes reference.

Conversely, for every structure or organization, one can point to a substructure and describe its performance in the form of a functional description. Consider:

S_1 : The function of the electron shell is to balance the nuclear charges.

S_2 : The electron shell balances the nuclear charges.

T_1 : The function of DNA is to code for proteins.

T_2 : DNA participates in the specification of proteins.

What are the differences in these sentences? We can interpret them as follows: In the case S_1 one is making reference to a perfectly defined structure, the atom, for which we already have an explicit formulation of its theory. Thus, although S_1 is comprehensible, it is totally dispensable in favor of S_2 , which is a statement that can be interpreted as a mere consequence of the total structure. For the second set, the dispensability of T_1 , in favor of T_2 , although thinkable, is more subtle. This arises because the sentences refer to a system that is included in a much larger one, the cell. One can certainly treat protein synthesis as an isolated system (i.e., in an *in vitro* experiment), but in the cell its condition as subsystem makes its functional description possible. Thus a functional description, when not dispensable, is symptomatic of the lack of a theory for the organization or structure of the system in which the subsystem, described in functional terms, occurs.

In general, the very common occurrence of functional descriptions in biology is in relation to the fact that normally the systems studied are subsystems of more inclusive ones. Within a given subsystem, considered isolated, functional descriptions disappear or are dispensable. Similarly, the dispensability of functional description is possible when the structure or organization of the system at large, with no possible further extensions, is given, and subsystems become consequences of the general structure, as in the case of the atom mentioned above, or of any well-defined machine. Thus we do not talk about the function of the state q_x of a Turing machine, except for pedagogical purposes.

There again, no matter how direct the causal connections may be between the changes of state of the components and the state in which they originate in the total system, the implications in terms of design alluded to by the notion of function are established by the observer and belong exclusively to his domain of description. Accordingly, since the relations implied in the notions of functions are not constitutive of the organization of an autopoietic system, they cannot be used to explain its operation.⁵

⁵*Eds.* These points rest on the possibility of achieving a purely operational explanation of living systems, which is not obvious. As Varela says, a nondispensable functional description shows our lack of a theory of the larger system in which the system of interest is embedded. But with multiple levels of complex embedding, nothing guarantees that we may ever have an operational account of the whole. Varela does not mention what conditions allow a valid functional description. Mossio et al. (2009b) suggest that one such condition is organizational differentiation, whereby the system produces varied material structures that contribute differently to self-maintenance. A related condition is near-decomposability in spatial and temporal scales, without which functional descriptions collapse (Di Paolo et al., 2018, 76). The fact that functional descriptions are so conditioned suggests they are not *just* an observer's choice for lack of knowledge.

What we will point out later, however, is that the communicative value of a functional description is not eliminated by the operational analysis of it as a particular instance. Clearly, an operational analysis will make possible and modify the precision of the functional description, but will not eliminate its communicative value simply because it does not depend on it. To regard living systems as machines is to point to their organization. To find that in the analysis of such machines functional descriptions occur frequently, and that it is not yet comfortable to dispose of them as operational devices, indicates the lack of a theory of the kind of machines living systems are. Only with such a theory will the idea of function lose its alleged operational value, and merely retain its value as a communicative tool.

8.2.3

An explanation can be characterized as a form of discourse that intends to make intelligible a phenomenal domain that has been recorded. This is the business of science. Further, when some domain is deemed explained, and thus rendered intelligible, it is so in reference to a social group of observers.

We should now distinguish between a *symbolic* (or communicative) and an *operational* (or causal) explanation. In both cases the recorded phenomena are reformulated or reproduced in conceptual terms that are deemed appropriate. The difference lies in the fact that in an operational explanation, the terms of such reformulations and the categories used are assumed to belong to the domain in which the systems that generate the phenomena operate. In a symbolic explanation, the terms of the reformulation are deemed to belong to a more encompassing context, in which the observer provides links and nexuses not supposed to operate in the domain in which the systems that generate the phenomena operate.

A characteristic feature of an operational explanation is that it proposes conceptual (or concrete) systems and components that can reproduce the recorded phenomena. This can happen through the

specification of the organization and structure of a system, as in the mechanistic framework adopted here. That is so because the organization of a machine, be it autopoietic or allopoietic, only states relations between components and rules for their interactions and transformations, in a manner that specifies the conditions of emergence of the different states of the machine, which then arise as a necessary outcome whenever such conditions occur. It follows, then, that the notions of purpose and function *have no explanatory value* in the phenomenological domain that they usually pretend to illuminate, because they do not participate as operational, causal elements in the reformulation of any of its phenomena.⁶ This does not preclude their being adequate for a communicative description. Accordingly, a prediction of a future state of a machine consists only in the accelerated realization of its succeeding states in an observer's mind, and any reference to an early state to explain a later one in functional or purposeful terms is an alternative form of his description, made in the perspective of his simultaneous mental observation of the two states, that induces in the mind of the listener an abbreviated realization of the machine. Therefore any machine, any part of one, or any process that follows a predictable course can be described by an observer as endowed with a project, a purpose, or a function, if properly handled by him with respect to an encompassing context.

Accordingly, if living systems are physical autopoietic machines, teleonomy becomes a descriptive term, which does *not* reveal any feature of their organization, but which reveals the consistency in their operation within the domain of observation. Living systems, as physical autopoietic machines, are *purposeless* systems.

8.3 Individuality

8.3.1

The elimination of the notion of teleonomy as a defining feature of living systems forces us to consider the organization of the individual as the central question for the understanding of the organization of living systems; likewise for any other autonomous systems.

In fact, a living system is specified as an individual, as a unitary element of interactions, by its autopoietic organization, which determines that any change in it should take place subordinated to its maintenance, and thus sets the boundary conditions that specify what pertains to it and what does not pertain to it in the concreteness of its realization. If the subordination of all changes in a living

⁶*Eds.* This seems at odds with the previous statement about explanations being dependent on a community of observers seeking to make a phenomenon (contextually) intelligible. Varela's point, perhaps rashly expressed here, is that by referring to phenomena outside the current operational domain of the system (e.g., what happened to the system in the past, or what happens in a wider context), a functional explanation escapes the causal discourse of the operational domain.

system to the maintenance of its autopoietic organization did not take place (directly or indirectly), it would lose that aspect of its organization that defines it as a unity, and hence it would disintegrate. Of course it is true for every unity, whatever way it is defined, that the loss of its defining organization results in its disintegration; the peculiarity of living systems, however, is that they disintegrate whenever their autopoietic organization is lost, not just that they can disintegrate. As a consequence, all change must occur in each living system without interference with its functioning as a unity in a history of structural change in which the autopoietic organization remains invariant. Thus ontogeny is both an expression of the individuality of living systems and the way through which this individuality is realized.

As a process, ontogeny, then, is the expression of the becoming of a system that at each moment is the unity in its fullness, and does not constitute a transition from an incomplete (embryonic) state to a more complete or final (adult) one.

8.3.2

The notion of development arises, like the notion of purpose, in a more encompassing context of observation, and thus belongs to another domain than that of the autopoietic organization of the living system. Similarly, the conduct of an autopoietic machine that an observer can witness is the reflection of the paths of changes that it undergoes in the process of maintaining its organization constant through the control of the variables that can be displaced by perturbations, and through the specification in this same process of the values around which these variables are maintained at any moment. The autopoietic machine has no inputs or outputs. Therefore, if there is any correlation between regularly occurring independent events that perturb it and the state-to-state transitions that arise from these perturbations, which the observer may pretend to reveal, then this correlation pertains to the history of the machine in the context of the observation, and not to the operation of its autopoietic organization.

8.3.3

This is not to say, however, that by defining the living system in a different context, the observer may not consistently use such regularities and define a different system with inputs that control outputs through certain internal transitions, giving no consideration to the autopoietic nature of the sources of those transitions. In a sense to

be developed later, this is a natural shift of context, from the autonomy of a system to its dependence on constraints or control from the environment in which it operates. That we perform such a shift to an alternative or dual perspective is obvious, and further, it seems that it is absolutely necessary to do so. The problem lies in the inadequate distinctions between the *different* domains in which such an alternative description lies, and thus in the confused extension of explanatory terms from one domain into the other. Such a confusion occurs, for example, when it is said that an organism has a representation of the environment within itself, and this supposed representation is allocated to some structural component—e.g., a receptor molecule in the cell membrane. This is, in the light of the preceding discussion, a category mistake that arises from an inadequate appraisal of the role of the observer. Such inadequacies have led to the widespread belief that statements such as “this organism picks up the information from the environment” are meaningful in some sense. In fact, because of the category mistakes it contains, it is not only misleading, but flatly incorrect, as we shall show later on in the book in some detail for the cognitive domains of the immune and nervous systems. Thus laboring on these points and keeping good track of what terms of explanation belong to which domain is not at all a futile exercise in logic and epistemology, but a very definite need if we are to recover the usefulness of concepts such as purpose and information for natural systems (cf. chapters 9 and 15).⁷

8.3.4

Notions such as specification and order⁸ used to characterize autopoiesis in the cell are referential notions; that is, they do not have meaning outside the context in which they are defined. Thus, when we speak about relations of specification, we refer to the specification of components in the context of that which defines the system as autopoietic. Any other element of specificity that may enter, however necessary it may be for the feasibility of the components, we take for granted, so long as it is not defined through the autopoietic organization. Similarly with the notion of order. Relations of order refer to the establishment of processes that secure the presence of the components in the concatenation that results in autopoiesis. No other reference is meant, however conceivable it may be within other perspectives of description.

Similarly, notions such as coding and transmission of information do not enter into the realization of a concrete autopoietic

⁷*Eds.* This is a key message of this chapter: the need to develop a practical awareness of the difference between operational and symbolic discourse to avoid mereological fallacies. As noted in section 2.1.3, there is a parallel here, with similar distinctions between phenomena pertaining to parts and phenomena pertaining to wholes (e.g., Dennett, 1969; Bennett and Hacker, 2003).

⁸*Eds.* See chapter 4.

system, because they do not constitute causal elements in it. Thus, the notion of specificity does not imply coding, information, or instructions; it only describes certain relations, determined by and dependent on the autopoietic organization, which result in the production of the specific components. The proper dimension is that of relations of specificity. To say that the system, or part of it, codes for specificity is a misnomer. This is because such an expression represents a mapping of a process that occurs in the space of autopoiesis onto a process that occurs in the space of human design, and it is not an operational reformulation of the phenomenon. The notion of coding is a communicative notion that represents the interactions of the observer, not a phenomenon operative in the domain of autopoiesis.

⁹*Eds. Adaptivity*, a related concept introduced by Di Paolo (2005), while not part of the definition of autopoiesis, is nevertheless fully operational. It relies on dynamical systems notions of rates and tendencies (e.g., Kelso and Engstrom, 2006) and the concept of a viability set derived from autopoiesis.

The same applies to the notion of regulation.⁹ This notion is valid in the domain of symbolic explanation, and it reflects the simultaneous observation and description of interdependent transitions of the system that occur in a specified order and at specified speeds towards certain states. The corresponding dimension in an autopoietic system is that of relations of production of order, but here again, only in the context of the autopoiesis and not of any particular state of the system as it would appear projected on our domain of communication.

Sources

(Maturana and Varela, 1975), (Varela and Maturana, 1972).

9 Symbolic Explanations

9.1 Descriptive Complementarity

9.1.1

In the previous chapter we argued that the notions of information and purpose are, from the point of view of an operational explanation, dispensable. This is because the living organization could be defined without resorting to such notions, and thus the explanation underlying the living phenomena need not include them as constitutive components. Further, we argued, such notions cannot enter into the definition of a system's organization, because they pertain to the domain of discourse between observers. Information and purpose may enter for pedagogical purposes. They do not enter into an operational explanation, for which autopoiesis is complete.

However, this question needs some further development. I still hold as valid the criticism against the naive use of information and purpose as notions that can enter in the operational definition of a system, on the same basis as material interactions (e.g., in Miller's 1965 definition of living systems).¹ But there are limitations to our one-sided presentation in the last chapter, stemming from the fact that we did not take our criticism far enough to recover a non-naive and useful role of informational notions in the descriptions of the living phenomena. It criticized without a corresponding *Aufhebung*.

9.1.2

The analysis in chapter 8 was based on the assumption that operational explanations are, in some sense, *intrinsically* preferable and sufficient. This seems to me wrong in two senses that I will try to make clear in what follows: (1) it gives the operational explanation an epistemological status not compatible with the very intention of

¹Eds. James G. Miller's approach to living systems was not as conceptually developed as the theory of autopoiesis, but it certainly did not use ideas of information and purpose naively. Miller explicitly states that he refers to information in the mathematical sense of negentropy. He defines living systems as open systems maintaining a steady state of negentropy, among other empirically motivated properties. His statements about purposes are actually an attempt at naturalizing the idea based on systemic and thermodynamic concepts.

the criticism leveled against the naive use of “information”; (2) it neglects the fact that informational terms, although belonging to a different category of explanation than the operational terms used in autopoiesis, could still be used as valid explanatory terms, and that furthermore, different modes of explanation could coexist.

There is, evidently, a need to overemphasize a neglected side of a polarity. Similarly, autonomy cannot in fact be conceived without a complementary consideration of how the system is also controlled in a dual context; in particular, autopoiesis and allopoiesis are complementary rather than exclusive characterizations for a system. What I will argue now is that an operational explanation for the living phenomenology needs a complementary mode of explanation to be complete, a mode of explanation that I have referred to as *symbolic* (cf. section 8.2.3; see also, Pattee, 1977). Further discussion of this view has to begin with a very brief consideration of what is at the base of the tendency to prefer purely operational explanations and to relegate informational terms to the category of “purely pedagogical.”

9.2 Modes of Explanation

9.2.1

A preference for (purely) operational explanation is still no more than a preference. Such preferences come from a community of inquiring individuals—a scientific community—who, through the inheritance of a tradition, come to agree on certain criteria of validity relative to certain values or intentions (Törnebohm, 1976; Radnitzky, 1973). The preference for causality in the common sense of contemporary science comes from a predominantly manipulative, operational, and technological orientation present in science over the last 150 years. Given such a preference, operational explanations came to be *the* explanations: non-operational explanations hold no power for manipulation and prediction.² Prediction, in fact, is the sign of a successful explanation in this kind of philosophy, so that causality and prediction are a triumphant duo that characterizes modern science, particularly in the Anglo-Saxon world.

This triumphant duo, however, has to be looked at in historical perspective. Alongside operational explanations, another, equally outstanding tradition has existed, which asserts the validity of finalistic or teleological explanations, where the terms of explanation are not “why”s but “what for”s. These two traditional modes of

²*Eds.* Even operational explanations depend on assumed understandings within a community. The question “Why X?” implicitly means “Why X instead of Y?” for one community and “Why X instead of Z?” for another. Alan Garfinkel (1981) provides clear examples of how prediction and manipulation are purpose dependent, so some operational explanations (say, at the “right” level of granularity), will be preferable to other operational explanations (cf. section 10.2.1).

explanation are best characterized with the German words *Erklärung* and *Verstehung*, usually translated as explaining and understanding (von Wright, 1971).³

Now, the intention behind a *Verstehung*-type explanation is not manipulation, but understanding, communication of intelligible perspective in regard to a phenomenal domain. Typical examples are teleological explanations in Aristotle and the vitalist explanations of the eighteenth century. To the extent that their main orientation is to understand and communicate this understanding, such explanations are fundamentally different in orientation from the operational explanation. It is a historical fact that western science has taken a very strong stand in preferring causal explanations since the time of Galileo, and in fact, made *Verstehung*-type explanations into an enemy, to be banned from science.

From our perspective, at the end of this twentieth century, having enough distance from the age of the Enlightenment, things look rather different. As I see it, there are four major developments that have contributed to altering the preference for purely operational explanation. First, the great renovation inside physics, the model for logical empiricism, after the constitution of quantum mechanics and its variegated epistemological problems, make both naive causality and naive objectivity completely inadequate. Secondly, the rise of biological science has introduced into science the need to consider phenomena of unbounded complexity relative to physical sciences. The epitome of this development is the history of genetics through the Watson-Crick model, intertwining structural components with the apparent need for a “coding” description. Thirdly, we have the extensive development, linked to the use of biological concepts, of cybernetics and systems theory in the area of design and prescription of systems, where the notions of communication and purpose are at the core of what is not only the main subject, but many of its practical consequences such as computers and complex systems of regulation in human services. Finally, and in much more subdued form for the world of science, we have the reawakening, in the European schools of thought, of the importance of the *Verstehung*-type explanation in human affairs.⁴

All these developments since the end of the nineteenth century forces us willy-nilly to a reevaluation of what we mean by our preference for operational explanations, and in fact, what we mean when we intertwine such explanatory modes, whether talking about a computer program or an animal dance.

³*Eds.* The distinction between the two kinds of explanations originates in the second part of the nineteenth century. Wilhelm Dilthey and others advocated for a methodological independence of the social and historical sciences from the causal and law like explanations of the natural sciences. *Verstehen*-type explanations sought a nonreductive and hermeneutical understanding of social and historical phenomena in terms of their meaning. Explanations of this type were influential in the development of a nonpositivistic sociology, as in the work of Max Weber.

⁴*Eds.* These points have driven some philosophers of science toward more encompassing and pluralistic perspectives. One relevant example focusing on biology is the case of William C. Wimsatt (2007), who has looked at the ontology of complex systems and concluded that, in light of our epistemic limitations, understanding them demands adopting multiple perspectives. Different perspectives become validated by a “practice of *Verstehen*” (237); that is, by adopting ways of seeing that render events meaningful from a particular point of view within the complex system.

9.2.2

What, I submit, is essential to understand in this relationship is that both forms of explanation refer to modes of description relative to some perspective of the observer, or rather, we should say, of an inquiring community. In the operational description the fundamental assumption is that phenomena occur through a *network of nomic* (lawlike) relationships that follow one another. In the symbolic, communicative explanation the fundamental assumption is that phenomena occur through a certain order or pattern, but the fundamental focus of attention is on certain *moments* of such an order, relative to the inquiring community. Thus these modes of explanation are exclusive and contradictory only to the extent that one assumes that laws of nature are comprehensible independently of an inquiring community, or that no nomic patterns are discernible in the world.⁵

⁵*Eds.* This statement subtly prefigures the “middle path” attitude of later work on enaction and its distancing from both idealism and naive realism (Varela et al., 1991). What counts as an intelligible law depends on a community of inquirers, but this by no means entails that the world is chaotic or arbitrary.

Both of the above demands are, of course, inessential and these alternative views of a recorded phenomenon need not be contradictory. If we can provide a nomic basis to a phenomenon, an operational description, then a teleological explanation only consists of putting in parenthesis or conceptually abbreviating the intermediate steps of a chain of causal events, and concentrating on those patterns that are particularly interesting to the inquiring community. Accordingly, Pittendrich introduced the term *teleonomic* to designate those teleological explanations that assume a nomic structure in the phenomena, but choose to ignore intermediate steps in order to concentrate on certain events (Ayala, 1970). Such teleological explanations introduce finalistic terms in an explanation while assuming their dependence on some nomic network; hence the name *teleo-nomic*.

9.3 Symbolic Explanation

9.3.1

As we discussed before, the connection between an operational description (such as autopoiesis) and a finalistic description lies in the observer who establishes the nexus. Thus, we concluded, purpose plays no causal role in autopoiesis, and thus, no role in the description of the system’s organization. The same conclusion was valid for the notions of message, information, and code. What is significant in both of these classes of notions, purpose and information, is that the observer chooses to *ignore* the operative connection between classes of events, and to concentrate on the ensuing

relationships. This is an important idea, and it is insufficient to discuss it as merely a pedagogical maneuver.

This possibility of choosing to ignore intervening nomic links is at the base of all *symbolic* descriptions.⁶ What is characteristic of a symbol is that there is a distance, a somewhat arbitrary relationship, between signifier and signified. This is, of course, very immediate in human discourse: Words and their contextual meaning have such a remote and involved historical and structural mode of coupling that any effort to follow such nomic connection is hopeless. Thus in order to understand language, we do not trace the sequence of causes from the waveform in the air to the history of the brain operations, but simply take it as a fact that we *can* understand. (And precisely because we cannot make every thing reducible to causal explanation, since we live and grow inside language, human life has the openness it has.)

Thus we come to the conclusion that purpose and symbolic understanding are interrelated as a duo, that is symmetrical to the duo of operational explanation and prediction. Under the term *symbol* we are subsuming here the varieties of its forms, such as code, message, information, and so on.

9.3.2

So far we have argued that operational and symbolic descriptions do not contradict each other, since they belong to different levels of descriptions among a community of observers. Unless we keep clear in our minds that by changing modes of explanation we are also changing the kind of framework of reference we are operating in, the whole issue becomes muddled. Teleonomic-symbolic terms get reduced to operative components, as, for example, in ascribing the specification of components to only one component (DNA), which is a (typically) useless form of reductionism. These two modes of explanation are distinct, yet they can be related without reducing one to the other.

The question we want to ask now is: Do we *need* both forms of explanation? Can't we just provide operational explanations of a phenomenon and be satisfied with them? In the case at hand, these questions would amount to whether the autopoietic characterization is enough to satisfy our need to explain the whole of the phenomenology of living systems. In a sense, we have already shown that, indeed, the autopoiesis of each individual suffices to generate all of the phenomenology of the autonomy of living systems, and that, through their coupling and complexification, we can see in

⁶*Eds.* We also choose what to ignore in operational explanations. This is standard scientific practice (cf. section 10.2.2). Most problems are demarcated by a series of applicable operational relations, and the demarcation works so long as some limiting assumptions hold. A structural engineer calculating safety factors for a bridge will not be preoccupied by quantum phenomena or gravitational waves. What symbolic explanations "ignore" are not just intervening nomic links, but boundaries across operational domains, as when linking phenomena at very different timescales.

them the foundation for evolutionary and historical phenomena. Thus, in principle, all of the biological phenomena can be reduced to autopoietic mechanisms.

This, however, is reminiscent of the statement that all of the history of the universe could be predicted if we only knew all the positions and momenta of all the particles of the universe so that we could calculate their future trajectories. These kinds of assertions are, above all, epistemological. What we are saying in the case of autopoiesis is that, if we could follow all the appropriate contingencies, the biological phenomenology would unfold from the autopoietic mechanism. What is obvious, however, is that this assertion, although it points to the sufficiency of autopoiesis as an operational explanation, says nothing about whether it is *cognitively possible or satisfactory*. Let us examine this in more detail.

From the little we know of studies in the origin of living systems and protocellular systems, the mere production of a boundary through a chemical dynamics is clearly a necessary but perhaps not a sufficient condition for a precursor to cellular systems. A fundamental issue here, as pointed out by Pattee (1972), is the reliability of component specification versus the variability available for selection. Selection and evolution cannot exist without reproduction. Autopoietic systems can become reproductive systems, as we discussed in chapter 5. However, their reproduction can become evolutionarily *interesting* only if (1) the process of specification of components is reliable, so that there is continuity of structures through time, and (2) they are flexible enough to generate a variety of components for selection to operate.

Living systems actually evolved through an appropriate combination of processes of specification and constitution, paradigmatically seen in the coupling between nucleic acids and proteins. Nucleic acids fulfill an essential role in specifying the protein components of cells, which are mostly responsible for processes of constitution and order. This is neatly seen in Eigen's (1971) work on the early evolution of living systems (cf. figure 4.1), where the minimum structure capable of generating a sequence of cell-like units takes the form of a "hypercycle" (i.e., organizational closure) in which there are "informational" components (nucleic acid) and "structural components" (proteins). Of course, the "informational" molecule is in no way different from any other molecule in its process of interaction among chemical species. The reason the name "informational" comes up at all is that we can change the time scale of our observation, consider the realization of these units through several generations, and

observe the continuity and reliability of their process of specification of components in an evolutionary process. In other words, we abstract or bracket in our descriptions a number of causal or nomic steps in the actual process of specification, and thus reduce our description to a skeleton that associates a certain part of a nucleic acid with a certain protein segment. Next we observe that this kind of simplified description of an actual dynamic process is a useful one in following the sequences of reproductive steps from one generation to the other, to the extent that the dynamic process stays stable (i.e., the kinds of dynamics responsible for bonding, folding, and so on). This seems to be the origin of the idea of genetic material as the central element of study for evolution and historical processes in biology. A symbolic explanation, such as the description of some cellular components as genes, betrays the emergence of certain *coherent patterns of behavior to which we choose to pay attention*.

9.3.3

In pointing at the coherence of behavior in chemical dynamics as being the base for symbolic description, we are saying nothing about how such coherent behavior actually arises. This is not a simple question, and is one that we will not consider in great detail here. It seems that Pattee's analysis in terms of the *nonholonomic constraints* in dynamical systems is the most adequate description (cf. Pattee, 1972, 1977). For example, the three-dimensional structure of an enzyme is a dynamic process constrained by the sequence of amino acids specified by the cell. These constraints determine a shape which is peculiar to the enzyme, and are at the base of its enzymatic capacity.⁷ The abbreviation in a symbolic description works thus: A gene codes an enzymatic recognition event. The physical basis of this description is through the nonholonomic (non-integrable) constraints introduced in the physico-chemical dynamics of folding by the amino acid sequencing.

As Pattee points out, in describing a dynamics via constraints, the observer is in fact changing perspective to a more encompassing context. Pragmatically this is expressed by labeling different levels in a hierarchy of controls, from one higher level to the next lower, e.g., genome to enzyme catalysis. In other words, *given* a certain behavior, we can immediately determine a symbolic description, the constraints on which it is based, and the hierarchical level responsible for introducing these constraints.

The natural evolution of such constraints from a homogeneous background of chemical dynamics is of course the upshot of this sort

⁷Eds. A nonholonomic constraint in dynamical systems is one that cannot be expressed as an integrable relation between variables. The cushions at the borders of a billiard table are a nonholonomic constraint. They "act" upon the billiard balls on certain conditions and can be seen as regulating the states of the system without entirely determining them. (The surface of the table is also a nonholonomic constraint.) Pattee saw in this idea the physical basis for how the cellular machinery regulates itself. Because of their relative stability, some molecular structures in the cell, such as DNA templates, can act as stable, repeatable constraints to other, faster metabolic processes without reducing their dynamical configurations, leading to a mutual coupling between structure and dynamics. See also Moreno and Mossio (2015), Juarrero (2023), and Thompson (2007, 54–57).

of analysis. We refer the reader to the discussions in (Pattee, 1972, 1977) and (Eigen, 1971).

9.3.4

Note that, in switching from one mode of description (the processes determining the autopoiesis of an individual) to a symbolic description (dealing with the evolutionary sequences of autopoietic structures), we perform a leap in time that betrays a radical change in perspective. What we rediscover is the classical duality between physiological time and evolutionary time. Both seem necessary if we are to have a satisfactory explanation of the phenomenology of living systems. If we do not accept the change from a causal description, then the actual handling of evolutionary phenomena, which depends on questions of reliability and reproduction, becomes literally impossible to comprehend. How are we to conceive and think at all about sequences of autopoietic units in purely operational terms, where all of the components participate in the specification of the unit, if we can hardly do so for a single individual? We must reduce our explanation to a symbol-like explanation embodied in the idea of genome, and proceed from that form of noncausal, symbolic description to cover the evolutionary phenomena by adding perhaps *new* causal notions such as natural selection in this symbolic domain.⁸ In other words, it is true that all historical and evolutionary phenomena are ultimately reducible to the coupling of autopoietic units to their environments. However, this is so on purely logical grounds. For the cognitive capabilities of the observer-community, purely operational explanations are in no position to satisfy the degree of detail we need for ontogenetic and phylogenetic explanations, and a change in explanatory mode is mandatory (see also Locker and Coulter, 1977). Thus, autopoiesis is, on logical grounds, necessary and sufficient to characterize living *systems*, as claimed before. What is incomplete here is that autopoiesis, though necessary, is *not* sufficient to give a satisfactory explanation of living *phenomena* on both logical *and* cognitive grounds.⁹

⁸*Eds.* If “new causal notions” become apparent in a symbolic domain, things may start to look like a *different* operational domain. The tension is with the idea, sometimes implicitly suggested here, that only one domain counts as operational. Pluralistic perspectives see the universe as “dappled,” forming a patchwork of lawful relations (e.g., Cartwright, 1999).

⁹*Eds.* Again, this may be the case because symbolic explanations bring new biological phenomena to the fore, as in correlations between selective pressures, ecological niches, and organic traits. Or it may also be the case because some biological phenomena, such as illness, fatigue, and aging, require other operational concepts besides autopoiesis, such as adaptivity and precariousness.

9.4 Complementary Explanations

9.4.1

To say that teleonomic-symbolic explanations are not really necessary is to succumb to a prejudice of our historical tradition that it is time to revise, because in actual practice we cannot do without both

operational and symbolic explanations. A preference for operational explanations seems to be rooted in the understanding that causes are “out there” and reflect a state of affairs independent of the describer. This is, by the very argument used here, untenable. Causes and “laws of nature” are modes of descriptions adopted by inquiring communities for some intentional purpose (such as manipulation and prediction), and they specify modes of agreement and thus of coupling with the environment. However, ultimately, an operational description is no more and no less than a mode of agreement within an inquiring community, and in no way has an intrinsically superior status to a symbolic explanation. They just have different consequences: a symbolic explanation generates a form of agreement in the inquiring community, and thus a coupling with the environment, that is not so dramatically visible in manipulation, but is more visible in more diffuse modes of relationships. A good example is the form in which Darwinian thought has modified our entire view of human affairs: not through manipulation, but through the form of agreement about issues that are central to our image, such as origin and descent. It is very unfortunate that operational explanations are normally identified with explanations *simpliciter*. Both modes of explanation are, ultimately, in the domain of discourse of the observer-community, and their only difference lies in the mode in which they generate agreement. It seems to me that there are tremendous advantages to maintaining this duality of explanations in full view. By staying with the purely operational descriptions, we are forced to use other descriptive modes in a rather sloppy and careless way, such as is typical in molecular biology. This kind of attitude is a remnant of the epoch of logical positivism with its insistence of *methodological monism*.

9.4.2

At the other extreme, the vitalist attitude, and more importantly the computer-gestalt attitude, which takes information as “stuff,” are equally misguided. The latter attitude is interesting, for it has taken the same kind of methodological flavor implicit in operational descriptions, and applied it to a domain where it simply does not work. This is typical in computer science and systems engineering, where information and information processing are in the same category slot as matter and energy. This attitude has its roots in the fact that systems ideas and cybernetics grew in a technological atmosphere that acknowledged the insufficiency of the purely

causalistic paradigm (who would think of handling a computer through the field equations of thousands of integrated circuits?), but had no awareness of the need to make *explicit* the change in perspective taken by the inquiring community. To the extent that the engineering field is prescriptive (by design), this kind of epistemological blunder is still workable. However, it becomes unbearable and useless when exported from the domain of prescription to that of description of natural systems, in living systems and human affairs. To assume in these fields that information is something that is transmitted, and that symbols are *things* that can be taken at face value, or that purpose and goals are transparent from the system itself as a program, is all, it seems to me, nonsensical. The fact is that information does not exist independent of a context of organization that generates a cognitive domain, from which an observer-community can describe certain elements as informational and symbolic. Information, *sensu stricto*, does not exist. (Nor, of course, do the “laws” of nature.)¹⁰

¹⁰ *Eds.* This understanding of information continues to be central to the enactive approach and strongly distinguishes Varela’s thought from informational realism, the idea that reality consists in objective informational entities interacting with each other.

Thus, by putting these two modes of explanations, historically antagonistic, into a dualistic perspective, we gain power of explanation. And also both modes of explanation are significantly modified. On the one hand, telic-symbolic explanations cannot be adduced without embedding them in a nomic causal substrate which can, in principle, account for them—that is, a network of processes that is abstracted in the process of defining a symbol. This is clearly seen in the transition from the name teleological to the name teleonomic: no goal or purpose without a frame of abstracted chains of events from which we are abstracting. On the other hand, the causal explanation is also modified, for it no longer holds its position as methodological king, and must make way for noncausal explanations as equally valid.¹¹ This amounts to no more and no less than a change in the authority images of our inquiring community; it has nothing to do with standards of science or romantic revolutions. To neglect this shift in authority implies, to say the least, a sloppy use of symbolic explanations in the natural sciences, and a split between natural sciences and human sciences, where the role of communication and understanding gains a central importance in preference to causal mechanisms for which we cannot possibly hope. In brief, then, the dual interplay between these two modes of explanation is productive when and only when the two are related to each other in a generative way by making explicit where the change of frame of reference occurs.

¹¹ *Eds.* This is a bit asymmetric. It seems reasonable to prefer symbolic explanations that can satisfy operational constraints (although this sidesteps the problem of what to do when the operational knowledge is incomplete). This, by the way, does not imply the symbolic explanations must be explicitly constructed as such nomic abbreviations. But the reciprocal effect on operational explanations assumed here seems rather weak. A symbolic account, in fact, can drive the development of new operational explanations by expressing the current understanding and interests of a community, defining the bounds of the systems in question, and specifying the suitable operational domain in which such explanations may be sought (Di Paolo, 1999, 19).

9.4.3

An elementary case of such dualistic operation is apparent in the understanding of the origin of life and the use of genetic material as an explanatory device in evolution and development. By way of another example, consider the interaction of hormone molecules with the receptor surface of a cell. This kind of interaction is best described by abstracting the actual process of interaction and the detailed description of the autopoietic dynamics, and phrasing it in terms of a symbol (or signal) with a regulatory effect, a description that emerges through a contracted account of the autopoietic dynamics of the individual cell. At the risk of being obnoxious, let me point out that there is *nothing* in the hormone molecule that is informational: its symbolic content is given by, first, the kind of dynamics determined by the autopoietic unity and its domain of interactions, and second, the observer who wishes to follow a certain coherence in the individual dynamic and thus chooses to contract a long and complex sequence of nomic chains.

To regard this cell-hormone interaction in any sense as “intrinsically” informational, or that the organism is “picking up information” from the environment, would be fundamentally wrong. But it seems equally wrong *not* to see in these kinds of events the beginning of symbolic interactions so prevalent in higher organisms and man, and the importance of their continuity with operational explanations.

9.5 Admissible Symbolic Descriptions

9.5.1

We now turn to a nagging question in the previous discussion: What is to count as a symbol, and when are we entitled to a symbolic explanation? To characterize what is meant by both these terms is important, and yet it is as difficult as characterizing a cause or an operational explanation. Clearly the criteria for both depend on the preferences of the observer community. But since from the vantage point of natural systems we have concluded that such symbolic explanations are not dispensable, I will try to be more precise as to how I use these terms here. That is to say, we ask: What is an *admissible symbolic description*?

Two main features characterize symbols in natural systems: (1) internal determination and (2) composition.¹²

¹² *Eds.* This account owes much to Pattee (1977). See also Pattee (1982). For an extension into the realm of language of the idea of symbols as constraints within the context of closure, see Rączaszek-Leonardi (2009).

1. *Internal Determination.* An object or event is a symbol only if it is a token for an abbreviated nomic chain that occurs *within the bounds of the system's organizational closure*. In other words, whenever the system's closure determines certain regularities in the face of internal or external interactions and perturbations, such regularities can be abbreviated as a symbol, usually the initial or terminal element in the nomic chain. A typical example is the genetic "code." A triplet of nitrogen bases stands for or "encodes" an amino acid in a protein sequence to the extent that there is a regular pattern in the chemical dynamics, which we can see repeated again and again. But such a dynamic pattern occurs entirely within the bounds of the cell's closure; the cell itself contains the "interpretation" for the symbol. We then chose the triplet as the symbol for the amino acids by abbreviating the long sequence of chemical steps in the autopoietic cycle and abstracting these steps from the internal recursion where such chemical reactions normally operate.

As a result, in our description we see a seemingly arbitrary relation between signifier and signified (e.g., triplet and amino acid) produced by ignoring the causal steps. To the extent that this ignoring is based on the regularities of the dynamics of an autonomous system, this symbolic description is admissible, and it plays a useful role in the study of autopoietic systems on a larger time scale. A similar analysis is, of course, possible for the hormone-cell interactions. Although in this case the initial element in the abstracted nomic chain is an external perturbation, it relates to causal events that occur entirely inside of the cell and are determined by it.

In the molecular examples of admissible symbols, their underlying causal chains are still apparent and accessible, and we can switch from one type of description to the other with a certain ease. However, in the case of the immune or nervous systems, and certainly in the case of the human language, we only have available to us the gross symbolic regularities; to try to make explicit the underlying nomic chains is, in general, fruitless. Still the symbolic process occurs within, and is determined by, the corresponding system's closure (cf. chapters 14, 15).

If we use a symbolic description for a system that cannot be construed as abbreviated nomic chains, or else where the regularities in behavior are not within and determined by the system's closure, then we shall say that it is an inadmissible symbolic

description. Thus, for example, the accidental collisions between bacteria in a culture generate no regular or recurrent pattern in the cell's dynamics, although they might in the culture at large. To describe such bodily encounters as symbolic would be inadmissible.

In summary, then, we can say that the internal determination of a symbol gives it the qualities of arbitrariness and interpretability, which are related and are both based on the system's closure.

2. *Composition.* A process that admits a symbolic description might or might not be of ontogenetic and phylogenetic interest and potential. In other words, among all the possible regularities that can emerge from a system's closure, only some of these might lead (through structural coupling or evolution) to a significant adaptive change in the cognitive domain of the system.

On empirical grounds, the regularities that have been fertile and preserved in evolution are those such that the symbols that stand for them can be seen as composable like a language—in other words, such that the individual symbols, as discrete tokens, can interact with each other in a syntax capable of generating new patterns in combination. Again, the typical example is the genetic symbols, which in a linear array are eminently composable. This is still somewhat possible, but not so clearly so, in the case of external signals and surface receptors. It is clearly impossible in the case of chemotaxis and gradient signals. The great symbolic systems of living beings—the genetic and nervous systems and human language—are all based on regularities whose symbols are composable through rules that generate a vast class of new phenomena out of a set of discrete elements.

Thus composability is a dimension of symbolic events that is independent of internal determination, and that addresses the question of *selection value* in development and evolution.¹³

The identity and autonomy of a system, in whichever domain, depends on its organizational closure. However, in order to fully understand how the cognitive domain of such a system can operate and be modified, we must look at the dynamic regularities that arise within the system and that can be treated as symbolic events. These are essential because the system's behavior can then be treated as if operating on the basis of the discrete number of regularities in the fashion of rules operating on the symbols of an alphabet. The cognitive domain of such a system can operate as if on a set of

¹³ We make no attempt here to characterize the difference (if there is one) between the syntax of human language, cognitive mechanisms, and genetic coding. This is not essential for our present purposes and deserves an independent discussion (cf. section 16.1.3).

discrete symbols that stand for a complex but regular dynamics. Evolutionarily, there has been strong pressure to develop autonomous systems such that composable regularities arise profusely. The nervous system is the paradigmatic case, and that is precisely why it enlarges an organism's cognitive domain in such a dramatic way (cf. chapter 15).

Thus, side by side with organizational closure, admissible symbolic descriptions are really what is needed in order to account for both existence *and* progressive change of autonomous systems in nature and culture. They are *complementary views*.

Sources

(Pattee, 1977), (Varela, 1981).

10 The Framework of Complementarities

10.1 Introduction

10.1.1

The world does not present itself to us neatly divided into systems, subsystems, environments, and so on.¹ These are divisions we make ourselves for various purposes. It is evident that different observer-communities find it convenient to divide the world in different ways, and they will be interested in different systems at different times—for example, now a cell, with the rest of the world its environment, and later the postal system, or the economic system, or the atmospheric system. The established scientific disciplines have, of course, developed different preferred ways of dividing the world into environment and system, in line with their different purposes, and have also developed different methodologies and terminologies consistent with their motivations.

Furthermore, throughout this book we have encountered again and again the fact that an observer-community may take alternative views of a system that, at first glance, appear exclusive, but that nevertheless are interdependent and mutually defining. Such was the case with autopoiesis and allopoiesis, and with causal and symbolic explanations, two instances that have been extensively discussed. It was evident through these discussions that keeping the interdependence of these views steadily in mind was a key to a more balanced understanding of natural systems—particularly in the case of autonomy. It is time to recast this issue of interdependence and complementarity of views in a more explicit form.²

In this chapter, we present a *conceptual and formal* framework within which a number of various preferred views on systems can be unified. Of particular interest to us here are the differences

¹*Eds.* With minor edits, this chapter is derived from Goguen and Varela (1979), with the exception of section 10.6 which is adapted from Varela (1976).

²*Eds.* These comments should be contrasted with the idea of nonintersecting phenomenal domains (section 6.3.4), which Varela recasts in terms of interdependence and codefinition both here and in his later work.

stemming from the study of natural systems (particularly biological and social systems) and artificial systems (such as engineering and computer systems). Contemporary systems theory has developed extensively through experience in the latter fields, but the insights derived from natural systems have remained by and large much less formally developed. In this book, I hold that the notions of cooperative interaction, self-organization, and autonomy—in brief, holistic notions—are basic to the study of natural systems. In the present framework these notions are not only made more explicit and applicable, but are also presented as complements to the more traditional notions of system theory, such as control and input-output behavioral description.

10.1.2

The next section discusses in general terms the role distinction plays in the creation and recognition of systems. The following three sections discuss certain dual perspectives on systems in some detail, including the autonomy/control, state-variable/input-output, holism/reductionism, and net/tree dualities. The fifth section develops the suggestion that such alternatives are complementary rather than antagonistic, into a suggestion that their interrelationship can often be expressed precisely as an adjoint functor relationship, in the sense of categorical algebra. The final section discusses the holism/reductionism relationship in some detail, in relation to the philosophy of science.

Future chapters will build on this foundation, to discuss the notion of autonomy in terms of a (mathematical) theory of self-reference or indefinite recursion and its applications.

10.2 Distinction and Indication

10.2.1

A *distinction* splits the world into two parts, “that” and “this,” or “environment” and “system,” or “us” and “them,” etc. One of the most fundamental of all human activities is the making of distinctions. Certainly, it is the most fundamental act of system theory, the very act of defining the system presently of interest, of distinguishing it from its environment.

Distinctions coexist with purposes. A particularly basic case is autonomy—a system defining its own boundaries and attempting to maintain them; this seems to correspond to what we think of as individuality. It can be seen in individuals (ego or identity maintenance)

and in social units (clubs, subcultures, nations). In such cases, there is not only a distinction, but an *indication*, that is, a marking of one of the two distinguished states as being primary ("this," "I," "us," etc.); indeed, it is the very purpose of the distinction to create this indication (Spencer-Brown, 1969; Varela, 1975a).³

A less basic kind of distinction is one made by a distinctior for some purpose of his own. This is what we generally see explicitly in science, for example, when a discipline "defines its field of interests," or a scientist defines a system that he will study.

In either case, the establishment of system boundaries is inescapably associated with what I shall call a *cognitive point of view*, that is, a particular set of presuppositions and attitudes, a perspective, or a frame in the sense of Bateson (1960) or Goffman (1974); in particular, it is associated with some notion of value, or interest. It is also linked up with the cognitive capacities (sensory capabilities, knowledge background) of the distinctior. Conversely, the distinctions made reveal the cognitive capabilities of the distinctior. It is in this way that biological and social structures exhibit their coherence.

10.2.2

The importance for system theory of cognitive coherence (or the cognitive point of view, or cognitive capability) is a theme that runs throughout this book. Because of the focus on system theory, we shall feel free to invoke the idea of an *observer*, or, observer-community: one or more persons who embody the cognitive point of view that created the system in question, and from whose perspective it is subsequently described.⁴

A simple but fundamental property of the situation involving a system and an observer is that she may choose to focus her attention either on the internal constitution of the system, or else on its environment, taking the system's properties as given. That is, an observer can make a distinction into an indication through the imposition of her value. If the observer chooses to pay attention to the environment, she treats the system as a simple entity with given properties and seeks the regularities of its interaction with the environment, that is, the constraints on the behavior of the system imposed by its environment.⁵ This leads naturally to the problem of controlling the behavior of the system, as considered in (engineering) control theory. On the other hand, the observer may choose to focus on the internal structure of the system, viewing the environment as background—for example, as a source of perturbations upon the system's autonomous behavior. From this viewpoint, the properties

³*Eds.* In this chapter and the following chapters, *distinction* and *indication* are most often co-implied. However, a distinction may be drawn to mark a difference between two entities, processes, ideas, and other elements, and regard their relation without focusing on either of them via an indication.

⁴*Eds.* The notion of the observer is generally invoked in the singular, but it should always be understood as implying a plural community of participants, both in terms of the development of perceptual and technical capabilities and in terms of common grounds, values, purposes, and interests that inform the act of observing, even when it is carried out by a single individual.

⁵Calling *S* "the system" rather than "the environment" already indicates a preference for marking *S*; that is, the language incorporates the preference. But we may speak of "marking the environment" to suggest that there are in fact two distinct possibilities.

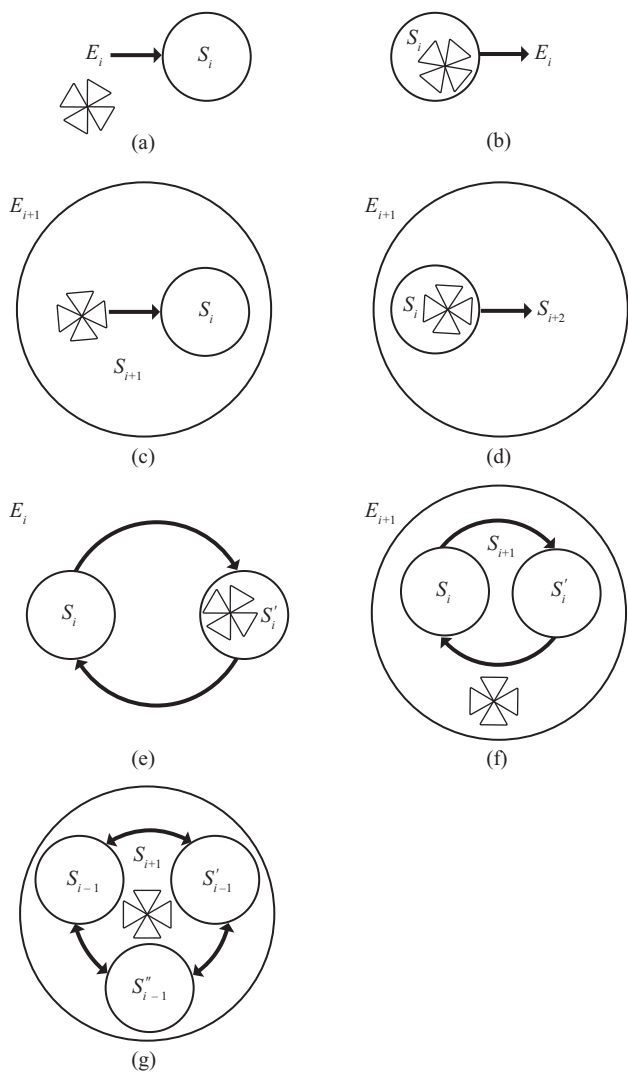


Figure 10.1 Various configurations of systems, subsystems, and marks: Each configuration represents a cognitive viewpoint, and the mark indicates its center. The arrows indicate the interactions. From (Goguen and Varela, 1979).⁶

⁶*Eds.* a: Control of system S_i by its environment E_i . b: Autonomy of S_i in E_i . c: Control of subsystem S_i in system S_{i+1} . d: Autonomy of S_i in S_{i+2} . e: Feedback control of system S_i by system S'_i . f: Communication between the coordination of two subsystems S_i , S'_i of system S_{i+1} . g: Coordination (ecology) of subsystems S_{i-1} , S'_{i-1} , S''_{i-1} .

of the system emerge from the interactions of its components. See figure 10.1.

Biology has iterated this process of indication, creating a hierarchy of levels of biological study (figure 10.2). The cell biologist emphasizes the cell's autonomy, and views the organism of which it is part as little more than a source of perturbations for which the cell compensates. But the physiologist views the cell as an element in a network of interdependences constituting the individual organism:

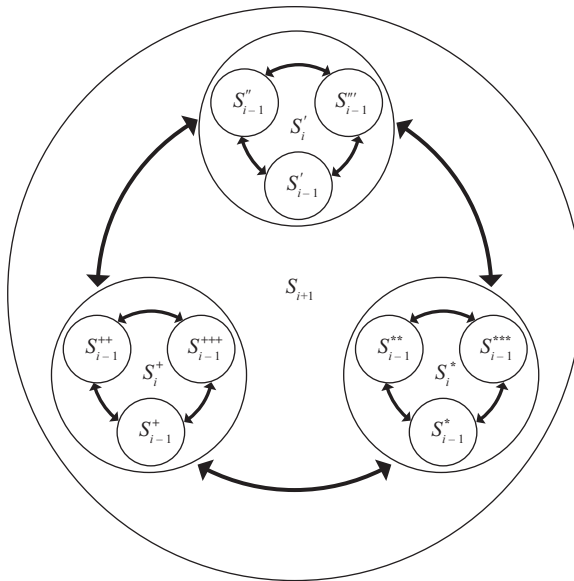


Figure 10.2

Diagrammatic evocation of a hierarchy of system levels. See text for further discussion. From (Goguen and Varela, 1979).

This corresponds to a wider view of the environment, namely the ecology in which the individual participates. A population biologist makes his distinctions at a still higher level, and largely ignores the cell. A similar hierarchy of levels can be found in the social sciences. It seems to be a general reflection of the richness of natural systems that indication can be iterated to produce a hierarchy of levels.

At a given level of the hierarchy, a particular system can be seen as an *outside* to systems below it, and as an *inside* to systems above it; thus, the status (i.e., the mark of distinction) of a given system changes as one passes through its level in either the upward or the downward direction. The choice of considering the level above or below corresponds to a choice of treating the given system as autonomous or controlled (constrained). Figure 10.1 illustrates a variety of configurations of systems, subsystems, and marks, and figure 10.2 illustrates the hierarchy of levels.

10.3 Recursion and Behavior

10.3.1

In system theory, the autonomy/control distinction appears more specifically as a recursion/behavior distinction. The behavioral view reduces a system to its input-output performance or behavior, and

reduces the environment to inputs to the system. The effect of outputs on the environment is not taken into account in this model of the system. The recursive view of a system, as expressed in the closure thesis, emphasizes the mutual interconnectedness of its components (von Foerster, 1974b; Varela, 1975a; Varela and Goguen, 1978). That is, the behavioral view arises when emphasis is placed on the environment, and the recursive view arises when emphasis is placed on the system's internal structure.

If we stress the autonomy of a system S_i (see figure 10.1), then the environmental influences become perturbations (rather than inputs) which are compensated for through the underlying recursive interdependence of the system's components (the S_{i-1} 's are the figures). Each such component, however, is treated behaviorally, in terms of some input-output description.

The recursive viewpoint is more sophisticated than the behavioral one, since it involves the simultaneous consideration of three different levels, whereas the behavioral strictly speaking involves only two. This is because the behavioral model, in taking the environment's view of the system, does not involve making any new distinctions. But expressing interest in how the system achieves its behavior through the interdependent action of its parts adds a new distinction, between the system and its parts.

10.3.2

The following may help to make this seem less abstract. The most traditional way to express the interdependence of variables in a system is by differential equations (cf. section 7.2.4). An autonomous system can be formally represented by equations of the form

$$\dot{x}_i = F_i(x, t) \quad \text{for } 1 \leq i \leq n, \quad (10.1)$$

where $x = (x_1, \dots, x_n)$ is the state vector of the system and $\dot{x}$ its rate of change. The autonomous behavior of the system is described by a solution vector $x(t)$ that satisfies (10.1). This involves treating everything as happening on the same level, and all variables as being observable; in effect, the environment is treated as part of the system (or ignored).

However, the effect of the environment on the system can be represented by a vector $e = (e_1, \dots, e_k)$ of parameters, giving

$$\dot{x}_i = F_i(x, e, t) \quad \text{for } 1 \leq i \leq n, \quad (10.2)$$

which explicitly takes account of two levels. Solutions to the system (10.2) are now also parametrized by e , that is, they are of the form $x(e, t)$.

The situation of (10.2) can be elaborated in two directions. In control theory, it is usual to assume that the internal variables x of the system are either unobservable or of no direct interest, and that we have instead direct access to (or interest in) an output vector y of variables that are functionally dependent on x . The variables e are usually taken to be under the control of the observer, and the question is posed, how to use those variables to obtain certain desired values of y . The equations are thus of the form

$$\dot{x}_i = F_i(x, e, t), \quad y = H(x). \quad (10.3)$$

Strictly speaking, the equations span three levels, and can be used, for example, to infer information about the system's internal state, but the emphasis ("mark") is on the environment, which is identified with the observer. Behavior appears as an input-output function $y(e, t)$, the observable results of applying the inputs (also called "controls") e to the system.⁷

⁷*Eds.* That is, this is what the observer is measuring.

An alternative elaboration of the situation of (10.2) views the vector e as not necessarily or particularly under the control of an observer, but rather as a source of perturbations upon (10.2). For example, the components e_j of e may be some coefficients, which are regarded as constants in the original equation (10.1). A natural question to pose is the stability of the system under such perturbations, that is, the relation of (10.1) to a perturbed system

$$\dot{x}_i = F_i(x, e, t) + \delta F_i(x, e, t) \quad (10.4)$$

in which δ (in a fairly intuitive notation) represents a "small change." It is known, for example, that changes in structural constants can cause the system to undergo a "catastrophic" change (in the sense of Thom 1972) into a new configuration.

The system (10.2) has in it nothing that intrinsically prefers the approach of either (10.3) or (10.4). This choice depends on the interest of the analyst.

Note that recursion plays a role in all these formulations, but is more obscure in the control-theory interpretation. On the other hand, the behavioral information, though still available, is more obscure in the stability interpretation (10.4). We are not, of course, claiming that either of these approaches is inherently better.

10.3.3

Historically speaking, some of the many possible approaches to systems have been much more developed than others. The most highly developed parts, in fact, center on the notions of control, input-output behavior, and state transition. This is

presumably because of the interest in applying these approaches in engineering.

The notion of autonomy, however, is particularly important for natural systems (biological and social systems), and the lack of a well-developed theory of autonomous systems is a serious difficulty. An engineer designing an artifact will choose the inputs of interest to him for this application with some assurance that the choice will be adequate. But a biologist studying a cell is forced to acknowledge the autonomy of the cell; if the biologist's preferences for input and output variables do not match the cell's internal organization, her cognitive-domain theory will be useless.⁸ Furthermore, the hierarchy of levels seems to particularly assert its importance for natural systems, so that it is generally necessary to take account of at least three levels. Even when the lowest level is very well understood, the role it plays at the next higher level, where it is interconnected with other systems, can be quite obscure. An enzyme biochemist may be able to describe a particular metabolic loop very effectively by a transfer function, but be quite unable to specify how it fits into the overall metabolic process of the cell as a coherent autopoietic whole.

This situation of being unable to understand how elements, even quite well-understood elements, coordinate or somehow function effectively together at the next higher level, is quite common in the study of natural systems, and is another source of our motivation for a better-developed theory of autonomous systems.

10.3.4

Some fragments of a theory emphasizing the autonomy of systems do exist, but are far less developed than the computer-gestalt, behavioristic approach. In fact, the dominance of control views in contemporary systems theory makes it closer to a theory of system components than to one of systems as unities (totalities). Let's mention briefly some existing approaches to representing, in formal terms, some of the characteristics of autonomy.

First and foremost, the idea of stability derived from classical mechanics has been extensively studied and used. As we said before (section 7.2.4), a set of interdependent differential equations can be used to represent the autonomous properties of a whole system. Rosen (1970), Iberall (1972), and Lange (1965) have applied this perspective to natural systems with various degrees of emphasis on autonomous behavior. More specific examples can be found in population biology (May, 1973), in molecular biology (Eigen and Schuster, 1978; Goodwin, 1976; Rössler, 1978; Bernard-Weill, 1975),

⁸*Eds.* This possibility alone should dispel any notions that thematizing the observer's activity of distinction and indication amounts to defending some kind of idealism. An observer may be free to introduce distinctions, but the world may still "resist" them, especially if the system under observation turns out to be autonomous.

and more recently in neurobiology (Katchalsky et al., 1974; Freeman, 1975). Some thought has been given to cooperative interactions in this area of hierarchical multilevel systems. The idea of hierarchy is often presented from the point of view of the interdependence of different levels of system descriptions (Pattee, 1972; Whyte et al., 1969; Mesarovic et al., 1970). Particular instances of hierarchical structure, including multilevel cooperation, can be found in Beer (1972), Kohout and Gaines (1976), and Baumgartner et al. (1976). Goguen (1970; 1972a) presents a general theory of hierarchical systems of interdependent processes. Its basic ideas are interconnection, behavior, and level, and its theoretical framework is categorical algebra. A last area in which the idea of a whole system is somewhat explicit is that of self-organizing systems. Work in this area, based on an information-theoretic approach, includes von Foerster (1960) and Atlan (1972; 1981).⁹

We do not intend to unite *all* these various threads of research together in a single framework. Rather, we emphasize the ways in which pairs of seemingly different points of view, such as autonomy/control, are complementary, in the sense of contributing to a better understanding of natural systems. But the idea of complementarity, fundamental though it seems, is still vague.¹⁰ The following develops an *explicit* definition.

10.4 Nets and Trees

10.4.1

If we retain interest only in the connectivity of a system, it is possible to *represent the recursion/behavior duality by a network/tree duality*. Intuitively, the nodes in these nets or trees represent the elements or components of a system, while their links represent interactions or interconnections. The reciprocal connectivity of a net suggests the coordination of a system's elements; a tree structure suggests the sequential subordination of a system's parts, each part having its own well-defined input-output behavior description¹¹. To be sure, in retaining only the basic connectivity of a system's organization, much is discarded in the net/tree representation. We intend to use this convenient general representation to study complementarity.

Now to the definition of nets and trees. Let there be a set $\{v_1, \dots, v_n\}$ of nodes (components or parts), which are to be interconnected by a set $E = \{e_1, \dots, e_p\}$ of edges (relations or processes).

Definition 10.1 A network is a directed graph G , that is, a quadruple $G = \langle |G|, E, \partial_0, \partial_1 \rangle$ where $|G| = \{v_1, \dots, v_n\}$ and $\partial_i : E \rightarrow |G|$ are the source

⁹Eds. Other formal or semiformal approaches concerned with questions of autonomy may be added to the list: Rosen's (M, R) systems (Rosen, 1991) and its reinterpretations in terms of hypersets (Chemero and Turvey, 2008) and lambda-calculus (Mossio et al., 2009a); Gánti's Chemoton model (Gánti, 2003); organizational perspectives (Longo and Montévil, 2013; Moreno and Mossio, 2015); Kauffman thermodynamic conception of autonomy (Kauffman, 2003); models based on formal chemistry (Fontana and Buss, 1994); and information-theoretic approaches (e.g., Bertschinger et al., 2008; Seth, 2010; Aguilera and Di Paolo, 2019).

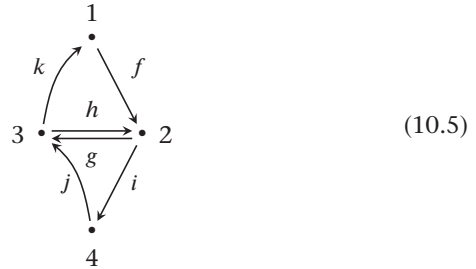
¹⁰Eds. For a brief overview of ideas of complementarity and a modern interpretation based on dynamical systems theory, see Kelso and Engstrom (2006).

¹¹Eds. For an example of network/tree representations, see figure 14.1.

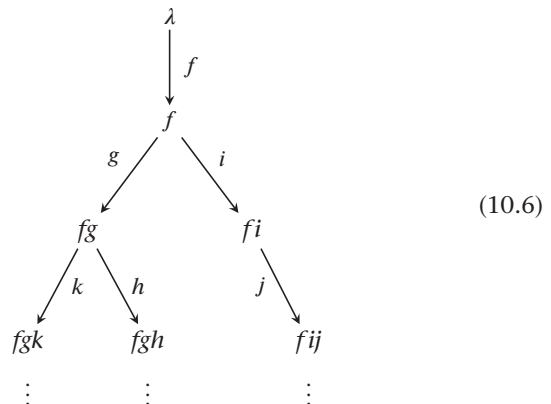
($i=0$) and target ($i=1$) functions, from the edges to the nodes of G . If $e \in E$, $\partial_0 e = v$, and $\partial_1 e = v'$, then we write $e : v \rightarrow v'$.

Definition 10.2 A path from v to v' in a graph G is a finite sequence $p = e_0 \dots e_n$ of edges that are adjacent, that is, satisfy $\partial_1 e_i = \partial_0 e_{i+1}$ for $1 \leq i \leq n$, with $\partial_0 e_0 = v$ and $\partial_1 e_n = v'$. If $\partial_0 p = v$ and $\partial_1 p = v'$, then we write $p : v \rightarrow v'$.

Example 10.3 Consider the graph G :



The nodes 1, 2, 3, 4 might represent four physical locations, each with a radio. Because of differences in transmitter and receiver strength, available frequencies, and terrain, communication is possible only along the channels indicated by arrows in (10.5). For example, there is a mountain between 1 and 4. Let us assume that node 1 is of particular interest—say our base. Then we are interested in the patterns of transmission which are possible starting from our base. For example, to reach node 3, if channel g is out, we can send a message via fij ; to verify its correctness, it could be sent back to node 1, via the path $fijk$; node 3 might also generate a reply, which would require a message to be sent to node 2, giving a path $ijkf$, and so on. Thus, we are interested in the set of all paths in G with source 1. This collection itself has a branching structure, because starting with a given path, it can sometimes be developed further by choosing alternative edges to get alternative paths. The collection of all choices can be represented by the following tree:



Notice that it is an infinite tree.

In some sense the tree (10.6) “unravels” or “unfolds” the graph (10.5) from node 1. To make this more precise we need to define trees, pointed graphs, and the idea of structure-preserving mappings of graphs called graph homomorphism. First we give the general construction.

Definition 10.4 A pointed graph G is a 5-tuple $G = \langle |G|, E, \partial_0, \partial_1, a \rangle$ such that $G = \langle |G|, E, \partial_0, \partial_1 \rangle$ is a graph and $a \in |G|$ is a vertex.

A pointed graph is reachable if for each vertex $v \in |G|$ there is a path $a \rightarrow v$ in G .

A graph G is loop-free if for all $v, v' \in |G|$ there is at most one path $v \rightarrow v'$.

A tree is a reachable loop-free pointed graph.

Definition 10.5 Let G be a pointed graph $G = \langle |G|, E, \partial_0, \partial_1, a \rangle$. Then the unfoldment $U_a(G)$ of G from a is the graph in which: $|U_a(G)|$ is all the paths $p : a \rightarrow v$ for $v \in |G|$; the edges of $U_a(G)$ are the pairs $\langle p, p e \rangle$ such that $p, p e \in U_a(G)$, and $e \in E$; $\partial_0 \langle p, p e \rangle = p$, and $\partial_1 \langle p, p e \rangle = p e$. The null path $a \rightarrow a$ is written $1_a : a \rightarrow a$, and is taken to be the point for $U_a(G)$.

Proposition 10.6 Let G be a pointed graph $G = \langle |G|, E, \partial_0, \partial_1, a \rangle$, and $U_a(G)$ the unfoldment of G from a . Then $U_a(G)$ is a tree.¹²

Proof. We must show that $U_a(G)$ is a pointed, reachable, and loop-free graph. By definition of $U_a(G)$, we know that it is a pointed graph, with point $1_a : a \rightarrow a$.

We now show that $U_a(G)$ is reachable. Consider a node $p : a \rightarrow v$ of $U_a(G)$, say $p = e_0 \dots e_n \neq 1_a$, with $e_i \in E$. Then we can show that

$$q = \langle 1_a, e_0 \rangle \langle e_0, e_0 e_1 \rangle \langle e_0 e_1, e_0 e_1 e_2 \rangle \dots \langle e_0 \dots e_{n-1}, e_0 \dots e_n \rangle$$

is a path from 1_a to p in $U_a(G)$. Clearly its source is 1_a , since

$$\partial_0 q = \partial_0 \langle 1_a, e_0 \rangle = 1_a$$

and its target is p , since

$$\partial_1 q = \partial_1 \langle e_0 \dots e_{n-1}, e_0 \dots e_n \rangle = e_0 \dots e_n = p$$

Moreover, q is a path, since its edges are adjacent, that is,

$$\begin{aligned} \partial_1 \langle e_0 \dots e_k, e_0 \dots e_{k+1} \rangle &= \partial_0 \langle e_0 \dots e_{k+1}, e_0 \dots e_{k+2} \rangle \\ &= e_0 \dots e_{k+1} \quad \text{for } 0 \leq k \leq n-1. \end{aligned}$$

So there is a path from 1_a to every node in $U_a(G)$.

Last we show that $U_a(G)$ is loop-free, that is, for every pair of nodes p, p' in $U_a(G)$ there is at most one path $p' \rightarrow p$ in $U_a(G)$ with target p . Consider again $p = e_0 \dots e_n$, and let's show first that there is exactly one edge with target p , e.g., exactly one path of length one. Any edge with target p is of the form $\langle r, r e_n \rangle$, with $r e_n = p$. Thus r must equal $e_0 \dots e_{n-1}$, and the unique edge is $\langle e_0 \dots e_{n-1}, p \rangle$. This says that if $p' \neq p$,

¹²Eds. Any network represented as a pointed graph can be unfolded as a (possibly infinite) tree.

a path $p' \rightarrow p$ must end with edge $\langle e_0, \dots, e_{n-1}, p \rangle$. Let now $p_k = e_0, \dots, e_k$. Then a path $p' \rightarrow p$ must be a composite of a path $p' \rightarrow p_{n-1}$ with edge $\langle p_{n-1}, p \rangle$. But we may reason in a similar way for the node p_{n-1} , and the path $p' \rightarrow p_{n-1}, p_{n-2}$, and $p' \rightarrow p_{n-2}$, and so on. Eventually we must find that $p' = p_k$ for some k , and the unique path $p' \rightarrow p$ is of the form

$$\langle p_k, p_{k+1} \rangle \langle p_{k+1}, p_{k+2} \rangle \dots \langle p_{n-1}, p \rangle.$$

if $p = p'$, the unique path $p \rightarrow p'$ is the null path at p . Thus $U_a(G)$ is loop-free, and the proof is complete. $\square$

Definition 10.7 Let $G = \langle |G|, E, \partial_0, \partial_1 \rangle$ and $G' = \langle |G'|, E', \partial'_0, \partial'_1 \rangle$ be graphs. Then a graph morphism is a pair $\langle |F|, F \rangle$ of functions $|F| : |G| \rightarrow |G'|$ and $F : E \rightarrow E'$, such that the source and target relationships are preserved, that is, such that $\partial'_0(F(e)) = |F| \partial_0(e)$, and $\partial'_1(F(e)) = |F| \partial_1(e)$; i.e., such that the diagram

$$\begin{array}{ccc} E & \xrightarrow{F} & E' \\ \partial_i \downarrow & & \downarrow \partial'_i \\ |G| & \xrightarrow{|F|} & |G'| \end{array}$$

commutes for $i=0$ and $i=1$. We abbreviate $\langle |F|, F \rangle$ to just F . A morphism of pointed graphs, from G to G' , is a graph morphism F such that $|F|a = a'$, where a, a' are selected vertices of G, G' respectively.¹³

¹³Eds. A morphism is a structure-preserving mapping. In this case, the links between sources and targets are preserved.

10.4.2

The relation between a graph G and its unfoldment is, from our perspective, very interesting. Given a node a in G , then $U_a(G)$ is a loop-free version of G .¹⁴ We could say $U_a(G)$ expresses G as a (possibly infinite) chain of subordinated choices, starting from the selected node. The unfoldment of G optimally “covers” G in a sense that is made precise through the “universal property” of $U_a(G)$, that any graph morphism $F : T \rightarrow G$ can be factored through a “covering morphism” $C_G : U_a(G) \rightarrow G$, defined as follows: for p a node of $U_a(G)$, let $|C_G|p = \partial_1 p$; and for $\langle p, p e \rangle$ an edge of $U_a(G)$, let $C_G \langle p, p e \rangle = e$.

We now show that C_G is a graph morphism. For $\langle p, p e \rangle$ an edge of $U_a(G)$, then $\partial_0 C_G \langle p, p e \rangle = \partial_0 e$, $|C_G| \partial_0 \langle p, p e \rangle = |C_G|p = \partial_1 p$, and $\partial_0 e = \partial_1 p$ because $p e$ is a path. Also, $\partial_1 C_G \langle p, p e \rangle = \partial_1 e$, and $|C_G| \partial_1 \langle p, p e \rangle = |C_G|p e = \partial_1 p e = \partial_1 e$. We now show that any other morphism from a tree can be factored through C_G .

Theorem 10.8 Let G be a pointed graph, let T be a tree, and let $F : T \rightarrow G$ be a pointed graph morphism. Then there is a unique pointed graph

¹⁴Eds. The loops that may exist in a network represented by G result in infinite branches in its tree version, as in tree 10.6.

morphism $\hat{F} : T \rightarrow U_a(G)$ such that

$$\begin{array}{ccc} G & \xleftarrow{C_G} & U_a(G) \\ & \nwarrow F & \uparrow \hat{F} \\ & & T \end{array}$$

commutes.

Sketch of Proof. Let $v \in |T|$. Then there is a unique path $p : r \rightarrow v$ in T , where r is its root, and $F_p : Fr = a \rightarrow Fv$ is a path in G . Now we define $|\hat{F}|v = F_p$, and for $e : v \rightarrow v'$, and edge in T , we define $\hat{F}e = \langle F_p, (F_p)Fe \rangle$, an edge in $U_a(G)$. Then

$$|C_G||F|v = |C_G|F_p = F\partial_1 p = |F|v,$$

and

$$C_G \hat{F}e = C_G \langle F_p, (F_p)Fe \rangle = Fe.$$

Thus

$$C\hat{F} = F.$$

We now verify that $\hat{F}$ is a pointed graph morphism:

$$\partial_0 \hat{F}e = \partial_0 \langle F_p, (F_p)Fe \rangle = Fp,$$

$$|\hat{F}|\partial_0 e = |\hat{F}|v = F_p, \quad \text{and} \quad \hat{F}r = F1_r = 1_a;$$

$$\partial_1 \hat{F}e = \partial_1 \langle F_p, (F_p)Fe \rangle = (F_p)Fe = F(p e) = |\hat{F}|\partial_1 e.$$

Thus $\hat{F}$ satisfies the required conditions. (For a detailed proof and the uniqueness argument, see Goguen, 1974, Theorem 7). $\square$

10.4.3

This theorem brings into focus the basic intuition that there is a mutual interdependence between a system's elements (as a graph) and the sequential subordination of their interconnections (as a tree). To express this more clearly let $\mathcal{G}r^*$ be a class of pointed graphs, and let $\mathcal{T}$ be the class of trees.¹⁵ Also, if G and G' are pointed graphs, let $\mathcal{G}r^*(G, G')$ denote the set of all pointed-graph morphisms from G to G' .

Since every tree is by definition a pointed graph, we have a mapping

$$\mathcal{T} \xrightarrow{F} \mathcal{G}r^*$$

which simply views trees as pointed graphs. We also have a mapping

$$\mathcal{G}r^* \xrightarrow{U} \mathcal{T}$$

which assigns to every pointed graph the tree $U(G)$ that covers it. These two mappings F, U are tightly interlocked. For any

¹⁵Eds. These classes, as sets of sets, are of a higher logical type.

$h \in \text{Mor}(G, FT)$, the set of morphisms from G to FT , we have

$$\begin{array}{ccc}
 G & \xleftarrow{C_G} & FU(G) \\
 & \searrow h & \uparrow F\hat{h} \\
 & & FT
 \end{array}
 \quad \text{for a unique} \quad
 \begin{array}{ccc}
 & & U(G) \\
 & & \uparrow \hat{h} \\
 & & T
 \end{array}$$

by Theorem 10.8. This says that there is a bijection

$$\varphi : \mathcal{G}r^*(FT, G) \rightarrow \mathcal{G}r^*(T, U(G))$$

defined by $\varphi(h) = \hat{h}$. We shall call (F, U, φ) a *complementarity* between $\mathcal{G}r^*$ and $\mathcal{T}$.¹⁶ This notion of net/tree complementarity effectively relates two levels of description of systems, in such a way that each necessitates the other. It is convenient at this point to see that similar notions of complementarity apply to other situations, and so we now turn to the general notion.

¹⁶Eds. For each graph-to-graph morphism between a tree interpreted as graph FT and as graph G , there is a unique corresponding tree-to-tree morphism between tree T and the unfolding of G into tree $U(G)$.

10.5 Complementarity and Adjointness

10.5.1

The net/tree complementarity is a particularly clear instance of the interdependence of apparent dualities. This section develops this idea in the general setting of category theory, which is becoming increasingly useful in systems theory (Goguen, 1972b; Arbib and Manes, 1975). Readers unfamiliar with this terminology may find a leisurely introduction in (Goguen et al., 1973, 1976) or (Arbib and Manes, 1975); we attempt to stay at a fairly intuitive level, although some technicalities are inevitable.

The intuitive idea of a *category* is that it embodies some structure by exhibiting the class of all *objects* having that structure, together with all the structure-preserving mappings or *morphisms* among them. (Somewhat more technically, categories assume there is an associative operation of composition on those morphisms whose source and target match.) The idea is due to Eilenberg and MacLane (1945).¹⁷

¹⁷Eds. For a general introduction to category theory, see Lawvere and Schanuel (2009).

For example, pointed graphs and pointed-graph morphisms constitute a category. If $\mathcal{C}$ is a category, and A, B are objects in $\mathcal{C}$, we shall let $\mathcal{C}(A, B)$ denote the set of all morphisms in $\mathcal{C}$ from A to B .

Usually, we are interested not only in objects from various categories, but even more in certain constructions performed on the objects of one category to yield objects of another category. For

example, unfoldment is a construction performed upon graphs that yields trees. This construction has a kind of consistency, in that it can also be extended to the morphisms; that is, a morphism of pointed graphs induces, in a natural way, a morphism between their unfoldments. This kind of consistency is expressed by saying that the construction is *functorial*, or is a *functor*. (More technically, this has to do with the preservation of the composition of morphisms.)

However, the unfoldment construction is natural in a much stronger sense: the “optimal” covering of a graph is its unfoldment; this is expressed by the universal property of Theorem 10.8, and the bijection φ the previous section. The concept of *adjunction* generalizes just this state of affairs to the following situation:

Let $\mathcal{A}$ and $\mathcal{B}$ be categories, let F be a functor from $\mathcal{A}$ to $\mathcal{B}$, and G a functor from $\mathcal{B}$ to $\mathcal{A}$. Then an *adjunction* is, in addition, a natural bijection

$$\varphi : \mathcal{B}(FA, B) \rightarrow \mathcal{A}(A, GB).$$

This says that every morphism $f : FA \rightarrow B$ determines a unique morphism $\varphi(f) : A \rightarrow GB$.¹⁸ (The precise sense of the “naturalness” of φ is that of natural transformation, due to Eilenberg and MacLane, 1945—which, however, we shall not define here. The idea of adjunction is due to Kan, 1958.)

The discussion at the end of the previous section shows that the net/tree complementarity is an instance of the concept of adjunction. *What we now propose is to explore the view that the precise concept of adjunction is an application of the general (and vague) concept of complementarity.*

10.5.2

Another example of this is Goguen’s (1972b) adjunction between minimal realization and behavior. Let $\mathcal{A}$ be the category of automata (in some fixed sense that we shall not explain in detail¹⁹), and let $\mathcal{B}$ be the category of input-output behaviors of such automata (with appropriate morphisms). Then there is a functor from $\mathcal{B}$ to $\mathcal{A}$ that constructs the minimal automaton $M(B)$ having behavior B ; and there is a functor Be from $\mathcal{A}$ to $\mathcal{B}$ that constructs behavior $Be(A)$ of an automaton A . Moreover, there is a natural bijection

$$\varphi : \mathcal{B}(Be(A), B) \rightarrow \mathcal{A}(A, M(B)).$$

that expresses the complementarity of the notions of internal state transition (as embodied in automata) and input-output behavior. Goguen (1972a) has shown that many other classes of

¹⁸Eds. Notice that the left adjoint functor F and the right adjoint functor G are literally *codefined* through φ , which is called the *adjunction* between F and G .

¹⁹Eds. See, for example, Arbib (1969).

systems exhibit such a complementarity with the input-output behaviors.

Here is still another example. If G is a graph, the collection of all paths (from all sources) in G forms a category whose objects are the nodes of G , and whose morphisms are the paths of G ; this category is denoted $\mathcal{Pa}(G)$, and called the *path category* of G . $\mathcal{Pa}$ is a functor from the category $\mathcal{Gr}$ of graphs to the category $\mathcal{Cat}$ of (small) categories. There is also a functor F from $\mathcal{Cat}$ to $\mathcal{Gr}$, which merely forgets the additional structure that categories have over graphs (namely, the possibility of composing morphisms), regarding the objects as nodes and the morphisms as edges. Again there is a natural bijection

$$\varphi : \mathcal{Cat}(\mathcal{Pa}(G), \mathcal{C}) \rightarrow \mathcal{Gr}(G, F\mathcal{C}).$$

expressing the complementarity of graphs and categories. Alternatively, $\mathcal{Pa}(G)$ is the free category generated by the graph G , and the adjunction (or corresponding “universal property”) expresses this relationship (Goguen, 1974).

Lawvere (1969), in a particularly fundamental paper, suggests that there is a complementary relationship between the traditional conceptual and formal viewpoints in the foundations of mathematics. This duality also appears as a semantic-syntax pair, in that Lawvere (1963) has shown an adjunction between a functor that associates with each algebraic theory its category of semantic models (i.e., its algebras), and a functor that extracts from each category the optimal syntactic theory of its algebraic component of structure.

The general system theory of Goguen (1970; 1972a) involves a hierarchy of levels, much as pictured in figure 10.2, with functors going outward if they regard a component at a lower level as a whole system at the next higher level, and functors going inward if they compute the behavior of the whole system, viewing the result as a single object at the lower level. There is a base level of given “objects” out of which systems can be constructed, and objects at level $i + 1$ are interconnections (that is, systems) of objects at level i . Goguen shows that each pair of an outward and an inward functor is an adjunction. The inward functor is in fact the fundamental categorical construction known as “limit.” Goguen also shows that the construction of interconnecting a system of systems (over some common subparts as “terminals”) to get a single system, is given by the dual concept of “colimit,” which also appears as an adjunction. This is not the place to give details, but the connection with themes of this book should be evident.

This general point seems particularly clear in the context of systems theory: There is no whole system without an interconnection of its parts; and there is no whole system without an environment. Such pairs are mutually interdependent: each defines the other.²⁰ What is remarkable about the notion of adjoint functor is that it captures the notion of complementarity in a very precise way, without imposing any particular model for the nature of the objects so related. It is also worth noting that there is a well-developed theory of adjunctions; for example, the composition of two adjoint pairs of functors is another adjoint pair. Of course, not all pairs of descriptive modes are complementary, and similarly, not all pairs of functors are adjoint. The so-called “adjoint functor theorem” provides some general conditions for when a given functor in fact does have an adjoint; and again, this may well find some application in general discussions about system theory. Much more work, including many further examples, will be needed to discover the proper domain of application, and the limits, of the adjointness idea.²¹

10.6 Excursus into Dialectics

10.6.1

In general, when different modes of description appear as opposites, it is more satisfactory to consider them as complementary instead. This is the case, quite rigorously, with the apparent dualities net/tree and recursion/behavior, as we have seen above. On a more intuitive level, there is a similar relationship for the pairs autonomy/control and operational/ symbolic discussed in earlier sections. As a matter of fact, we may go one step further to duality and dialectics as a broad philosophical idea.²² Accordingly, I would like to go into a brief excursus to discuss trinities.

By trinity I mean the consideration of the ways in which pairs (poles, extremes, modes, sides) are related yet remain distinct—the way they are not one, not two (Varela, 1976). The key idea here is that we need to replace the metaphorical idea of “trinity” with some built-in injunction (heuristic, recipe, guidance) that can tell us *how* to go from duality to trinity:

* = the it / the process leading to it.

The slash in this *star* (*) statement is to be read as: “consider both sides of the /,” that is, ‘consider both the it and the process leading

²⁰ *Eds.* A collection of parts does not a system make. Systems and parts express a two-way relation between component interactions and integration of the whole. In invoking both terms together (*parts* and *whole*), the two concepts become codefined. The same logic reappears with the question of what makes a system distinct, and what therefore belongs to what it is not and yet affects it, i.e., its environment. Again, just drawing any boundary will not express these relations. Adjointness between functors is one way of expressing such mutually defined concepts in formal terms.

²¹ We have not discussed at all the notion of complementarity in physics, or whether the present framework is applicable. To do so is completely beyond my competence.

²² *Eds.* Dialectical themes emerge frequently in Varela’s thought, often focusing on matters of complementarity, codefinition, and mutual historical transformation between pairs of “opposites.” Apart from his early engagement with dialectics (Varela, 1976), dialectical ideas appear in his treatment of the organism as a knotted relation between identity and knowledge dialectics (Varela, 1991b), as well as in references to the work of biologist Richard Lewontin on the organism as the subject and object of evolution (Lewontin, 1983; Varela et al., 1991). Varela’s dialectics remain descriptive and do not take the form of a method for critically expanding categories. Some developments in the enactive approach are more explicitly dialectical in this sense (Di Paolo, 2018; Di Paolo et al., 2018).

to it.” Thus the slash here is to be taken as a compact indication of a way of transiting to and from both sides of it.

We can now transcribe the familiar relationship between nets /trees into a star form:

*= network / trees constituting the network.

because the duality is connected with processes in both directions quite explicitly. The totality (the net) is seen as emerging or resulting from part-by-part approximation of the trees (the process leading to it).

Similarly we may consider a more generally appealing star:

*= whole / parts constituting the whole.

By a whole, a totality here we mean a simultaneous interaction of parts (components, nodes, subsystems) that satisfies some criteria of distinction. Thus a star of a more operational flavor is

*= stability / approximation in time.

Let us formulate a number of other interesting dualities in this complementarity framework, informally called star. To this end, take any situation (domain, process, entity, notion) that is autonomous (total, complete, stable, self-contained), and put it on the left side of the /. Put on the other side the corresponding process (constituents, dynamics). For example:

being/becoming	environment/system
space/time	context/text
reality/recipe	semantics/syntax
simultaneous/sequential	autonomy/control
arithmetic/algebra	symbolic/operational
analog/digital	

In each of these cases the dual elements can be seen as complementary: they *mutually specify each other*. There is, in this sense, no more duality, since they are related.²³

10.6.2

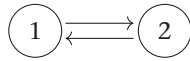
Notice that this separation of duality is no “synthesis” (in the Hegelian sense), since there is really nothing “new,” but just a more direct appraisal of how things are put together and related through our descriptions, not losing track of the fact that every “it” can be seen on a different level as a process.²⁴

More generally, we may see that this view of complementarity signifies a departure from the classical way of understanding dialectics. In the classical (Hegelian) paradigm, duality is tied to the idea of

²³ *Eds.* The star * operation is an invitation to seek the mutual dependence of concepts assumed to be distinct without erasing the distinction: a *potential* unity of opposites. It may be a good heuristic to assume that such a dependence exists when confronted with recalcitrant distinctions (e.g., mind-matter). But it may not always be the case. Ideally, the * is an invitation to make further dialectical moves that query the posited interdependence and clarifies it. Some of the examples in Varela’s list do not seem to be obvious dialectical pairs (e.g., “reality/recipe,” “simultaneous/sequential”), while others are more clearly so (“autonomy/control,” “environment/system”). The test resides in whether a deep understanding of one of the terms demands a reference to, or implies its opposite, as in the pair “boundary/crossing.”

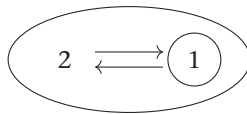
²⁴ *Eds.* In this sense, Varela’s is a sort of *contemplative* or *interpretative* dialectics.

polarity, a clash of opposites.²⁵ Graphically,



The basic form of these kinds of duality is symmetry: Both poles belong to the same level. The nerve of the logic behind this dialectics is negation; pairs are of the form *A/not-A*.

In this chapter, dualities are adequately represented by *imbrication* of levels, where one term of the pair *emerges* from the other. Graphically,



The basic form of these dualities is asymmetry: Both terms extend across levels. The nerve of the logic behind this dialectics is self-reference, that is, pairs of the form: it / process leading to it.

Pairs of opposites are, of necessity, on the same level and stay on the same level for as long as they are taken in opposition and contradiction. Pairs of the star form make a bridge across one level of our description, and they specify each other. When we look at natural systems, nowhere do we actually find opposition except from the values we wish to put on them.²⁶ The pair predator/prey, say, does not operate as excluding opposites. But both generate a whole unity, an autonomous ecosystemic domain, where there are complementarity, stabilization, and survival values for both. So the effective duality is of the star form: ecosystem/species interaction.

We may generalize this to say that there is an *interpretative rule* for dualities:

For every (Hegelian) pair of the form A/not-A there exists a star where the apparent opposites are components of the right-hand side.

It is, I suspect, only in a nineteenth-century social science that the abstraction of the dialectics of opposites could have been established. This also applies to the observer's properties. We have maintained all along that whatever we describe is a reflection of our actions (perceptions, properties, organization). There is mutual reflection between describer and description. But here again we have been used to taking these terms as opposites: observer/observed, subject/object as Hegelian pairs. From my point of view, these poles are not effectively opposed, but moments of a larger unity that sits on a metalevel with respect to both terms. In other words, it is

²⁵ *Eds.* This is a gross oversimplification of Hegelian dialectics as a science and practice of thought, which actually includes such relations of imbrication and emergence that Varela presents as a contrast, as well as the transformation of internal relations between opposites, moving the original categories from the abstract to the concrete. This move is open-ended (especially in materialist dialectics), thus establishing a deep inner connection between the logical and the historical (see, e.g., Ilyenkov, 2018)

²⁶ *Eds.* This point is oddly in tension with the epistemology of distinctions and indications developed in this chapter and the following chapters. We don't find *anything* in nature in the absence of distinctions, so exclusions are naturally implied. The act of indicating one half of a distinction entails the nonindication of the other half. We must surmise that here, Varela is thinking of contradiction in the plain sense of Aristotelian logical negation, in contrast to broader concepts of tension, as in the material relation between a flow and a barrier, for example.

possible to apply the interpretive rule here as well. Briefly stated, this interpretation could be phrased as: conversational pattern / participants in a conversation. I am here using “conversation” in a general and loose sense. Species interaction achieving a stable ecosystem can be thought of as the biological paradigm for a conversational domain. But human interactions can be similarly treated, as participants engaged in dialogue, whether with each other, with the environment, or with ourselves. This is the process underlying the conversational patterns that constitute the autonomous unity to which we belong and which we construct. We shall return to human cognition and conversational pattern in chapters 15 and 16. I only wanted to point out that the star framework could be applied to the observer’s properties as well, to see knowledge as an “it” generated through a process.

10.7 Holism and Reductionism

10.7.1

If we think of the philosophy of science, the duality holism/reductionism comes to mind as analogous to the material previously discussed in this chapter.

Most discussions place holism/reductionism in polar opposition (Smuts, 1926; Laszlo, 1972). This seems to stem from the historical split between empirical sciences, viewed as mainly reductionist or analytic, and the (European) schools of philosophy and social science that grope toward a dynamics of totalities (e.g., Kosík, 1967; Radnitzky, 1973). In the light of the previous discussion, both attitudes are possible for a given descriptive level, and in fact they are complementary. On the one hand, one can move down a level and study the properties of the components, disregarding their mutual interconnection as a system. On the other hand, one can disregard the detailed structure of the components, treating their behavior only as contributing to that of a larger unit. It seems that both these directions of analysis always coexist, either implicitly or explicitly, because these descriptive levels are mutually interdependent for the observer. We cannot conceive of components if there is no system from which they are abstracted; and there cannot be a whole unless there are constitutive elements.

10.7.2

It is interesting to consider whether one can have a measure for the degree of wholeness or autonomy of a system. One can, of course,

always draw a distinction, make a mark, and get a “system,” but the result does not always seem to be equally a “whole system,” a “natural entity,” or a “coherent object” or “concept.” What is it that makes some systems more coherent, more autonomous, more whole, than others?²⁷

A first thing to notice is, that in the hierarchy of levels, “emergent” or “immanent” properties appear at some levels. For example, let us consider music as a system or organization of notes (for the purpose of this example, we do not attempt to reduce notes to any lower-level distinctions). Then harmony only arises when we consider the simultaneous or parallel sounding of notes, and melody only arises when we consider the sequential sounding of notes. That is, harmony and melody are emergent properties of a level of organization above that of the notes themselves. Similarly, form can only emerge at a still higher level of organization, relating different melodic units to one another. These properties—form, melody, and harmony—are *systems’ properties*, arising from hierarchical organizations of notes into pieces of music; they are not properties of notes (Goguen, 1977). It also appears that “life” is an emergent property of the biological hierarchy of levels: it is nowhere to be found at the level of atoms and molecules, but it becomes clear at the level of cells through the autopoietic organization of molecules. Language can be seen as an emergent property at a still higher level of this hierarchy (Maturana, 1978). In general, organizational closure can be viewed as providing the mechanism through which emergent properties of new units arise, and thus as the “hinges” for the hierarchy of levels in natural systems. Thus, one point of view toward wholeness is that it co-occurs with interesting emergent properties at some level.²⁸

Another point of view toward wholeness, is that it can be measured by the *difficulty of reduction*: Because it is very hard to reduce the behavior of organisms to the behavior of molecules, we may say that organisms are whole systems. Similarly, it is very difficult (if not impossible) to reduce the effects of melodies to the effects of notes. One must consider properties of patterns of notes or molecules.

A third point of view is that a system is whole to the extent that its parts are *tightly interconnected*, that is, to the degree that it is difficult to find relatively independent subsystems. This is clearly related to the previous views. An interesting corollary of this view is that a system with a strongly hierarchical organization will be less whole than a system with a strongly heterarchical organization; that is, nets are more whole than trees. More precisely, given that the graph of connections of the parts of a system has no isolated subsystems, the

²⁷ *Eds.* Some methods quantify the degree of integration of a system relative to the segregation between components. Such is the case of Integrated Information Theory (Oizumi et al., 2014). These methods have been applied to the question about integration of a system relative to its environment (Aguilera and Di Paolo, 2019, 2021). See also Bertschinger et al. (2008).

²⁸ *Eds.* Describing differences in scale in terms of *levels* and *hierarchies* is intuitive but sometimes misleading. Often, there is merely a conventional basis for positing such distinctions. However, Varela is hinting here at the possibility of using organizational closure to approach the issue in a principled manner. For another principled proposal in terms of critical transitions, see Longo et al. (2012). Interestingly, if closure, or autonomy, serves as the criterion for distinguishing a phenomenal domain with emergent properties, the neat notion of a hierarchy of levels is just one of many topological possibilities. There is no obvious hierarchy between the nervous and the immune systems within an organism or between a conversation and its participants. For more on emergence in relation to the enactive approach see Thompson (2007, appendix B).

²⁹ Eds. Herbert Simon (1969) formalizes these ideas with his notion of nearly decomposable systems (i.e., modular systems with connections between parts that do not alter how these parts operate, even if they do influence the result of their operation). A *tightly* interconnected system would be one that is not nearly decomposable. These are often described as softly assembled systems, exhibiting long-range correlations and nonadditive interactions among components (Van Orden et al., 2003; Kello and Van Orden, 2009).

more treelike it is, the less whole it is, while still being (presumably) a system. The extreme is probably a pure linear structure, without any branching at all.²⁹

A fourth point of view is that a system seems more whole if it is more *complex*, that is, more difficult to reduce to descriptions as interconnections of lower-level components. It is necessary in this discussion to take account of the (very modern) point of view that the more complex a system is to describe, the more *random* it is (Kolmogorov, 1968). Thus, for example, the wholeness of a living system is, in everyday encounters, construed as unpredictability. The more difficult it is to reduce a system to a simple input/output control, the more likely it is we will deem it alive. In this sense complete autonomy is logically equivalent to complete randomness. (Another example: a piece of music that is too complex, relative to our cultural expectations and inherent capacities, will sound random, chaotic, perhaps meaningless, but it will also sound whole. Here the extreme is white noise, Goguen, 1977.)

This viewpoint toward wholeness involves measurement relative to some standard interpreting system, an observer-community. But given such a standard, this viewpoint can be deduced from the preceding ones. For surely, it is difficult to describe a system, it will also be difficult to reduce it to lower levels, and its parts will seem to be tightly interconnected. Quite possibly, its very complexity will appear as an emergent property. As Atlan has recently remarked in a fundamental paper, the point when randomness becomes “information” for a system depends strictly on the observer’s position (Atlan, 1981). Different cognitive viewpoints might well be better able to process what now seems like a very complex system, and thus see it as less whole. Once again, the relativity to cognitive capacity appears.

10.7.3

These descriptive levels haven’t been generally realized as complementary largely because there is a difference between publicly announced methodology and actual practice in most fields of research in modern science. A reductionist attitude is strongly promoted, yet the analysis of a system cannot begin without acknowledging a degree of coherence in the system to be investigated; the analyst has to have an intuition that he is actually dealing with a coherent phenomenon. Although science has publicly taken a reductionist attitude, *in practice* both approaches have always been active. It is not that one has to have a holistic view as opposed to

a reductionist view, or *vice versa*, but rather that the two views of systems are complementary.

Similar conclusions apply to the understanding of autopoiesis in relation to allopoiesis, or to symbolic descriptions as opposed to causal. Neither choosing one pole *against* the other nor treating them at the *same* level seems adequate; rather they must be acknowledged as distinct, but interdependent cognitive perspectives of the observer-community.

There is a strong current in contemporary culture advocating autonomy, information (symbolic descriptions), and holism as some sort of cure-all and as a radically “new” dimension. This is often seen in discussions about environmental phenomena, human health, and management. In this book we take a rather different view. We simply see autonomy and control, causal and symbolic explanations, reductionism and holism as complementary or “cognitively adjoint” for the understanding of those systems in which we are interested. They are intertwined in any satisfactory description; each entails some loss relative to our cognitive preferences, as well as some gain.³⁰

³⁰ *Eds.* This attitude is characteristic of Varela’s later work as well. It is often remarked that the enactive approach rejects reductionism. It should be equally noted that it also rejects holism. Or, rather, that it rejects the “ism” in both of them, while acknowledging the need to deploy both attitudes contextually, as required by the kind of questions that we are asking.

Sources

(Goguen and Varela, 1979), (Varela, 1976).

11

Calculating Distinctions

11.1 On Formalization

This chapter, and the next two as well, deal with further ways to formalize the systemic features and processes that concern us in this book. That is, we seek a mathematical format within which we can capture some of the intuitions pointed out so far.

The decision to pursue such formal representations is based on the view that mathematical precision, when possible, makes a conceptual framework more useful and points out its limitations. I agree with this view.¹ Throughout these chapters, the formalisms developed will continue to give more insight into systemic autonomy and its mechanisms, as well as into where this approach is most immature. This was also the intention of the last chapter, in considering the notion of descriptive complementarity.

There are *two* essential topics to be discussed. First, in this chapter and the following one, I shall discuss a formalism to represent the act of *distinction*, a fundamental notion that runs through most of the present book.

Secondly, I shall deal with the question of circularity or *self-reference*, which is the nerve of the kind of dynamics we have been considering in living systems and autopoiesis—in organizational closure in general—and which is embedded in the participatory epistemology underlying this approach. We formalize the idea of self-referential processes in two stages. First the question is looked at in the context of the calculus of indication as self-indication or reentry of indicational forms. In chapter 13, the ideas developed in the bare ground of indication are enriched with computational capacities, and we develop the more operational concept of eigen-behavior, e.g., the self-determined states of a recurrent process. This

¹To abridge Chomsky: “The search for a rigorous formulation ... has a more serious motivation than mere concern for logical niceties or the desire to purify well established methods of ... analysis. Precisely constructed models ... can play an important role, both negative and positive, in the process of discovery itself. By pushing a precise but inadequate formulation to an unacceptable conclusion, we can often expose the exact source of this inadequacy, and consequently, gain a deeper understanding ...” (Chomsky, 1957, 5).

amounts to a characterization of autonomy through the invariances that emerge under organizational closure.

Throughout this presentation the reader should keep in mind that in dealing with these issues, we are often treading new ground, and grappling with problems that are both philosophical and technically difficult. Thus, all I have to say here should be treated as signposts in a terrain that needs way more study and research. We shall find uneven development of these ideas and important gaps in their applicability.

11.2 Distinctions and Indications

11.2.1

I have presented the arguments that make it apparent that the act of distinction lies at the foundations of any description. The most fundamental operation is that of distinguishing the “it” to be studied from its background. A distinction emerges out of an observer-community that decides the sense in which a distinction is performed. Thus we have physical boundaries, functional groupings, conceptual categorization, and so on, in an infinitely variegated museum of possible distinctions.

The act of distinction reveals a twofold aspect of the observer-community. On the one hand, it reveals the way in which such a distinction is accomplished: the *criteria* of distinction. On the other hand, it reveals the *intention* in selecting such criteria of distinction—the relative value of the distinction. These two aspects of the act of distinction have independent consequences.

The criteria of distinction used by an observer-community establish the kinds of entities to be studied, and thus the phenomenologies that are considered relevant. Once a class of entities is specified through a criterion of distinction, a phenomenology is concomitantly born, and this is all that is necessary for the existence of a phenomenological domain. This epistemological aspect of the distinction we considered at some length in the discussion of autopoiesis (chapters 6 and 10).

A distinction cannot exist without its concomitant value. The distinction thus becomes an *indication*, i.e., an indication is a distinction that is of value.² The indicative aspect of any distinction is of central importance because it is what enables the distinction to be changed throughout the history of an observer-community. This is so to the extent that values arise out of the continual self-interpretation of inquiring communities, which reinterpret the

²Eds. *Value* should here be understood in the broad sense of *focus or interest*. In distinguishing a system and its environment, we may place a “mark” on either side to signal our wish to focus on one or the other.

traditional or given indications in a partially redundant, but also partially innovative, way. Indications evolve within the uninterrupted process of the hermeneutical circle (Gadamer, 1960).

11.2.2

Another way of understanding the central role of distinctions in cognitive operations is in the language of the previous chapter. A distinction sets out a very fundamental kind of duality: *outside/inside*. It constructs a fundamental relativity across a distinction in whatever *indicative space* or domain we wish to consider. The interplay of this basic duality, by moving in and out across the border established by the distinction, creates the tension and basic forms of interdependence, which all of the dualities previously discussed map into (Varela, 1976). Typically, the whole/parts polarity is a form of distinction across levels, where we are either inside a system, or considering the system in its environment. A complementary view consists of explicitly relating these two views, star-fashion, across a distinction. Similarly, the complementary view of the outside/inside duality consists of finding the ways of relating these two poles, of making their codependent origination explicit. Thus, in exploring the *laws that govern the imbrication and crossing of indicational boundaries*, we are also examining a property common to all descriptions generated by an observer-community.³

11.3 Recalling the Primary Arithmetic

11.3.1

The first to recognize the importance of the act of distinction was George Spencer-Brown. In his lucid book, *Laws of Form*, he proposed a formalism for what he called, for the first time, a *calculus of indications* (Spencer-Brown, 1969).⁴ In what follows I shall recapitulate these ideas. The reader is encouraged to read the original book; my presentation is inevitably very much colored by my own interests.

Spencer-Brown found the inspiration for his work in logic. He was trained in logic and mathematics at Cambridge and Oxford; he was a student of Bertrand Russell's. In the early sixties he began to suspect that what had been hitherto taken for granted and solid ground in logic, could be seen at a very different light.⁵ In fact, since Frege and Russell's *Principia* it has been taken as a dogma that one could not find a simpler ground for logic than the notion of true and false as applied to the form of simple statements. In 1919, Russell posed this question in relation to logical propositions:

³Eds. As noted in section 10.2.1, distinctions and indications tend to be conflated in much of the present analysis. This means that the focus or value is always placed on one side of the distinction (the side indicated by a mark). Yet, as Varela suggests here, it is conceivable to focus on the *relations* between the sides of a distinction to investigate complementarity, adjointness, and dialectics as suggested in chapter 10. This alternative view, with its emphasis on codependent origination, informs much of the later enactive work and is actually complementary to the formalization presented in this and chapter 12.

⁴The book was first published in England in 1969, by George Allen & Unwin of London. It was published again in the United States by Julian Press of New York in 1972, and in a paperback edition in 1974 by Bantam Books. Reviews have appeared in *Journal of Symbolic Logic* 42:317 (1977) and *Nature* 223, 1392 (1969). (Eds. See Beer 1969, Gould 1977).

⁵Eds. For a recent assessment of the significance and circumstances surrounding Spencer-Brown's work, as well as biographical sketches, see (Kauffman et al., 2023).

The problem is: “What are the constituents of a logical proposition?” I do not know the answer, but I propose to explain how the problem arises ... We may accept as a first approximation, the view that *forms* are what enter into logical propositions as their constituents. And we may explain (though not formally define) what we mean by the “form” of a proposition as follows: The “form” of a proposition is that, in it, that remains unchanged when every constituent of the proposition is replaced by another. (Russell, 1919, 198–199).

What Russell is saying, and what since has been the royal route of mathematical logic, is that the basic building blocks of our formal discourse are these invariant patterns (“forms”), which must be taken as initials for a representation. Such simple patterns are well known, of course, as logical postulates, the initials of a Boolean algebra or any algebra of logic.

It is within the underlying epistemology of this approach that Spencer-Brown reframed this foundational question in a very different light:

A principal intention of this essay is to separate what are known as algebras of logic from the subject of logic, and to re-align them with mathematics.

Such algebras, commonly called Boolean, appear mysterious because accounts of their properties at present reveal nothing of any mathematical interest about their arithmetics. Every algebra has an arithmetic, but Boole designed his algebra to fit logic, which is a possible interpretation of it, and certainly not its arithmetic.⁶ Later authors have, in this respect, copied Boole, with the result that nobody hitherto appears to have made any sustained attempt to elucidate and to study the primary, non-numerical arithmetic of the algebra in everyday use which now bears Boole’s name.

When I first began, some seven years ago, to see that such a study was needed, I thus found myself upon what was, mathematically speaking, untrodden ground. I had to explore it inwards to discover the missing principles. They are of great depth and beauty, as we shall presently see. (Spencer-Brown, 1969, xi)

What is this untrodden ground that Spencer-Brown was envisioning? Again in his own words:

The theme of this book is that a universe comes into being when a space is severed or taken apart. The skin of a living organism cuts off an outside from an inside. So does the circumference of a circle in a plane ... The act is itself already remembered, even if unconsciously, as our first attempt to distinguish different things in a world where, in the first place, the boundaries can be drawn anywhere we please. At this stage the universe cannot be distinguished from how we act upon it, and the world may seem like shifting sand beneath our feet.

⁶*Eds.* Broadly, *algebra* is concerned with abstract relations between patterns of formal operations, and *arithmetic* is concerned with techniques of concrete calculation.

Although all forms, and thus all universes, are possible, and any particular form is mutable, it becomes evident that the laws relating such forms are the same in any universe. It is this sameness, the idea we can find a reality which is independent of how the universe actually appears, that lends such fascination to the study of mathematics. That mathematics, in common with other art forms, can lead us beyond ordinary existence, and can show us something of the structure in which all creation hangs together, is no new idea. But mathematical texts generally begin the story somewhere in the middle, leaving the reader to pick up the thread as best he can. Here the story is traced from the beginning. (Spencer-Brown, 1969, v)

Spencer-Brown's vision, then, amounts to a subversion of the traditional understanding on the basis of descriptions. It views descriptions as based on a primitive *act* (rather than a logical value or form), and it views this act as being the most simple yet inevitable one that can be performed. Thus it is a nondualistic attempt to set foundations for mathematics and descriptions in general, in the sense that subject and object are interlocked. From this basic intuition, he builds an explicit representation and a calculus for distinctions.

11.3.2

The key idea in Spencer-Brown's representation of indications is that all distinctions in their fundamental sense are alike, and all domains in which distinctions are performed are also alike. This gives rise to the notion of *primary* distinction and *indicational space*. We erase every qualitative difference of the criteria of distinctions, and simply reduce them to their essential quality: generating a boundary in whatever domain. Similarly, the value of the distinction is simply identified with the *name* of the content of the distinction, and so every value is treated alike. In this fashion all distinctions are similar (primary), and all indication are alike (a name). This sets the stage to represent indications in a simple fashion, and to consider calculations among them.

Definition 11.1 *Draw a distinction. Call the parts of the space shaped by the distinction the states of the distinction. Call the space and states the form of the distinction.*

Definition 11.2 *Let a state distinguished by the distinction be marked with a mark  of distinction. Call the state the marked state. Call  a cross. Call the concave side of the mark its inside, and let any mark be intended as an instruction to cross the boundary of the primary distinction. Let the crossing be to the state indicated by the mark.*

Definition 11.3 *Call the state not marked with a mark the unmarked state. Let a space with no mark indicate the unmarked state.*

Definition 11.4 *Call any arrangement of marks considered with regard to one another (that is, considered in the same form) an expression. Call a state indicated by an expression the value of the expression. Call expressions of the same value equivalent.*

Definition 11.5 *By the previous definition the forms $\sqcap$, $\sqcup$, are expressions. Call them the simple expressions. Let there be no other simple expressions.*

It should be noticed how, by condensing all distinctions into a primary one, and all indications to the same name or token, the only explicit symbol of this calculus, $\sqcap$, acquires a double sense. On the one hand it represents the act of distinction, of crossing the boundary in an indicational space. On the other hand, it is a value, the content of a distinction: It marks the outside part of this distinction. Likewise with the (non)symbol “ $\sqcup$ ”. On the one hand it expresses the relation between crosses in an expression that are contained in the same form; it is an implicit operation of continence. On the other hand, it is a value of an expression.

This condensed meaning of the symbols in the calculus is essential if one is to see the two basic actions that marks in an expression may have: They can either cross or be contained. If two crosses are contained in the same space, their value is that of distinguishing twice; thus, their value is the marked state. If a cross crosses another, we undo a distinction; hence their value is the unmarked state. Hence we can state the following two basic axioms that embody these fundamental relationships between crosses (Spencer-Brown, 1969):⁷

Axiom 11.6 *Form of Condensation*

$$\sqcap \sqcap = \sqcap.$$

Axiom 11.7 *Form of Cancellation*

$$\sqcap \sqcup = .$$

Call the calculus determined by taking these two primitive forms of equations as initials the *calculus of indications*.

The two axioms embody the two fundamental properties that are present whenever we draw a distinction in some indicational space, the duality inside/outside. First, we see that the distinction itself marks the outside, so that a mark on the outside (the right side

⁷Eds. It is important to remember that all distinctions are treated equally: just as a distinction (i.e., there is no distinguishing A from non-A and then distinguishing something different, B from non-B). When sharing the same space (on the line), the components of an expression should be read as a sequence of operations. Thus, to perform the operation of distinction twice results in the same marked state as performing it once. In contrast, a cross that operates on a whole expression creates another level in the space. When the operation of distinction is thus applied to a marked state, this is equivalent to undoing the original distinction, resulting in the unmarked state.

method, one regards the deepest spaces as sending signals of value up through the expression to be combined into a global valuation. To do this, let m stand for the marked state and n for the unmarked state. Thus $mm = m$, $mn = nm = m$, and $\overline{m} = n$, $\overline{n} = m$. Now use these labels as signals in the following example:

$$\begin{array}{c} \text{Diagram 1} \end{array} \rightarrow \begin{array}{c} \text{Diagram 2} \end{array} \rightarrow \begin{array}{c} \text{Diagram 3} \end{array} \rightarrow \begin{array}{c} \text{Diagram 4} \end{array} \rightarrow \begin{array}{c} \text{Diagram 5} \end{array} n.$$

Here $\begin{array}{c} \text{Diagram 1} \end{array}$ has the value $n =$. This procedure starts from the deepest spaces and labels those values that are unambiguous until a value for the whole expression emerges. Here is one more example:

$$e = \begin{array}{c} \text{Diagram 1} \end{array} \rightarrow \begin{array}{c} \text{Diagram 2} \end{array} m \quad m \quad n \quad m \quad n \quad m \quad n \quad \begin{array}{c} \text{Diagram 3} \end{array} m.$$

Hence $e = \overline{}$.

It should be noted that these methods are quite compatible. They reflect the dual nature of an expression as (self-)operator or operand. Viewed as an operator, the expression filters its own inner signals, creating a pattern or waveform that culminates in the evaluation. Even at this level, the relation between a form (or figure) and its dynamic unfoldment (or vibration) can be seen. This reflects the complementarity invariance/change, or space/time, which is to encountered at many points in science. For our consideration of circularity (self-reference), keeping this complementarity in mind is the key to resolving what have usually appeared to be vexing paradoxes. We shall return to this topic in the next chapters; for the rest of the chapter, however, we concentrate on the static and finite calculations on forms.

11.3.3

We can turn now to consider certain general theorems that characterize the calculus of indications.⁹

Theorem 11.8 *A form consisting of a finite number of crosses can be simplified to a simple expression.*

Proof. Consider any arrangement e in a space s . Find the deepest space in s . It can be found with finite search, since the number of crosses is finite. Call this space s_d .

Now s_d is either contained in a cross or not contained in a cross. If s_d is not contained in a cross, then s_d is s and there is no cross in s ,

⁹*Eds.* Theorems 11.8 to 11.10 establish the results that show that the two axioms lead to a consistent arithmetic (i.e., it is possible to simplify any finite expression in a finite number of steps, and this simplification is unique).

and so e is already simple. If s_d is in a cross c_d , then c_d is empty, since if c_d were not empty, s_d would not be deepest.

Now c_d either stands alone in s or does not stand alone in s . If c_d stands alone in s , then e is already simple. If c_d does not stand alone in s , then c_d must stand either: (case 1) in a space together with another empty cross (if the other cross were not empty, s_d would not be deepest) or (case 2) alone in the space under another cross. Case 1: In this case c_d condenses with the other empty cross. Thereby, one cross is eliminated from e . Case 2: In this case c_d cancels with the other cross. Thereby, two crosses are eliminated from e .

Thus, there will be a time when e is simplified. □

Theorem 11.9 *If any space contains an empty cross, the value indicated in the space is the marked state.*

Proof. Let e be any expression containing an empty cross. Then e is of the form

$$e = p_1 \overline{\quad} p_2.$$

By theorem 11.8, both parts p_1, p_2 reduce to a simple expression e_1, e_2 :

$$e = e_1 \overline{\quad} e_2.$$

But e_1, e_2 are either the marked or the unmarked state. Thus in any case, by the axioms,

$$e_1 \overline{\quad} e_2 = \overline{\quad},$$

and e indicates the marked state. □

Theorem 11.10 *The simplification of an expression is unique.*

Proof. Count the number of crossings from s_0 to the deepest space in e . If the number is d , call the deepest space s_d .

By definition, the crosses covering s_d are empty, and they are the only contents of s_{d-1} . Being empty, each cross in s_{d-1} can be seen to indicate only the marked state. Follow the following procedure:

1. Make a mark m on the outside of each cross in s_{d-1} . We know of course that

$$m = \overline{\quad}.$$

Thus no value in s_{d-1} is changed, since

$$\begin{aligned} \overline{\quad} &\rightarrow \overline{\quad} m \\ &= \overline{\quad} \overline{\quad} \\ &= \overline{\quad}. \end{aligned}$$

Therefore, the value of e is unchanged.

2. Next consider the crosses in s_{d-2} . Any cross in s_{d-2} is empty or covers one or more crosses already marked with m . If it is empty, mark it with m , so that the considerations in 1 apply. If it covers a mark m , mark it with n . We know that

$$n = .$$

Thus no value in s_{d-2} is changed. Therefore, the value of e is unchanged.

3. Consider the crosses in s_{d-3} . Any cross in s_{d-3} is empty or covers one or more crosses already marked with m or n . If it does not cover a mark m , mark it with m . If it covers a mark m , mark it with n . In either case, by considerations 1 and 2, no value in s_{d-3} is changed, and so the value of e is unchanged.

The procedure in subsequent spaces to s_0 requires no additional consideration. Thus, by the procedure, each cross in e is uniquely marked with m or n . Therefore, by dominance of m relative to n , a unique value of e in s_0 is determined. By the procedure leaves the value of e unchanged. Therefore the simplification of an expression is unique. $\square$

Corollary 11.11 *The value of an expression constructed by taking steps from a given simple expression is distinct from the value of an expression constructed from a different simple expression.*

Proof. Each step in the construction is reversible by simplification. But the simplification is unique; thus the corollary follows. $\square$

Theorem 11.12

Let p stand for any expression.¹⁰ The in any case,

$$\overline{p|p} = .$$

Proof. Let $p = \overline{\quad}$. Then

$$\begin{aligned} \overline{p|p} &= \overline{\overline{\quad}\overline{\quad}} && \text{by substitution (S)} \\ &= , && \text{A11.7} \end{aligned}$$

Let $p = .$ Then

$$\begin{aligned} \overline{p|p} &= \overline{\quad} && \text{S} \\ &= . && \text{A11.7} \end{aligned}$$

By theorem 11.8, there is no other case of p , and thus the theorem is proved. $\square$

¹⁰*Eds.* Theorems 11.12 and 11.13 prove two useful results that can be applied to any expression, which form the initials for the algebra of indications described in section 11.4.

Theorem 11.13 *Let p, q, r stand for any expressions. Then in any case,*

$$\overline{pr} \mid \overline{qr} \parallel = \overline{p} \mid \overline{q} \parallel r.$$

Proof. Let $r = \neg$. Then

$$\overline{pr} \mid \overline{qr} \parallel = \overline{p} \mid \neg \mid \overline{q} \parallel \neg \quad S$$

$$= \neg \mid \neg \parallel \quad T11.9$$

$$= \neg; \quad A11.7$$

and

$$\overline{p} \mid \overline{q} \parallel r = \overline{p} \mid \overline{q} \parallel \neg \quad S$$

$$= \neg. \quad T11.9$$

Let $r = \neg$. Then

$$\overline{pr} \mid \overline{qr} \parallel = \overline{p} \mid \overline{q} \parallel \quad S$$

and

$$\overline{p} \mid \overline{q} \parallel r = \overline{p} \mid \overline{q} \parallel \quad S$$

There is no other case of r (T11.8), and the theorem is proved. $\square$

11.4 An Algebra of Indicational Forms

11.4.1

We have presented the foundations for indicational forms, and have examined how this calculus behaves. Now we are in a position to take the particular patterns established in the last theorem as initials to construct a new calculus, or indicational *algebra* in which we deal only with the valid pattern that can be established among variables, regardless of their value (Spencer-Brown, 1969):

Initial 11.14 (Position)

$$\overline{p} \mid p \parallel = .$$

Initial 11.15 (Transposition)

$$\overline{pr} \mid \overline{qr} \parallel = \overline{p} \mid \overline{q} \parallel r.$$

By taking these two initials as valid, we can calculate other valid equivalences between indicational forms.

Proposition 11.16

$$\neg \mid p \parallel = p.$$

Proof.

$$\overline{p} \mid = \overline{p \mid p \mid p \mid} \quad \text{I11.14}$$

$$= \overline{p \mid p \mid p \mid p \mid} \quad \text{I11.15}$$

$$= \overline{p \mid p \mid} \quad \text{I11.14}$$

$$= \overline{p \mid p \mid p \mid p \mid} \quad \text{I11.14}$$

$$= \overline{p \mid p \mid} p \quad \text{I11.15}$$

$$= p. \quad \text{I11.14}$$

□

Proposition 11.17

$$\overline{pq} \mid q = \overline{p} \mid q$$

Proof.

$$\overline{pq} \mid q = \overline{p \mid q \mid} q \quad \text{P11.16}$$

$$= \overline{p \mid q \mid} q \quad \text{P11.16}$$

$$= \overline{p \mid q \mid q \mid} \quad \text{I11.15}$$

$$= \overline{p \mid q \mid} \quad \text{I11.14}$$

$$= \overline{p} \mid q. \quad \text{P11.16}$$

□

We now list several other algebraic propositions that are demonstrable. We omit their full proof.

Proposition 11.18

$$\overline{\mid} p = \overline{\mid}.$$

Proposition 11.19

$$\overline{p \mid q} \mid p = p.$$

Proposition 11.20

$$pp = p.$$

Proposition 11.21

$$\overline{p \mid q} \mid \overline{p \mid q} = p.$$

Proposition 11.22

$$\overline{\overline{p} \mid \overline{q} \mid r} = \overline{p} \mid \overline{q} \mid \overline{r}.$$

Proposition 11.23

$$\overline{\overline{p} \mid \overline{q} \mid \overline{s} \mid r} = \overline{p} \mid \overline{q} \mid \overline{s} \mid \overline{r}.$$

Proposition 11.24

$$\overline{\overline{q} \mid \overline{r} \mid \overline{p} \mid \overline{r} \mid \overline{s} \mid r \mid \overline{t} \mid r} = \overline{r} \mid \overline{p} \mid \overline{q} \mid \overline{r} \mid \overline{s} \mid t.$$

This primary algebra, and some of its results as we have listed above, has now to be compared with respect to the primary arithmetic from which it was derived. In other words, we have to ask whether the algebra is *complete* with respect to the arithmetic, so that if we consider an equivalence to be case in the arithmetic, then such an equivalence must be derivable from the initials of the algebra.¹¹ More precisely,

Theorem 11.25 *The primary algebra is complete. That is, $p = q$ can be proved in the arithmetic if and only if $p = q$ can be derived from I11.14 and I11.15.*

In order to proceed with the proof, it is first necessary to obtain some standard or canonical form to represent expression through algebraic reductions.

Lemma 11.26 *Let e be any expression in the primary algebra. Then e can be reduced to an expression containing not more than two appearances of any given variable. More precisely, suppose that x is a variable in e . Then there are expressions A, B, C , containing no appearances of x , such that*

$$e = \overline{Ax} \mid \overline{x} \mid B \mid C.$$

The proof of this lemma is fairly standard, so we shall omit it here. We can now turn to the proof of completeness.

Proof of Theorem 11.25. Because of the rules of algebraic manipulation, it is immediate that if an equivalence $p = q$ is derivable from I11.4 and I11.15, then it is valid in the arithmetic. Thus assume, conversely, that $p = q$ is a valid arithmetic expression. We show now that $p = q$ is derivable from I11.14 and I11.15.

The proof proceeds by induction on the number n of variables contained in p, q . If $n = 0$, then p, q are arithmetical expressions, and the assertion is trivially true. Assume that we have established the theorem for expressions containing less than n variables. Consider now expressions p, q containing a total of n distinct variables.

¹¹*Eds.* Since the algebra derives from the arithmetic via the expressions (I11.14) and (I11.15), any equivalence that one finds through algebraic manipulations has a corresponding equivalence in the arithmetic. But the converse is not obvious; there might be arithmetical results that cannot be proven algebraically. Theorem 11.25 shows that this is not the case, so the algebra is complete.

By the lemma, we can reduce p, q to their canonical form with respect to a variable x :

$$p = \overline{A_1 x} \mid \overline{B_1 x} \parallel C_1, \quad (11.1)$$

$$q = \overline{A_2 x} \mid \overline{B_2 x} \parallel C_2, \quad (11.2)$$

since this reduction is proved with algebraic steps only. Thus by substitution we find that

$$\overline{A_1} \mid A_3 = \overline{B_1} \mid B_3, \quad (11.3)$$

$$\overline{A_2} \mid A_3 = \overline{B_2} \mid B_3 \quad (11.4)$$

to be arithmetically true. However, these two equations contain less than n variables, and thus they are, by hypothesis, demonstrable.

Then we have the following steps:

$$\begin{aligned} p &= \overline{A_1 x} \mid \overline{B_1 x} \parallel C_1 && \text{by (11.1)} \\ &= \overline{\overline{A_1} \mid x} \mid \overline{\overline{B_1} \mid x} \parallel C_1 && \text{P11.24} \\ &= \overline{x A_1 \mid C_1} \mid \overline{x \mid \overline{B_1} \mid C_1} \parallel && \text{I11.15} \\ &= \overline{x A_2 \mid C_2} \mid \overline{x \mid \overline{B_2} \mid C_2} \parallel && \text{by (11.3), (11.4)} \\ &= q. && \text{by (11.2)} \end{aligned}$$

Thus $p = q$ with n variables is derivable from I11.14 and I11.15 if $p = q$ with less than n variables is derivable. This completes the induction step and the proof. $\square$

11.4.2

This concludes our summary of G. Spencer-Brown's calculus of indications. As we have seen, it is a formalism of great simplicity and elegance, which allows a representation of the act of indication and its basic laws. We have presented the basic notation, rules, and coherence of the calculus, and the algebra of indicational forms. There is more in the original presentation, and the reader is again encouraged to read it.

It seems helpful to close this chapter with a recapitulation of the motives for presenting this calculus in the context of autonomous systems and cognitive processes. There are two main reasons for pursuing an indicational calculus

First of all, in discussing the autonomy of living systems, we realized how important the act of distinction is in characterizing any phenomenal domain. In fact, a criterion of distinction is all that is necessary to establish a phenomenal domain in which the unities

distinguished are seen to operate. In this regard, a rigorous representation of indications serves a double purpose. On the one hand, it gives a foundation for systemic descriptions; I would say that it can be regarded as the foundation of systems theory, just as much as mathematicians can regard set theory as a foundation of their field. On the other hand, to start from the foundation of indication is faithful to the epistemology that pervades this presentation, in which the observer-community is always a participant. An indication reveals, in this sense, the interlocking between describer and described. Spencer-Brown puts it very beautifully:

We now see that the first distinction, the mark, and the observer are not only interchangeable, but, in the form, identical, (Spencer-Brown, 1969, 76).

In this sense, an indicational calculus can also be taken as a foundation for mathematics itself, which was, of course, Spencer-Brown's intention. It is relatively easy to see how indicational forms underlie the well-known results of propositional calculus; a brief discussion of this relation of indications to logic can be found in appendix B. But this is an issue I shall not discuss at this point, because it would take us off the track.

A second reason for pursuing indicational calculation is that in this simple, but fundamental level of systemic description, the questions of circularity and self-computation become more transparent and can be more clearly represented. In fact, at this level the whole question of self-formation can be fully resolved in a way that is not possible when we get closer to actual operations in natural systems. As it is normally the case, such simplified solutions are a guide to the more intricate, empirically bound descriptions. The next chapter deals with circularities of indicational forms.

Sources

(Spencer-Brown, 1969), (Kauffman and Varela, 1980).

12 Closure and Dynamics of Form

12.1 Reentry

12.1.1

We have chosen the calculus of indications as our basic ground for systemic descriptions. We wish to consider in this chapter the indicational forms of those systems exhibiting autonomy.¹ When describing a system, we have seen that all indications are relative to one another, as they all stand in relation to some indicational space or domain. So far we have considered only the most fundamental of these relations: containment. That is, we have only been concerned with the inside/outside relationship between crosses. This gives rise to expressions which, if they were geometrical forms, would be like Chinese boxes. When considering autonomous systems, and because of the closure thesis, we have seen that their organization contains “bootstrapping” processes that exhibit indefinite recursion of their component elements. This would amount to a form that reenters its indicational space, that informs itself. In the geometrical analogy it would be like a Klein bottle, where inside and outside become hopelessly confused.

One very simple way of describing this reentry is to say that a form, say f , is identical with parts of its contents,

$$f = \phi(f) \quad (12.1)$$

where ϕ is some indicational expression containing f as a variable. In another language, we are dealing with self-referential expressions: f says that ϕ is the case for itself. For example, consider

$$f = \overline{\overline{f}} \quad \text{P11.16}$$

which is a consequence in the primary algebra. Yet another demonstrable example in the primary algebra is

¹*Eds.* This chapter draws from Kauffman and Varela (1980) with elements from Varela (1975a) and Varela and Goguen (1978) integrated into it. See also Kauffman (1978a,b).

which would render the calculus of indications inconsistent, and thus useless, by confusing every form. Naturally, (12.2) is not a demonstrable proposition in the primary algebra, as there is no arithmetic value that will satisfy it: if f is marked, then it is unmarked; if f is unmarked, then it is marked. This paradoxical or oscillatory nature of some reentrant forms cannot be accommodated in the calculus without a much more careful consideration of what is involved.

12.2 The Complementarity of Pattern

12.2.1

An adequate appraisal of reentrant forms in all their diversity amounts to considering one basic descriptive complementarity: pattern/dynamic, or space/time. A hint of this was presented in section 11.3.2 in discussing the ways in which an expression is evaluated, where the dual nature of a mark is revealed. Viewed as an operator, the expression can be seen as filtering its own inner signals to create a pattern that is its evaluation. In the forms considered so far in the calculus of indications, this dynamic aspect is obscured because the filtering process always results in a steady value. In reentrant forms, however, we have a richer expression of this dynamic quality of forms.

In the simple example

$$f = f \sqcap \quad (12.2)$$

we may look at f in two different ways: as a *pattern* or form, or as a *dynamic* in time. On the one hand, we may simply use our notational convention and write

$$f = \square \quad (12.3)$$

as a *value* or pattern, not reducible to marked or unmarked. On the other hand, temporally we may view (12.2) as a *prescription* for recursive action:

$$f \rightarrow f \sqcap ;$$

thus

$$\sqcap \rightarrow \sqcap \sqcap \rightarrow \sqcap \sqcap \sqcap \rightarrow \dots,$$

and this regenerates constantly an oscillation that is identical to parts of itself:

$$f = \dots \sqcap \sqcap \sqcap = f \sqcap .$$

Thus the vibration yields a coherent form (e.g., $\square$) while the associated recursive dynamics unfolds the vibration into a temporal oscillation (e.g., $\neg$, $\neg$, $\neg$, $\neg$, ...). Notice however that, in this process, the deepest space in f has become *indeterminate*, and it is not evident how calculations are to be carried out.

What is involved here, then, is a cognitive peculiarity of us, as observer-communities. *Either* we behold a closing circle as a complete figure *or* we travel through the circle ceaselessly; these two processes are mutually interlocked, yet cannot be expressed simultaneously. If we allow condensation for each step of this recursive action, we can interpret (12.2) as a waveform

$$\dots \neg \rightarrow \neg \rightarrow \neg \rightarrow \neg \rightarrow \dots$$

or

$$\dots \rightarrow \neg \rightarrow \neg \rightarrow \neg \rightarrow \dots$$

more graphically represented as

$$\dots \neg \neg \neg \neg \neg \neg \neg \dots$$

or

$$\dots \neg \neg \neg \neg \neg \neg \neg \dots$$

³Eds. Expressions in the primary arithmetic that are evaluated by the signal method (in section 11.3.2) produce a waveform that eventually settles on one of the two states. In the case of reentry, the waveform never settles.

⁴Eds. And autonomy.

⁵Eds. Time in this context remains abstract. There is no notion of duration or rates of change involved, and the conception implicitly entails a discrete external measure indicating the point at which the next step in the waveform is realized. The analogy with the living cell, valid at an abstract level, should thus be qualified.

⁶Eds. It should be noted that Hans Jenny was studying wave phenomena in material systems and in real time. For a classic approach to patterns, structures, and oscillatory phenomena in biology, see Winfree (2001).

where the upswing indicates the appearance of a marked state.³

We have to pay attention to the fact that the double nature of self-reference, its blending of operand and operator, cannot be conceived of outside of time as a process in which two states alternate. True as it is that a cell is both the producer and the produced that embodies the producer, this duality can be pictured only when we represent for ourselves also a cyclical sequence of processes in time. Both aspects are evident in the idea of autopoiesis⁴: the invariance of a unity and the indefinite recursion underlying the invariance. Therefore we find a peculiar equivalence of self-reference and time, insofar as self-reference cannot be conceived outside time, and time comes in whenever self-reference is allowed.⁵

At an even more fundamental level, one can consider reentry as one kind of periodicity of descriptions in *any* domain. This theme of periodicity as the complementarity of invariance/dynamic has been elegantly stated by Hans Jenny:

Since the various aspects of these phenomena are due to vibration⁶, we are confronted with a spectrum which reveals patterned, figurate formations at one pole and kinetic-dynamic processes at the other, the whole being generated and sustained by its essential periodicity. These aspects,

however, are not separate entities but are derived from the vibrational phenomenon in which they appear in their “unitariness”. ... The three fields—the periodic as the fundamental field with the two poles of figure and dynamics—invariably appear as one. They are inconceivable without each other ... nothing can be abstracted without the whole ceasing to exist. We cannot therefore label them one, two, three, but can only say that they are threefold in appearance and yet unitary ... Hence we cannot say that we have a morphology and a dynamics generated by vibration, or more broadly by periodicity, but that all these exist together in true unity ... It is therefore warrantable to speak of a basic or primal phenomenon which exhibits this threefold mode of appearance. (Jenny, 1967, 121–122).

We are not going to pursue this beautiful theme in all its ramifications here. However, we will consider how this cognitive complementarity is encountered in the domain of indications, where patterns become indicational forms, and dynamics are recursive actions brought about by the reentry of the form into itself (Kauffman and Varela, 1980).

Once again, for the simple case

$$f = f \sqcap$$

we may regard the signals as moving outward like ripples on a pond, so that a timelike vibration by a yields a pattern of the form

$$\begin{array}{c} \sqcap \\ \sqcap \\ \sqcap \\ a \dots \end{array}.$$

That is, we may suppose that each time $a = m$, a mark appears, so that if time is represented as $t = 0, 1, 2, 3, \dots$, then

$$a = \begin{cases} m, & t \text{ is odd} \\ n, & t \text{ is even,} \end{cases}$$

and the outward expression grows in the pattern $\sqcap, \sqcap \sqcap, \begin{array}{c} \sqcap \\ \sqcap \end{array}, \dots$

In space we would see something like

$$\begin{array}{c} \sqcap \\ \sqcap \\ \sqcap \\ \dots \end{array} = f,$$

where the deepest space is now indeterminate due to its vibration. Here, the form is maintained by the vibration (or *growth*) at its center. Since the deepest space is indeterminate, calculation has failed. Form and dynamic have become one with the vibration. Nevertheless it must be noted that part of the vibration has been remembered as the external spatial pattern of the form. This pattern is maintained by the central vibration against the dynamical

pressure toward simplification (via calculation). Viewed entirely spatially, this temporal form becomes an *infinite* expression consisting of a descending sequence of marks. Thus its interior repeats itself. This *description* $f = \overline{f}$ can be seen as self-reference where f in-forms itself. This is the spatial context.

Temporally, we may view $f = \overline{f}$ as a *prescription* for recursive action (e.g., $f \rightarrow \overline{f}$), and this regenerates the waveform. Thus vibration yields (self-referential spatial) form, while the associated recursive dynamic (to the self-reference) unfolds the vibration into a temporal oscillation.

12.2.2

The complementarity between dynamic and pattern of reentrant forms can now be studied formally. The path is clearly staked out in the primary algebra. For if we apply the primary algebra to $f = \overline{f}$ we have seen that $f = \overline{\overline{f}}$. Thus in the primary calculus the marked state is fixed or purely spatial, while everything else is included in the unmarked state. $f = \overline{f}$, being vibratory, has been shunted out into the unmarked state, and it has been so shunted by the initial of position

$$= \overline{p} \mid p \mid . \quad \text{I11.14}$$

This says that if we wish to articulate reentrant forms, we must find ways to limit the cancellation permitted by the forms of position. In what follows, I will propose two different ways to do this: (1) by extending the initial arithmetic to include an oscillatory or *autonomous state*, and (2) by adopting an algebra with different initials where oscillatory terms can be *constructed* containing many terms of the form $\overline{p} \mid p \mid$. These two extensions of the primary algebra are explored in the remainder of this chapter.

12.3 The Extended Calculus of Indications

12.3.1

Let the calculus of indications, and the context from which it is seen to arise, be valid, except for the modifications introduced hereinafter.⁷

Definition 12.1 *Let there be a third state, distinguishable inform, distinct from the marked and unmarked states. Let this state arise autonomously, that is, by self-indication. Call this third state appearing in a distinction the autonomous state.*

⁷Eds. The idea of developing an extended calculus to accommodate operations with undefined results within a given domain is common in mathematics and often leads to practical applications. Such is the case of complex numbers, which may themselves be considered somewhat analogous to reentrant forms that have no solutions within the set of real numbers. If we recursively evaluate the value of x in the equation $x = -1/x$ starting with $x = 1$, we generate an oscillatory waveform $x = 1, -1, 1, -1, \dots$. Instead of expressing x as waveforms, we can assign it a fixed imaginary value: the imaginary unit i . Section 12.3 performs an analogous job for the reentrant form.

Definition 12.2 Let the autonomous state be marked with the mark $\square$, and let this mark be taken for the operation of an autonomous state, and be itself called the self-cross to indicate its operation.

Definition 12.3 Call the form of a number of tokens $\sqcap$, $\sqcup$, $\square$, considered with respect to one another, an arrangement. Call any arrangement intended as an indicator an expression. Call a state indicated by an expression the value of the expression.

Let v stand for any one of the marks of the states distinguished or self-distinguished: $\sqcap$, $\sqcup$, $\square$. Call v a marker.

Definition 12.4 Note that the arrangements $\sqcap$, $\sqcup$, $\square$ are, by definition, expressions. Call a marker a simple expression. Let there be no other simple expressions.

Let the following initials be valid, and be used to determine a calculus out of them. Call this calculus the *extended calculus of indications*.

Initial 12.5 (Dominance)

$$\sqcap v = \sqcap. \quad \text{I12.5}$$

Initial 12.6 (Order)

$$\sqcap \sqcap = \sqcap. \quad \text{I12.6}$$

Initial 12.7 (Constancy)

$$\square \sqcap = \square. \quad \text{I12.7}$$

Initial 12.8 (Number)

$$\square = \square \square. \quad \text{I12.8}$$

Theorem 12.9 The value indicated by an expression consisting of a finite number of crosses and self-crosses can be taken to be the value of a simple expression; that is, any expression can be simplified to a simple expression.⁸

⁸Eds. In analogy to theorems 11.8 to 11.10, theorems 12.9 to 12.11 establish the consistency of the extended arithmetic.

Proof. Let α be any expression, and let s be its indicative space. Being finite, α must have a reachable space which is the deepest in it. Call it s_d . s_d is either (1) contained in a cross, or (2) not contained in a cross.

1. If s_d is in a cross c_d , then c_d is either empty or contains a finite number of self-crosses, for otherwise s_d would not be deepest.
2. If s_d is not contained in a cross, then s_d either contains a finite number of self-crosses or it does not. In either case it is already simple, since the self-crosses can be condensed by I12.8.

Now, c_d either (3) stands alone in s , or (4) does not stand alone in s .

3. If c_d stands alone in s , then x is already simple, since it is either a cross or a self-cross, according to I12.7, I12.8
4. If c_d does not stand alone in s , the c_d must stand either (4a) in a space together with a marker (otherwise, s_d would not be deepest), or (4b) alone in the space under another cross.

In either case the initials apply and two markers are eliminated from α , and the expression reduced in depth by 1. Thus there will be a time when α has been simplified to a marker. $\square$

Theorem 12.10 *If any space pervades an empty cross, the value indicated by the space is the marked state.*

Proof. Evident. $\square$

Theorem 12.11 *The simplification of an expression is unique.*

Proof. Let α be any expression in a space s . Find the deepest space s_d . By hypothesis the crosses covering s_d are either empty or contain a self-cross (perhaps after condensation), and they are the contents of s_{d-1} , perhaps together with self-crosses. Mark m outside each empty cross in s_{d-1} , mark an a next to every cross covering a self-cross, and an a next to every self-cross in s_{d-1} . We know that

$$\begin{aligned} \square &\rightarrow \square m = \square \square = \square, \\ \square &\rightarrow \square a = \square \square = \square, \\ \square &\rightarrow \square a = \square \square = \square \square = \square = \square. \end{aligned}$$

Thus no value in s_{d-1} is changed.

Consider next the markers in s_{d-2} . Mark every self-cross with an a . Any cross in s_{d-2} either is empty or covers some marker already marked with m or a . If it is empty, mark it with m . If it covers a mark m , mark it with n ; if it covers no m but an a , mark it with a . We know

$$n = \square, \quad a \rightarrow a a = \square \square = \square = \square = a,$$

so that no value in s_{d-2} is changed.

Continue the procedure to subsequent spaces up to $s_0 = s$. By the procedure each marker is uniquely marked with m , n , or a . Therefore a unique value of x is determined. But the procedure leaves α unchanged, and the rules of the procedure are taken from the initials. Therefore, the value of α is uniquely determined by the procedure is the same as the value determined by simplification. Thus the simplification of an expression is unique. $\square$

Corollary 12.12 *The value of an expression constructed by taking steps from a given simple expression is distinct from the value of an expression constructed from a different simple expression.*

Proof. Every step in the construction is reversible by simplification. But the simplification is unique according to the preceding theorem. Thus the corollary follows. $\square$

The preceding results show that the three values of the calculus are not confused, that is, the calculus is consistent. Indeed, its consistency is seen, by the form of the proofs, to follow closely that of the calculus of indications.

Theorem 12.13 *Let p, q be any expressions.⁹ Then in any case*

$$\overline{p|q} \mid p = p.$$

Proof. Let $p = \neg$. Then

$$\begin{aligned} \overline{p|q} \mid p &= \overline{\neg|q} \mid \neg & S \\ &= \neg & T12.10 \\ &= p. & S \end{aligned}$$

Let $p = \vee$. Then

$$\begin{aligned} \overline{p|q} \mid p &= \overline{\vee|q} & S \\ &= \vee & T12.10 \\ &= & I12.6 \\ &= p. & S \end{aligned}$$

Let $p = \square$. Then

$$\begin{aligned} \overline{p|q} \mid p &= \overline{\square|q} \mid \square & S \\ &= \square|q \mid \square. & I12.7 \end{aligned}$$

Take $q = \neg$:

$$\begin{aligned} \overline{\square|q} \mid \square &= \overline{\square|\neg} \mid \square & S \\ &= \square & I12.5, I12.7 \\ &= p; & S \end{aligned}$$

Take $q = \vee$:

$$\begin{aligned} \overline{\square|q} \mid \square &= \overline{\square|\vee} \mid \square & S \\ &= \square & I12.7, I12.8 \\ &= p; & S \end{aligned}$$

⁹Eds. In analogy to theorems 11.12 and 11.13, theorems 12.13 to 12.15 prove the results that will form the initials for the extended algebra of indications.

Take $q =$:

$$\begin{aligned} \overline{\overline{q}} \mid \square &= \overline{\overline{\square}} \mid \square & S \\ &= \overline{\square} & \text{I12.7, I12.8} \\ &= p; & S \end{aligned}$$

There is no other case of q . There is no other case of p . Thus the theorem follows. $\square$

Theorem 12.14 *Let p be any expression. Then in any case*

$$p \mid \overline{\square} \mid p = p \mid \square.$$

Proof. Evident. $\square$

Theorem 12.15 *Let p, q, r be any expressions. Then in any case*

$$\overline{\overline{pr}} \mid \overline{\overline{qr}} \mid = \overline{\overline{p}} \mid \overline{\overline{q}} \mid r.$$

Proof. Evident. $\square$

Let the results of the three preceding theorems be taken as initials to determine a new calculus. Call this calculus the *extended algebra*.

Initial 12.16 (Occultation)

$$\overline{\overline{p}} \mid \overline{\overline{q}} \mid p = p. \quad \text{I12.16}$$

Initial 12.17 (Transposition)

$$\overline{\overline{pr}} \mid \overline{\overline{qr}} \mid = \overline{\overline{p}} \mid \overline{\overline{q}} \mid r. \quad \text{I12.17}$$

Initial 12.18 (Autonomy)

$$p \mid \overline{\square} \mid p = p \mid \square. \quad \text{I12.18}$$

From these initials the following propositions are demonstrable; full proofs can be found elsewhere (Varela, 1975a), except for P12.24, which is left here as an exercise.

Proposition 12.19 $p = \overline{\overline{p}} \mid .$

Proposition 12.20 $p \mid p = p.$

Proposition 12.21 $p \mid \overline{\square} \mid = \overline{\square} \mid .$

Proposition 12.22 $\overline{\overline{p}} \mid \overline{\overline{q}} \mid \overline{\overline{r}} \mid = \overline{\overline{pr}} \mid \overline{\overline{q}} \mid \overline{\overline{r}} \mid .$

Proposition 12.23 $\overline{\overline{p}} \mid \overline{\overline{qr}} \mid \overline{\overline{sr}} \mid = \overline{\overline{p}} \mid \overline{\overline{q}} \mid \overline{\overline{r}} \mid \overline{\overline{p}} \mid \overline{\overline{r}} \mid .$

Proposition 12.24 $\boxed{\boxed{p}} = \boxed{p} \boxed{p} \boxed{}.$

Proof. We first note that by I12.18

$$\boxed{\boxed{}} = \boxed{}.$$

Now

$$\boxed{p} \boxed{p} \boxed{} = \boxed{p} \boxed{\boxed{\boxed{}}} \boxed{} \quad \text{P12.19}$$

$$= \boxed{p} \boxed{\boxed{}} \boxed{p} \boxed{\boxed{}} \quad \text{I12.17}$$

$$= \boxed{p} \boxed{\boxed{\boxed{}}} \boxed{p} \boxed{\boxed{\boxed{}}} \quad \text{I12.17}$$

$$= \boxed{p} \boxed{\boxed{}} \boxed{p} \boxed{} \quad \text{P12.19, I12.16}$$

$$= \boxed{p} \boxed{\boxed{}} \boxed{} \quad \text{I12.18}$$

$$= \boxed{}.$$

□

Proposition 12.25 $\boxed{p} \boxed{p} \boxed{p} \boxed{} = \boxed{p} \boxed{}.$

Proposition 12.26 $\boxed{p} \boxed{r} \boxed{q} \boxed{r} \boxed{} = \boxed{p} \boxed{r} \boxed{q} \boxed{r} \boxed{r} \boxed{r} \boxed{}.$

It is interesting to note how some of the results valid in the primary algebra are also valid in this algebra. In fact, only the following are found to be invalid:

$$\boxed{p} \boxed{p} = \phantom{\boxed{p}},$$

$$\boxed{ab} \boxed{b} = \boxed{a} \boxed{b},$$

$$\boxed{a} \boxed{b} \boxed{a} \boxed{b} = a,$$

$$\boxed{\boxed{b} \boxed{r} \boxed{a} \boxed{r} \boxed{x} \boxed{r} \boxed{y} \boxed{r}} = \boxed{r} \boxed{ab} \boxed{rxy}.$$

These consequences of the primary algebra have a direct dependence on the validity of I11.14, the form of position, which is exactly the key difference, as it is reflected at the algebraic level, between the two calculi.

Theorem 12.27 *The extended algebra is complete.*¹⁰

Proof. We must show that if $\alpha = \beta$ can be proved true in the arithmetic, it is also algebraically demonstrable. The proof is done by induction on the number of variables of the equation $\alpha = \beta$ containing an aggregate of less than n variables. Let now $\alpha = \beta$ contain n variables. Let us write α and β in their canonical forms with respect to a variable p :

¹⁰Eds. As noted for the analogous case in section 11.4.1, it remains to be shown whether any equivalence in the extended arithmetic can be derived from the initials of the extended algebra.

$$\alpha = \overline{\alpha_1 p} \parallel \overline{\alpha_2 p} \parallel \alpha_3, \quad (12.4)$$

$$\beta = \overline{\beta_1 p} \parallel \overline{\beta_2 p} \parallel \beta_3, \quad (12.5)$$

These identities are demonstrable, since Lemma 11.23 can be proved using results valid in both calculi. We now have, by hypothesis,

$$\overline{\alpha_1 p} \parallel \overline{\alpha_2 p} \parallel \alpha_3 = \overline{\beta_1 p} \parallel \overline{\beta_2 p} \parallel \beta_3.$$

Substituting values for p , we find

$$\overline{\alpha_1} \parallel \alpha_3 = \overline{\beta_1} \parallel \beta_3, \quad (12.6)$$

$$\overline{\alpha_2} \parallel \alpha_3 = \overline{\beta_2} \parallel \beta_3, \quad (12.7)$$

$$\overline{\alpha_1 \square} \parallel \overline{\alpha_2 \square} \parallel \alpha_3 = \overline{\beta_1 \square} \parallel \overline{\beta_2 \square} \parallel \beta_3, \quad (12.8)$$

having at most $n - 1$ variables and therefore demonstrable. By (12.8),

$$\overline{\alpha_1 \square} \parallel \overline{\alpha_2 \square} \parallel \alpha_3 \square = \overline{\beta_1 \square} \parallel \overline{\beta_2 \square} \parallel \beta_3 \square \quad (12.9)$$

is also demonstrable by substitution. Thus

$$\alpha_3 \square = \overline{\alpha_1 \square} \parallel \overline{\alpha_2 \square} \parallel \alpha_3 \square \quad \text{I12.18}$$

$$= \overline{\beta_1 \square} \parallel \overline{\beta_2 \square} \parallel \beta_3 \square \quad \text{by (12.19)}$$

$$= \beta_3 \square, \quad \text{I12.18}$$

so that

$$\alpha_3 \square = \beta_3 \square \quad (12.10)$$

is demonstrable.

Now

$$\square \alpha = \overline{\alpha_1 p} \parallel \overline{\alpha_2 p} \parallel \alpha_3 \quad \text{by (12.4)}$$

$$= \overline{p \parallel \alpha_1} \parallel \overline{p \parallel \alpha_2} \parallel \overline{p \parallel p} \parallel \square \alpha_3 \quad \text{P12.26}$$

$$= \overline{p \parallel \alpha_1} \parallel \overline{\alpha_2 p} \parallel \overline{p \parallel p} \parallel \square \alpha_3 \quad \text{P12.19}$$

$$= \overline{p \parallel \alpha_1} \parallel \overline{\alpha_2 p} \parallel \overline{\square \alpha_3} \parallel \overline{p \parallel p \parallel \alpha_3} \parallel \square \quad \text{I12.17}$$

$$= \overline{p \parallel \alpha_1} \parallel \overline{\alpha_3 \square} \parallel \overline{\alpha_2 \parallel \alpha_3 \square p} \parallel \overline{\square p \parallel p \parallel \alpha_3} \parallel \square \quad \text{I12.17}$$

$$= \overline{p \parallel \beta_1} \parallel \overline{\beta_3 \square} \parallel \overline{\beta_2 \parallel \beta_3 p \square} \parallel \overline{\beta_3 \square p \parallel p} \parallel \square \quad \text{by (12.6), (12.7), (12.10)}$$

$$= \overline{p \parallel \beta_1} \parallel \overline{p \parallel \beta_2} \parallel \overline{p \parallel p} \parallel \beta_3 \square \quad \text{I12.17, P12.19}$$

$$= \overline{p \parallel \beta_1} \parallel \overline{p \parallel \beta_2} \parallel \beta_3 \square \quad \text{P12.26}$$

$$= \beta \square \quad \text{12.11}$$

showing that

$$\alpha \sqsupset = \beta \sqsupset \quad (12.11)$$

is demonstrable. Since by hypothesis $\alpha = \beta$ is true, although perhaps not demonstrable, it is also true, although perhaps not demonstrable, that $\overline{\alpha} = \overline{\beta}$, by substitution. An exactly similar argument to the preceding one about this new identity will show that

$$\overline{\alpha} \sqsupset = \overline{\beta} \sqsupset \quad (12.12)$$

is demonstrable.

Now,

$$\alpha = \overline{\alpha \sqsupset} \alpha \quad \text{I12.16}$$

$$= \overline{\beta \sqsupset} \alpha \quad \text{by (12.4)}$$

$$= \overline{\alpha \beta \sqsupset} \alpha \quad \text{I12.17}$$

$$= \overline{\alpha \beta \sqsupset \beta \sqsupset} \quad \text{by (12.11)}$$

$$= \overline{\alpha \sqsupset} \beta \quad \text{I12.17}$$

$$= \overline{\beta \sqsupset} \beta \quad \text{by (12.12)}$$

$$= \beta. \quad \text{I12.16}$$

□

12.3.2

Let us now consider the extension to equations of higher degree. Let any expression in the calculus be permitted to reenter its own indicative space at an odd or an even depth. Consider the expression

$$f = \overline{f \sqsupset} f, \quad (12.13)$$

where f reenters its own space at an odd and an even depth. In this case the value of f cannot be obtained by fixing the values of the variables that appear in the expression.

For example, let $f = \sqsupset$; then

$$= \overline{f \sqsupset} f \quad \text{S; by (12.13)}$$

$$= \overline{\sqsupset} \quad \text{S}$$

$$= . \quad \text{P12.19}$$

Now let $f = \sqsupset$:

$$\sqsupset = \overline{\sqsupset \sqsupset} \sqsupset \quad \text{S; by (12.13)}$$

$$\begin{aligned}
 &= \overline{\square} \square & S \\
 &= \square. & \text{I12.18, P12.20}
 \end{aligned}$$

By allowing reentry we have introduced a degree of indeterminacy which we must try to classify.

Definition 12.28 *Let the number of times reentry occurs in an expression determine a way to classify such expression. Call an expression with no reentry of first degree, those expressions with one reentering variable of second degree, and so on.*

Thus

$$f = \overline{f p} \quad (12.14)$$

is of second degree, while

$$f = \overline{f p} \overline{f q} \quad (12.15)$$

is of third degree.

To escape ambiguity in writing it is therefore necessary to adopt the convention that any variable whose value is the autonomous state can be taken to be a second-degree expression. Thus if $p = \square$, then this equation is of second degree, and by the preceding convention we have also

$$p = \overline{p}$$

Alternatively, any self-cross represents a reentrant expression, because we may write

$$\square = p$$

and thence

$$p = \overline{p}.$$

In this way we may look at a self-cross alternatively as a value in the arithmetic or as a basic form of a higher-degree equation, and thus they provide the connection between the arithmetic and reentrant expressions.

Definition 12.29 *Let α be an expression of any degree. A solution of α is any simple expression (if one exists) to which α can be shown to be equivalent.*

According to the definition, any first-degree expression will have one and only one solution. For higher degree we have seen that more than one solution are possible. But we have so far no assurance that any such solution exists in all cases of reentrant expressions.

Theorem 12.30 *Every expression has at least one solution in the extended calculus.¹¹*

¹¹*Eds.* In other words, there are no unsolvable expressions in the extended calculus, although some reentrant expressions may have more than one solution.

Proof. By Lemma 11.26 we only need to prove the result for expressions of the form

$$f = \overline{f \mid A \mid f B} \mid C.$$

where A, B, C contain no appearances of f —i.e., for expression of degree ≤ 3 .

Consider the case $C = \top$. Then it must be that $f = \top$.

Consider the case $C = \square$. Let A, B take on all possible values, and record the solutions for f as entries in the following table:

$\begin{array}{c} \diagdown \\ A \end{array} \quad B$	$\top$	$\square$
$\top$	$\square$	$\square$
$\square$	$\left\{ \begin{array}{c} \top \\ \square \end{array} \right\}$	$\left\{ \begin{array}{c} \top \\ \square \end{array} \right\}$

Consider the case $C = \square$. We obtain the table

$\begin{array}{c} \diagdown \\ A \end{array} \quad B$	$\top$	$\square$
$\top$	$\square$	$\square$
$\square$	$\left\{ \begin{array}{c} \top \\ \square \end{array} \right\}$	$\left\{ \begin{array}{c} \top \\ \square \end{array} \right\}$

Thus, an expression of any degree is equivalent to at least one simple expression. This completes the proof of the theorem. $\square$

12.4 Interpreting the Extended Calculus

12.4.1

I have presented an extension of the calculus of indications to encompass occurrences of self-referential situations, through the introduction of a third state in the form of indication, seen to arise autonomously by self-indication.

¹²*Eds.* That is, solved to create a basic expression.

The principal idea behind this approach can be stated thus: We choose to view the form of indication and the world arising from it as containing the two obvious dual domains of indicated and void states, and a third (not so obvious, but distinct) domain, of a self-referential autonomous state, which other laws govern and which *cannot* be reduced¹² by the laws of the dual domains. If we do not incorporate this third domain explicitly in our field of view, we force ourselves to find ways of avoiding it (as has been traditional) and to confront it, when it appears, in paradoxical forms.

We have shown that a third value can be introduced in a (Boolean) arithmetic while preserving consistency, and even more, providing a complete algebra to represent every arithmetic form. Although departing from the calculus in permitting reentrant expressions, these new forms are seen to fit into the calculus without contradiction, and thus it indeed serves as the basis of a rigorous foundation for higher-degree equations. Thus we have arrived at one satisfactory result that we were looking for.

Although a self-cross represents the paradigm for self-reference, it is the reentry of any expression into its own indicative space that permits us to recover all the basic forms of circularity. The results proven, however, show that (as is clear to the intuition) all the variety of reentrant expressions can be made equivalent to the basic values of the arithmetic. The connection of these expressions with the calculus hinges critically on the autonomous value, itself simultaneously a state in the form and a reentrant expression. Many such reentrant expressions can be shown to be equivalent to a self-cross, that is, shown to behave essentially as the basic paradigm of self-reference; however, as seen in the proof of theorem 12.30, not all reentrant expressions take on an autonomous value, as some of them are equivalent to a mark or a blank. Thus although some reentrant expressions may appear to be self-referential, in fact, at a closer inspection, they are not. The calculus not only shows that all self-referential situations can indeed be treated on an equal footing as belonging essentially to one class, but also shows a way to decide when an apparently self-referential situation is truly such.¹³

¹³*Eds.* That is, when after calculations, it is shown to reduce to a self-cross rather than to the marked or unmarked state of the primary arithmetic.

When restricted to the calculus itself, we can contemplate the behavior of self-reference; when allowed reentry, we can contemplate the unity in the diversity of self-referring situations. By moving farther from the arithmetic to free reentry we permit diversity to appear; by confining ourselves to the calculus we simplify back to the basic forms and regain uniqueness.

12.4.2

When Spencer-Brown introduces reentry and arrives at an expression equivalent to its content, $f = \overline{f}$ (what we call a self-cross), he notes its disconnection with his arithmetic and thus chooses to interpret it as an *imaginary* state in the form seen in *time* as alternation of the two states of the form. This interpretation is, in my opinion, one of his most outstanding contributions. He succeeds in linking time and description in a most natural fashion.¹⁴

However, we have seen that this interpretation could not be extended consistently to equations of higher degree; we took the alternative path of introducing a third value. What for the calculus of indications is contradictory with the arithmetic, here is a constitutive part of it, and we do not need any interpretation of a self-cross other than as an embodiment of self-reference or autonomy.

We may interpret a self-cross, a value in the extended arithmetic, as an alternation of the other values in time. Conversely we may take the states, marked and unmarked, as timeless constituents of a self-cross occurring as an oscillation in time. Either point of view reattaches time directly to our dealing with self-referential forms. We may note that, by considering a self-cross as an oscillation in time, we may also consider other reentrant expressions as modulations of a basic frequency. This is one of the applications that is clear for higher-degree expressions (Spencer-Brown, 1969, 67). To what extent a reentrant expression can be characterized by a certain frequency remains to be investigated: some of this is discussed below (see Kauffman and Varela, 1980).

12.4.3

The extended calculus can be interpreted for logic in much the same manner as the primary calculus (appendix B), and we need not repeat the process here. In fact the key difference between the two calculi, in this interpretation, is the same as between a two- and a three-valued logic. The adoption of a third value leads necessarily to the abandonment of the law of the excluded middle (*tertium non datur*), which, in the primary calculus, takes the form¹⁵

$$\overline{\overline{p}} \mid p = .$$

This form is not valid in the extended calculus, and it can be shown to be the source of contradictions when reentrant expressions are allowed in the primary calculus. We find a similar but not identical form in the extended calculus in

¹⁴ *Eds.* As noted in section 12.2.1, *time* in this context concerns sequential (and implicitly clocked) ordering of operations, not other aspects of temporality such as durations or rates of change.

¹⁵ *Eds.* In the logical interpretation of the calculus of indication, this expression reads that it is false that p and not- p are both true (Spencer-Brown, 1969, appendix 2).

$$\overline{\overline{p}} \mid p \mid \square = \square.$$

Of course, the abandonment of such a classical principle has a number of consequences, but these are not as serious as one might expect. Ackermann (1950) and Fitch (1950), for example, have presented contradiction-free logical systems leaving out *tertium non datur*, and have been able to show that such a logic is rich enough to permit the construction of most of classical mathematics. Thus a three-valued logic, although it forces us to abandon logical principles that appear basic to our common discourse, can nevertheless be reconstructed so as to deal in some other way with the common forms of discourse (and thus with basic mathematics). For the extended algebra, which is interpretable as one of these logics, similar conclusions are valid Varela (1979a, see also appendix B).

The consequences of introducing more than two values in a calculus or a logical system have been a current field of investigation since Łukasiewicz. Such additional values are usually interpreted in terms of probability or necessity (Gaines, 1976).¹⁶ Günther (1967) has been alone in pointing out that another possible interpretation of many-valued logics is as a basis for a “cybernetic ontology,” that is, for systems capable of self-reference, and precisely one additional value, he claims, must be taken as time.¹⁷ I follow here Günther’s suggestion that a third value might be taken as time. But I have shown that this third value can be seen at a level deeper than logic, in the calculus of indications, where the *form* of self-reference is taken as a third value in itself, and in fact confused with time as a necessary component for its contemplation. In the extended calculus, self-reference, time, and reentry are seen as aspects of the same third value arising autonomously in the form of distinction.

12.5 A Waveform Arithmetic

I wish to return now to the question of “imaginary” Boolean values, as Spencer-Brown calls them. In the extended calculus of indications we have allowed time to enter to the extent that it is embodied in the autonomous value (cf. section 12.4.2). The intention in the following further expansion on the calculus of indications is to unfold all the details contained in this autonomous state. We shall do so by developing an algebraic structure in which the full extent of the pattern/dynamic (spatial/temporal) complementarity can be accommodated (Kauffman and Varela, 1980).¹⁸

¹⁶Eds. For an account of many-valued logics, including fuzzy logic, see Malinowski (1993).

¹⁷Günther’s work is not easy to read, and I have found his papers on time of 1967, more illuminating than the other ones. For a more complete bibliography, see von Foerster (1974a, 487). It is no accident that Günther found the origin of his interests in Georg Wilhelm Friedrich Hegel.

¹⁸Eds. Sections 12.5 to 12.9 are dedicated to investigating the deep relation between reentrant forms, oscillation patterns, and algebraic operators and the equilibria that they generate. Reentrant forms generate oscillatory patterns, and they may be constructed so that they have any arbitrary period. The central theme is the complementarity between operators and patterns of reentry and oscillations and dynamics.

Consider again $f = \overline{f}$. By taking successive replacement of f in its equivalent form we have a sequence

$$f, \overline{f}, \overline{\overline{f}}, \overline{\overline{\overline{f}}}, \dots$$

Now, if we let f take the two possible initial values, $\overline{\quad}$ and $\overline{\overline{\quad}}$, and apply condensation, we get two sequences:

$$\overline{\quad}, \overline{\overline{\quad}}, \overline{\overline{\overline{\quad}}}, \dots$$

and

$$\overline{\overline{\quad}}, \overline{\overline{\overline{\quad}}}, \overline{\overline{\overline{\overline{\quad}}}}, \dots$$

These two sequences can be looked at as the successive values of two basic *waveforms*:¹⁹

$$\begin{aligned} i &= \dots \overline{\quad} \overline{\overline{\quad}} \overline{\overline{\overline{\quad}}} \overline{\overline{\overline{\overline{\quad}}}} \dots, \\ j &= \dots \overline{\overline{\quad}} \overline{\overline{\overline{\quad}}} \overline{\overline{\overline{\overline{\quad}}}} \overline{\overline{\overline{\overline{\overline{\quad}}}}} \dots \end{aligned}$$

As a first step toward making this temporal expression a well-defined object, note that in the example provided above, $i = \overline{i}$ if we interpret $\overline{\quad}$ as *ordinary inversion plus a half-period shift*. That is, suppose we have a periodic pattern $x = \dots ababab \dots$. Then we *define*

$$\overline{x} = \dots \overline{b} \overline{a} \overline{b} \overline{a} \overline{b} \overline{a} \overline{b} \overline{a} \overline{b} \overline{a} \dots$$

Thus if

$$i = \dots \overline{\quad} \overline{\overline{\quad}} \overline{\overline{\overline{\quad}}} \overline{\overline{\overline{\overline{\quad}}}} \dots,$$

then

$$\begin{aligned} \overline{i} &= \dots \overline{\overline{\overline{\overline{\overline{\quad}}}}} \overline{\overline{\overline{\overline{\quad}}}} \overline{\overline{\overline{\quad}}} \overline{\overline{\quad}} \overline{\quad} \overline{\overline{\quad}} \overline{\overline{\overline{\quad}}} \overline{\overline{\overline{\overline{\quad}}}} \dots, \\ \overline{\overline{i}} &= i. \end{aligned}$$

Note also that $ij = \overline{\quad}$, since at any given time either i or j is marked. Now the sequence $\dots ababab \dots$ can be looked at in a different way, somewhat similar to the construction of complex numbers from the real field. That is, we think of the ordered pair (a, b) as representing the essential features of the sequence. This shift in perspective, in fact, permits a more detailed algebraic treatment of the intuitions described above.

Definition 12.31 Let B be any algebra (or arithmetic) satisfying the initials for the primary algebras. Define

$$\hat{B} = \{(a, b) | a, b \in B\}$$

and define

$$(a, b) = (\overline{b}, \overline{a}), \quad (12.16)$$

$$(a, b)(c, d) = (ac, bd). \quad (12.17)$$

¹⁹Eds. This temporal interpretation was proposed by Spencer-Brown, who pictured waveforms in analogy to flip-flop switching electronic circuits. He called them "imaginary Boolean values." (Spencer-Brown, 1969, p. 99).

Let

$$\begin{aligned} i &= (\overline{\square}, \overline{\square\square}), \\ j &= (\overline{\square\square}, \overline{\square}). \end{aligned}$$

Identify

$$a \equiv (a, a).$$

The smallest collection of such pairs representing waveforms is, in fact, one containing no variables, but only the constant elements $\{\overline{\square}, \overline{\square\square}, i, j\}$. Call it the *waveform arithmetic* V . By the previous definition we see that

Proposition 12.32 *Let P denote the primary arithmetic. Then $V = \hat{P}$.*

As required, the basic waveforms i, j arise out of the static forms of the primary arithmetic. Next, we note the following

Proposition 12.33 *In $\hat{B}$ occultation and transposition are valid, and*

$$\overline{i} = i, \quad \overline{j} = j, \quad ij = \overline{\square}.$$

Proof. These verifications are straightforward and are left to the reader. $\square$

This proposition shows that in the waveform arithmetic and in the related algebras, occultation and transposition can be used as initials to construct them. The next section develops this idea.

12.6 Brownian Algebras

Let $a, b, \dots, p, q, \dots$ be a collection of variables, and let the cross be defined as before. Take the following two initials as valid:

Initial 12.34 (Occultation)

$$\overline{p} \overline{q} = p. \quad \text{I12.14}$$

Initial 12.35 (Transposition)

$$\overline{pr} \overline{qr} = \overline{p} \overline{q} r. \quad \text{I12.15}$$

Call any algebra satisfying I12.34 and I12.35 a *Brownian algebra*. We can now derive some forms of equations valid in these algebras. For the missing proofs, see Kauffman and Varela (1980).

Proposition 12.36

$$\overline{\overline{a}} = a.$$

Proof.

$$\overline{\overline{a}} = \overline{\overline{\overline{a}} \overline{a}} = \overline{\overline{a}} \overline{a} = a. \quad \text{I12.34}$$

$$= \overline{\overline{a} \overline{a} \overline{a} \overline{a}} \quad \text{I12.34}$$

$$= \overline{a \overline{a} \overline{a}} \quad \text{I12.35}$$

$$= \overline{\overline{a} \overline{a}} \overline{aa} \quad \text{I12.35}$$

$$= \overline{aa} \quad \text{I12.34}$$

$$= a \quad \text{I12.34}$$

□

Proposition 12.37

$$aa = a.$$

Proposition 12.38

$$\overline{\overline{a}} = \overline{a}.$$

Proposition 12.39

$$\overline{\overline{a} \overline{b} \overline{c}} = \overline{ac} \overline{b} \overline{c}.$$

Proposition 12.40

$$\overline{a \overline{r} \overline{b} \overline{r}} = \overline{ab} \overline{r}.$$

Proposition 12.41

$$\overline{ab} \overline{b} = \overline{a} \overline{b} \overline{bb}.$$

Proposition 12.42

$$\overline{a \overline{b} \overline{a} \overline{b}} = \overline{a} \overline{bb}.$$

Proposition 12.43

$$\overline{a \overline{br} \overline{cr}} = \overline{a} \overline{b} \overline{c} \overline{a} \overline{r}.$$

Proposition 12.44

$$\overline{\overline{ax} \overline{a} \overline{y} \overline{a} \overline{az}} = \overline{ax} \overline{a} \overline{y} \overline{a} \overline{a} \overline{xyz}.$$

It is worth while comparing these consequences with those valid in the primary algebra and in the extended algebra. In all cases cancellation of the terms $\overline{p} \overline{p}$ or those involving $\overline{\overline{p}}$ yields the corresponding result in the primary algebra. What is interesting in the Brownian algebras is that the two initials adopted surprisingly yield an algebra, closely related to the primary algebra, that automatically avoids the cancellation of $\overline{p} \overline{p}$.²⁰ In the extended algebra we had to add an extra initial, I12.18, which explicitly introduced a reentering value, and thus avoided the pre-emptying of these self-interfering forms $\overline{p} \overline{p}$. In the Brownian algebra we have not

²⁰Eds. In other words, the Brownian algebra is a positive framework that automatically avoids the law of the excluded middle.

assumed values such as $\overline{\overline{\square}}$, but we see that we are implicitly allowing them, and thus allowing all of their variety. Let us explore this further.

12.7 Completeness and Structure of Brownian Algebras

We have already seen that the arithmetic V generated by $\overline{\square}$, $\overline{\overline{\square}}$, i , j with $\overline{i} = i$, $\overline{j} = j$, and $ij = \overline{\square}$ satisfies I11.14 and I11.15. Hence V is a model for a Brownian algebra. All of our consequences hold for expressions in V . In fact we shall show that any Brownian algebra is *complete* with respect to this arithmetic.

Theorem 12.45 *Let α and β be two algebraic expressions. Then $\alpha = \beta$ is a consequence of I11.14 and I11.15 if and only if $\alpha = \beta$ is true in the arithmetic V .*

The proof of this result requires some preliminary work as outlined below. The first result we need is an algebraic reduction of forms.

Proposition 12.46 *Let α be any expression in the Brownian algebra. Then α can be reduced to an expression containing no more than four appearances of a given variable. More precisely, suppose that x is a variable in α . Then there are expressions A, B, C, D involving no appearance of x such that*

$$\alpha = \overline{Ax} \overline{Bx} \overline{Cxx} \overline{D}.$$

Proof. First note, by using P12.39, that any expression is equivalent to an expression no more than two crosses deep. Hence we find that

$$\alpha = \overline{xa_1} \overline{b_1} \dots \overline{xa_n} \overline{b_n} \overline{xc_1} \overline{xd_1} \dots \overline{xc_m} \overline{xd_m} \overline{xe_1} \dots \overline{xe_p} \overline{f},$$

where $a_1 \dots a_n, b_1 \dots b_n, c_1 \dots c_m, d_1 \dots d_m, e_1 \dots e_p, f$ are expressions in which x does not appear.

Note that $\overline{xa} \overline{b} = \overline{x} \overline{b} \overline{a} \overline{b}$ by P12.40. Similarly $\overline{xc} \overline{xd} = \overline{x} \overline{xd} \overline{xc} \overline{d}$. Thus the proposition follows at once from these facts and repeated applications of P12.40. $\square$

The next result involves evaluating an expression at $\overline{\square}$, $\overline{\overline{\square}}$, i , and j . Recall that these generate the waveform arithmetic V . We should note that while our initials for Brownian algebra do *not* assume the existence of elements such as i and j , we can assume that such elements are present in the Brownian algebra under discussion.

Proposition 12.47 Let $\alpha(x) = \overline{x A} \overline{x} \overline{B} \overline{x x} \overline{C} \overline{D}$ where where A, B, C, D are expressions involving no appearance of the variable x . Then

$$\alpha(\overline{\quad}) = \overline{A} \overline{D},$$

$$\alpha(\overline{\quad}) = \overline{B} \overline{D},$$

$$\overline{\alpha(i)} \overline{\alpha(j)} = \overline{D},$$

$$\alpha(i)\alpha(j) = \overline{A} \overline{B} \overline{C} \overline{D}.$$

Proof. The first two equations are obvious. For the next note that

$$\alpha(i) = \overline{i A} \overline{i B} \overline{i C} \overline{D} \quad (\text{since } \overline{i} = i)$$

$$= \overline{i \overline{A} \overline{B} \overline{C}} \overline{D} \quad (\text{using P12.40 twice}).$$

Similarly,

$$\alpha(j) = \overline{j \overline{A} \overline{B} \overline{C}} \overline{D}.$$

Hence (letting $E = \overline{A} \overline{B} \overline{C}$) we have

$$\begin{aligned} \overline{\alpha(i)} \overline{\alpha(j)} &= \overline{i E} \overline{D} \overline{j E} \overline{D} \\ &= \overline{i E} \overline{j E} \overline{D} && \text{I12.35} \\ &= \overline{i E j E} \overline{D} && \text{P12.46} \\ &= \overline{\quad} \overline{E} \overline{D} && (ij = \overline{\quad}) \\ &= \overline{D}. \end{aligned}$$

Finally,

$$\begin{aligned} \alpha(i)\alpha(j) &= \overline{i E} \overline{D} \overline{j E} \overline{D} \\ &= \overline{i E} \overline{j E} \overline{D} \\ &= \overline{i \overline{j} \overline{E}} \overline{D} && \text{P12.40} \\ &= \overline{ij} \overline{E} \overline{D} \\ &= \overline{E} \overline{D} && (ij = \overline{\quad}) \\ &= \overline{\overline{A} \overline{B} \overline{C}} \overline{D} \\ &= \overline{A} \overline{B} \overline{C} \overline{D}. \end{aligned}$$

This completes the proof of the proposition. $\square$

Proof of Theorem 12.45. We are given two algebraic expressions α and β such that $\alpha = \beta$ can be proved as a theorem about the arithmetic V . That is, $\alpha = \beta$ is true when all variables are replaced by choices of $\overline{\quad}$, $\overline{\overline{\quad}}$, i, j . We wish to show that under these conditions $\alpha = \beta$ is demonstrable from the initials. The proof will proceed by induction on the total number n of variables in the two expressions.

If $n=0$, then $\alpha = D$ and $\beta = D'$, where D and D' are constants. Hence $D = \overline{\quad}$ or $\overline{\overline{\quad}}$, and $D' = \overline{\quad}$ or $\overline{\overline{\quad}}$. By hypothesis $D = D'$ and there is nothing to prove.

Thus we may assume that $n > 0$ and that the theorem is true for all smaller n . Let x be a variable appearing in one or both of the expressions α, β . By proposition 12.46 we can assume that

$$\alpha = \overline{x A} \overline{x} \overline{B} \overline{x x} \overline{C} \overline{D}$$

and

$$\beta = \overline{x A'} \overline{x} \overline{B'} \overline{x x} \overline{C'} \overline{D'},$$

where $A, B, C, D, A', B', C', D'$ are expressions involving no appearance of x .

By the evaluations of proposition 12.47 and the hypotheses of this theorem it then follows that the following formulas are demonstrable:

$$\overline{A} \overline{D} = \overline{A'} \overline{D'}, \quad (12.18)$$

$$\overline{B} \overline{D} = \overline{B'} \overline{D'}, \quad (12.19)$$

$$D = D', \quad (12.20)$$

$$\overline{A} \overline{B} \overline{C} \overline{D} = \overline{A'} \overline{B'} \overline{C'} \overline{D'}. \quad (12.21)$$

We now apply P12.44 to demonstrate $\alpha = \beta$:

$$\begin{aligned} \alpha &= \overline{x A} \overline{x} \overline{B} \overline{x x} \overline{C} \overline{D} \\ &= \overline{\overline{x A} \overline{x} \overline{B} \overline{x x} \overline{A} \overline{B} \overline{C}}} \overline{D} && \text{P12.36, P12.44} \\ &= \overline{\overline{x A} \overline{D} \overline{x} \overline{B} \overline{D} \overline{x x} \overline{D} \overline{A} \overline{B} \overline{C} \overline{D}}} && \text{I12.35} \end{aligned}$$

Now substitute, using Equations (12.18)–(12.21), reverse steps, and conclude that $\alpha = \beta$.

This completes the induction step and the proof of theorem 12.45. $\square$

Some technical comments are in order here. We have actually proved that any *free* Brownian algebra is complete with respect to V .

That is, given a set S , we can form an algebra $B(S)$ by regarding the elements of S as variables with no special relations. We then form a set of *expressions* $E(S)$ by the following rules:

1. s is an expression for each $s \in S$.
2. $\overline{\quad}$ and $\mid$ are expressions.
3. If X and Y are expressions, then $\overline{X\mid}$, $\overline{Y\mid}$, and XY are also expressions.

Initials 12.34 and 12.35 generate an equivalence relation on $E(S)$. We let $B(S) = E(S)/\sim$ and say $\alpha = \beta$ if $\alpha \sim \beta$ for $\alpha, \beta \in E(S)$. We call $B(S)$ the *free Brownian algebra on the set S* .

We can now examine not only the contents and consequences of a given Brownian algebra, but also the relations *between* Brownian algebras. This is in line with the mathematical doctrine that the structure-preserving maps between objects of a class are at least as important as the objects themselves. Thus, we go on now to the definition of homomorphisms between Brownian algebras.

Definition 12.48 *Let B, B' be Brownian algebras. Then a homomorphism $h : B \rightarrow B'$ is a set mapping such that*

$$\begin{aligned} h(s) &= s, \\ h(\overline{\mid}) &= \overline{\mid}, \end{aligned}$$

and

$$\begin{aligned} h(xy) &= h(x)h(y), \\ h(\overline{x\mid}) &= \overline{h(x)\mid} \end{aligned}$$

for all elements $x, y \in B$.

It is “well known” that in a free algebra any homomorphism is determined by its values $h(s)$ for $s \in S$. In the case at hand, a homomorphism between a Brownian algebra B and V amounts to assigning the variables $s \in S$ a value in the waveform arithmetic. This can be best stated in the following theorem, which is simply a reformulation of theorem 12.45 in this context.

Theorem 12.49 *Let $B(S)$ be a Brownian algebra in the set S . Then for $\alpha, \beta \in B(S)$, $\alpha = \beta$ if and only if $h(\alpha) = h(\beta)$ for every homomorphism $h : B(S) \rightarrow V$.*

We now show that the waveform arithmetic reveals a great deal about the overall structure of Brownian algebras. We first need to define the Cartesian-product construction of algebras (not to be confused with the $\wedge$ -construction).

Definition 12.50 Let B and B' be Brownian algebras. Then the product algebra $B \times B'$ is defined by taking the Cartesian product of the underlying sets and defining operations by

$$\overline{(a, b)} = (\overline{a}, \overline{b}) \quad (12.22)$$

$$(a, b)(c, d) = (ac, bd). \quad (12.23)$$

Similarly, if A is an indexing set and we have algebras B_a , $a \in A$, then we can form a product of all of these and denote it by $\prod_{a \in A} B_a$.

Theorem 12.51 Let $B(S)$ be any free Brownian algebra on the set S . Let $A = \{h : B(S) \rightarrow V\}$ be the set of homomorphisms of $B(S)$ to the waveform arithmetic V . Let V_n denote (a copy of) V , corresponding to each homomorphism $h \in A$.

Then there is an injective homomorphism $\Phi : B(S) \rightarrow \prod_{h \in A} V_h$.

Proof. Define $\Phi : B(S) \rightarrow \prod_{h \in A} V_h$ by

$$\Phi(x) = \prod_{h \in A} h(x) \quad \text{for each } x \in B(S).$$

Since $x = y$ in $B(S)$ if and only if $h(x) = h(y)$ for all $h \in A$, we see that $x = y$ in $B(S)$ if and only if $\Phi(x) = \Phi(y)$. This says that Φ is injective. $\square$

Theorem 12.51 follows, in fact, from even deeper results in the language of De Morgan algebras (see Kauffman, 1978a). But from our point of view, this result is quite significant, since it shows that any Brownian algebra can be regarded as a *subalgebra* of tuples of elements from the waveform arithmetic, e.g., of $\prod_{h \in A} V_h$. The latter, as we know, is entirely generated by self-reflective elements, that is, by solutions of $x = \overline{x}$. Thus, the waveforms associate with the simple reentrant form $x = \overline{}$ stand at the base of all our considerations. The “real” logical or indicational values such as $\overline{}$ are seen as combinations of such synchronized “imaginary” waveforms ($ij = \overline{}$). This principle remains true in the general context of all algebras satisfying occultation and transposition.

In this regard it is worth noting that the self-reflexive elements of a Brownian algebra are *irreducible* in the sense that given x such that $x = \overline{x}$, then $x \neq yz$, where $y \neq z$ and $\overline{y} = y, \overline{z} = z$. That is, we have the following proposition:

Proposition 12.52 Let $B(S)$ be a Brownian algebra containing elements x, y, z satisfying $x = \overline{x}, y = \overline{y}, z = \overline{z}$. Then

$$xy = xz \Rightarrow y = z.$$

Proof. Note that this result will follow at once if we prove that for any α, β , if $\alpha x = \beta x$ and $\overline{\alpha} x = \overline{\beta} x$ and $x = \overline{x}$, then $\alpha = \beta$. To prove this:

$$\alpha = \overline{\alpha} \overline{x} \alpha \quad \text{I12.34}$$

$$= \overline{\beta} \overline{x} \alpha \quad (\text{hypothesis})$$

$$= \overline{\beta} \overline{x} \alpha \quad (x = \overline{x})$$

$$= \overline{\beta \alpha} \overline{x \alpha} \quad \text{I12.35}$$

$$= \overline{\alpha \beta} \overline{x \beta} \quad (\text{hypothesis})$$

$$= \overline{\alpha} \overline{x} \beta \quad \text{I12.35}$$

$$= \overline{\beta} \overline{x} \beta \quad (\text{hypothesis})$$

$$= \beta. \quad \text{I12.34}$$

This completes the proof. $\square$

12.8 Varieties of Waveforms and Interference Phenomena

In the algebra $\hat{B}$ associated with a Brownian algebra B , we find the periodic elements from B . Thus for $a, b \in B$, we have the correspondence $(a, b) \leftrightarrow \dots ababab \dots$. So far, however, we have encountered only waveforms of period 2, namely, those of the waveform arithmetic V . There is no reason to stick to patterns of period 2 (high frequency) in the context we have developed so far. Let us now discuss explicit constructions for waveforms of arbitrary period.

Definition 12.53 *Let p be any even integer, $p \in \mathbb{Z}$, and $p = 2k$ for some $k \in \mathbb{Z}$. Let B be a given Brownian algebra. Define $\mathcal{S}_p(B)$ to be the set of sequences in B of period p . That is,*

$$\mathcal{S}_p(B) = \{a = (a)_n, n \in \mathbb{Z} \mid a_n \in B \text{ and } a_{n+p} = a_n \text{ for all } n\}.$$

This collection of sequences can be transformed into an algebra by extending the operations of containment and crossing in the following way:

$$ab = (ab)_n = a_n b_n, \quad (12.24)$$

$$\overline{a} = (\overline{a})_n = \overline{a_{n-k}}, \quad k = p/2. \quad (12.25)$$

Thus crossing is accomplished by combining ordinary inversion for B with a half-period shift (whence p must be even). This extends the previous use of crossing for the $\wedge$ -construction.

By way of illustration, consider sequences of period 4 and the special case $B = P$, i.e., $\mathcal{S}_4(P)$. Take one such sequence a :

$$a = \overline{\quad} \overline{\quad} \overline{\quad} \overline{\quad} = \overline{\quad}, \quad \overline{\quad}, \quad \overline{\quad}, \quad \overline{\quad}, \dots,$$

whence

$$\overline{a} = \text{---} \text{---} \text{---} \text{---} = \text{---} \text{---} \text{---} \text{---}, \dots$$

These two waveforms may be combined:

$$\overline{a} \mid a = \text{---} \text{---} \text{---} \text{---} = \text{---} \text{---} \text{---} \text{---}, \dots = a.$$

It is immediate to verify that occultation and transposition hold in $\mathcal{S}_p(B)$. Thus $\mathcal{S}_p(B)$ is a Brownian algebra. The next result gives a more precise idea of the structure of this algebra, by showing that is indistinguishable from tuples of forms of period 2.

Proposition 12.54 *Let B be a Brownian algebra. Then we have following isomorphism of algebras:*

$$\mathcal{S}_p(B) \cong \prod_{l=1}^k \hat{B}_l,$$

where $\hat{B}_l$ ($l = 1, \dots, k$) denotes (a copy of) B corresponding to each of the integers 1 through k , and $k = p/2$.

Proof. Let

$$\bar{\mathcal{S}}_p(B) = \{(\alpha, \beta) \mid \alpha = (\alpha_1, \dots, \alpha_k), \beta = (\beta_1, \dots, \beta_k)\}, \quad \alpha_i, \beta_j \in B.$$

Define $(\alpha, \beta)(\alpha', \beta') = ((\alpha_1 \alpha'_1, \dots, \alpha_n \alpha'_n), (\beta_1 \beta'_1, \dots, \beta_n \beta'_n))$ and $\overline{(\alpha, \beta)} = ((\overline{\beta_1}, \dots, \overline{\beta_k}), (\overline{\alpha_1}, \dots, \overline{\alpha_k}))$. Then we may map $\mathcal{S}_p(B)$ to $\bar{\mathcal{S}}_p(B)$ by $h : \mathcal{S}_p(B) \rightarrow \bar{\mathcal{S}}_p(B)$, where $h(a) = ((a_1, \dots, a_k)(a_{k+1}, \dots, a_p))$. This is clearly an isomorphism. On the other hand we have $g : \bar{\mathcal{S}}_p(B) \rightarrow \prod_{l=1}^k \hat{B}_l$ by $g(\alpha, \beta) = ((\alpha_1, \beta_1), (\alpha_2, \beta_2), \dots, (\alpha_k, \beta_k))$, and this is also an isomorphism. Hence the composition $g \circ h : \mathcal{S}_p(B) \rightarrow \prod_{l=1}^k \hat{B}_l$ is the desired isomorphism. $\square$

If $p=2$, then this result shows that $\mathcal{S}_2(B) \cong \hat{B}$ as expected. When $B = P$, the primary arithmetic, then $\mathcal{S}_p(B) \cong \prod_{l=1}^k V_l$ (where $v_l \cong v$). For the example of $a \in \mathcal{S}_4(P)$ discussed above, one obtains $\mathcal{S}_4(P) \cong V \times V$, and $a \leftrightarrow (i, \overline{\text{---}})$.

By combining two sequences in the same indicational space we produce an interference between the waveforms they represent. This is what is involved in the extension of crossing in (12.25) above. Interference follows quite naturally for waveforms of the same period (but arbitrary frequency). We are left with the question of interference of waveforms of widely different periods, so that we may handle the resulting patterns in an adequate way. One approach, presented below, is to concentrate on the *least* possible period resulting from the interference.

Let $\text{lcm}(p, q)$ denote the least common multiple of the integers p, q . For even integers, the lcm always exists, and we may define a

mapping

$$\mu : \mathcal{S}_p(B) \times \mathcal{S}_q(B) \rightarrow \mathcal{S}_r(B),$$

where $r = \text{lcm}(p, q)$, and

$$\mu[(a, b)_n] = a_n b_n.$$

Thus two sequences of different periods interfere to form a new sequence whose least period divides the least common multiple of the initial periods. By choosing to look at interference this way we are simply stressing the *high-frequency components of the interference pattern*. This suggests the construction of an algebra of sequences of varying periods. This algebra, although it can be defined, is somewhat different from the Brownian algebras studied so far. We call such structures *generalized Brownian algebras*.

Definition 12.55 *A generalized Brownian algebra is an algebra satisfying the initials*

$$aa = a \quad (\text{iteration}), \quad (12.26)$$

$$\overline{a} = a \quad (\text{reflection}), \quad (12.27)$$

$$\overline{a|b|}c = \overline{ac|bc|} \quad (\text{transposition}). \quad (12.28)$$

It is easy to verify that many forms of equations are consequences of these initials. Occultation does *not* follow from them, as the models to be discussed will show.

Definition 12.56 *Let B be a Brownian algebra, and let $\mathcal{S}(B)$ denote the set of periodic sequences in B with period of the form $2k$ where k is odd. If $a, b \in \mathcal{S}(B)$, then we write $a = b$ when $a_n = b_n$ for all n and $p(a) = p(b)$ ($p(a)$ = the period assigned to a). Operations are defined as follows:*

$$\left. \begin{aligned} (ab)_n &= a_n b_n, \\ p(ab) &= \text{lcm}(p(a), p(b)); \end{aligned} \right\} \quad (12.29)$$

$$\left. \begin{aligned} (\overline{a})_n &= \overline{a_{n-k}}, \quad k = p(a)/2, \\ p(\overline{a}) &= p(a). \end{aligned} \right\} \quad (12.30)$$

Since $\text{lcm}(\text{lcm}(x, y), z) = \text{lcm}(x, \text{lcm}(y, z))$, the operation (12.29) is associative. This has to be made explicit at this point. Associativity has always been implicitly assumed.

Lemma 12.57 *$\mathcal{S}(B)$ is a transposition algebra whenever B is a Brownian algebra.*

Proof. See Kauffman and Varela (1980). □

an algebraic version of the sequence. That is, if $x \in B$, then $\alpha = (x, \overline{x})$ and $\beta = (\overline{x}, x)$ belong to $\hat{B}$ and represent two phase-shifted versions:

$$\begin{aligned}\alpha &: \dots x \overline{x} \overline{x} x \overline{x} x \overline{x} \dots, \\ \beta &: \dots \overline{x} x \overline{x} x \overline{x} x \overline{x} \dots,\end{aligned}$$

We are given $T : B \rightarrow B$ and obtain the corresponding mapping $\hat{T} : \hat{B} \rightarrow \hat{B}$ defined by the same formula. Note that $\hat{T}(\alpha) = \alpha$ and $\hat{T}(\beta) = \beta$. Thus the sequences generated by T become the fixed points of $\hat{T}$. This correspondence holds more generally, as shown in the next result.

Theorem 12.58 *Let B be any algebra (or arithmetic) satisfying all the initials for the primary algebra. We shall say that B is primary. Then an algebraic mapping $T : B \rightarrow B$ will generate sequences of period at most 2.*

Proof. Since T is an operation in the primary algebra, it has a canonical structure with respect to the variables that it operates on; in fact,

$$T(x) = \overline{ax} \overline{bx} \parallel c$$

for some $a, b, c \in B$ not containing x . Now, if $c = \overline{}$, there is nothing to prove, for T would be a constant transformation. So let us assume $c = $, and consider T of the general form

$$T(x) = \overline{ax} \overline{bx} \parallel .$$

In this case simple calculation shows

$$T^2(x) = \overline{abx} \overline{\overline{a} \overline{b} \overline{x}} \parallel ,$$

and

$$T^3(x) = T(x)$$

by using P12.39, P12.41, and I12.34. Induction on n will show

$$T^{n+2}(x) = T^n(x)$$

for all n . □

Theorem 12.59 *Let B be primary, and T an algebraic mapping. Let $\hat{T}$ be the corresponding mapping on $\hat{B}$. Then there exists a $z \in \hat{B}$ such that $\hat{T}(z) = z$.*

Proof. In particular we may take $z = (T(x), T^2(x))$ or $z = (T^2(x), T(x))$ for any $x \in B$. To see that this pair will be a fixed point for $\hat{T}$, first note that for any $z \in \hat{B}$, $z = (\alpha, \beta)$,

$$\begin{aligned}T(z) &= \overline{az} \overline{bz} \parallel \\ &= \overline{(a\alpha, a\beta)} \overline{b(\beta, \alpha)} \parallel \end{aligned}$$

$$\begin{aligned}
&= (\overline{\alpha\beta}, \overline{a\alpha})(\overline{b\alpha}, \overline{b\beta}) \\
&= (\overline{\alpha\beta}, \overline{b\alpha})(\overline{a\alpha}, \overline{b\beta}).
\end{aligned}$$

Thus it suffices to show that

$$\overline{aT^2(x)} \overline{bT(x)} = T(x)$$

and

$$\overline{aT(x)} \overline{bT^2(x)} = T^2(x)$$

This is a straightforward computation. $\square$

Thus the algebraic structure of B reflects the properties of periodic sequences that are generated from B . More specifically, sequences from B become algebraic fixed points in B .

What we see emerging here is rather beautiful harmony between oscillations, reentrant forms, algebraic operations, and their fixed points. So far we have seen that the reentry of a form to its own indicational space, as in $x = \overline{x}$, gives rise to a fundamentally new arithmetic V , where we have the waveforms i, j . These are fixed points for the cross $\overline{\quad}$ in the new algebra V . We can attempt to generalize this situation by showing how every reentrant form will give rise to an oscillation: The fixed points of the operator represent the spatial view of the oscillation, while its associated sequences represents the temporal context. Now, in order to do this, we have to be able to construct an algebra where infinite expressions are defined, so that we are assured every algebraic expression actually has a fixed point in the *same* algebra, rather than in a larger one (i.e., i is in V , not in P). We provide such a construction in the next section, and in doing so we shall see that the correspondence between pattern (reentry) and oscillation (sequences) will be partially lost. Although every reentrant form will oscillate, a given waveform can be generated through many alternative operators when they are allowed to reenter.

In the remainder of this section we will concentrate on procedures to generate an oscillation of arbitrary period from an operator. We have proved that operators of one variable will generate only period 2. In order to obtain sequences of period other than 2, it is necessary to use recursion on more than one variable. For example, let $T : P \times P \rightarrow P \times P$ be defined by $T(x, y) = (\overline{y}, \overline{x} \overline{y})$, that is, we have the set of two equations

$$\begin{aligned}
x &= \overline{y}, \\
y &= \overline{x} \overline{y}.
\end{aligned}$$

Then

$$\begin{aligned} T(\sqcap, \sqcap\!\!\sqcap) &= (\sqcap, \sqcap), \\ T(\sqcap, \sqcap) &= (\sqcap\!\!\sqcap, \sqcap\!\!\sqcap), \\ T(\sqcap\!\!\sqcap, \sqcap\!\!\sqcap) &= (\sqcap, \sqcap\!\!\sqcap), \end{aligned}$$

and hence $T^3 = T$, and T produces two entrained oscillations of period 3:

$$\begin{aligned} y^\nabla &= \dots, \sqcap, \sqcap, \sqcap, \sqcap, \sqcap, \dots = \text{---}\sqcap\text{---}\sqcap\text{---}\sqcap\text{---}\dots, \\ x^\nabla &= \dots, \sqcap, \sqcap, \sqcap, \sqcap, \sqcap, \dots = \text{---}\sqcap\text{---}\sqcap\text{---}\sqcap\text{---}\dots \end{aligned}$$

We say that T produces an oscillation of period 3 and *dimension 2*. Notice that we may also represent this two-dimensional waveform by the spatial pattern of reentry of the operation which generates it,

$$y^\nabla = x \sqcap \sqcap,$$

and thus

$$x^\nabla = \sqcap \sqcap \sqcap.$$

In general, we now show that it is a fairly simple matter to determine operators T that produce a given waveform.

Definition 12.60 Let P denote the primary arithmetic, let $X = \{x_1, x_2, \dots\}$ be a set of variables, and let $\bar{P}^n = (P/X) \times (P/X) \times \dots$ be the n -fold Cartesian product of (P/X) with itself. An algebraic operator $\mathbf{T} : \bar{P}^n \rightarrow \bar{P}^n$ is a function $\mathbf{T}(x_1, \dots, x_n) = (T_1(x_1, \dots, x_n), \dots, T_n(x_1, \dots, x_n))$, where each $T_k(x_1, \dots, x_n)$ is an expression in the primary algebra involving the variables $x_1, \dots, x_n$. $\mathbf{T}$ might be thought of as a set of n equations:

$$\begin{aligned} x_1 &= T_1(\mathbf{x}), \\ &\vdots \\ x_n &= T_n(\mathbf{x}) \end{aligned}$$

We say that $\mathbf{T}$ is periodic if there exists an integer p such that

$$\mathbf{T}^{p+n} = \mathbf{T}^n$$

for all $n > N$, where $p, n, N \in \omega$, the set of non-negative integers, and N is some specified integer.

With this definition we have immediately the following interesting result:

Theorem 12.61 Every operator is periodic.

Proof. $\bar{P}^n$ has cardinality 2^n (with respect to arithmetic values). Hence for any fixed $\mathbf{x}$ the set $\{\mathbf{T}^n(\mathbf{x}) | n = 1, 2, \dots\}$ is finite. Thus the sequence $\mathbf{T}(x), \mathbf{T}^2(x), \dots$ must be eventually periodic for each x . Since there are a finite number of such x , the least common multiple of the corresponding periods is necessarily a period for $\mathbf{T}$. $\square$

Lemma 12.62 *Let $a = \{a_n | n = 1, 2, \dots\}$ be any periodic sequence of values from P . Let p be the least period of a , and choose n so that $2^{n-1} < p < 2^n$. Then there exist an operator $\mathbf{T} : \bar{P}^n \rightarrow \bar{P}^n$ of least period p and a starting value $\mathbf{x}$ such that*

$$a_n = \pi(\mathbf{T}^n(\mathbf{x})) \quad \text{for } n = 1, 2, 3, \dots$$

Here $\pi : \bar{P}^n \rightarrow P$ by $\pi(\alpha_1, \dots, \alpha_n) = \alpha_1$.

Thus the sequence a can be seen as the first component of an n -dimensional entrained oscillation.

Proof. We shall give an algorithm for producing the requisite operator. The following notation is convenient. Let $\mathbf{b} = (b_1, b_2, \dots, b_n) \in P^n$, and let $\Delta(\mathbf{b}) = \epsilon_1 \epsilon_2 \dots \epsilon_n$, where $\epsilon_i = 1$ if $b_i = \sqcap$ and $\epsilon_i = 0$ if $b_i = \sqcup$. Regard $\Delta(\mathbf{b})$ as an integer expressed in the binary system. Let $\Omega(\mathbf{b})$ be the corresponding decimal integer. Let $\sigma(\mathbf{b})$ be the following operator:

$$\sigma(\mathbf{b})(x_1, \dots, x_n) = \overline{b_1(x_1) b_2(x_2) \dots b_n(x_n)},$$

where

$$b_i(x) = \begin{cases} \sqcap, & \text{if } b_i = \sqcap \\ x, & \text{if } b_i = \sqcup \end{cases}.$$

Note that $\sigma(\mathbf{b}) : \bar{P}^n \rightarrow P$ and

$$\sigma(\mathbf{b})(\mathbf{x}) = \sqcap \Leftrightarrow \mathbf{x} = \mathbf{b}.$$

For computations it is often useful to use $\Omega(\mathbf{b})$ or $\Delta(\mathbf{b})$ as the name of $\sigma(\mathbf{b})$.

Now choose $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p \in P^n$ so that

$$\mathbf{b}_i \neq \mathbf{b}_j \quad \text{if } i \neq j \quad (12.31)$$

$$\pi(\mathbf{b}_k) = a_k, \quad k = 1, 2, \dots, p. \quad (12.32)$$

This can be done, since $2^{n-1} < p \leq 2^n$.

Let $T_k(x_1, \dots, x_n) = \sigma(\mathbf{b}_{\alpha_1})\sigma(\mathbf{b}_{\alpha_2}), \dots, \sigma(\mathbf{b}_{\alpha_l})$, where $\{\alpha_1, \dots, \alpha_l\}$ is the set of indices α such that the k th coordinate of $\mathbf{b}_{\alpha+1}$ is marked (we view α modulo p , so that $\mathbf{b}_{p+1} = \mathbf{b}_1$). Finally, let $\mathbf{T}(\mathbf{x}) = (T_1(\mathbf{x}), T_2(\mathbf{x}), \dots, T_n(\mathbf{x}))$.

It is easy to verify that $\mathbf{T}(\mathbf{b}_k) = \mathbf{b}_{k+1}$ and $\mathbf{T}(\mathbf{b}_p) = \mathbf{b}_1$. Thus $\mathbf{T}$ produces the desired periodic sequence. This completes the proof of the lemma. $\square$

In order to illustrate the foregoing lemma, suppose that we wish to produce the period-5 oscillation



That is, $a_1 = \overline{}$, $a_2 = \overline{\overline{}}$, $a_3 = \overline{}$, $a_4 = \overline{\overline{}}$, $a_5 = \overline{\overline{}}$. Then $n = 3$. We may choose $\mathbf{b}_1, \dots, \mathbf{b}_5$ as follows:

	x_1	x_2	x_3	$\Omega(\mathbf{b})$
$\mathbf{b}_1$	$\overline{}$	$\overline{\overline{}}$	$\overline{\overline{}}$	4
$\mathbf{b}_2$	$\overline{\overline{}}$	$\overline{\overline{}}$	$\overline{\overline{}}$	0
$\mathbf{b}_3$	$\overline{}$	$\overline{\overline{}}$	$\overline{}$	5
$\mathbf{b}_4$	$\overline{\overline{}}$	$\overline{\overline{}}$	$\overline{}$	1
$\mathbf{b}_5$	$\overline{\overline{}}$	$\overline{}$	$\overline{\overline{}}$	2

Hence $\Omega(\mathbf{b}_1) = 4, \Omega(\mathbf{b}_2) = 0, \Omega(\mathbf{b}_3) = 5, \Omega(\mathbf{b}_4) = 1, \Omega(\mathbf{b}_5) = 2$.

Hence

$$T_1 = 20 = \overline{x\ y\ z}\ \overline{xyz} = \overline{xz},$$

$$T_2 = 1 = \overline{xyz},$$

$$T_3 = 05 = \overline{xyz}\ \overline{x\ y\ z}.$$

Thus $\mathbf{T}(x, y, z) = (\overline{xz}, \overline{xyz}, \overline{xyz}\ \overline{x\ y\ z})$.

In general there will be many operators $\mathbf{T}$ such that $a_n = \pi \mathbf{T}^n(x)$. We can easily generalize this lemma to embed any entrained oscillation in a higher-dimension one without repetitions, thereby obtaining an operator $\mathbf{T}$ whose projected iterates give rise to this entrainment.

In the next section we look more closely at the formal structure of this category of operators.

12.10 Reentrant Forms and Infinite Expressions

12.10.1

We have already remarked that certain reentrant or self-referential expressions in the primary algebra can only have solutions in the larger context of Brownian algebras, where complex values are allowed—that is, where oscillations are not discarded. We also pointed out that a temporal form might be viewed as an *infinite* expression, consisting of a descending cascade of marks and variables. Thus $f = \overline{f}$ could be seen as something like

$$f = \overline{\overline{\overline{}}},$$

where the deepest space is indeterminate. This extension of the primary algebra into infinite forms to express their dynamic vibration has been unnecessary until now, because we have adopted the course of expressing oscillation in terms of sequences, their algebraic characteristics, and their generation from algebraic operators. Now we need to relate more explicitly the dynamic quality of a vibration to its self-referential spatial form. In the fundamental case of $f = \overline{f}$, we need to make sense of the spatial quality of f 's reentry, or, in symbols,

$$f = \overline{\square}.$$

What can this possibly mean?

Consider the following example, due to Spencer-Brown. Let $T(x) = \overline{xa} \overline{b}$. Then iterating T we have, as expected,

$$\begin{aligned} T^2(x) &= \overline{\overline{xa} \overline{b}} \overline{a} \overline{b} \\ &= \overline{xa} \overline{b} \\ &= T(x). \end{aligned}$$

Thus

$$\overline{xa} \overline{b} = \overline{\overline{xa} \overline{b}} \overline{a} \overline{b} = \overline{\overline{\overline{xa} \overline{b}} \overline{a} \overline{b}} \overline{a} \overline{b} \overline{a} \overline{b} = \dots,$$

and if we allow this process to proceed indefinitely, we are led to contemplate the infinite expression

$$x^\nabla = \overline{\dots a \overline{b} \overline{a} \overline{b}}.$$

This form contains a copy of itself, and thus reenters its own indicational space

$$x^\nabla = \overline{x^\nabla a} \overline{b}.$$

By going to an infinite expression, we have eliminated x as a variable, and obtained a form or spatial pattern, which embodies the operation. In other language, x^∇ is the fixed point of T , for obviously

$$T(x^\nabla) = \overline{x^\nabla a} \overline{b} = \overline{\dots a \overline{b} \overline{a} \overline{b}} = x^\nabla.$$

The equation $T(x^\nabla) = x^\nabla$ is an expression of the direct identity of these expressions;²¹ it is not a statement that one can be calculated from the other. In general, by going into a suitable structure, where infinite expressions are allowed, we are assured that every operation will have a fixed-point solution, for we can form

$$x^\nabla = T(T(T(\dots))) = \lim_{n \rightarrow \infty} T^n,$$

²¹ *Eds.* In other words, it defines the eigenbehavior of T ; see chapter 13.

the infinite concatenation of this operator. In this universe of infinite forms, in other words, we are free to express reentry of forms. What needs to be examined is what is the relation between these spatial reentries and their temporal quality: Given a pattern, how does it vibrate?

In order to explore this question we first have to construct a universe of infinite expressions, and see more precisely how reentry is expressed in them. We have two basic clues from the above discussion. First, in order to construct an infinite expression we may make it grow step by step, and never allow the process to come to a halt. This means introducing the idea of an *order* in the class of expressions, so that at each successive step the next expression is a better approximation to the infinite one. The sequence of approximations, in the limit, defines the infinite expression. This is a process somewhat reminiscent of the idea of order, approximation, and limit in calculus; the order being introduced here, however, is quite different from numerical approximation. Secondly, in order to construct a reentry, it is enough to consider *equations* among these infinite expressions, or, in other words, fixed points for operators in this extended domain.

For example, for the operation $T(x) = \overline{xa} \mid b \mid$ we may consider a sequence of approximate expressions thus;

$$\perp \sqsubseteq \overline{\perp a} \mid b \mid \sqsubseteq \overline{\overline{\perp a} \mid b \mid} \overline{a} \mid b \mid \sqsubseteq \dots,$$

where we start with an undefined expression $\perp$, and successively add more and more components. Here $\sqsubseteq$ indicates the order relation "being better defined than."²² Then

$$x^\nabla = \lim_{n \rightarrow \infty} T^n(\perp),$$

Let us pursue this construction more systematically. The original inspiration for it is due to the stupendous work of Dana Scott (1971), which has had a strong impact on theoretical computer science. In the next chapter, I shall develop these ideas more rigorously (Goguen et al., 1977; Goguen and Varela, 1979), and thus the presentation that follows is mainly intended as a *sketch* for the role this idea plays in this domain of indicational forms.

12.10.2

More specifically, let us denote by B the collection of all forms that can be constructed in the primary algebra of indications. Let B_n denote the collection of forms of depth n or less. Naturally

$$B_n \subseteq B_{n+1},$$

²²Eds. In fact, $f \sqsubseteq g$ means that g is at least as determined as f , and possibly more determined; see definition 12.63.

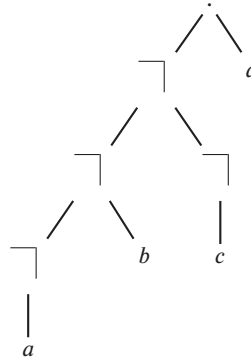
where $\subseteq$ is used in its usual set-theoretic sense of inclusion. We assume that algebraic expressions contain variables from the initial list X . We also assume that an arbitrary, but fixed, assignment of values for the variables is given. Now for the announced order relation in B :

Definition 12.63 *Let f, g be any expressions in B . Then we say that g is at least as determined as f , and write $f \sqsubseteq g$, if the contents of f coincide exactly with part of (or are equal to) the contents of g when compared starting at the shallowest depth of both expressions.*

To see this more clearly, let us agree to rewrite a form as a *tree*, by spreading the form in two dimensions. Given, say,

$$f = \overline{\overline{a} \mid b \mid c} \mid d,$$

rewrite it as



where “.” denotes the containment operation.

With this convention we can now reformulate definition 12.63 by saying that $f \sqsubseteq g$ if when we take both trees and superimpose them starting at their roots, the branches of f will coincide exactly with part of the branches of g . Thus, at some points f will stop where g will continue to branch further. This partial coincidence can be made more precise by saying that, at the points where f stops and g continues, f has an *undetermined* value. In this sense, f is less determined than g , or f *approximates* g by a lesser degree of determination. We assume then, the existence of some undetermined value $\perp$, a *bottom*, which approximates everything.

Definition 12.64 $\perp \sqsubseteq f$ for every f in B .

Proposition 12.65 $f \sqsubseteq g$ and $g \sqsubseteq f$ if and only if $f = g$.

Note that not always can two expressions be compared: $\sqsubseteq$ is not total in B . For example, if the roots of two expressions' trees are not the same, they cannot be compared. We want, however, to be able to construct an expression that a pair of forms could approximate.

An expression h is said to be an *upper bound* for f, g if and only if both $f \sqsubseteq h$ and $g \sqsubseteq h$. Now we can define a way to construct a least upper bound for pairs of forms.

Definition 12.66 *Let f, g be any two expressions. The join of f and g , written $f \sqcup g$, is that upper bound of f, g obtained by superimposing f and g at their roots, identifying identical branches, and dropping bottoms where some determined expression occurs. $f \sqcup g$ is undefined whenever f, g cannot be thus superimposed.*

What we are doing here is taking the intuitive idea of order in B , and making it explicit as a “poset,” a partially ordered set. The order and the join are related very closely:

Proposition 12.67 $f \sqcup g = g \Leftrightarrow f \sqsubseteq g$.

Consider now some sequence of expressions $f_1, f_2, \dots, f_n = \langle f_i \rangle_{i \in \omega}$. We can inductively extend the idea of the join over such a collection.

Definition 12.68 *The join of a sequence $\langle f_i \rangle$, where it exists, is a form f such that*

$$f = \bigsqcup_{i=1}^n f_i = \left(\bigsqcup_{i=1}^{n-1} f_i \right) \sqcup f_n.$$

This allows to consider a *limit* by extending joins over *infinite* sequences:

Definition 12.69 *The limit of a sequence $\langle f_i \rangle_{i \in \omega} = f_1, f_2, \dots, f_n, \dots$, when it exists, is a form f such that*

$$f = \lim_{n \rightarrow \infty} \bigsqcup_{i=1}^n f_i = \bigsqcup_{i=1}^{\infty} f_i = \bigsqcup_{n \in \omega} f_n.$$

Nothing has assured us that such a limit has any meaning as a form. We have begun to deal with infinite forms, and our intuition about them is quite different from the finite ones. Remember that we had

$$B_n \subseteq B_m \quad \Leftrightarrow \quad n < m,$$

and, in fact, the “poset” B is

$$B_n = \bigcup_{i=1}^n B_i$$

for a (perhaps large but) finite n .

We can similarly consider the collection of infinite forms

$$B_{\infty} = \bigcup_{n=1}^{\infty} B_n.$$

Call B_{∞} the class of *continuous forms*. What is this structure? What does it look like? Surely B_{∞} has still a coherent partial ordering. Furthermore, we do know what a join looks like for a finite collection of forms. But what does an f in B_{∞} look like? Take any chain

$$f_1 \sqsubseteq f_2 \sqsubseteq \dots \sqsubseteq f_n \sqsubseteq \dots$$

and its limit

$$\bigsqcup_{n \in \omega} f_n = f.$$

Since for every n

$$\bigsqcup_{i=1}^n f_i = f_n,$$

the limit is not as abstract as it seems, but is the member of the infinite sequence as we “watch” it grow for unbounded values of n . Once a sequence is well specified, $\bigsqcup_{n \in \omega} f_n$ is also well specified in the sense of being an effective construction for an unending form. Symbols like $\bigsqcup_{n \in \omega} f_n$ do not denote objects that we can display graphically, but they are a well-defined mathematical construction.

This gives us an idea of what the elements of B_∞ look like. In fact, every element in B_∞ can be *defined* as the limit $\bigsqcup_{n \in \omega} f_n$ of a sequence

$$f_1 \sqsubseteq f_2 \sqsubseteq \cdots \sqsubseteq f_n \sqsubseteq \cdots$$

where we can take any f and chop it at some depth i :

$$\text{Chop}_i(f) = f_i,$$

and, of course,

$$f_i = \bigsqcup_{j=1}^i f_j,$$

so that for any f we can construct a sequence $\langle f_i \rangle$ that approximates f with any desired degree of accuracy. Thus, we have

Proposition 12.70 *B_∞ is a chain-complete “poset,” that is, every chain has a least upper bound.*

This is quite nice, because there is a neat correspondence between sequences and elements in B_∞ . We do not need to assume anything at all about $\sqcup$, for we can construct its elements: They are limits of sequences. In B_∞ , therefore, we have as nice a structure as we had in B_n .

For any two forms, f, g , in B_∞ ,

$$f \sqcup g = \bigsqcup_{n \in \omega} (f_n \sqcup g_n),$$

$$f \sqsubseteq g \Leftrightarrow f_i \sqsubseteq g_i.$$

B_∞ is also a “poset.”

The operations of crossing and containment can be naturally extended to B_∞ , by the convention of looking at every finite form as an infinite sequence of identical forms. That is, for any f in B_n , write its sequence as $\{f_1 = f_2 = \cdots = f_n = f\}$, so that

$$f = \bigsqcup_n f_n.$$

Then for any form in B_∞ crossing and containment can be extended thus:

$$fg = \bigsqcup_n (f_n g_n),$$

$$\overline{f} = \bigsqcup_n (\overline{f_n}).$$

For example, we have, as expected, that

$$\overline{\bigsqcup_n f_n} = \bigsqcup_n \overline{f_n} = \overline{\bigsqcup \bigsqcup \dots} = \overline{\bigsqcup},$$

since every f_n is in some B_n .

12.10.3

This is as much as we need to know about infinite indicational forms. Let us consider now reentry in these terms.

As we have seen, a reentrant expression takes the form of fixed point of

$$g = \Phi(f), \quad (12.33)$$

where Φ is some algebraic expression. Even more generally, we may have multiple reentry and a system of interrelated equations

$$g_1 = \Phi_1(f_1), \dots, g_n = \Phi_n(f_n). \quad (12.34)$$

Now, an equation like (12.33) is really a mapping between forms

$$\Phi : B_\infty \rightarrow B_\infty,$$

and similarly, for (12.34),

$$\langle \Phi_1, \dots, \Phi_n \rangle : (B_\infty)^n \rightarrow (B_\infty)^n$$

reentry arises as the fixed points of these maps of B_∞ onto itself.

Definition 12.71 *Let $B_\infty(X_n)$ be the class of continuous forms on $X_n = \{x_1, \dots, x_n\}$ variables. Then a system of equations in $B_\infty(X_n)$ is a function*

$$\Phi : X_n \rightarrow B_\infty^n(X_n),$$

where $B_\infty^n(X_n)$ denotes the n -fold Cartesian product of $B_\infty(X_n)$ with itself and we write

$$x_i = \Phi_i(\mathbf{x})$$

as the i th equation of Φ .

Consider now any algebraic expression Φ . Let $\Phi^n = \Phi(\Phi^{n-1})$, and $\Phi^0(f) = f$. Consider the chain, for some f ,

$$\Phi^0(f) \sqsubseteq \Phi^1(f) \sqsubseteq \dots \sqsubseteq \Phi^n(f) \sqsubseteq \dots.$$

Surely $\Phi^n(f) \sqsubseteq \Phi^{n+1}(f)$. Then we have

Proposition 12.72 *For any algebraic expression Φ , and a given f , $\langle \Phi^n(f) \rangle$ is a chain and has a limit $\bigsqcup_{n \in \omega} \Phi^n(f) = \Phi_f$.*

Next we introduce the notion of continuity.

Definition 12.73 A function $\phi : B_\infty \rightarrow B_\infty$ is continuous if for every chain

$$\phi\left(\bigsqcup_n f_n\right) = \bigsqcup_n \phi(f_n),$$

(i.e., it preserves upper bounds).

Note that if ϕ is continuous then ϕ is also monotonous, i.e.,

Proposition 12.74 If ϕ is continuous, then $f \sqsubseteq g$ implies $\phi(f) \sqsubseteq \phi(g)$.

Proof. ϕ continuous implies

$$\phi(f \sqcup g) = \phi(f) \sqcup \phi(g);$$

$f \sqsubseteq g$ implies $f \sqcup g = g$. Then $\phi(f \sqcup g) = \phi(g) = \phi(f) \sqcup \phi(g)$, and hence $\phi(f) \sqsubseteq \phi(g)$. $\square$

As is clear from the definition of the join,

Proposition 12.75 Crossing and containment are continuous, i.e.,

$$\overline{\bigsqcup_n f_n} = \bigsqcup_n \overline{f_n} \quad \text{and} \quad \left(\bigsqcup_n f_n\right)\left(\bigsqcup_n g_n\right) = \bigsqcup_n (f_n g_n).$$

Furthermore, since an algebraic expression Φ is composed only of repeated application of crosses, we obtain

Proposition 12.76 Every algebraic expression is continuous.

With all of this, we can finally state the result that we were seeking all along:

Theorem 12.77 Let Φ be a system of equations in $B_\infty(X_n)$. Then there is $a_\Phi \in B_\infty$, which is a minimum fixed point for Φ , $\Phi(a_\Phi) = a_\Phi$. a_Φ is called the solution of Φ over $B_\infty(X_n)$. In fact $a_\Phi = \lim_{n \rightarrow \infty} \Phi_n(\perp, \dots, \perp) = \bigsqcup_{n \in \omega} \Phi^n(\perp, \dots, \perp)$.

Proof.

$$\Phi(a_\Phi) = \Phi\left(\bigsqcup_{n \in \omega} \Phi^n(\perp)\right) = \bigsqcup_{n \in \omega} \Phi^{n+1}(\perp) = \bigsqcup_{n \in \omega} \Phi^n(\perp) = a_\Phi.$$

$\square$

This theorem assures us that this new universe B_∞ is large enough to handle *all* kinds of reentry. In fact, it handles too much. We have no idea how complex the equation Φ could be; perhaps it is infinite. Thus, for the present purposes we will narrow our scope, and consider only those infinite expressions in B_∞ that correspond to *finite* reentry (e.g., where the reentry can be drawn on a sheet of paper).

Definition 12.78 A finite system of equations over $B_\infty(X_n)$ is a system of equations Φ such that

$$\Phi : X_n \rightarrow B^n(X_n),$$

where $B^n(X_n)$ designates the class of indicational expressions of the primary algebra generated by the set on X_n of variables.

Thus we can focus on the forms in B_∞ that arise from (finite) operations in B .

Definition 12.79 *The set R_B of rational expressions of dimension n is the subset of B_∞ , satisfying*

$$R_B = \{\Phi_i | \Phi_i : X_n \rightarrow B^n(X_n), n > 0, 1 \leq i \leq n\}.$$

Thus elements of R_B are single components of an n -dimensional reentrant form.

R_B is a rather enchanted land for self-referential forms, where every operation immediately gets an associated value: a reentrant form computed through the fixed-point construction. In this sense, operands (i.e., elements of R_B) and operators (i.e., certain maps $R \rightarrow R$) correspond to one another.²³

Let us examine now the relation of R_B to the temporal context. With every $a \in R_B$ let us associate a sequence $(a_n)_{n \in \omega}$. This can surely be done, since by definition, for every a there is an algebraic map T such that

$$a = \lim_{n \rightarrow \infty} T^n(\perp),$$

and thus a has the associated sequence

$$(a_n) = (\perp, T(\perp), T^2(\perp), \dots).$$

At every *finite* n , each term of this sequence is finite, and thus we can reduce them algebraically by choosing an *initial* value for the indeterminate term, $\perp$. We immediately see, by theorem 12.61, that each sequence will be periodic. If we change the initial value, it will generate the same periodic sequence with an inversion and period shift. Thus, up to phase shift, *every rational expression oscillates*.

For example, consider the reentrant form presented before:



Since it must belong to R_B (being infinite), we know that it must be the fixed point for an equation

$$x^\nabla = \overline{\overline{x^\nabla} \mid \boxed{}},$$

and this reveals that it is, in fact, the first component of a system

$$(x^\nabla, y^\nabla) = (\overline{y^\nabla} \mid \overline{\overline{x^\nabla} \mid y^\nabla}),$$

that is, $x^\nabla = a_{\Phi_1}$ with

$$\Phi(x, y) = (\Phi_1(x, y), \Phi_2(x, y)) = (\overline{y} \mid \overline{\overline{x} \mid y}).$$

Whence

$$(x^\nabla, y^\nabla) = \lim_{n \rightarrow \infty} \Phi^n(\perp, \perp).$$

²³This is meant to revise the notion in mathematics that formal domains cannot be reflexive, that is, type-free. This idea has been fully propounded and explored in combinatorial logic and topology by Dana Scott (1972; 1973), see also Wadsworth (1976). For further discussion of the notion of rational elements of continuous algebras see Wright et al. (1976) and Goguen et al. (1977). Obviously what we say here is very informal and expository, and the interested reader is encouraged to look at the aforementioned papers for a detailed discussion.

From this we can immediately obtain a sequence

$$(x_n^\nabla) = (\perp, \top, \overline{\perp}, \overline{\top}, \dots).$$

Let us evaluate it at $\perp \rightarrow \top$:

$$(x_n^\nabla) = (\perp, \top, \overline{\perp}, \overline{\top}, \dots) = \text{---} \square \text{---} \square \text{---} \square \text{---} \dots,$$

and then at $\perp \rightarrow \overline{\top}$:

$$(x_n^\nabla)' = (\overline{\top}, \perp, \overline{\top}, \perp, \dots) = \text{---} \square \text{---} \square \text{---} \square \text{---} \dots$$

giving the same waveform with a different phase.

The inverse process, however, is much more complex. For a given sequence we can find *many* operators that will generate it, and therefore several elements of R_B can be associated with it. For example, $i \in V$ can be produced by $T(x) = \overline{x}$, but also entrained with other oscillations as in $T(x, y) = (\overline{y}, \overline{x})$. Notice also that the reentrant form $\square$ will correspond to both i and j , depending on how $\perp$ is evaluated. Phases are irrelevant, as they should be in the static world of forms.

Much remains to be explored between the correspondences of sequences and reentrant infinite expressions. Another topic of great interest to investigate is whether R_B is a Brownian algebra, or, in general, how it behaves under quotient from some set of initials. This would give an idea of the *arithmetic* intrinsic to R_B . It is obvious that we are only skimming the surface.

12.11 Autonomous Systems and Reentrant Forms Reconsidered

12.11.1

It seems important, at this point, to regain some perspective on what we have been pursuing in these past two chapters on indicational forms.

Our intention was to present, in a mathematical format, two of the key ideas on which our discussion on autonomy and the phenomenology of autonomous systems seems to be based. First, there is the central role of the act of distinction, whereby the unities are differentiated and phenomenal domains are born. Secondly we have stressed the constitutive role of recursive, self-referential processes of natural systems. What we have done is construct a formalism where these questions can be investigated in detail, so that difficulties and implications are brought to the surface.

What is at stake is the need to examine closely the nature of whole autonomous systems, and the correlated necessary appearance of

circular interrelations of processes. I do not feel that the full impact of this cognitive issue has been realized. When Wiener brought the feedback idea to the foreground, not only did it become immediately recognized as a fundamental concept, but it also raised major philosophical questions as to the validity of the cause-effect doctrine. The picture seemed closer to a circular causation, where one can deal only with the ensuing totality and its manifested stability. In other words, the nature of feedback is that it gives a mechanism, which is independent of particular properties of components, for constituting a stable unit. And from this mechanism, the appearance of stability gives a rationale to the observed purposive behavior of systems and a possibility of understanding teleology.²⁴ Since Wiener, the analysis of various types of systems has borne this same generalization: Whenever a whole is identified, its interactions turn out to be circularly interconnected, and cannot be taken as linear cause-effect relationships if one is not to lose the system's characteristics.

In the ideal land of pure indicational forms, the texture of this circular interdependence can be appreciated more fully, and the difficulties in finding a precise expression for it are also apparent. However, these fundamental considerations seem to me necessary if we are not to betray some deep intuitions about natural systems and their organizations. It is surprising that there has not been more attention paid to the key role of closure.

I contend that the reluctance to concede a central role to circularity *per se* in system's organization is basically a heritage from positivism, or what I would like to call a *Fregean* viewpoint. The basic assumption here is that we can look at a system and identify initial or atomic elements with which a larger system can be constituted, and so on until an output is reached. The idealized form of this logic is the Whitehead-Russell theory of types, where some atomic elements are given, and do not affect operations of higher types. The mental picture is that of a tree with roots and branches. But, this view is awkward for describing whole systems, where the picture is more that of a closed network with roots and branches intertwining, and where the describer is eminently present.²⁵ It resembles the network of language that the late Wittgenstein was concerned with. No type distinctions are possible in such a network. This kind of logic is the basis of what I wish to call a *Brownian* approach to systems.

Surely, a lot of contemporary cybernetics and systems theory do recognize implicitly the relevance of circularity and of the observer's

²⁴ *Eds.* See Rosenblueth et al. (1943). Also, see Jonas (1966) for a criticism of the cybernetic concept of purpose.

²⁵ *Eds.* There is a hint here of the critique of arborescent thinking in Western science by Deleuze and Guattari (1987), where different phenomena (branches) are manifestations of a single root principle. Closure and reentry seem to imply a more rhizomatic approach, where levels are entwined.

viewpoint. This is fine, and can take care of itself. My point is, however, that when such notions are formulated explicitly, there is usually a return to a Fregean attitude, and this is what is involved in postulating inputs and outputs, or fixed reference points, or finiteness in the recursion, where, again, there is openness of organization and complete distance from the observer-community. This reflects, as I said before, the historical fact that the most sophisticated tools in systems theory have been generated in the context of engineering and computer science. There, the goal of the *design* is the motivating force, and hence the input-output approach is quite suitable. The system is quite definitionally open and “out there.”

In contradistinction, in dealing with natural systems, the whole idea of input-output becomes muddled. Who and how are we to select a fixed set of input and output spaces? It is more accurate to talk about environmental perturbations/compensations (p/c's). And this is quite different. For then, we explicitly start with a system's stability, coming from the closure of its organization, which is the basis for its capacity to confront a perturbation and compensate it. Of course, we may take a *fixed* set of such p/c's and treat the system *as if* it were organizationally open. However useful this has been and is in engineering and design, it misses a deeper insight about the system's organization proper (and for those systems of which humans are a part of, this has proven to be disastrous).

I am saying then that in our dealing with natural systems, we have rested upon a philosophical and methodological position adequate only for the domain of design, and that this is a conceptual inconsistency.²⁶ A science of organizations, and first of the organization of natural systems, must effect a transition from a Fregean to a Brownian foundation. The closure thesis is a methodological guideline. It is in order to make sense of this general guideline that a more rigorous *foundation* is needed. I am claiming that the study of descriptive complementarity, indicational forms, and their reentry and dynamics is a step in that direction.²⁷

12.11.2

The strategy followed in this chapter can be summarized as follows. From the basic notion of indication and the primary algebra of indications (of G. Spencer-Brown) we moved into two complementary directions in order to bring out the interlock between closure and dynamism immanent in a form. First, we developed the notion of a Brownian algebra, where waveforms can be represented through sequences. Second, we expanded indicational forms to infinitary

²⁶ Eds. Even in the domain of design, one finds advocates of perspectives based on autonomy and closure (e.g., Winograd and Flores, 1986; Escobar, 2018).

²⁷ Eds. We cannot deny that rich though the Brownian approach is, it seems quite far from replacing input-output formalisms, either in the study of natural systems or in engineering. Varela did not pursue further developments of this formal approach in his later work, though he never abandoned its epistemological perspective, which in fact predated it. Apart from cultural and practical reasons (it seems hard to imagine how to model complex metabolic networks in terms of reentrant forms), one reason why the formalism may prove hard to apply is that it does not easily accommodate within itself a concept of *adaptive interactions* between systems. See the discussion in section 13.11.

indicational algebras where reentry can be expressed properly. The relations between dynamics (oscillations) and pattern (reentry) can then be established through the qualities of operators that characterizes a form. These two complementary views of the temporal and spatial properties of form were introduced, in a minimal arithmetic, in the extended calculus of indications.

In those calculi, “antinomic” forms are allowed to appear without restrictions, and thus we have found a way to *construct* from an antinomic situation, which, formerly, we might have avoided rather than faced. By not avoiding the antinomies, we have found a wider domain where all the preceding forms can be lodged. A similar case at the numerical level is to be seen in the construction of the complex numbers, starting from the antinomic form of $x^2 = -1$, not solvable in the real domain because it needs a number that is neither positive nor negative. This antinomy is solved by admitting this behavior within a larger arithmetic containing a new value $i = \sqrt{-1}$, and thus extending the real domain to the complex domain. In analogy, we have presented a similar construction at a more fundamental level. By allowing an antinomic form (from the point of view of logic), we have constructed a new, larger domain akin to the complex plane, where new forms can be lodged, including those of the preceding primary domain found to be in conflict with the introduction of reentrant expressions. Again, rather than avoiding the antinomy, by confronting it we enter a new domain.

This intercrossing of phenomenal domains at the point of self-reference is of course encountered repeatedly in nature; its typical process is the emergence of autopoietic systems. It is therefore very interesting that a similar situation is found when we trace our description to this fundamentally simple level. It appears that, regardless of what it is that we are describing, successively larger levels are connected and intercross at the point where the constituents of the next lower level act on themselves, where reentrant forms appear. Self-reference is the hinge upon which levels of serial inclusiveness intercross. Rather than recording any particular instance, these calculi provide a record of the *general* form of this situation.

There is little doubt that the results presented here on the closure and dynamics of forms are of interest in themselves, and there is much else that should be explored. We shall not do so in this book, since our intention is to provide some ground for understanding the autonomy of natural systems. Thus we shall turn now to other levels of descriptive tools, which are the natural continuation of simple

indicational forms. The basic idea pursued in the next chapter is to examine the closure and dynamics of forms when the mark is *diversified*, so that it becomes a variety of possible processes. To be sure, by taking this step in the direction of empirical events, much of the richness of results will be missed, and our presentation will become more fragmented and incomplete; but to stay solely with indicational forms would be to renounce our basic source of insights from natural systems. As will be apparent, many of the approaches and insights developed for the reentering of indications can, with appropriate modifications, be applied to more concrete systemic processes.

Sources

(Kauffman and Varela, 1980), (Spencer-Brown, 1969), (Varela, 1975a), (Varela and Goguen, 1978).

13

Eigenbehavior: Some Algebraic Foundations of Self-Referential System's Processes

13.1 Introduction

This chapter is concerned with representing organizational closure in operational terms. To this end we shall go beyond what was presented in the last chapter to construct two key notions: infinite trees of operators and solutions of equations over them.¹ The idea of a solution of an equation over the class of infinite trees is an appropriate way to give more precise meaning to the intuitive idea of coordinations and simultaneity of interactions. The self-referential and recursive nature of a network of processes, characteristic of the autonomy of natural systems, is captured by the *invariant* behavior proper to the way the component processes are interconnected. Thus the complementary descriptions behavior/recursion (cf. chapter 10) are represented in a nondual form. The (fixed-point) invariance of a network can be related explicitly to the underlying recursive dynamics; the component processes are seen as unfoldment of the unit's behavior.

¹*Eds.* Parts of this chapter are drawn from Varela and Goguen (1978) and a manuscript that Varela coauthored with Joseph A. Goguen entitled "Some Algebraic Foundations of Self-referential System's Processes." To the best of our knowledge, this manuscript has remained unpublished.

13.2 Self-Determined Behavior: Illustrations

I propose the name *eigenbehavior* for an expression in the mathematical sense described below that is intended to represent the autonomy of some concrete system.

The name seems justified on several counts. First, the prefix "eigen" carries from German the connotation of "proper" and "self," and eigenbehavior is properly or self-determined behavior, i.e., autonomy. Second, this compound is a generalization consistent with the standard use of "eigenvalue" and "eigenvector" in linear algebra to denote certain fixed points of linear maps. Thirdly, in at

least two fields the term eigenbehavior has been proposed to denote, in particular instances, exactly what from our point of view is a solution to some system's closure. Niels Jerne (1974b) introduced the idea as a qualitative characterization for the moment-to-moment stable state of the totality of cellular interactions that specifies the immune network in living organisms. (We shall elaborate on this in chapter 14.) Von Foerster's (1977) paper is entitled "Objects: tokens for eigenbehavior," and discusses the closure of the sensory-motor interactions in a nervous system, giving rise to perceptual regularities as objects. Our usage, then, not only is linguistically appropriate, but also extends previous usage to a more general systemic and mathematical content.

Even in a very general, informal sense, the notion of eigenbehavior is quite interesting. Let us consider a few illustrations of it before going into the more detailed treatment.

Eigenbehaviors can be characterized as the fixed points of certain transformations. Consider an operation α , from a domain A to itself, $\alpha : A \rightarrow A$. A fixed point for α is a value $v \in A$ such that $\alpha(v) = v$. Fixed points, in general, have several interesting properties. First, in a naive sense, a fixed point is self-referential or recursive: v says something about itself, namely, that it is invariant under the operation α . Second, fixed points are uniquely characterized with respect to all the other values taken by the operation α . Consider for example the case where α is the function $\cos : R \rightarrow R$. Then it is easy to verify that $x^\nabla = 0.739085$ (rad) is a fixed point, and in fact the only one among the continuum of values taken by $\cos$. Third, fixed-point values can be expressed through repeated or indefinite iterations of the operations to which they are related; that is, they can be "unfolded" in terms of their defining operations. For example, we may express x^∇ by an indefinite iteration of the operation $\cos$, i.e., $x^\nabla = \cos(\cos(\cos(\dots)))$. Note that we may disregard the value on which the iteration was initiated; it can be any number in the domain R . Now to some examples.

A rather witty illustration of such eigenbehaviors, due to von Foerster, can be described in the linguistic domain. Take the following sentence form:

S : "This sentence has ... letters."

Let $S(n)$ be the number of letters in s when we insert the verbal name of the number n in the empty slot. Thus $S(3) = 27$, since, "three" has 5 letters, which we add to the 22 constant letters of S . By trial and error we find that $S(33) = 33$ is the only fixed point. Only for

"This sentence has thirty-three letters" does the sentence have the mentioned number of letters.

Even for a fairly simple process, the resulting eigenbehaviors can be surprisingly complex. Let me illustrate this fact. Consider an urn containing one white ball and one black ball. Let us perform the following experiment: draw one ball from the urn, at random, and whatever its color, replace it and add another ball of the same color to the urn. Repeat the above procedure many times, so that the number of balls grows very large. We then ask the question: What will be the percentage of, for example, black balls in the urn? The answer is surprising: the percentage can approach any value between 0 and 100, but in each experiment it will converge to only one stable value (Blackwell and Kendall, 1964). In other words, after an initial period of fluctuation (initial stages of approximation) the ratio will settle to a certain value and will stay close to it (eigenbehavior), although if we repeat the experiment (consider another organism of the same type) the stable value will be a different one. This experiment is illustrated in figure 13.1. It is obvious that the outcome of the first few draws has a much more significant influence on the final value of the run than do later draws.²

We may now consider a more concrete illustration, in terms of the ideas already developed in the previous chapter. Consider an electrical circuit used for computer logic, the flip-flop. One reason to choose this example is that, being used as a logical block, it can

²Eds. This process is known as Pólya's urn scheme. George Pólya demonstrated the strong convergence of the process to a limit that is itself a random variable between 0 and 1 (Pólya, 1930). The model is a striking example of nonergodicity or path dependence and has seen applications in various fields, including the modeling of positive returns in economics and technological development (Arthur, 1994). It highlights the important point that eigenbehaviors can be history dependent (i.e., the result of a history of enactments), which is a key idea in the enactive approach (Varela et al., 1991; Di Paolo, 2005; Di Paolo et al., 2022).

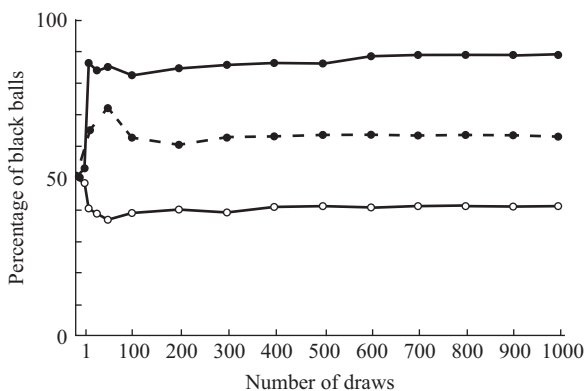
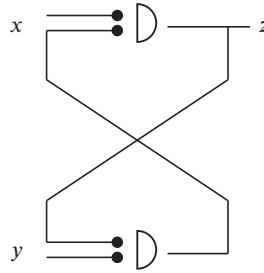


Figure 13.1

Recursive behavior of the urn example described in the text. Three separate experiments are plotted, each up to 1000 draws. In all of them an initial stage of fluctuations is followed by a stable behavior, which differs in each case. It can be shown that there is equal probability for the behavior converging to any percentage of black balls.

be interpreted as an indicational form (cf. appendix B). In fact the standard diagram for the flip-flop,



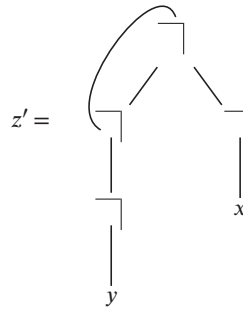
can be readily transposed into its corresponding eigenbehavior

$$z = \overline{\overline{z}} \mid \overline{y} \mid x, \quad (13.1)$$

or

$$z' = \overline{z} = \overline{\overline{\overline{z}} \mid \overline{y} \mid x} = \overline{\overline{y} \mid \overline{x}},$$

with its tree



Now, z is the limit of an approximation

$$z = \bigsqcup_{n \in \omega} z_n,$$

where

$$z_1 = \overline{\overline{z_0} \mid \overline{y_1} \mid x_1},$$

and, in general,

$$z_n = \overline{\overline{\overline{z_{n-1}} \mid \overline{y_n} \mid x_n}}. \quad (13.2)$$

Clearly

$$z_n \sqsubseteq z_{n+1}.$$

This also specifies as sequences for x and y

$$x = \bigsqcup_n x_n,$$

$$y = \bigsqcup_n y_n.$$

All of this makes sense because, in an actual flip-flop, the expression (13.1) is, of course, interpreted in time as a discrete step-by-step recursive function (13.2), for a given sequence of inputs x_i, y_i . In fact, we could have done that all along in B_∞ , by interpreting z (in time) as a finite sequence, starting with some z_0 , and under some finite of x_i 's and y_i 's, the following algebraic expression is valid (as can be easily verified by induction):

$$z_n = z_0 \overline{\alpha(n)} \beta(n).$$

with

$$\alpha(n) = y_1 \overline{y_2} \dots \overline{y_n},$$

$$\beta(n) = \overline{y_2} \dots \overline{y_n} x_1 \overline{y_3} \dots \overline{y_n} x_2 \dots \overline{y_n} x_{n-1} \overline{x_n}.$$

This is a recursive expression that algorithmically determines z_n for every n , and this is what is normally done in representing these kinds of logical circuits with feedback.

We can see, however, that this approach fits hand in glove with our approximation to an infinite expression (13.1), which embodies the self-referential quality of this reentrant circuit. The eigenbehavior represents, formally and intuitively, the basic structure of the flip-flop as a logical *design*, rather than describe it as an *ad hoc* sequential expression. The time/recursive expression shows how it can actually be operated; its reentrant forms show what it is and what it *means*.³

What we see emerging from this example is that an eigenbehavior traps the intuitive idea of the global coordination or meaning of a unit, through the way in which it arises in its underlying processes. This has been standard lore in mathematical physics, where invariant transformations and fixed-point topological properties of differential dynamics are a royal road to representations of physical laws. However, these tools have been mostly concerned with numerical and differentiable representations, and there has been little development of the corresponding notions for non-numerical and informational processes. These only seem necessary when considering the phenomena proper to complex, natural systems and engineering design as well. In fact, the initial development of the ideas on continuous algebras came from the work of Scott (1971), dealing with the *semantics* of programming languages. These notions extend rather naturally to the semantics (i.e., behavior) of recursive processes in natural systems (Goguen and Varela, 1979).

³*Eds.* We may take this to indicate that the eigenbehavior is not simply the stable pattern that results from an operation, but rather a way of characterizing the kind of operation that it is.

13.3 Algebras and Operator Domains

13.3.1

The next few sections are strictly concerned with the mathematical grounds necessary to represent self-referential system processes in the spirit described above. Thus the reader will be faced again with a considerable number of mathematical ideas, most of which are likely to be unfamiliar. I ask patience for this lengthy development, but I am convinced that this is the sort of precision that lends some of the intuition behind this view of system's autonomy a possibility of being discussed, tested, and applied.

Four main steps follow. First, we develop some notions that are required for the representation of infinite trees: namely, operator domain, finite trees of operators, and their role in the class of algebras of operators (or Σ -algebras). Second, we present the extension of Σ -algebras to the infinite case, through order-theoretic notions and approximations. This yields the class of *continuous algebras*, and we study the role on infinite trees among them. Third, we discuss the notion of eigenbehavior as solutions of equations in continuous algebras, and we construct the set of rational (infinite) trees, which characterize recursive processes.

Throughout the presentation of these ideas, there are some difficult turns of which the reader should be forewarned, or else the technical details may seem unnecessarily complicated. The first subtle point is that, in discussing algebras of operators, we shall do so by trapping their “abstract” quality, that is, the fact that an operator name can designate many different processes in different situations. This quality of abstractness is expressed here as equivalence “up to an isomorphism” of different algebras. A second possible difficulty arises when variables are introduced into Σ -algebras and trees. The transition from simple expressions to expressions with variables seems, at first glance, simple and harmless. Thus it is surprising that when rigor is demanded, delicate steps are needed to make it come out right. In the case at hand, we end up constructing two objects (later called $T_{\Sigma}(X)$ and $T_{\Sigma(X)}$) which may seem mysterious. Third, the illusion that, with these tools, all our problems are gone is dispelled when we realize that the collection of infinite trees is rather unknown territory. This leads to a first classification of trees—those that we shall describe as rational—but this does not exhaust their complexity.

13.3.2

Previously (chapter 10) we have used trees and nets to describe the connection properties of systems. But such a view does not take account of the operational capabilities of the components that are so interconnected. One step in this direction is to label each mode with a function that describes the *operation* of the associated component.

In this respect, it is important to avoid confusion between an operation and its name; for example, a careful distinction will permit us to use the same name for several operations, occurring in several situations, but having a similarity of function that it is desirable to capture. Thus, we first introduce an abstract symbol system for naming operations. The most basic quality an operation can have is the number of arguments it takes, and we include this quality in the basic notion.⁴

Definition 13.1 *An operator domain (or signature) is a family Σ_k of disjoint sets, indexed by the natural numbers $k \in \omega$. Σ_k is the set of “operator symbols of rank k ,” and elements of Σ_0 are symbols for “constants” (which take no arguments).*

For an ordinary arithmetical operations, the following signature Σ would be appropriate: $\Sigma_0 = \mathbb{Z}$, the positive and negative integers; $\Sigma_1 = \{-\}$, the unary negation operator, as in the expression $-(1+1)$; $\Sigma_2 = \{+, \times\}$, the usual binary addition and multiplication.

An operator domain gives a basic syntax for operators, but says nothing about their semantics, that is, their meaning or interpretation. If Σ is an operator domain, then a Σ -algebra is exactly a set of elements together with a particular function for each symbol in the operator domain; that is, it gives a concrete interpretation of the abstract operation symbols. More precisely, an operation symbol $\sigma \in \Sigma_n$ is interpreted as a function $\sigma : A^n \rightarrow A$ of n arguments on a set A , and a constant symbol $\sigma \in \Sigma_0$ is interpreted as an element of A . This leads to

Definition 13.2 *Given an operator domain Σ , a Σ -algebra A is a set A , called the carrier, plus, for each $\sigma \in \Sigma_n$ with $n > 0$, a function $\sigma_A : A^n \rightarrow A$, and for each $\sigma \in \Sigma_0$, an element σ_A of A .*

Given an operator domain Σ , we can consider *expressions* compounded from its symbols, of the general form $\sigma(t_1, \dots, t_n)$, with σ of rank n , and each t_i , either a constant symbol, or else itself a compound expression.⁵ More precisely now,

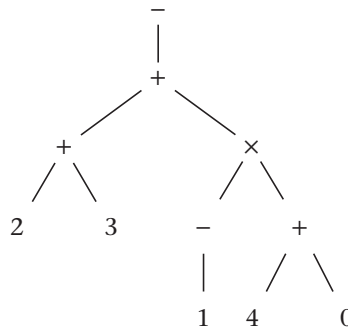
⁴*Eds.* This will be achieved by introducing families of operators according to the number of arguments that they take. An operator of rank 0 is simply a constant, operators of rank 1 are so-called unary operators, operators of rank 2 are binary and operate on two arguments, and so on. The interpretation of an operator domain is given by specifying each operator as a function within a corresponding algebra. This function takes a number of arguments equal to the operator's rank, and the result of the operation is a constant of rank 0.

⁵*Eds.* An expression is a syntactically correct composition of operators from domain Σ . The set of all expressions T_Σ is defined recursively as the minimal set that contains all the Σ constants and any expression formed by taking operator σ from the operator domain and applying it to elements of T_Σ which, of course, can include other expressions. This recursiveness motivates representing expressions as trees, eventually showing a structure-preserving homeomorphism between the two.

Definition 13.3 Let Σ be an operator domain. Then the set T_Σ of all (well-formed) Σ -expressions is (recursively) the least set of expressions such that:

1. $\Sigma_0 \subseteq T_\Sigma$, and
2. if $\sigma \in \Sigma_n$, if $n > 0$, and if $t_i \in T_\Sigma$ for $i = 1, \dots, n$, then $\sigma(t_1, \dots, t_n) \in T_\Sigma$.

It is possible, and suggestive, to view these Σ -expressions as trees whose nodes are labeled with symbols from Σ . Let Σ be the operator domain mentioned above. Then $-(+(+(2, 3), \times(-1, +(4, 0))))$ is a Σ -expression, which is $-((2 + 3) + ((-1) \times (4 + 0)))$ in the more usual infix notation, and can be viewed as the tree



in which the various subexpressions correspond nicely to subtrees.

This suggests the following

Definition 13.4 Let Σ be an operator domain. Then a Σ -tree t is a tree $\langle |t|, E, \partial_0, \partial_1, r \rangle$ (see definition 10.1) plus a function $t : |t| \rightarrow \Sigma$ such that if the number of edges out of $a \in |t|$ is n , then $t(a) \in \Sigma_n$

That is, a node with n child nodes must be labeled with an operator symbol of rank n , as was the case above.

The reader may now wish to prove that there is in fact a bijective correspondence between Σ -trees and Σ -expressions. There are quite a number of equivalent infix notations for binary operators besides those mentioned in the example; there are also Polish prefix and postfix notations. For example, the above tree would be given as $- + + 2 3 \times (-1) + 4 0$ and $2 3 + (-1) 4 0 + \times + -$, respectively, in prefix and postfix notations. Again, one can establish bijective correspondences among any two of these notational systems. Moreover, the above mentioned notations far from exhaust all the possibilities.

Something is going on here: There seems to be an *abstract* underlying notion of Σ -tree or Σ -expression, which expresses the independence of the basic concept from any particular choice of how to represent it; and all representations are in some way

isomorphic. This abstract quality of Σ -expressions is quite deep, and to make it more precise we begin by making T_Σ into a Σ -algebra, by defining operations as follows:

1. for $\sigma \in \Sigma_0$, $\sigma_T = \sigma$ in T_Σ , and
2. for $\sigma \in \Sigma_n$ and $t_i \in T_\Sigma$, $\sigma_T(t_1, \dots, t_n) = \sigma(t_1, \dots, t_n)$ in T_Σ ,

where we have written σ_T for σ_{T_Σ} .

Next, we use a fundamental insight from category theory, that it is important to consider not only the “objects,” but also, and perhaps more significantly, their relationships with one another, as expressed in the “structure-preserving” mappings between them. In the case of Σ -algebras, “structure-preserving” is given by

Definition 13.5 *Given an operator domain Σ and Σ -algebras A, A' , a Σ -homomorphism from A to A' is a function $h : A \rightarrow A'$ such that*

1. if $\sigma \in \Sigma_0$, then $h(\sigma_A) = \sigma_{A'}$, and
2. if $\sigma \in \Sigma_n$, $h(\sigma_A(a_1, \dots, a_n)) = \sigma_{A'}(h(a_1), \dots, h(a_n))$.

A Σ -homomorphism h is “structure-preserving” in the sense that if we do an operation σ in the algebra A , and then apply h , we get the same result as if we apply h to the arguments, and then do σ in A' .

We will use Σ -homomorphisms to characterize the property of being “abstractly the same as Σ -expressions,” by introducing the following general notion.

Definition 13.6 *A Σ -homomorphism h is said to be an isomorphism in $\mathcal{C}$ if it has an inverse in $\mathcal{C}$, that is, a Σ -homomorphism g such that both compositions gh and hg are identities. Σ -algebras related by Σ -isomorphism are said to be isomorphic.*

For example, it is possible to make the set of all Σ -trees (see definition 13.4) into a Σ -algebra (call it T'_Σ) in such a way that the bijection between Σ -trees and Σ -expressions is actually a Σ -isomorphism between T_Σ and T'_Σ . This isomorphism makes precise the sense in which Σ -trees and Σ -expressions are “abstractly the same.” Furthermore, all the other abstractly equivalent representations also give isomorphic Σ -algebras. What we now want is a more genuinely abstract way to characterize this notion. The following is the key.

Definition 13.7 *A Σ -algebra T is initial in a class $\mathcal{C}$ if there is a unique homomorphism, $h_A : T \rightarrow A$, from T to A , for all A in $\mathcal{C}$.*

A remarkable general property of initial algebras is that, if they exist, they are uniquely defined *up to isomorphism* by the class $\mathcal{C}$ on algebras within which they are initial. In algebra, the property of

being “defined uniquely up to isomorphism” is said to embody the idea of abstraction; that is, initiality defines an algebra “abstractly;” this has the practical meaning of being independent of the manner of representation of elements, capturing exactly the “abstract algebraic structure” and nothing extra. The following result expresses this, and thus shows that initiality captures the notion of being “abstractly the same.”

Proposition 13.8 *If T, T' are both initial in a class $\mathcal{C}$ of Σ -algebras, then T and T' are isomorphic in $\mathcal{C}$. If T' is isomorphic in $\mathcal{C}$ to an initial algebra T , then T' is also initial in $\mathcal{C}$.*

Proof. See (Goguen et al., 1977, proposition 1.1). $\square$

What the above does not guarantee us is the existence of initial Σ -algebras. Let $\mathcal{Alg}_\Sigma$ denote the class of all Σ -algebras, together with their Σ -homomorphisms. The following result was first proved by Birkhoff (1935).

Theorem 13.9 *T_Σ is initial in $\mathcal{Alg}_\Sigma$.*

Proof. It will help our understanding of what is going on here to have an idea of what the unique homomorphism $h_A : T_\Sigma \rightarrow A$ looks like. If $\sigma \in \Sigma_0$, then by the definition of homomorphism, we have to have $h_A(\sigma) = \sigma_A$. Now assume that we have defined h for trees of depth $< n$, and let t be a tree of depth n . Then t is of the form $\sigma(t_1, \dots, t_n)$, with all t_i of depth less than n . The definition of Σ -homomorphism then forces that $h_A(t) = \sigma_A(h_A(t_1), \dots, h_A(t_n))$, and we are assuming that $h_A(t_i)$ are already well defined. Thus, $h_A(t)$ is well defined, and by induction on n , h is defined. $\square$

This function $h_A : T_\Sigma \rightarrow A$ can be interpreted as assigning to each Σ -tree in T_Σ its “natural” interpretation in A , that is, the element that the compound Σ -expression t in fact denotes in A .

Examples 13.10

1. Let Σ be the operator domain of the example above. Then T_Σ contains trees such as that drawn above. Now let A be $\mathbb{Z}$, with the operation symbols in Σ interpreted in their usual way. Then for t the tree above, $h_A(t)$ is the result of actually performing the arithmetic operations that are only symbolically indicated in t , thus $h_A(t) = 1$.
2. Let Σ be the operator domain with $\Sigma_0 = \{0\}$, $\Sigma_1 = \{s\}$, $\Sigma_k = \emptyset$, $k > 1$, where 0 is “zero” and s is “successor.” Then the Σ -algebra of natural numbers ω is initial in $\mathcal{Alg}_\Sigma$. This provides a characterization that is different from the usual Peano postulates.

MacLane and Birkhoff (1967) prove these are equivalent characterizations.

13.4 Variables and Derived Operators

For the developments to follow, we give an algebraic explication, which can be used in a Σ -term, of the concept of a “variable.” We are not assuming that this is an already defined idea. In fact, this is a somewhat mysterious idea, and hope that the present discussion may contribute to its clarification.⁶

Previously, we dealt with single operators of various ranks, acting on a Σ -algebra A . We would like to be able to define compound operators, such as $x + zy(x - y)$, formed from operators in Σ and “variables” x, y , etc. The notion of a “freely generated” Σ -algebra is the key to a rigorous development of this topic.

Let $X = \{x_1, \dots, x_n\}$ be a set of symbols disjoint from Σ , and called “variables.” We first form a new signature $\Sigma(X)$ by adjoining the elements of X as new constants: $\Sigma(X)_0 = \Sigma_0 \cup X$; and $\Sigma(X)_k = \Sigma_k, k > 0$. Then $T_{\Sigma(X)}$ is the initial $\Sigma(X)$ -algebra, and it differs from T_Σ in that its leaf nodes may carry elements of X . Because the operator symbols in $\Sigma(X)$ include those of Σ , we can think of $T_{\Sigma(X)}$ as a Σ -algebra, simply by ignoring the X part of the operator domain. More specifically, we define a new Σ -algebra, $T_\Sigma(X)$, with carrier that of $T_{\Sigma(X)}$, and with operators those named by Σ in $T_{\Sigma(X)}$.

If A is a Σ -algebra, then each element t of T_Σ has a definite interpretation $h_A(t)$ in A . However, elements of $T_{\Sigma(X)}$ or $T_\Sigma(X)$ do not have definite interpretations in A , because the elements of X do not designate definite elements of A . However, if we *assign* values in A to the variable symbols in X , using a function $C : X \rightarrow A$, then we should be able to get definite values for each element of $T_{\Sigma(X)}$; these values will, of course, in general depend upon the values assigned to the variables. In this explication, variables are constants without fixed values, but which can be assigned any desired value.

The following result shows how terms in $T_\Sigma(X)$ get values in a Σ -algebra A once the variables in X are given values in A . The advantage of using $T_\Sigma(X)$ rather than $T_{\Sigma(X)}$ is that $T_\Sigma(X)$ is a Σ -algebra, so that we can talk about Σ -homomorphisms.

Proposition 13.11 *$T_\Sigma(X)$ is the free Σ -algebra generated by X , in the sense that if $C : X \rightarrow A$ is any function mapping X into the carrier of a Σ -algebra A , then there is a unique Σ -homomorphism $\hat{C} : T_\Sigma(X) \rightarrow A$ such that following diagram commutes:*

⁶*Eds.* Variables are introduced as symbols that do not belong to Σ and play the role of undefined constants (there are no variable operators of higher rank in this formalism). An algebra may be constructed in which some of nodes in the expressions are now undefined variable symbols. To assign values to these expressions, it is necessary to map variables into constants in Σ_0 .

$$\begin{array}{ccc}
 X & \xrightarrow{i_X} & T_\Sigma(X) \\
 & \searrow C & \downarrow \hat{C} \\
 & & A
 \end{array}$$

where i_X is the inclusion of X into $T_\Sigma(X)$.

Proof. For details see (Goguen et al., 1977, proposition 23). The following describes just the construction of $\hat{C}$ from C . Since A is a Σ -algebra, we can make A into a $\Sigma(X)$ -algebra by letting x name $C(x)$ in A , i.e., $x_A = C(x)$. Then there is a unique $\Sigma(X)$ -homomorphism $\hat{C} : T_{\Sigma(X)} \rightarrow A$. Since $\hat{C}$ is a $\Sigma(X)$ -homomorphism, it is also a Σ -homomorphism. □

This result says in effect that an element t of $T_\Sigma(X)$ defines a function $t(x_1, \dots, x_n)$ on a Σ -algebra A , since giving $x_1, \dots, x_n$ values in A by a function $C : X \rightarrow A$ also determines a value $\hat{C}(t)$ for t in A . Thus, elements of $T_\Sigma(X)$ are themselves operators, derived from the more basic operators in Σ , and we shall call them “derived operators,” or “ Σ -trees in n variables.”

Definition 13.12 Let $X_n = \{x_1, \dots, x_n\}$; let A be a Σ -algebra; let $\langle a_1, \dots, a_n \rangle \in A^n$; and define $a : X_n \rightarrow A$ by $a(x_i) = a_i$, for $1 \leq i \leq n$. Then for every $t \in T_\Sigma(X_n)$ we define its corresponding derived operator on A , $t_A : A^n \rightarrow A$, by $t_A(a_1, \dots, a_n) = \hat{a}(t)$, where $\hat{a} : T_\Sigma(X_n) \rightarrow A$ is the unique homomorphism extending $a : X_n \rightarrow A$ guaranteed by proposition 13.11.

The following diagram may help visualize these relationships:

$$\begin{array}{ccccc}
 X_n & & \xrightarrow{a} & & A \\
 & \searrow & & \nearrow \hat{a} & \\
 & & T_\Sigma(X_n) & &
 \end{array}
 \quad
 \begin{array}{ccc}
 & & A^n \\
 & \xleftarrow{t_A(a)} &
 \end{array}$$

Given a Σ -tree t in n variables, we let a vary in $\hat{a}(t)$ while keeping t fixed. This amounts to “evaluating” the rank- n term t in A with variables x_i given value $a_i \in A$.

13.5 Infinite Trees

13.5.1

In this section we extend the previous ideas on Σ -algebras and Σ -trees (or Σ -terms) to the case of infinite trees (or terms). As discussed

in chapter 10, infinite trees arise as unfoldings of circular situations, and are the basis of an autonomy/control complementarity.

This extension into infinity requires, however, some careful development of additional concepts. These concepts make possible the rigorous discussion of indefinite recursion. The latter requires appropriate notions of approximation and limit. The reader will have to bear with me through some rather technical material before its fruits can be seen. We shall apply some notions of order and continuity to obtain a characterization of infinite trees similar to that given for finite trees in the previous section.⁷ This material follows (Goguen et al., 1977), but is simpler, less general, more detailed, and better illustrated (Goguen and Varela, 1979).

The fundamental concept is that of a partially ordered set or “poset.” We define the order-theoretic concepts of greatest importance to us in this context.

Definition 13.13 *A poset is a set P together with a partial order $\sqsubseteq$, that is, a reflexive, antisymmetric, and transitive relation on P .*

A poset P is strict iff it has an element $\perp \in P$ such that $\perp \sqsubseteq p$ for all $p \in P$; such an element $\perp$ is called minimum or bottom for P .

An upper bound for a subset S of P is any $x \in P$ such that $a \sqsubseteq x$ for all $a \in S$. We let $(a \sqcup b)$ denote the least upper bound of $\{a, b\}$, and let $\bigsqcup S$ denote the least upper bounds (l.u.b.) of an arbitrary subset S of P .

A subset S of P is directed iff every finite subset of S has an upper bound in S .

Let $S \subseteq P$; then S is a chain iff for all $a, b \in S$, either $a \sqsubseteq b$ or $b \sqsubseteq a$. P is (ω) -chain-complete iff every (countable) chain S in P has a least upper bound in S .

Note that any two minimum elements of a poset P are in fact equal.

The natural numbers ω are a poset with the usual order. Every subset $S \subseteq \omega$ is directed, since every finite subset of numbers in S has an upper bound in S , namely the maximum of the set of numbers. Also, every subset $S \subseteq \omega$ is a chain. But ω is not chain-complete. For example, ω itself is a countable chain having no least upper bound in ω .

Let A, B be sets, and consider the set $[A \multimap B]$ of all partial functions from A to B , that is, maps for which not all a 's in A have values in B ; their domains may have “holes,” as suggested by the notation $\multimap$. Elements of $[A \multimap B]$ correspond to subsets f of $A \times B$ satisfying the following “functional” property: If $\langle a, b \rangle \in f$ and $\langle a, b' \rangle \in f$, then $b = b'$. Then $[A \multimap B]$ is a poset with the order relation of set inclusion;

⁷*Eds.* The procedure will involve generating an encoding for trees using strings of natural numbers. This will facilitate the investigation of partial orders in sets of trees, which will allow the claim that a sequence of finite trees progressively approximates trees encoded by infinite strings (i.e., infinite trees). On this basis, the set of finite and infinite partial trees can be given the structure of an algebra (see section (13.6)), which is the final step toward defining equations (see section 13.7) and the iterative methods for finding their eigenbehaviors.

least upper bounds are set unions. The latter will exist in $[A \Rightarrow B]$ iff the set union is still a functional set; this, of course, is not always the case, but if $\langle f_i \rangle_{i \in I}$ is a chain in $[A \Rightarrow B]$, then $\bigcup_{i \in I} f_i$ exists and is a functional set. Thus $[A \Rightarrow B]$ is ω -chain-complete.

In line with the category-theoretic doctrine that structure-preserving maps are at least as important as the corresponding objects, we next introduce the notion of order-preserving maps for posets.

Definition 13.14 *Let P, P' be posets. Then a map f from P to P' is monotonic iff for all $p_0 \sqsubseteq p_1$ in P , $f(p_0) \sqsubseteq f(p_1)$ in P' .*

If P, P' are strict posets, then $f : P \rightarrow P'$ is strict iff $f(\perp) = \perp$.

Let P, P' be posets, and let $\langle p_i \rangle_{i \in I}$ be an ω -chain in P . Then $f : P \rightarrow P'$ is ω -chain-continuous iff

$$f\left(\bigsqcup_{i \in I} p_i\right) = \bigsqcup_{i \in I} f(p_i) \quad \text{in } P'.$$

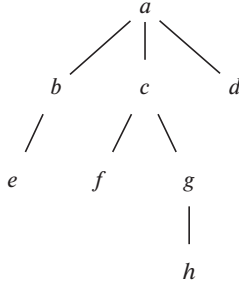
This last definition says that a map is continuous if and only if the same value results from taking the least upper bound of a chain and then looking at its image, as from mapping each member of the chain and then taking the least upper bound of the images. This notion of continuity is reminiscent of the one found in elementary calculus.

13.5.2

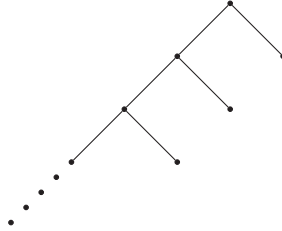
We are interested in putting a partial orders on sets of Σ -trees. A clue to how to do this is provided by our example above: If we can make a set of Σ -trees into a set of partial functions, of the form $[A \Rightarrow B]$, then the set will have a natural partial order; it also seems reasonable to guess that $B = \Sigma$ will work. But it is not clear what A should be. Here we use an elegant representation of nodes by strings of natural numbers. The basic idea is that the string shall encode the sequence of choices of branches required to get from the root to the node in question. Thus, the root is represented by the empty string λ . An $(n+1)$ st child of the root is represented by the string consisting of the single integer $n+1$; in particular, the first child is represented by 0, and the second by 1. More generally, if u is a string of non-negative integers, representing a node, then the $(n+1)$ st child of u is represented by the string un , consisting of u followed by n . The set of all possible node representations is then the set ω^* of all finite strings of non-negative integers; this is the set A we wanted above.

Before going on to Σ -trees and formal definitions, let us see how this encoding of nodes to strings works on simple examples.

Consider the tree



in which a, b, c, d, e, f, g, h are the names of the nodes. These are represented by the following strings (in the same order): $\lambda, 0, 1, 2, 00, 10, 11, 110$. One advantage of this approach is that it extends to infinite trees without difficulty. For example, the infinite tree structure



has as its set of node representations $\lambda, 0, 1, 00, 01, 000, 001, \dots$; to be precise, it is $\{0^n | n \in \omega\} \cup \{0^{n-1}1 | n \in \omega\}$.

Clearly, not any set of strings can be the set of representations of the nodes of a tree. Those sets that can be are captured in the following. At the same time, we show how to handle Σ -trees.

Definition 13.15 A full tree domain is a subset D of ω^* such that, for all $u \in \omega^*$ and $n \in \omega$,

1. $un \in D$ implies $u \in D$,
2. $un \in D$ implies $ui \in D$ for all $i \in \omega$ with $i < n$.

Let $\langle \Sigma_n \rangle_{n \in \omega}$ be an operator domain, and let Σ denote the set $\bigcup_n \Sigma_n$. Then a full Σ -tree is a partial function $t : \omega^* \rightarrow \Sigma$ such that the domain of definition of t is a tree domain D such that

3. if $u \in D$ but $ui \notin D$ for all $i \in \omega$, then $t(u) \in \Sigma_0$,
4. if $ui \in D$ and $i \in D$ for all $i \in \omega$, then $t(u) \in \Sigma_n$, for some $n > i$.

We shall say that a tree t is finite iff its domain is finite. Let FF_Σ denote the set of all finite full Σ -trees, and let FT_Σ denote the set of all full Σ -trees, finite or not.

Now the set FF_Σ of all full finite Σ -trees can be given operations $\sigma_F : \text{FF}_\Sigma^n \rightarrow \text{FF}_\Sigma$ for each $\sigma \in \Sigma_n$ in a way analogous to those given earlier for T_Σ , and it can be shown that the resulting Σ -algebra is an initial Σ -algebra. Therefore it is isomorphic to T_Σ , and we have another example of a different representation of the same structure. Notice that the carrier of FF_Σ is the set of all finite elements of $[\omega^* \twoheadrightarrow \Sigma]$ satisfying conditions 1-4 above. We shall hereafter identify FF_Σ and T_Σ , both as Σ -algebras and as sets.

There is, however, a problem with our plan to use this approach to get an order structure on Σ -trees: the order relation on full Σ -trees is not very interesting. In fact, we have the following

Proposition 13.16 *For $t, t' \in [\omega^* \twoheadrightarrow \Sigma]$, define $t \sqsubseteq t'$ to mean that, as sets of ordered pairs (that is, as subsets of $\omega^* \times \Sigma$), t is a subset of t' . Then, if t, t' are full Σ -trees and $t \sqsubseteq t'$, either $t = t'$ or $t = \emptyset$, the empty tree.*

Proof. If t is a full Σ -tree, D is its domain, and $u \in D$, then $t(u) \in \Sigma_n$ iff u has exactly n children, namely, $u0, u1, \dots, u(n-1)$.

Let D, D' be the domains of t, t' and assume $t \neq \emptyset, t \sqsubseteq t'$ and $t \neq t'$. Then there is some $u \in D' - D$. Write $u = vw$, choosing v of maximum possible length such that $v \in D$ (this is possible, because at worst $v = \lambda$, and both v and w are uniquely determined by giving the length of v). By conditions 1 and 2 of definition 13.15 (and induction), $v \in D'$ and $t'(v) \in \Sigma_n$ with $n > 0$. By condition 3 of definition 13.15, $t(v) \in \Sigma_0$. But $t \sqsubseteq t'$ implies $t(v) = t'(v)$. We saw that this is impossible, so the assumption that $t \neq t'$ must have been wrong. $\square$

What we really want are finite approximations to the infinite Σ -trees. These are easy to obtain, if we relax the requirement that if $t(u) \in \Sigma_n$ then u has exactly n children, by letting some of the child nodes be “undefined.” The following shows how we do this.

Definition 13.17 *A (partial) tree domain is a subset D of ω^* such that for all $u \in \omega^*$ and $n \in \omega$,*

1. $un \in D$ implies $u \in D$.

Let $\langle \Sigma_n \rangle_{n \in \omega}$ be an operator domain. Then a (partial) Σ -tree is a partial function $t : \omega^ \twoheadrightarrow \Sigma$ such that the domain of definition D of t is a partial tree domain, and*

2. *if $ui \in D$ and $i \in \omega$, then $t(ui) \in \Sigma_n$ for some $n > i$.*

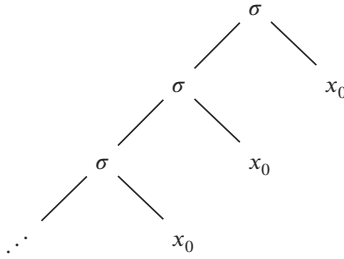
(Thus, a partial tree satisfies conditions 1 and 4 of definition 13.15, but not necessarily 2 and 3.) If t, t' are partial Σ -trees, then $t \sqsubseteq t'$ iff $t \subseteq t'$ as sets of ordered pairs. Let CT_Σ denote the set of all partial Σ -trees (both finite and infinite).

The following table should help in keeping track of the notation for various kinds of Σ -trees:

	Full	Partial (or full)
Finite	$\text{FF}_\Sigma \cong T_\Sigma$	F_Σ
Finite or infinite	FT_Σ	CT_Σ

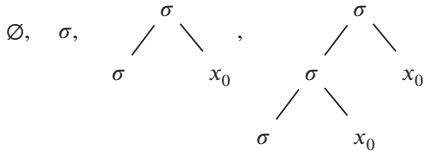
Note that $F_\Sigma \subseteq \text{CT}_\Sigma$. Hereafter, we shall feel free to drop the word “partial” and refer to elements of CT_Σ and F_Σ as “ Σ -trees.” To avoid confusion, elements of T_Σ and FT_Σ will be referred to as “full Σ -trees.”

We illustrate that the ordering relation $\sqsubseteq$ on CT_Σ is nontrivial. Let t be the following full Σ -tree:



where $\sigma \in \Sigma_2$ and $x_0 \in \Sigma_0$. This tree has as its domain D that given in the previous example. We now construct a sequence $t^{(0)}, t^{(1)}, t^{(2)}, \dots$ of finite partial Σ -tree approximations to t : let $D^{(n)} = \{u \in D \mid u \text{ has length } < n\}$, and let $t^{(n)}$ be the restriction of t to $D^{(n)}$.

For example, $D^{(0)} = \emptyset$, $D^{(1)} = \{\lambda\}$, $D^{(2)} = \{\lambda, 0, 1\}$, $D^{(3)} = \{\lambda, 0, 1, 00, 01\}$; and $t^{(0)}, t^{(1)}, t^{(2)}, t^{(3)}$ look like



Clearly $t^{(n)} \sqsubseteq t^{(n+1)} \sqsubseteq t$ for all $n \in \omega$. Moreover $\bigcup_n t^{(n)} = t$.

That this situation generalizes nicely, is shown by the following

Proposition 13.18 CT_Σ is a chain-complete poset.

Proof. We recall from a previous example that chains of partial functions have least upper bounds that are partial functions. Now, let $\langle t_i \rangle_{i \in I}$ be a chain of partial functions $\omega^* \rightarrow \Sigma$, each satisfying conditions 1 and 2 of definition 13.17. Then it is not hard to show that the union $\bigcup_{i \in I} t_i$, also satisfies 1 and 2, and is therefore in CT_Σ . Therefore, it is the least upper bound of $\langle t_i \rangle_{i \in I}$. $\square$

In particular, CT_Σ is ω -chain-complete, by restricting to chains $\langle t_i \rangle_{i \in I}$.

Proposition 13.19 *Let $t \in \text{CT}_\Sigma$, and let D be the domain of t . Let $D^{(n)} = \{u \in D \mid (\text{length of } u) < n\}$, and let $t^{(n)}$ be the restriction of t to $D^{(n)}$. Then $\langle t^{(n)} \rangle_{n \in \omega}$ is an ω -chain of finite Σ -trees with least upper bound t .*

13.6 Continuous Algebras

The set CT_Σ of all finite and infinite (partial) Σ -trees can be given the structure of a Σ -algebra, in very much the same way as was done earlier for the finite full Σ -trees. This makes CT_Σ into a kind of ordered algebra and leads toward the characterization of the algebra of Σ -trees as initial in various classes of ordered algebras.

Definition 13.20 *Let Σ be an operator domain. We make CT_Σ into a Σ -algebra as follows, making quite explicit use of the ordered-pair representation of partial functions:*

1. for $\sigma \in \Sigma_0$, let $\sigma_{\text{CT}} = \{\langle \lambda, 0 \rangle\}$;
2. for $\sigma \in \Sigma_n$, $n > 0$, and $t_1, \dots, t_n \in \text{CT}_\Sigma$, let $\sigma_{\text{CT}}(t_1, \dots, t_n) = \{\langle \lambda, \sigma \rangle\} \cup \bigcup_{i < n} \{\langle i u, \sigma' \rangle \mid \langle u, \sigma' \rangle \in t_{i+1}\}$,

where we have written σ_{CT} rather than $\sigma_{\text{CT}_\Sigma}$.

Informally, a compound expression $\sigma(t_1, \dots, t_n)$ is obtained by taking a node labeled σ as root, and attaching the roots of the trees $t_1, \dots, t_n$ below it (as children), in order.

Now, CT_Σ already has a partial ordering. We make use of the algebraic operations given above to provide a characterization of the order structure of CT_Σ analogous to that provided by proposition 13.16 for FT_Σ .

Proposition 13.21 *For $t, t' \in \text{CT}_\Sigma$, $t \sqsubseteq t'$ iff: $t = \emptyset$; or $t = t' = \sigma_{\text{CT}}$ for $\sigma \in \Sigma_0$; or there is some $\sigma \in \Sigma_n$ with $n > 0$, and $t_1, \dots, t_n, t'_1, \dots, t'_n \in \text{CT}_\Sigma$ such that $t = \sigma_{\text{CT}}(t_1, \dots, t_n)$, $t' = \sigma_{\text{CT}}(t'_1, \dots, t'_n)$, and $t_i \sqsubseteq t'_i$ for $1 \leq i \leq n$.*

Proof. See (Goguen et al., 1977). □

Not only is CT_Σ an ω -complete poset and a Σ -algebra, but these structures are related, in the following particularly felicitous way.

Proposition 13.22 *For each $\sigma \in \Sigma$, the operation σ_{CT} on CT is ω -chain-continuous.*

Proof. Let $\sigma \in \Sigma_n$. The result is trivial if $n = 0$, so we may assume $n > 0$.

Let $\langle t_{ij} \rangle_{j \in \omega}$ be ω -chains in CT_Σ for $i = 1, \dots, n$. Let $t_i = \bigcup_{j \in \omega} t_{ij}$; these l.u.b.'s exist by proposition 13.19, and in fact, they are set unions.

Now, it is "well known," for the n -fold Cartesian-product poset $\text{CT}_\Sigma \times \dots \times \text{CT}_\Sigma$ ordered by $\langle t_1, \dots, t_n \rangle \sqsubseteq \langle t'_1, \dots, t'_n \rangle$ iff $t_i \sqsubseteq t'_i$ for $i = 1, \dots, n$, that the l.u.b.'s can be computed componentwise: that is, for $t_{ij} \in \sigma_{\text{CT}}$ for $i = 1, \dots, n$ and $j \in \omega$,

$$\bigsqcup_{j \in \omega} \langle t_{1j}, \dots, t_{nj} \rangle = \langle \bigsqcup_{j \in \omega} t_{1j}, \dots, \bigsqcup_{j \in \omega} t_{nj} \rangle$$

(which is $\langle t_1, \dots, t_n \rangle$).

What we want to show is that

$$\sigma_{\text{CT}}(\bigsqcup_{j \in \omega} t_{1j}, \dots, \bigsqcup_{j \in \omega} t_{nj}) = \bigsqcup_{j \in \omega} \sigma_{\text{CT}}(t_{1j}, \dots, t_{nj}).$$

So let us calculate, using definition 13.20.

$$\begin{aligned} & \sigma_{\text{CT}}(t_1, \dots, t_n) \\ &= \{ \langle \lambda, \sigma \rangle \} \cup \bigcup_{i < n} \{ \langle i u, \sigma' \rangle \mid \langle u, \sigma' \rangle \in \bigsqcup_{j \in \omega} t_{i+1,j} \} \\ &= \{ \langle \lambda, \sigma \rangle \} \cup \bigcup_{i < n} \left(\bigcup_{j \in \omega} \{ \langle i u, \sigma' \rangle \mid \langle u, \sigma' \rangle \in t_{i+1,j} \} \right) \\ &= \bigcup_{j \in \omega} \left(\{ \langle \lambda, \sigma \rangle \} \cup \bigcup_{i < n} \{ \langle i u, \sigma' \rangle \mid \langle u, \sigma' \rangle \in t_{i+1,j} \} \right) \\ &= \bigsqcup_{j \in \omega} \sigma_{\text{CT}}(t_{1j}, \dots, t_{nj}). \end{aligned}$$

□

Definition 13.23 *An ordered Σ -algebra is a Σ -algebra whose carrier is a strict poset, and whose operations are monotonic. A homomorphism of an ordered Σ -algebra is a strict monotonic Σ -homomorphism. Let $\mathcal{P}alg_{\Sigma}$ denote the class of all ordered Σ -algebras, together with all strict monotonic homomorphisms among them.*

An ω -continuous Σ -algebra is a Σ -algebra whose carrier is a strict ω -complete poset whose operations are ω -continuous. A homomorphism of ω -continuous Σ -algebras is a strict ω -continuous Σ -homomorphism. Let ωalg_{Σ} denote the class of all ω -continuous Σ -algebras, together with all strict ω -continuous Σ -homomorphisms among them.

We have shown that CT_{Σ} is an ω -continuous Σ -algebra, and thus, an ordered Σ -algebra. The result we are aiming for is that CT_{Σ} is initial in ωalg_{Σ} . The proof uses the following two results, the first of which is certainly of independent interest.

Proposition 13.24 *F_{Σ} is initial in $\mathcal{P}alg_{\Sigma}$.*

Proof. Let $\Sigma(\perp)$ denote the signature Σ enriched by the new constant symbol $\perp$. Now, we can make any ordered Σ -algebra A into a $\Sigma(\perp)$ -algebra, by letting $\perp$ in $\Sigma(\perp)$ denote $\perp$ in A , which exists because A is strict. Also, note that a strict Σ -homomorphism is the same thing as a $\Sigma(\perp)$ -homomorphism.

The reader may want to verify the following lemma, upon which this proof relies: F_{Σ} , as a $\Sigma(\perp)$ -algebra, is isomorphic to $T_{\Sigma(\perp)}$. Then, for any ordered Σ -algebra A , there is a unique $\Sigma(\perp)$ -homomorphism $h_a : F_{\Sigma} \rightarrow A$, and we shall be done if we can show that it is monotone.

Let $t \sqsubseteq t'$ in F_{Σ} . Since $F_{\Sigma} \subseteq \text{CT}_{\Sigma}$, we can apply proposition 13.22 to get that either: (1) $t = \perp$, or (2) $t = t' = \sigma_F$ for $\sigma \in \Sigma_0$, or (3)

$t = \sigma_F(t_1, \dots, t_n)$ and $t' = \sigma_F(t'_1, \dots, t'_n)$ with $t_i \sqsubseteq t'_i$ (for $i = 1, \dots, n$) and $\sigma \in \Sigma_n$, noting that each t_i and t'_i must be in F_Σ , not just in CT_Σ .

In case (1), $h_A(t) = \perp$, since h_a is strict, so certainly $h_A(t) \sqsubseteq h_A(t')$.

In case (2), obviously $h_A(t) \sqsubseteq h_A(t')$ since $h_A(t) = h_A(t')$.

Case (3) is the interesting one, and the proof proceeds by induction on the cardinality of the domain of definition of t . Cases (1) and (2) above are in fact the basis of the induction. For the inductive step, we assume that $h_A(t_i) \sqsubseteq h_A(t'_i)$, and calculate

$$\begin{aligned}
 h_A(t) &= h_A(\sigma_{\text{CT}}(t_1, \dots, t_n)) && \text{[form of } t \text{ from (3)]} \\
 &= \sigma_A(h_A(t_1), \dots, h_A(t_n)) && [h_A \text{ is a homomorphism}] \\
 &\sqsubseteq \sigma_A(h_A(t'_1), \dots, h_A(t'_n)) && [\sigma_A \text{ is monotone}] \\
 &= h_A(\sigma_F(t'_1, \dots, t'_n)) && [h_A \text{ is a homomorphism}] \\
 &= h_A(t'). && \text{[form of } t' \text{ from (3)]}
 \end{aligned}$$

Thus $h_A(t) \sqsubseteq h_A(t')$. $\square$

Proposition 13.25 *The operations of F_Σ are ω -chain-continuous.*

Proof. The proof of proposition 13.24 goes through when restricted to trees with finite domains. $\square$

Now the main result.

Theorem 13.26 *CT_Σ is initial in ωalg_Σ .*

Proof. The proof is based on the fact that $\omega\text{alg}_\Sigma \subseteq \mathcal{P}\text{alg}_\Sigma$, so that for any A in ωalg_Σ , there is a unique strict monotone Σ -homomorphism $h_A : F_\Sigma \rightarrow A$. The work of the proof is to extend this to an ω -continuous Σ -homomorphism $\bar{h}_A : \text{CT}_\Sigma \rightarrow A$, using the approximation suggested by proposition 13.21: For $t \in \text{CT}_\Sigma$, we have $t = \bigsqcup_{n \in \omega} t^{(n)}$, with each $t^{(n)} \in F_\Sigma$; we then define (when no confusion will arise, we write $\bigsqcup_n$ for $\bigsqcup_{n \in \omega}$)

$$\bar{h}_A(t) = \bigsqcup_n h_A(t^{(n)}),$$

knowing that this l.u.b. exists because A is ω -continuous. It remains to show that (1) the extension is unique, (2) is ω -continuous, and (3) is a Σ -homomorphism.

1. Suppose $h' : \text{CT}_\Sigma \rightarrow A$ extends $h_A : F \rightarrow A$ and is ω -continuous. Then

$$h'(t) = h'(\bigsqcup_n t^{(n)}) = \bigsqcup_n h'(t^{(n)}) = \bigsqcup_n h_A(t^{(n)}) = \bar{h}_A(t).$$

2. We first show that $\bar{h}_A$ is monotonic. Let $t_0 \sqsubseteq t_1$ in CT_Σ . Then $t_0^{(n)} \sqsubseteq t_1^{(n)}$ for all $n \in \omega$, so that $h_A(t_0^{(n)}) \sqsubseteq h_A(t_1^{(n)})$ by monotonicity of h_A ; therefore

$$\bar{h}_A(t_0) = \bigsqcup_n h_A(t_0^{(n)}) \sqsubseteq \bigsqcup_n h_A(t_1^{(n)}) = \bar{h}_A(t_1),$$

as desired.

Now assume that $t = \bigsqcup_i t_i$, for $\langle t_i \rangle_{i \in \omega}$ an ω -chain in CT_Σ . We want to show that $\bigsqcup_i \bar{h}_A(t_i) = \bar{h}_A(\bigsqcup_i t_i; L) = \bar{h}_A(t)$.

The key lemma is the following: For each $n \in \omega$, there is some $j \in \omega$ such that $t^{(n)} \subseteq t_j$. To show this, it suffices to show that for any sets A, B , if $t' \in [A \Rightarrow B]$ is finite, and if $t' \subseteq \bigsqcup_i t_i$ for $\langle t_i \rangle_{i \in \omega}$ a chain in $[A \Rightarrow B]$, then there is some $j \in \omega$ such that $t' \subseteq t_j$.

Now let $b \in \text{CT}_\Sigma$ be an upper bound of the chain $\langle \bar{h}_A(t_i) \rangle_{i \in \omega}$, i.e., $\bar{h}_A(t_i) \subseteq b$ for all $i \in \omega$. Then (by the lemma of the above paragraph), for each $n \in \omega$, there is some $j \in \omega$ such that $\bar{h}_A(t^{(n)}) \subseteq \bar{h}_A(t_j) \subseteq b$. Therefore, $\bigsqcup_n \bar{h}_A(t^{(n)}) = \bar{h}_A(t) \subseteq b$. It now follows that $\bar{h}_A(t) = \bigsqcup_i \bar{h}_A(t_i)$; i.e., that $\bar{h}_A$ is ω -continuous, as desired.

3. We now show that $\bar{h}_A$ is a Σ -homomorphism. First, observe that for each $\sigma \in \Sigma_n$, $t \in \text{CT}_\Sigma$ for $i = 1, \dots, n$, and $k > 0$.

$$\sigma_{\text{CT}}(t_1, \dots, t_n)^{(k)} = \sigma_{\text{CT}}(t_1^{(k-1)}, \dots, t_n^{(k-1)}),$$

while $t^{(0)} = \perp$. Now let us compute:

$$\begin{aligned} & \bar{h}_A(\sigma_{\text{CT}}(t_1, \dots, t_n)) \\ &= \bigsqcup_k h_A(\sigma_{\text{CT}}(t_1, \dots, t_n)) && \text{[definition of } \bar{h}_A] \\ &= \bigsqcup_k h_A(\sigma_{\text{CT}}(t_1^{(k-1)}, \dots, t_n^{(k-1)})) && \text{[above observation]} \\ &= \bigsqcup_k \sigma_A(h_A(t_1^{(k-1)}), \dots, h_A(t_n^{(k-1)})) && \text{[} h_A \text{ is a } \Sigma\text{-homomorphism]} \\ &= \sigma_A(\bigsqcup_k h_A(t_1^{(k-1)}), \dots, \bigsqcup_k h_A(t_n^{(k-1)})) && \text{[} \sigma_A \text{ is a } \omega\text{-continuous]} \\ &= \sigma_A(\bar{h}_A(t_1), \dots, \bar{h}_A(t_n)) && \text{[definition of } \bar{h}_A] \end{aligned}$$

(where some subscripts k range over $k > 0$ rather than $k \in \omega$).

□

This completes our general discussion of infinite trees and initial continuous algebras.

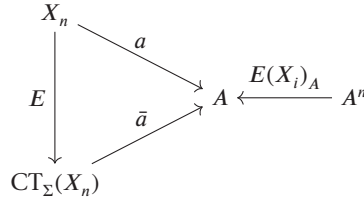
13.7 Equations and Solutions

We are now ready to give the definitions of solutions of an equation over a continuous algebra. This will enable us to formalize the idea of an eigenbehavior, discussed at the outset of this chapter in a general way.

Definition 13.27 A system of n equations in CT_Σ is a function $E : X_n \rightarrow \text{CT}_\Sigma(X_n)$. We write $x_i = E(x_i)$ as the i th equation.

For any A in alg_Σ , $E_A : A^n \rightarrow A^n$ is the derived operator of E over A , $E_A = \langle E(x_1)_A, \dots, E(x_n)_A \rangle : A^n \rightarrow A^n$

This definition can be represented in the diagram



so that $(E_A(a))_i = E(x_i)_A(a) = \bar{a}(E(x_i))$.

Proposition 13.28 *Let E be a system of n equations in CT_Σ , A in ωalg_Σ . Then E_A has a minimum fixed point $|E_A| \in A^n$, called the solution of E over A , or the eigenbehavior of E over A .*

Proof. Define $|E_A| = \bigsqcup_{k \in \omega} E_A^k(\perp, \dots, \perp)$. Then $|E_A|$ is a fixed point, since

$$\begin{aligned}
 E_A(|E_A|) &= E_A\left(\bigsqcup_{k \in \omega} E_A^k(\perp, \dots, \perp)\right) \\
 &= E_A\left(\bigsqcup_{k \in \omega} E_A^k(x_i)(\perp), \dots, \bigsqcup_{k \in \omega} E_A^k(x_n)(\perp)\right) \\
 &= \left(\bigsqcup_{k \in \omega} E_A^{k+1}(x_i)(\perp), \dots, \bigsqcup_{k \in \omega} E_A^{k+1}(x_n)(\perp)\right) \\
 &= |E_A|.
 \end{aligned}$$

Consider now $a \in A^n$ such that $E_A(a) = a$. Then clearly $|E_A|^{(0)} \sqsubseteq a^{(0)}$. Assume, for induction, that $|E_A|^{(k)} \sqsubseteq a^{(k)}$. Then, since E_A is monotonic,

$$E_A(|E_A|^{(k)}) = |E_A|^{(k+1)} \sqsubseteq E_A(a^{(k)}) = a^{(k+1)}.$$

Thus, $|E_A| \sqsubseteq a$, and $|E_A|$ is the minimum fixed point. $\square$

This proposition shows how to construct a fixed point of E through the indefinite iteration of the trees forming the system of equations.

Call $|E|$ the solution for the case $A = CT_\Sigma$, and write $|E|_A = h_A^n(|E|)$. We now show that we can either solve an equation and then interpret it, or first interpret and then solve.

Proposition 13.29 *For any system $E : X_n \rightarrow CT_\Sigma(X_n)$ of equations and any $A \in \omega alg_\Sigma$, $|E|_A = |E_A|$.*

Proof. By induction on k it is clear that

$$h_A^n(E^k(\perp, \dots, \perp)) = E^k(\perp_A, \dots, \perp_A).$$

Then

$$\begin{aligned}
 |E|_A &= h_A^n(|E|) \\
 &= h_A^n\left(\bigsqcup_{k \in \omega} E^k(\perp, \dots, \perp)\right)
 \end{aligned}$$

$$\begin{aligned}
&= \bigsqcup_{k \in \omega} h_A^n(E^k(\perp, \dots, \perp)) && \text{(continuity of } h) \\
&= \bigsqcup_{k \in \omega} E_A^k(h_A(\perp), \dots, h_A(\perp)) \\
&= \bigsqcup_{k \in \omega} E_A^k(\perp_A, \dots, \perp_A) \\
&= |E_A|.
\end{aligned}$$

□

We have then that every system of equations has an eigenbehavior over CT_Σ ; conversely, this is a way in which we could hope to characterize infinite trees of CT. Could we not associate with an infinite tree the equation(s) for which it is a solution? We would like such equational elements of CT_Σ to be well behaved, in the sense of being describable and having adequate composition properties. But there is no assurance that this is the case: The problem of dealing with equational elements of CT_Σ is quite complex (Wright et al., 1976). We can, however, say something more precise about a *small* part of CT_Σ , namely those infinite trees that are solutions for *finite* systems of equations, that is, systems E such that $E : X_n \rightarrow F_\Sigma(X_n)$. An ordered algebra is said to be *equationally complete* if every finite system of equations has a solution over A .

Let R_Σ denote the set of equational elements of CT that are solutions for finite systems, i.e., the collection of eigenbehaviors

$$R_\Sigma = \{|E|_i \mid E : X_n \rightarrow F_\Sigma(X_n), n > 0, 1 \leq i \leq n\}.$$

The reason for this definition of equational completeness and the use of the “ R ” in R_Σ is the following characterization of the trees in R_Σ , which we state without proof (see Goguen et al., 1977, propositions 5.3, 5.4).

Proposition 13.30 *If $t \in R_\Sigma$, then for each $\sigma \in \Sigma_n$, $t^{-1}(\sigma) \subseteq \omega^*$ is a regular subset of $\{0, 1, \dots, k\}^*$ for some k .*

Thus the elements of R_Σ can be described and compared by means of some computable procedures. This is, of course, *not* the case for all the other elements in CT_Σ : Some infinite trees might not even be finitely or even recursively describable, and we have no idea how they behave under composition, quotient, and so on.

By contrast the elements of R_Σ are very well behaved: R_Σ is a subalgebra of CT_Σ (Goguen et al., 1977, proposition 5.5). (Accordingly, Scott calls elements in R_Σ “algebraic,” and elements in $CT_\Sigma - R_\Sigma$ “transcendental.”) For our present purposes we need only be concerned with the construction of R_Σ as an equationally complete

subalgebra, since finite systems of equations are certainly the ones that are needed in most (if not all) the concrete applications. This is so to the extent that equations embody the ways in which the system components interconnect. We may assume this situation to be always captured in a finite description (i.e., in trees of F_{Σ}).

13.8 Reflexive Domains

Let us pause for a moment to reconsider what these algebraic developments mean in the broader context of the investigation proposed in these pages. Paying attention to the autonomy of natural systems led us to the closure thesis—that is, to consider the complementarity between the recursive underlying dynamics of a unity, and the way in which such dynamics generates a coherent pattern, a behavior of a unity affording a criterion of distinction. In order to carry this characterization one step further, we decided to make precise what we mean by complementarity, and by a coherent behavior and its underlying processes. That is the spirit of the notion of organizational closure, and of complementarity as adjunction, developed so far. Further precision of these ideas hinges upon the construction of appropriate calculi, where elements or operands are on the same descriptive level with operators or processes, and where the products of processes become effectively interrelated with the processes that generate them.

Let us formulate these notions more formally thus: Consider a descriptive domain of elements D of some kind (stable levels of reactants, coherent pattern of behavior, meaning of a conversation, and so on). By the closure thesis, these belong to an autonomous system if they arise out of processes acting on the very same elements, that is, some appropriate class of operations or processes, which we may denote $[D \rightarrow D]$. We need to keep the distinction between elements (criteria of distinction) and processes (underlying dynamics), but in such a way that they are effectively related, that is, in such a way that they are seen as the same, except for the means we choose to observe them, in a star fashion. One way of formulating this complementarity is to demand a correspondence

$$D \overset{R}{\leftrightarrow} [D \rightarrow D]. \quad (13.3)$$

When R is an isomorphism, we call D a *reflexive domain*. It can be understood as a descriptive realm which can operate on itself (act on itself).⁸

⁸*Eds.* In other words, there is a one-to-one correspondence between processes in D and transformations or operations between processes in D .

Now, functions or operations that can operate on themselves have been a headache in mathematics for a long time. If we simply ask whether (13.3) is true, in general, for various kinds of D 's and of functions on the D 's, the answer is *no*. Such reflexive domains cannot exist without inconsistencies. However, the condition (13.3) becomes possible if we *restrict* the kinds of domains and their operations (see appendix B). We started this characterization on the simplest possible grounds: those of indicational forms. We succeeded in expressing pattern/dynamics in an explicit form. In order to carry the overall distinction into diversified operations, we presented the development of continuous algebras, where a special kind of descriptive domain (i.e., a continuous algebra) and special operations in them (i.e., continuous) could yield a correspondence as well. In fact, under these restrictions, we had

$$\text{CT}_{\Sigma} \underset{E}{\overset{S}{\rightleftarrows}} [\text{CT}_{\Sigma} \rightarrow \text{CT}_{\Sigma}].$$

S relates equations to their eigenbehavior (minimum fixed-point solution). E relates infinite trees to the equations of which they are a solution. In this case, the correspondence (13.3) is not an isomorphism since E is a one-to-many map. Thus CT_{Σ} is close to, but not identical with, a reflexive domain.

So far, this approach has been used in detail only in the semantics of programming languages, where it originated, as we have said, with the work of Scott. The motivation for Scott's work was to consider the relation between the meaning of a program (i.e., its criteria of distinction), and its computational behavior (i.e., its underlying closure). In this sense, a program is looked at as an autonomous object, as a *text*. This is, of course, not to say that the computer itself is looked as an autonomous object: We are talking about the coherence of recursive programs. These are not so distant from other texts, proper to natural languages, that arise as coherent objects (cf. chapter 16; Becker, 1980; Linde, 1980). In the present interpretation, Scott's work, as elaborated by the theory of continuous algebras, means that this insight into the coherence of a programming text *can be generalized to the coherence of other autonomous units, providing us with precise formal tools to represent them*.

To be sure, these algebraic foundations have limitations, but they still contain a large class of possible models. For each particular system under study it is necessary to specify in detail which operator domain is to be considered, and what is its order structure. Once this is established, all the results from the theory of continuous algebras

become available, since our treatment was “abstract” through the notion of initiality. In other words, this means that we begin to have available a range of mathematical tools, beyond those of differential dynamics (cf. section 13.10), within which we can include any process whatsoever that can be made precise enough to define an operator domain satisfying the appropriate restrictions of order and continuity. I hasten to warn the reader that beyond the cases of text coherence, in programming languages (e.g., Stoy, 1977) and planning discourse (Linde and Goguen, 1978), this theory has not yet been applied in any detail. The ground is entirely open. In the sections that follow I shall try to give a glimpse of the flavor such applications can have, without pretending to be exhaustive.

13.9 Indicational Reentry Revisited

We can now deal more adequately with the issue raised in chapter 12, in relation to infinite indicational expressions. We gave there an informal construction the class B_∞ of continuous forms; we shall briefly review it here, in a more rigorous form. At the same time it will serve as an exercise for in the applications of continuous algebras just presented.

Let Σ be the following operator domain: $\Sigma_0 = (1, 0)$; $\Sigma_1 = \{\sigma_1\}$; $\Sigma_2 = \{\sigma_2\}$; $\Sigma_k = \emptyset$, $k > 2$. Let B denote the set of forms in the indicational arithmetic formed by crossing and containment of the primary values *marked* ($\overline{\quad}$) and *unmarked* ($\quad$). Thus B is a collection of trees formed out of the carrier $\{\overline{\quad}, \quad, \text{cross}, \text{containment}\}$. Now make B into a Σ -algebra thus:

1. for $\sigma \in \Sigma_0$ let $1_B = \overline{\quad}$, $0_B = \quad$;

2. for $\sigma_1 \in \Sigma_1$, and $t \in T_\Sigma$ let

$$\sigma_1(t) = \overline{t} = \text{cross}(t);$$

3. for $\sigma_2 \in \Sigma_2$, and $t, t' \in T_\Sigma$ let

$$\sigma_2(t, t') = tt' = \text{containment}(t, t').$$

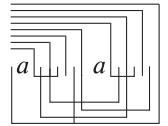
Consider now the initial Σ -algebra T_Σ . There is a unique homomorphism

$$\text{ind} : T_\Sigma \rightarrow B$$

assigning to each tree in T_Σ an interpretation as an indicational form. For instance,

$$\text{ind}[\sigma_2(\sigma_1(0), \sigma_1(1))] = \overline{\overline{\quad} \quad} = \text{ind}[\sigma_2(1, 1)].$$

Expressions in the primary indicational algebra are easily obtained by interpreting derived operators in $T_\Sigma(X_n)$, where



which satisfies the following set of equations (Kauffman, 1978b) where a is a constant value:

$$\begin{aligned} x_1 &= \overline{a x_2}, \\ x_2 &= \overline{a x_1}, \\ x_3 &= \overline{x_1 x_2 x_4}, \\ x_4 &= \overline{x_2 x_3}. \end{aligned}$$

The solution to this set of equations is an infinite tree constituted by four interdependent infinite trees; f represents the limit of the four interdependent variables. Since we may look at this limit as an unfolding tree, f can also be interpreted as an oscillation in *time* given by the sequences of expressions which the unfolding determines. In this case, the reader may want to verify that the period of f will be one-half the period of the constant a .

As a sobering note, the following is worth noting. It seems natural to consider equivalence classes in R_B , introduced by a set of initials such as occultation and transposition, which would make R_B into a Brownian algebra. This question, however, is surprisingly complicated, because we have little idea about how to work with elements in R_Σ in general, and with R_B in particular. As a result of this lack of knowledge, we cannot have an idea of, for example, how many arithmetical values are available in R_B . Is it just four, as in V ? (For further discussion on this current research, the reader should see Wright et al., 1976, and Courcelle 1978).

⁹These ideas were developed jointly with J. A. Goguen. A full account will appear elsewhere. *Eds.* See note in section 13.1.

13.10 Double Binds as Eigenbehaviors⁹

I wish to repeal once more that the main purpose of the detailed discussion of the notion of eigenbehavior over continuous algebras is to give meat and precision to the invariants that characterize autonomy. These algebraic ideas can be *directly* applied only to the extent that we have a fairly detailed idea of the kinds of operations that are appropriate for some specific systems. The more difficult it is to find precise operational descriptions for the processes present in a system recursion, the more removed that case will be from this particular representation of autonomy.

For example, if we are dealing with the recursions of numerical and logical systems, eigenbehaviors apply directly. In these cases

eigenbehavior can be interpreted as the *meaning* or semantics of a process. Consider for example the following recursive process:

$$f(x) = [\text{if } x = 0, \text{ then } 1; \text{ else } x \cdot f(x - 1)].$$

This process is a mixture of Boolean and numerical operators, and, up to the value of x , it defines the factorial function $!_x$. Clearly, however, the recursion involved in the factorial function (process) need not be limited to some fixed x , and in fact, it seems that the meaning of this function should be independent of any specific value of x . In this context, this can clearly be accomplished by taking a function defined by

$$T^i(x) = [\text{if } x \leq i, \text{ then } f(x); \text{ else } \perp].$$

and thus we have a chain

$$T^0 \subseteq T^1 \subseteq T^2 \subseteq \dots$$

with a fixed point

$$! = \bigsqcup_{k \in \omega} T^k(\perp),$$

which is the factorial function. Thus “factoriality” appears as the fixed point of this particular recursion.

To be sure, in the above example we have precise knowledge of the operations involved (Boolean and numerical). When dealing with natural systems the autonomous quality (the semantics or invariance) will be, by necessity, less precise, but also much more rich and interesting. One illustration of this situation is the possibility of gaining a clearer understanding of autonomous units realized in one class of communicational injunctions, the pathological double binds.

The term “double bind” was introduced in the behavioral sciences by Bateson (1960) to describe the mechanism underlying some forms of schizophrenia. The basic insight of this theory was that the etiology of the disease was correlated with some regular pattern of communication within the social matrix.¹⁰ Most frequently this meant communication with the person’s family. In a simplified form, this pattern of interaction can be stated as a game following a set of injunctions:

1. start playing the game,
2. produce a certain behavior B ,
3. produce the logical opposite of the behavior B (not B),
4. do not leave the game.

A typical instance is that of a child and his mother. By the child’s dependency, the first injunction is satisfied. The mother then demands, in an overt verbal form, a behavior such as “love me.”

¹⁰Eds. While Bateson’s concept of the double bind was originally received with interest by mental health practitioners and has proved helpful for clarifying relational complexities in clinical situations, empirical evidence for its influence on the etiology of schizophrenia has not been abundant. The idea is considered hard to approach empirically (see. e.g., Abeles, 1976).

Yet in a covert, body-level communication, she rejects the child's response by conveying the message "if you love me you are no good." Again the fourth injunction is fulfilled by the simple inability of the child to exit to another relation. The result is that the child may cut himself off from contact and construct a separate reality. The pathogenic double bind is completed.

Similar sorts of double bind are very common. The "be spontaneous" variety is perhaps the most familiar. Whenever a behavior is demanded as spontaneous, the very nature of the request makes the demands impossible. Confusion ensues as to what behavior to adopt.

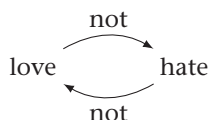
What all such situations have in common is the generation of a *punctuation* of human behavior (Wilden, 1972) in a certain context, that is, the parceling of discrete units of behavior, and the generation of injunctions by communication that operate on these behavioral states in a determined fashion. This can be represented in an operator domain:

$$\Sigma_0 = \text{behavioral states,}$$

$$\Sigma_k = \text{injunctions, } k = 1, \dots, n.$$

Whenever a social and cultural context has produced such a punctuation, "grammars" of communicative behavior will ensue. In many instances, such behavior will take the form of a finite tree, with exit points into different contexts, or to a different punctuation. Binds arise when trees become infinite, that is to say, when loops arise. Such loops, in our context, can be defined as an eigenbehavior for the equation that defines the infinite tree. The interest in these cases lies in their eigenbehaviors, since they are directly perceived or experienced as undesirable states.

In the double bind mentioned above the states are $\Sigma_0 = \{\text{love, hate}\}$, and operations $\Sigma_1 = \{\text{not}\}$, so that the loop can be represented thus:



In general, in a Bateson-like double bind (2-bind), one has a set of 2 states $\Sigma_0 = \{b_1, b_2\}$, and a tree constituted thus:

$$b_1 \rightarrow \text{not } b_1 = b_2 \rightarrow \text{not } b_2 = b_1 \rightarrow \dots$$

with the eigenbehavior

$$b^\nabla = \text{not}(\text{not}(\text{not}(\dots))) = \text{not}(b^\nabla).$$

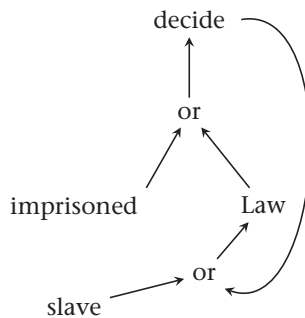
Note that the theory predicts that this eigenbehavior is different from the other states in Σ_0 , the initial social punctuation, and hence pathological by the standards of that social context. Such a new state is expressed or has a personal meaning as alienation.

In general then, we define an *n-bind* in human communication as an infinite tree of operations on a set of *n* behavioral states, whose eigenbehavior is a new state experienced as undesirable.

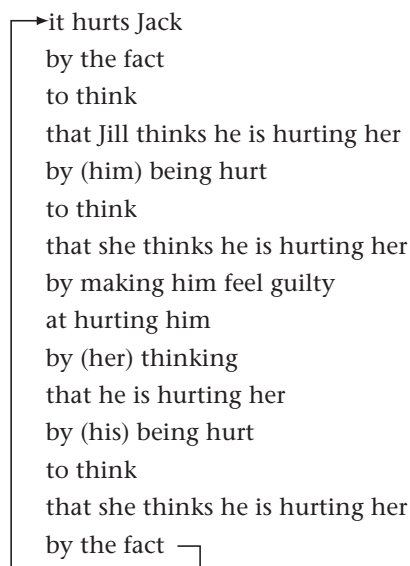
Negation on 2 states is but one way of producing binds. Consider, for example, the following situation:

1. You must make a decision.
2. Either you are imprisoned or you obey the Law.
3. If you obey the Law, either you live as a slave or you must make a choice.

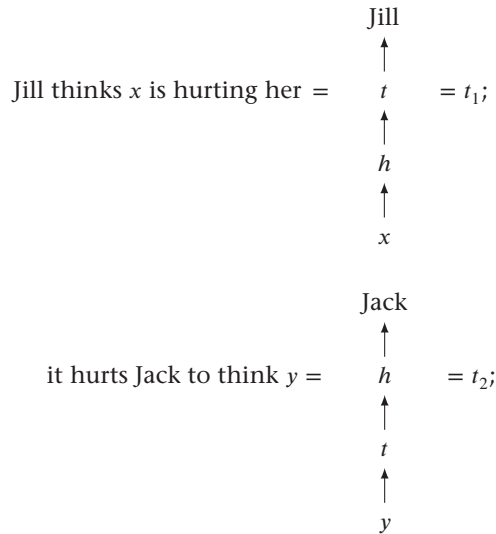
In this case we have a 3-bind based on the *or* operation, in a loop



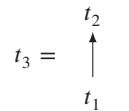
A more realistic form of equation would be required to represent one of the many eigenbehaviors described by Laing (1970):



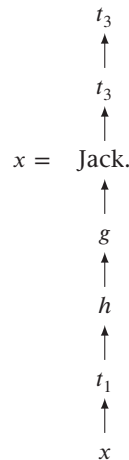
Consider here $\Sigma_0 = \{\text{Jack, Jill}\}$, $\Sigma_2 = \{\text{hurt, think, make guilty}\} = \{h, t, g\}$. Then we have the trees:



and



The double bind between these two persons arises as the eigenbehavior of the equation



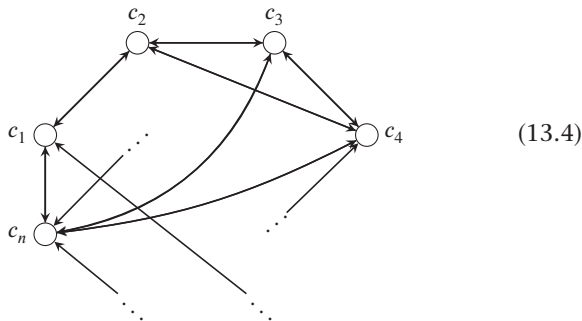
There are, of course, very many questions that we cannot pursue here. For example: What kind of experience would correspond to binds involving more than one equation? How much can the meaning of the states and injunctions (operations) change and still produce the same resulting experience—that is, is there some set of equational constraints valid in human communicational patterns?

13.11 Differentiable Dynamical Systems and Representations of Autonomy

13.11.1

Let us remember that a cellular autopoietic system can be defined as a network of chemical reactions that satisfies two conditions: (1) the chemical species produced are precisely those that constitute the chemical productions producing the chemical species (i.e., closure of the network), and (2) the chemical species produced specify a boundary, physically demarcating the network of productions as a unit in space. The notion of autopoiesis describes the necessary requirements for a class of systems to generate the living phenomenology (chapters 2–5), but it says little on how to represent this organization. Let us consider now some representations of the cellular case, where closure is accomplished through chemical transformations.

Imagine a set of chemical species, $c_1, \dots, c_n$, where there is reciprocal interaction among any subset of them:



The operations acting on the c_i 's are production and destruction, and the way to follow what happens is to observe the change in mass of every c_i . Thus let us consider the following operator domain Σ :

$$\Sigma : \begin{cases} \Sigma_0 = \{c_1, \dots, c_n\} = \{\text{concentration of } n \text{ chemical species}\} \\ \Sigma_2 = \{p, d, +\} = \{\text{production, destruction, sum of masses}\} \\ \Sigma_k = \emptyset, \quad k \neq 0, 2. \end{cases}$$

We can now consider some specific chemical network, for example,

$$\begin{aligned} x_1 &= p(c_3, x_1) + d(x_1, x_2), \\ x_2 &= p(x_1, x_2) + d(x_2, c_4), \end{aligned} \quad (13.5)$$

with the solution

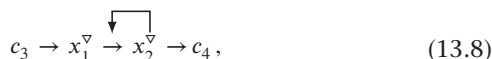
$$x^\nabla = \langle x_1^\nabla, x_2^\nabla \rangle$$

$$= \left\langle \begin{array}{c} \text{+} \\ \diagup \quad \diagdown \\ \cdots p \quad \cdots d \\ \diagdown \quad \diagup \\ c_3 \quad x_2^\nabla \end{array}, \begin{array}{c} \text{+} \\ \diagup \quad \diagdown \\ \cdots p \quad \cdots d \\ \diagdown \quad \diagup \\ x_1^\nabla \quad c_4 \end{array} \right\rangle. \quad (13.6)$$

This equation and solution can be rewritten in the more traditional format of chemical reactions:



i.e., x_2 catalyzes its own production. For the eigenbehavior we can write simply



where c_3, c_4 are given constant concentrations of species c_3, c_4 , and x_1^∇, x_2^∇ are the concentrations of reactants c_1, c_2 .

This network corresponds to one form of the well-known Lotka-Volterra reaction scheme. A simple autopoietic network can be thought of having the general equations

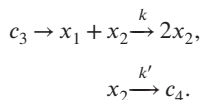
$$\begin{aligned} x_1 &= p(x_1, x_n) + d(x_1, x_2), \\ x_i &= p(x_{i-1}, x_i) + d(x_i, x_{i+1}), \quad i = 2, \dots, n, \end{aligned} \quad (13.9)$$

with an eigenbehavior of n interrelated trees $x^\nabla = \langle x_1^\nabla, \dots, x_n^\nabla \rangle$.

Let us perform the following reinterpretation of this operator domain. First, take the variable x_i as *time*-dependent, real-valued variables. Secondly, interpret the operations p, d as *differential* operators in the variables: production with a positive sign, destruction with a negative one. With this further enrichment of the operator domain Σ , we can rewrite (13.5) in its differential form

$$\begin{aligned} \frac{dx_1}{dt} &= c_4 - kx_1x_2, \\ \frac{dx_2}{dt} &= kx_1x_2 - k'x_2, \end{aligned} \quad (13.10)$$

where constant k, k' represent the rates of the two reactions; that is, we introduce the time dependency in (4),



Thus in this case, the eigensolution x^∇ can be related to the differentiable representation by linearizing (13.10) around the steady

state and getting the two interrelated solutions—or eigenvalues $\langle \bar{x}_1, \bar{x}_2 \rangle$:

$$\frac{\bar{x}_1}{\bar{x}_2} = \frac{k'}{k' \bar{x}_2}. \quad (13.11)$$

Much study has been devoted to Lotka-Volterra systems of this kind. Although simple, they exhibit a remarkable variety of damped and unstable oscillations depending on the values of the affinities (k, k') , and the perturbations the system is undergoing (fluctuations in c_3, c_4) (see e.g., Glansdorff and Prigogine, 1971).

We have taken this Lotka-Volterra example to this point because it contains, in a nutshell, an important feature and an important limitation of the present approach that should be made clear. As we mentioned in chapters 7 and 10, the classical notion of *stability* in differentiable dynamics is the only well-understood and accepted way of representing autonomous properties of systems. The work of Thom (1972), Eigen and Schuster (1978), Rössler (1978), Lewis (1977), Bernard-Weil (1975), Rosen (1970) and Goodwin (1976) provides excellent examples of the fertility of this approach for the case of molecular self-organization.

These descriptions look for relevant variables to characterize the coherent, invariant behavior of a unit. Once a set of relevant variables has been identified, a dynamical relation is adopted for the system. This framework for the system's representation has behind it considerable experience from mathematical physics. In the systemic framework, the criterion of distinction for the unit to be studied is given by the invariances resulting from the differentiable description, such as steady states, oscillations, and phase transitions.

An underlying assumption is, however, that there is a collection of interdependent variables, and it is the reciprocal interaction of these component variables that brings about the emergence of an autonomous unit. This is to say that in the instances cited before, the differentiable dynamic description becomes a *specific* case of organizational closure. By adopting the differentiable framework one can mine the richness of the experience behind it. At the same time one finds the limitations imposed by it: More often than not, autonomous systems cannot be represented with differentiable dynamics, since the relevant processes are not amenable to that treatment. This is typical for informational processes of many different kinds, where an algebraic-algorithmic description has proven more adequate.

¹¹Eds. For this reason, protocell models in synthetic biology (e.g., Mavelli and Ruiz-Mirazo, 2007) are both mathematical and computational, attempting to capture differentiable and/or stochastic dynamics in a piecewise fashion modulated by computational decision events to handle self-referential and structural conditions, such as osmotic crises, membrane breakdown, cell division, and others.

Accordingly, the fertility of the differentiable representation of autonomy and organizational closure is mostly restricted to the molecular level of self-organization.¹¹ This is beautifully seen in the work of Eigen and his notion of the hypercycle, recently examined in great detail by Eigen and Schuster (1978); see figure 4.1. The basic idea here is that a unit of survival in molecular evolution is a closed circuit of reactions with certain structural and dynamic characteristics. Eigen obtains several time invariances for this chemical closure, which serve to illuminate features of the early evolution of life. Also in a differentiable framework, Goodwin (1970; 1976) discusses pathways of metabolic transformation with a view to cellular unity.

13.11.2

There are two comments that are in order at this point. First, it must be noted that Eigen and Goodwin's work is *not* equivalent to a formalization of autopoiesis. This is so because starting from the need to use the differentiable approach, they concentrate on the network of reactions and their temporal invariances, but disregard on purpose the way in which these reactions do or do not constitute a unit in space. Their unit is characterized (is distinguishable) through the time invariances of their dynamics. That is to say, they concentrate on aspect 1 of autopoiesis, but not on aspect 2. This is just as well, for there is much to investigate in just this aspect of recursive chemical networks. It is interesting, however, that the invariances of these systems also reflect space boundaries in some cases, or at least it seems that this could be so in the case of hypercycles. This is considered more explicitly in the well-known ideas of Thom (1972), where a three-dimensional form is associated with a class of dynamics.

¹²Eds. It seems odd that the autonomy perspective would prohibit the modeling of material fluxes between cell and environment, since these are essential enabling conditions for metabolism. These exchanges do not imply an input-output epistemology by themselves. Autopoiesis is often assumed to be a case of self-organization, and Varela has suggested that underlying self-organizing behavior, we can always find a kind of systemic closure (Varela, 1983), but there are important differences of a different sort. Moreno and Mossio (2015, 114) argue that dissipative structures are relatively simple, and this is why they can arise spontaneously and self-organize when the right conditions are met. Autonomous systems, in contrast, arise from the ongoing operation of historically linked processes of production and distinction, which are too complex to arise spontaneously.

A second comment at this point is that a clear distinction should be made between models such as hypercycles, and the analysis of molecular systems through generalized thermodynamics and dissipative structures (Nicolis and Prigogine, 1977). This is so because a dissipative structure takes a complementary view of a unit, namely, it considers the unit as an open, or allopoietic, unit, characterized by the *fluxes* through its boundary. It corresponds to an input-output description in contrast with a recursion description, since the organization of the system takes fluxes explicitly into account in the definition of the environment. In this case, the units distinguished are, strictly speaking, *different* units than the ones distinguished through the closure of some interdependent variables.¹² This is, of course, not to say that there is more merit in one or the other

approach. In fact, as discussed in chapter 10, they have to be viewed as *complementary* characterizations of a system. In the case of dissipative structures, the general allonomous, input-output description is enriched with the differentiable dynamic machinery, and through dynamic variables very detailed results can be obtained. Thus for example, it is possible, in certain cases, to relate explicitly a certain state of flux to the emergence of a spatial boundary, as in the Zhabotinsky reactions.¹³

It is still a matter of investigation how well the differentiable-dynamics approach can accommodate, in a useful way, the spatial *and* the dynamic view of a system. Both on the closure side (e.g., Eigen) or on the input-output side (e.g., Prigogine), there are some striking results showing spatial patterns arising out of recursive, non-linear reaction schemata. Thus, one can only say that this form of representation has so far provided the most promising approach to coordination, autonomy, and closure at the cellular and molecular level.

But it is in going beyond the molecular level, where we can't rely on a strong physico-chemical background of knowledge, that the insufficiency of the differentiable framework appears, and thus the need to have a more explicit view of the autonomy/control complementarity, and an extension of differentiable descriptions to operational-algebraic ones. A typical borderline case, which we shall examine, is the immune system. A further case where both approaches have been tried is the nervous system. For example, Freeman (1975) prefers a differentiable view that characterizes the time invariances of cell masses, while some (e.g., Arbib, 1972) prefer a more algebraic view, emphasizing cooperation and competition of processes.

We cannot give here an account of how all of these results hold together. In this book I am concerned with emphasizing *one* aspect of systems that has been *neglected*: autonomy. I have offered a characterization of what this means in general, and have provided a representation for some key notions. Thus, for example, autopoiesis, as a case of closure, is not exhausted either in the (possible) algebraic eigenbehavior representation, or in the differentiable-dynamic one. The clear distinction between a class of organizations and its representation must be maintained. Going beyond the differentiable framework of representation was necessary in the past for the allonomy, control viewpoint. Likewise, for the characterization of autonomy, it seems necessary to go beyond the differentiable framework, while keeping its unique insights for some cases (such

¹³ *Eds.* For an overview of Belousov-Zhabotinsky chemical reactions, see Cassani et al. (2021).

molecular organization). The algebraic framework presented above is a step in that direction, though nothing more than a step.

13.11.3

Is there anything useful to say about the relationship between the present algebraic approach to represent autonomy and the classical differentiable one? In some sense one can see that the latter is a specific case of the former, since we deal with some specific collection of operations (differentiation, addition, and multiplication of numerical variables, and so on), and eigenbehavior reduces to the classical notion of stability. This only says to me that the general framework presented here is capable of including this classical picture, and thus lends some credibility to its more encompassing character. However, there are very many detailed questions about the transition from algebraic to differentiable—from eigenbehavior to stability—that are left entirely untouched here, and where more work is needed. Clearly, both approaches cover somewhat *non-overlapping* aspects of systemic descriptions. Thus, it is necessary to have a way of dealing with plasticity and adaptation. Natural systems are under a constant barrage of perturbations, and they will undergo changes in their structure and eigenbehavior as a consequence of them. There is no obvious way of representing this fundamental time-dependent feature of system-environment interactions in the present algebraic framework. In contrast, the question of plasticity is a most natural one in differentiable dynamics because of the topological properties underlying this form of representation; hence the notions of homeorhesis and structural stability in all their varieties. To what extent can the experience gained in the differential approach be generalized? How can notions such as self-organization and multilevel coordination be made more explicit in this context? Is category theory a more adequate language to ask these questions? These and many more are open questions.¹⁴

¹⁴ *Eds.* An attempt to address some of these questions from a mathematical framework, looking at systems that must evolve by adapting to viability constraints, is found in the *viability theory* of Jean-Pierre Aubin (1991).

I offer table 13.1 to summarize some of the current tools available to present autonomy. In this table I have put the two sides of the autonomy/ control complementarity, into correspondence with the two sides of the closure/interactions complementarity. Thus we compare the point of view for characterization of a system with the point of view for its representation. The terms included in the table are simple evocations of notions that are currently more or less well developed. Clearly the lower half of the table is far better developed with regard to mathematical representation. In this book I am concerned with the upper half.

Table 13.1

	Representation	
	Closure	Interaction
Characterization		
Autonomy	identity connectivity indefinite recursion eigenbehavior stability	perturbations- compensations cognitive domain resilience ontogenesis
Control	coordination of parts hierarchical levels finite recursion signal flow state transitions	black box dissipative structures input-output

13.11.4

As the reader can well see by now, there is much to say and do in relation to the representations of autonomy. Let me risk being obnoxious in repeating that what I have done here is simply to stake out descriptions which embody, in a mathematical framework, some key ideas that are pursued in this book: closure, autonomy, distinction, recursion. In no way should these formalisms be confused or identified with the intuitions behind them; rather they should be considered only as a vehicle to sharpen precision and reveal inadequacies.

We shall not say anything else about formal representations. Let them stand in their open-ended, incomplete state. In the next and final part of this book, I turn to an altogether different aspect of autonomy, namely, that of the knowledge processes associated with the establishment of a unity. In table 13.1 it is the upper right-hand corner. In this corner, we look at a unit as autonomous, but in its coupling and interactions with an environment. In this larger view of the autonomous unit, the organizational closure results in a classification of environmental perturbations, and hence in the establishment of a cognitive domain. We now turn to analyze this in detail.

Sources

(Goguen et al., 1977), (Varela and Goguen, 1978).



Cognitive Processes

[The] minimal characteristics of mind are generated whenever and wherever the appropriate circuit structure of causal loops occurs. Mind is a necessary, an inevitable function of the appropriate complexity, wherever that complexity exists.

—G. Bateson, *Steps to an Ecology of Mind* (1972, 490)

That the world becomes picture is one and the same process whereby, in the midst of beings, man becomes subject.

—M. Heidegger, *The Age of the World Picture* (2002, 69)

Introduction to Part III

Part III elaborates the cognitive implications of autonomy. “Cognitive,” in this context, refers to a description of the domain of interactions that an autonomous system can participate in without losing its autonomy. In some cases, it may seem that the application of this term is at best a metaphor. But this is not how Varela intends it, and his usage follows the logic of his operational definition of autonomy. The chapters in this part offer a more concrete exploration of the themes in parts I and II of the book. It is also representative of three of Varela’s major research concerns: immune networks, neurodynamics and enaction, and a naturalized, nondualistic epistemology that is capable of including our participation as living, embodied, and experiencing beings.

Chapters 14 and 15 demonstrate the power and versatility of the closure thesis, showing us examples of autonomous systems that are themselves embedded and participate in the autopoiesis of the organism and the immune and nervous systems, respectively.

In chapter 14, Varela introduces the concept of immune networks, originally proposed by Niels Jerne as an alternative approach to the function and organization of the immune system in response to the paradoxes and unanswered questions of the otherwise well-established clonal selection theory. The key conceptual difference between these two visions consists in a different conception of the self-nonself discrimination of immune reactions. In clonal selection theory, those cell clones that produce antibodies that match the body’s own antigens are eliminated during a critical period during ontogenesis. The immune self, in this sense, is defined by this negative initial process of learned tolerance. Jerne’s immune network aimed to explain discrimination as a distributed property of an ongoing dynamic interaction of the immune system with itself, as well as the rest of the body and external antigens. Because each antibody is also an antigen, the immune system is self-reacting. Lymphocytes form a network that keeps antibody production at a relatively stable level (or “eigenbehavior,” as Jerne himself called these states). Unlike the definition of the self as that which the system is tolerant of, the network theory provides a positive definition of the self as a network of self-assertion. In collaboration with immunologist Nelson Vaz, Varela elaborated on this idea as a relational frame of reference from which external stimuli could be seen as immunologically relevant or not. To know itself is a precondition for the organism to recognize something as foreign.

Chapter 14 is a first expression of the idea of the closure of the immune system and puts perhaps too much emphasis on self-assertion, bordering on a kind of internalism. This leads to the claim, later revised, that a stimulation that does not engage with the operation of the closed network is simply *nonsensical* or *noise*. Subsequent work sought to correct for this bias. In collaboration with immunologist António Coutinho, geneticist/philosopher John Stewart, and others, Varela was involved in a series of theoretical, modeling, and empirical developments of the idea of immune networks during the 1980s and early 1990s (see, e.g., Coutinho, 2003). These developments included empirically driven clarifications, extensions, and amendments of the original idea, though with no radical break with the central notion of an active autonomous core of immune self-interaction.

Chapter 15 defends the view that the nervous system is also organizationally closed, and hence that it is an autonomous system according to the closure thesis. Varela explores the neuroanatomy and connectivity of the nervous system in order to present the thesis that it is organized as a closed system of ongoing production of neural activity. The key contrast is with the hypothesis (which has been adopted as dogma) that the nervous system receives and processes environmental information via internal representations that serve to decide the organism's next move. Adopting the perspective of autonomy, instead, avoids this mixing of symbolic and operational discourse and offers a picture of a self-asserting plastic network that is perturbed in its dynamics by environmental processes. As a closed network, the nervous system has no input or output. It cannot discriminate between endogenous or externally induced changes in neurodynamics. This closure, however, is partly achieved *through* the environment because of the environmentally mediated correlations between motor and sensory activity, thus avoiding a kind of internalism (which some statements in the chapter may suggest). Varela expands on several important implications of this central idea, showing it at play in examples such as locomotion and perceptual size constancy, with a nod toward Jean Piaget's equilibration framework as suggesting a path for explaining more complex sensorimotor and cognitive performance in a nonrepresentational manner. This idea was later picked up in enactive research (see Varela et al., 1991; Di Paolo et al., 2017).

Varela's involvement in neurobiology was a constant in his research career from his doctoral studies until the end of his life. He was successful in translating several of his theoretical ideas

into empirical research. The key connecting theme of his work in neuroscience emerges from the implications for action, perception, cognition, and experience of the operational closure of the nervous system. This is the idea that recursive neurodynamic patterns modulate sensorimotor engagements, and these in turn perturb autonomous neural activity, thus breaking with notions of perception as input-driven and the brain as processing information gathered by the senses. This break is central to the enactive approach.

Varela became particularly interested in looking at temporal dynamics, both in single cells and neural populations. About the time of publication of this book, he collaborated in studies on perceptual framing and alpha rhythms in cortical activity (Varela et al., 1981) and later, with Wolf Singer, on neurodynamics during binocular rivalry (Varela and Singer, 1987). The implications of these findings for perceptual awareness are interpreted in enactive terms in Varela et al. (1991).

Toward the end of the 1990s, new recording techniques and novel methods for nonlinear data analysis enabled Varela and his team to perform more sophisticated studies of the two-way relation between large-scale neurodynamics and perceptual experience, including the mutual modulation between perceptual discrimination and epileptic activity (e.g., Le Van Quyen et al., 1997, 1998; Martinerie et al., 1998). One key study demonstrated what Varela called the “emergence of a cognitive moment” and how it depends on the coordination of many brain regions, rather than specific locations (Rodriguez et al., 1999). The mechanisms for this kind of large-scale integration rely on patterns of synchronization and desynchronization, coherence and decoherence, which Varela called the “neuronal glue” (Varela, 1995, p. 82). These global patterns correlate with significant moments in the flow of an act, such as the moment of recognition of a face in a projected image or the moment of initiating the motor act to signal this recognition.

The notion of closure is here interpreted in terms of the mutual modulation between endogenous brain activity (states of preparation, expectation, attention, and emotional tone) and sensorimotor processes. This integration is constantly in the making, to the point that it is hard to distinguish between so-called “bottom-up” and “top-down” neural processes (Varela et al., 2001). Such observations put more flesh on the claim in chapter 15 that, as an autonomous system, the nervous system cannot tell whether a change in activity is internally or externally produced, not because

there is no difference between these cases, but because the brain is always integrating them into patterns of coherence and decoherence. Thus, there is the difficulty (and some would say impossibility) of assuming that neural activity has any representational function.

Varela's interests in the fine temporal structure of action and perception were also manifest in his other writings and a key theme of his more phenomenological investigations (Varela, 1991a, 1999b; Varela and Depraz, 2005).

Viewing the immune and nervous systems as autonomous highlights some of the central messages of this book and Varela's later work—in particular, the inescapable connection between the emergence of an identity and that of a domain of significance, or a cognitive domain. Chapter 16 picks up this thread and explores its broader implications, some of which already have appeared elsewhere in the book. This chapter looks at a broader horizon, imagining new niches that the idea of autonomy may inhabit and, in some cases, prefiguring those areas in which this idea has indeed flourished in enactive and embodied cognitive science, philosophy, and ethics.

To observe an autonomous system *as* autonomous is to become aware that the system has a sort of *perspective* from which environmental processes affect its ongoing constitution. Varela calls this realm the “cognitive domain” of the system. At this stage, there is no suggestion that an autonomous system must be understood as an intentional agent, the way that we typically think of living organisms, so the term “cognitive” is used just in this narrow, technical sense. Later work in the enactive approach (Weber and Varela, 2002; Di Paolo, 2005; Thompson, 2007; Thompson, 2022, and others) provides specific arguments for what conditions must occur in order for autonomous systems to be considered as agents in the full sense of the word.

The perspective of the autonomous system, however, is not simply a receptacle to register data about the external environment. Throughout the book, Varela is at pains to show that this idea cannot make operational sense. This leads to his tireless critique of the notion of information (in its semantic sense) as a sure way to confuse symbolic and operational discourse and his preference for a hyphenated version, *in-forma-tion*, in an attempt to capture a codependent sense of coherence instead of correspondence.

In distinguishing engagements with the environment based on their effects on autonomy, a system's perspective is in fact the result of a historically changing relation of codefinition between

its own structure and that of its surroundings. Implicit in the idea of autonomy is a critique not only of representationalism, but also of objectivism. This critique might resemble some forms of constructivism, and scientists such as Piaget and Heinz von Foerster have deeply influenced both Varela and constructivist thinkers. But Varela's epistemology is not exactly constructivist, and the differences become clearer in his later work.

The urgent issue under the surface of all this discussion is that as embodied, living, and historically situated observers, we too are an entanglement of autonomies. What is true about autonomous systems in general is true about us as well. What does this mean for human forms of knowing? Varela explores this question in broad strokes (and admits these explorations are incomplete) by looking especially at the social dimensions of autonomy. Not only do we grow and change in the context of a community that shapes the way that we realize our identity, our ongoing becoming, and hence the way that we can understand the world, but also the social domain is a space where new autonomous processes may emerge. Prefiguring work on the interactive turn in social cognition and participatory sense-making (De Jaegher and Di Paolo, 2007; De Jaegher et al., 2010), as well as broader calls for an interactional "figure-ground reversal" in cognitive science (Dingemanse et al., 2023), Varela takes a proposal by Pask (1975, 1976), considers conversations as coherent totalities, and suggests that it is profitable to understand them as autonomous processes. This shows that Varela, far from being against investigating social processes using ideas of autonomy, was only skeptical of their careless transplantation from one domain to another (see, e.g., Varela, 1981, 1989).

The autonomy perspective invites a reconsideration of the traditional notion of the subject, a theme that becomes central in later enactive thinking. In the context of these reflections, Varela suggests that we must approach the question of knowledge with a naive mind once again, highlighting the operations of distinctions that we enact in our experience as we engage with the world. We do not have access to our cognitive domain, in the sense that we can access something external to us, because we are always *within* it. This is the fundamental circularity that animates the enactive perspective (Varela et al., 1991). This does not condemn our knowledge to some kind of internalism. The cognitive domain is one of actual and potential *interactions*, not of representations in the head. Knowing then becomes an ever-turning circling among sciences (and other epistemic practices) and human experience (understood in terms

of both lived experience and practical activity). Varela calls this an *epistemology of participation*, which again resonates with recent enactive work on engaged/engaging epistemology (De Jaegher, 2021).

Varela closes these reflections with an ethical note suggesting that to be participant knowers involves us in the implications of our acts of knowing. Knowers and the known become inter-dependent. In contrast with the epistemology of representation and control, a participatory epistemology makes it impossible to disregard our responsibility and our role in the world that we have come to know.

14

The Immune Network: Self and Nonsense in the Molecular Domain

14.1 Organizational Closure and Structural Change

14.1.1

The intention of this last part of the book is to show how the mechanisms of identity of an autonomous system correlate with the establishment of *cognitive interactions* with its environment. In other words, I shall argue that mechanisms of knowledge and mechanisms of identity are two sides of the same systemic coin.¹

Instead of embarking at this point in a discussion of what this means (more explicitly than in chapters 2–6), I shall adopt the strategy of discussing two cases in detail, the immune and the nervous systems. The central idea is to look at the organizational closure of these two systems, pointing to the invariances that permit their distinction and characterization as units. The discussion, however, relates the closure of these two systems to a complementary feature: their *structural plasticity*—that is to say, how the specific components that realize their closure can be modified and changed under perturbations from the environment.

The very rich capacity for structural plasticity of these two systems is at the core of their cognitive performance, and is why they are taken here as examples. This is not to say that the same kind of events do not take place in other autonomous systems. The lessons from the immune and nervous network are generalized in chapter 16, where the argument is unfolded more completely.

The presentations in this part rely on the two key notions of structural coupling and cognitive domain.² Also, the exposition is based on empirical results about the structure of the immune and nervous systems. The reader unfamiliar with this biological background will have to bear with me through a number of details which, at this

¹*Eds.* This is a continuing theme in Varela's thought. The deep entwinement among identity and sense-making, interaction, and experience is a hallmark of enactive thinking (Varela et al., 1991). Varela revisits this topic, with explicit reference to the examples in this book, in Varela (1991b, 1997) and Weber and Varela (2002).

²*Eds.* Throughout this discussion, it is important to keep in mind that the term “cognitive” should be understood with reference to the interactions between the autonomous system in question (e.g., the immune system) and its own environment (e.g., the molecular ecology of the organism). It should not be confused with the cognitive domain of the organism as a whole.

³*Eds.* Varela's work on the immune system presented here was developed in collaboration with immunologist Nelson Vaz. See Vaz (2011). Varela continued this work throughout the 1980s and 1990s in collaboration with immunologist António Coutinho (e.g., Varela and Coutinho, 1989, 1991) and others (e.g., Calenbuhr et al., 1995; Detours et al., 1994; Stewart and Varela, 1989, 1990, 1991; Varela and Stewart, 1990). During this period, a combination of simulation models and empirical studies led to changes and extensions (but not radical rejection) of the views presented in this chapter. For general accounts, see Varela et al. (1993) and Coutinho (2003).

⁴*Eds.* For a historical overview of Niels Jerne's work and the genesis of the idiotypic network theory, see Civello (2013).

stage, are as necessary for the general argument as the mathematical proofs of the previous part.

14.1.2

In the remainder of this chapter, I present a conceptual framework to accommodate important recent developments in cellular immunology and immunogenetics, which have rapidly rendered obsolete a number of fundamental notions about the nature of immunological events (Vaz and Varela, 1978).³ This task is, at the same time, very difficult and very necessary, and can only be partially attempted here. A minimal acquaintance with the basic constituents of the immune system is assumed (see, e.g., Eisen, 1974).

We expand on the ideas initially formulated by Niels Jerne (1974b; 1974a), who first stressed the need to view the immune system as a network of interconnected events, i.e., in its closure, rather than as the activity of isolated clones of lymphocytes.⁴ This view is gaining increasing acceptance among immunologists (Raff, 1977).

The main point of the presentation is to see how the very nature of the circular interconnectedness of the lymphoid system provides a source of stability and, at every moment, its own identity.

At the outset, it is important to make explicit that I do not intend to postulate detailed cellular or molecular mechanisms underlying immune events. I will, however, propose in general terms how the complexity of the lymphoid system may be organized in a closed network of interactions, which is the essence of the immunological self.

14.2 Self-versus-Non-Self Discrimination

14.2.1

The inability of the organism to undertake immune responses against its own components, which nevertheless may function as immunogens to other organisms, was acknowledged by Paul Ehrlich in 1900. Ehrlich coined the term *horror autotoxicus* to designate the indifference of the organism to the immunogenicity of its own substances, but failed to propose an explanation for such phenomena.⁵ Burnet (1959), however, was the first to insist that this immunological *tolerance* to autocomponents demanded a scientific explanation. He then postulated the existence of a mechanism of "self-discrimination" through which the organism "learns" to discriminate between its own structure ("self") and the structure of foreign (nonself) materials.

⁵*Eds.* See Ehrlich (1900). For an account of Paul Ehrlich's pioneering selection theory of antibody formation, see, for instance, Silverstein (1999).

Burnet's idea had the great merit of making immunologists aware of the need to explain the mechanism of antibody formation in accordance with the ideas on protein synthesis. Antibody molecules are synthesized like other proteins, using messenger RNA, and not antigen, as templates. Burnet postulated that the cells able to make antibodies existed before any contact with the antigen, and that antigen molecules simply "selected" the cells with antibodies that happened to fit their antigenic determinants, among a large population of different antibody types. He further postulated that each cell clone was able to make only one (or very few) types of antibodies, and thus the theory was called the clonal selection theory of antibody formation. Both the clonal aspect (one cell, one antibody) and the selective aspect (preexistence of the genetic determination) of Burnet's ideas have been extensively confirmed experimentally, and will not be discussed further.⁶

A central problem in the immune system, however, is to explain the origin of the immense variety of lymphocyte clones, which endow every vertebrate organism with an apparently unlimited versatility in the performance of specific immune responses. In other words, the central problem is to understand how such a diversity of lymphocytes is generated.

At this point, the attitude held about the concept of self-versus-nonsel self discrimination becomes critically important. Whereas it is perfectly possible to imagine that the organism directly inherits from its ancestors the genes coding for the proteins that function as specific antigen receptors (such as antibodies and other membrane proteins), it is impossible to imagine that the organism inherits a specific refractoriness to response against its own constituents. Every vertebrate organism is perfectly able to recognize as foreign antigens substances present on the cells and tissues of other organisms of the same species, including its own parents. How could the organism lack exactly those genes coding for receptors against its own components if the composition of "self" was unpredicted before fertilization? The inescapable alternative to this paradox is that genes coding for receptors against self components are inherited among all the other genes, and the process of their neutralization has to be resolved somatically, during the ontogenesis of each organism.

The concept of self-versus-nonsel self discrimination was formulated on the assumption that all responses of lymphocytes against autologous components are deleterious, and that in a healthy organism there should be no possibility of lymphocytes reacting to self antigens. Since the clonal selection theory postulated (correctly) that

⁶*Eds.* For reflections on the clonal selection theory, see Tauber (1994), Podolsky and Tauber (1997), and Cohn et al. (2007).

each cell clone was able to form only one type (or a few types) of antibody, the question was perceived as one of eliminating the (forbidden) self-reactive clones. Burnet postulated that these clones were eliminated by simple contact with the specific antigen during critical periods of the ontogenesis, and therefore raised the theoretical possibility of “fooling” the immature lymphoid system, making it recognize as self materials that were actually nonself, such as allogeneic cells introduced into newborn animals. Owen had previously observed that dizygotic twins in cattle, which may share a common placental circulation, are living immunological “chimeras,” in the sense that they possess in the blood an otherwise incompatible mixture of cells from both individuals.⁷ Burnet proposed ways of producing these immunological tolerance experimentally. A few years later, Medawar and co-workers and Hašek and co-workers demonstrated that mice and chickens, respectively, could be made tolerant to tissue transplants of otherwise incompatible donors if exposed to a sufficient number of cells of the donor during the perinatal period.⁸ When this happened, the panorama of immunology was drastically changed: Now, the contact of the organism with antigens could not only make it specifically immune to the antigen, but also make it specifically tolerant to the antigen.

⁷Eds. See Owen (1945) and Martin (2015).

⁸Eds. See Medawar (1956) and Hašek et al. (1961).

Burnet had also postulated that antigenic materials that existed in secluded regions of the organism—such as the lens of the eye—would have escaped recognition as self and should be fully antigenic to the organism. Indeed, reactions against these materials have been found to occur in human autoimmune diseases. Moreover, a vast array of autoimmune pathological reactions were subsequently induced in experimental animals. These findings lent further support to the concept of self-versus-nonself discrimination.

With the support of these experimental findings, the existence of some mechanism leading to the destruction of self-reactive clones was imperative. Therefore, virtually all theories formulated to explain the diversity of immune responses included a mechanism for the elimination of the responsiveness to self.

There are now, however, serious difficulties in reconciling a large number of experimental findings with these postulates. To begin with, immunological tolerance may be induced, by a variety of methods, in adult, fully immunocompetent organisms (Katz and Benacerraf, 1974). Some autoantigens that are targets of autoimmune diseases, such as thyroglobulin, have been found in the serum of both normal human newborns and adults, and therefore are not, as formerly believed, secluded from contact with the lymphoid

system. Thus, immune responses against autologous components that are essential parts of the lymphoid system itself, such as antibody molecules themselves, are constantly going on, and probably constitute an essential aspect of the operation of the lymphoid system (see below).

14.2.2

There is, however, a more fundamental weakness in the theories that propose to explain the nature of immune responsiveness by considering only the origin and fate of individual clones of lymphocytes. These theories neglect the need to harmonize the activities of these clones with *one another* in the organism as a whole. This neglect of a holistic view, this desire for a simple causality, fails to incorporate two of the most important developments in immunology: the genetic control of immune events, and the importance of cellular interactions in regulating the quality and magnitude of immune events.

1. *Genetic Control of Immune Events.* There is now clear evidence that genetic controls place clear-cut restrictions on the immunological versatility of individual organisms. Marked differences in the immune responsiveness to a variety of different antigens are found among inbred strains of laboratory animals. By appropriate crosses between “low-responder” and “high-responder” populations, the number of genes controlling specific immune responsiveness—or Ir-genes—could be estimated; in many instances, single dominant Ir-genes were found to control the elicitation of specific immune responses (McDevitt and Landy, 1972).

The products of Ir-genes are not antibodies: Ir-genes and the genes coding for the heavy and light chains of immunoglobulins exist in different chromosomes. Furthermore, although the Ir-gene product controls the recognition of immunogenicity of the antigen molecules, it exerts no control over the specificity of the antibodies formed, once the molecule is recognized as immunogenic; and, in what appeared to be a most curious coincidence, Ir-genes were found to map precisely onto the particular chromosomal region that codes the most important transplantation histocompatibility antigens of the species—a region known as the major histocompatibility complex (MHC) (Bodmer, 1972; Snell et al., 1976). There is much evidence in the recent literature suggesting that the products of Ir-genes may be the so-called Ia-antigens (or I-region-associated antigens) (Katz and Benacerraf,

⁹Thus, compatibility at the MHC is necessary for the cooperative interaction of T-cells with B-cells or macrophages, and killer T-cell activity is much more efficient against chemical or viral-induced modifications of the membrane of syngeneic (MHC-compatible) than allogeneic (MHC-incompatible) cells (Gershon, 1974; Paul and Benacerraf, 1977). There is significant evidence for a similar situation in immune responses against other membrane-bound proteins of either internal or external origin. The requirement for MHC compatibility in cell cooperation events, however, is not an absolute one: T-cells collected from organisms made tolerant to the MHC-antigens of the cooperating cells, or selectively depleted *in vitro* of "suppressor" T-cells, cooperate well with the allogeneic partner, and cooperation between MHC-incompatible cells occurs perfectly well in allophenic (or tetraparental) mice that are prepared by the fusion of eight-cell-stage mouse embryos (McDevitt et al., 1976). Thus, the ability to cooperate is crucially dependent on the animal's past history.

¹⁰Thus, unlike B-cells, specific T-cells cannot be removed from a cell suspension by passage through columns containing the antigen bound to an insoluble support; nor can they be destroyed by incubation with highly radioactive antigen in so-called "antigen-suicide" experiments, unless the antigen is present on the surface of B-cells.

1975), which may function as strong transplantation antigens. In other words, genes controlling the immune responsiveness of the individual were found to be tightly associated with some of the most important genes characterizing its individuality in terms of its antigenicity for other individuals.

2. *Cellular Interactions in Immune Events.* The lymphocytes that give rise to antibody-secreting plasma cells (B-cells) are not the cells which evolve in the thymus (T-cells). The properties of T-cells actually proved to be quite different from those of B-cells. Reciprocal interactions occur between B- and T-cells, which may have net "helper" (stimulatory) or "suppressor" (inhibitory) effects, and similar interactions occur between lymphocytes and macrophages.

Although a degree of immunological specificity as delicate as that involving antibodies governs the intervention of T-cells in immune events, the receptors used by T-cells are not antibodies and do not bind to the same antigenic determinants as antibodies. The "determinant" to which T-cells seem to respond somehow includes molecules of the membranes of other cells, such as B-cells or macrophages, which are coded by genes of the MHC.⁹

A further point of interest is that since T-cells contain few, if any, antibody molecules on their membranes, they are able to bind only trivially small amounts of antigen as compared to B-cells.¹⁰ Thus, when foreign antigen molecules penetrate the body of previously immunized organisms, they are bonded through antigen-antibody reactions to B-cells and macrophages, but the activity of these cells is regulated by T-cells.

14.2.3

In order to make a proper description of the operations of the lymphoid system, it will be necessary to change some of the fundamental attitudes derived from the clonal selection theory, which have permeated the whole field of immunology. These changes in attitude will stem *not* from the description of precise cellular or molecular mechanisms underlying immune events, but rather from a change in our *interpretation* of the meaning of immune responsiveness, involving a change in our referential standards: from an antigen-centered immunology to an organism-centered immunology. In addition to the study of the origins, functions, and fates of individual clones of lymphocytes, it is necessary to understand how

the activities of these clones may be harmonized with those of *other* clones in the organism as a whole. We must replace the notion of the lymphoid system as a collection of unconnected lymphocyte clones carrying receptors directed outward (toward unpredictable encounters with foreign materials), with the notion of a network of interacting lymphocytes, where the receptors are directed *inward*, making the activities of the whole lymphoid system curl and close onto itself.

The acknowledgment that the immune responses we study experimentally as isolated events are actually loops in a complex web of interdependent cellular events is important, but it adds little to our understanding of how the system operates. Actually, such added complexity may result initially in less, rather than more, understanding. There is a more fundamental point to be perceived, however, in the operation of the lymphoid network: Its cellular interactions are not only complex, they are *self-determined*. We don't "turn on" the lymphoid system of an animal when we expose it to an antigen; the lymphocytes are already operating before we intervene. We may prefer to think that we turn on specific clones of lymphocytes—and this, to an extent, actually happens. However, the activity of all lymphocyte clones in the organism is harmonized and interconnected in the organism as a whole, and literally the whole lymphoid system participates in all immune responses.

Of course, experimental immunizations with well-defined antigens are very important in providing a frame of reference in which to analyze immune events. However, most of the time, in order to ascribe meaning to our measurements, we adopt the input-output perspective of the engineer. We consider our "normal" animals as *tabula rasa*, as an empty arena where nothing "specific" is happening, until we drop in our antigens; that is the sign for the show to begin. Our antigens are as pure and structurally defined as possible (inputs), and all efforts are made to evoke responses as strong as possible (outputs), although these responses may reach levels that are never met under natural circumstances. Most of the time, we are concerned with the mechanisms controlling the operation of specific clones of lymphocytes. Compelled by the evidence that the operation of these clones is under the influence of complicated interactions among different cell types, immunology is now trying to elucidate the molecular mechanisms of these interactions, and to understand how they are genetically controlled.

The three most general conclusions deriving from these efforts have been:

1. that immune events occur under stringent genetic control;
2. that immune events have a cascading quality, in the sense that the activation of a restricted number of cells has the potential of sequentially activating an ever growing population of other cells (this is typically illustrated in the idiotypic network proposed by Jerne, but may be expanded in other ways);
3. that previously unsuspected “suppressive” events operate in immune responses; many of these “suppressive” activities cannot be measured or well defined by the standard immunological methods of assaying cellular activity, such as antibody formation and the quantitation of cell proliferation.

Accordingly, it is necessary to sketch a new perspective in which

1. the genetic characteristics of the organism are related in a coherent way to the performance of its immunological activities;
2. the cascading properties of the immune system are understood as essential expressions of its mode of operation and its search of new states of self-determined behavior;
3. the “suppressive” influences are integrated as physiological components of the system.

To develop this new conceptual framework, we must understand initially that we may only talk about immune events in a *relational* or referential way. The antigenic properties of a molecule, its immunogenicity or its tolerogenicity, are *not* inherent properties of the molecule such as its size or shape; they can only be defined in reference to a particular organism, in comparisons that take into account not only the genetic background of the organism but also its previous immunological history. It is the behavior of the organism that declares whether the contact with the molecule was immunologically relevant or not.

Since there are certain genes in the organism that place clearcut limitations in its immunological versatility and define the boundaries of the domain of interactions of its lymphoid system, it is obvious that once the nature of the products coded for by these genes is elucidated and their function in immune responses is understood, the immunological activities of any particular organism will be defined *in reference to these substances*. In other words, these substances constitute the *immunological structure* of the organism, its immunological “self.” The presence of foreign materials in the

organism can only acquire immunological relevance by interaction with these components of this immunological “self.” All immune events will be understood as self-referential, performed in reference to the immune structure.

This is a positive definition of “self,” which is in direct opposition to theories that define “self” in a negative way, because the organism is visualized as not responsive to self antigens. However, if the organism doesn’t know itself, how can it detect the presence of something foreign? If the organism doesn’t use its own structure as a reference for discriminations, what other points of reference are available to it? To posit that a certain cohesive organization of the lymphoid system, in which components are always interacting with each other, not only defines the immunological “self” structure but also determines *in the same* process the spectrum of stimuli that the organism may perceive as immunologically relevant, is far simpler, and more to the point, than to propose that the organism must “learn” to discriminate between self and nonself. Thus, responses to self are always going on, and are the only way the self may recognize nonself. In the last analysis, all a foreign molecule can do after it penetrates the organism is to change the ways in which the cells of the organism interact with each other. What is not previously specified in the repertoire of interactions among the components of the lymphoid system simply does not enter its realm of operation, and is merely nonsensical to the system. Thus, the central distinction in the operation of the lymphoid system is not between “self” and “nonself,” but rather between what can and what cannot interact with the immunological structure: a distinction between identity and “nonsense,” or immunological “noise.”

14.3 The Lymphoid Network

14.3.1

The fundamental idea proposed here is that *the immune system may be viewed as an autonomous unit*—that is, as a network of cellular interactions that at each moment determines its own identity. It is basically necessary to understand how this autonomy is achieved and maintained. This idea hinges decisively, according to the closure thesis, on an adequate consideration of the interactions between components of the lymphoid system beyond the level of clonal specificity. Such interactions we refer to as *connectivity*, giving rise to a lymphoid network comprising the totality of the lymphoid tissue.

14.3.2

It is important to realize that the connectivity of the immune system is rooted in one of its most pervasive properties: the degeneracy of immunological specificity.

The elicitation of specific immune responses against an enormous variety of different antigens, including synthetic molecules that never existed in nature before, has always been interpreted as a manifestation of an almost endless versatility of the lymphoid system, which any theory of immune responsiveness must explain. More recently, it has become clear that the versatility of any individual organism, although wide, is not endless, but rather is very precisely defined by its genetic background.

In addition to these inherited limitations, it is also important to understand that the specificity of immune events is actually rather degenerate. There are no one-to-one relationships in immune events. It is extremely rare to obtain the formation of homogenous types of antibodies, even against a single defined antigenic determinant and in inbred organisms. More often, the responses consist in the production of a highly heterogeneous population of antibody molecules, which is ever changing during the development (or maturation) of the immune response, and which constantly changes its average binding affinity with the antigen (Siskind and Benacerraf, 1969).¹¹ This degeneracy is not restricted to reactions mediated by antibodies.¹²

Furthermore, it is important to bear in mind that most forms of immunological stimulation met by the organism under natural circumstances consist of extremely complex materials, many of which are part of living microbial organisms. An additional point of interest is that once organisms have responded to a particular form of antigenic stimulation, this response will bias the type of responses they will make to cross-reactive antigens—a phenomenon known as the “original antigenic sin.”

Thus, the specificity of immune events can only be understood by taking into account the great heterogeneity of the responding lymphocyte population. Acknowledging that the specificity of immunological reactions is highly degenerate is not only a more faithful representation of immunologic phenomenology, but is also essential to understanding the sources of the structural plasticity of the lymphoid system in undertaking an immense variety of “specific” reactions.

It is important, therefore, not to confuse the specificity of immune responses (which is highly degenerate) with the (very

¹¹These changes do not always occur in the same way: when two different individuals from the same inbred population are tested with the same antigen, they produce different populations of antibodies, which may share some antibody types but differ in many others. Careful immunochemical analysis shows that the same antigen-binding site on an antibody molecule may react with very different antigenic determinants (Richards et al., 1975).

¹²For example, non-cross-reacting forms of egg-white lysozyme may induce cross-tolerance, and urea-denatured ovalbumin, which fails to react with antiovalbumin antibodies, is still able to interact with “ovalbumin-specific” T-cells (Ishizaka, 1976). On the other hand, it is quite clear that T-cells are exquisitely able to discriminate between different types of antigen molecules, which cannot be discriminated by antibodies. Thus, it is impossible to say that T-cells only recognize coarse differences between antigen molecules: They simply seem to recognize details of the antigen molecules that are not the antigenic determinants to which antibodies are directed.

highly) specific ways in which immune events may be used in the experimental or medical analysis of biological phenomena. The products of immune reactions, of course, are invaluable reagents for a number of highly specific procedures.

14.3.3

The recently developed interest in immunological networks stems mainly from the work of Jerne, who postulated the existence of a network resulting from the existence of idiotypic determinants in antibody molecules. There is now solid evidence demonstrating that structures associated with the variable regions of antibody molecules, where the combining sites with antigen are expressed, may function as (idiotypic) antigenic determinants for other antibody molecules formed by the same organism. It has been conclusively demonstrated that these anti-antibodies are formed as a secondary consequence of antigen stimulation, and that they may stimulate or inhibit the activity of T- and/or B-cells responding to the specific antigen (Klusens and Köhler, 1974).

These findings should make clear the inadequacy of experimental approaches to the problem of regulation of immune events that examine merely the activity of individual clones of lymphocytes. It is misleading to view the activity of individual clones in isolation, because once the first (antigen-binding) antibodies are formed, they generate anti-antibodies (anti-idiotypic), which in their turn would generate anti-anti-antibodies, and the process would grow in an ever-branching tree involving the whole lymphoid system. It is obvious, therefore, that the system exhibits closure, which modulates the magnitude of this process; in other words, at a certain point this cascade of stimulatory reactions must start "biting its own tail" and leave the system in a new state of dynamic equilibrium. Two very important aspects of Jerne's consideration of these questions are apparent. First is the notion that the disturbance caused by antigen penetration influences many more lymphoid cells than those able to interact directly with the antigen. Different antigens may cause disturbances of different magnitude, depending on the relevance that the primary disturbance they cause has to the operation of the network. A second important notion is that the individual clones of cells that are primarily stimulated by the antigen are not primarily "designed" to react with the antigen, but rather to react with endogenous components of the network, which Jerne very properly denominated the "internal image" of the antigen. This stresses the fact that the responsiveness of the lymphoid system is directed

inward, using the structure (i.e., the network of interactions) as the only available point of reference for its comparisons.

Jerne also pointed out the possibility of anti-idiotypic antibodies reacting with antibodies other than those specifically generated by the antigen, which he calls the “unspecific parallel set” of antibodies. This is also a consequence of the fact that the receptors of the network operate on the recognition of endogenous components; the fact that they recognize exogenous materials (antigens) is fortuitous. Thus, it is possible that within a set of antibodies carrying the same idiotypes, only a subset may be able to be stimulated by an exogenous antigen. However, when this stimulated subset starts producing antibodies (antigen-specific), in their turn these antibodies stimulate the production of anti-idiotypic antibodies, which may interact with the whole original set carrying the same idio type.

14.3.4

Antibodies are initially expressed as structural glycoproteins of B-cell membranes. After they are secreted by plasma cells into the body fluids, they either interact with the membranes of other cells of the organism to perform a meaningful role in the network, or remain idle until they are catabolized. Thus, antibodies are links between different types of cells in the organism, and they perform meaningful functions only when interacting with cell membranes. In addition, antibodies may affect the behavior of cells through the activation of enzyme systems, such as the complement system.

Some antigens are able to stimulate B-cells directly, whereas others can only do it in the presence of T-cells. All T-independent antigens studied so far are polymeric molecules, expressing many copies of the same antigenic determinant; this allows the interplay of cooperative forces between the multiple determinants on the molecule and the multiple antibody binding sites clonally expressed on B-cells (Coutinho and Möller, 1975).¹³ Whereas the presence of T-cells is not required for the initiation of this type of response, their progression is modulated by suppressor T-cells, and they should not be seen as clonal activities that may be undertaken independently of the network.

Nonpolymeric antigen molecules, although expressing different types of antigenic determinant of their surfaces, rarely express more than one or two copies of the same determinant. Thus, their attachment to B-cells hinges on only one or two antibody binding sites; only reactions with a very high antigen-binding affinity would be expected to be effective under these circumstances.¹⁴ This creates

¹³In addition, all T-independent antigens have been found to function as polyclonal B-cell mitogens: they stimulate not only specific, but also nonspecific B-cells to antibody production. However, since specific B-cells may bind these antigens more efficiently, they are stimulated by lower concentrations than unspecific B-cells. This type of antigen stimulation involves only the production of IgM antibodies, and lacks the adaptive quality characteristic of T-dependent responses, in that repeated contacts with the antigen result in repeated primary-type responses.

¹⁴However, in addition to their endogenously synthesized antibodies, B-cells may express on their membranes receptors for the Fc portion of other antibodies and for activated components of the complement system (Nussenzweig, 1974).

another dimension in the connectivity of the network of interactions in the lymphoid system, which allows for the formation of antibodies to antigenic determinants that occur as a single copy on monomeric antigen molecules. This would occur because, once the molecule is bound through one of its determinants to antibodies on the surface of a particular B-cell, antibodies to different antigenic determinants on the same molecules could bind to the molecule, and then be bound to the Fc receptors, strengthening the binding of the antigen to the cell. The presence of receptors for complement components on the B-cell may further potentiate this process. Conversely, under other conditions, complement may remove antigen-antibody complexes from the surface of B-cells.

This process is obviously a cyclic one. Once B-cells forming antibodies against one of the determinants of the molecule are activated, they will produce antibodies which potentiate—or, in other conditions, interfere with—the formation of antibodies to the other determinants on the molecule by corresponding specific B-cells. The potentiating and inhibitory effects that antibodies may alternatively exert on antibody formation (Uhr and Möller, 1968) may be partially explained by these mechanisms.

These considerations are useful in illustrating how the activity of one clone of B-cells may influence the activity of other clones of B-cells in the network, even when these cells are totally unrelated, except for the fact that they form antibodies to determinants that happen to occur together on the same antigen molecule. As opposed to the idiotypic interactions described above, this aspect of the network operation affects exclusively the responses to monomeric (T-dependent) antigens.

14.3.5

A third form of connectivity to be considered is the one occurring between lymphocytes themselves.¹⁵

It should be clear however that the interconnectivity exhibited by lymphoid cells through the formation of interacting populations of serum antibodies is basically an expression of the interconnectivity of the cells themselves, and that the antibodies are messengers that spread the word of their clone of origin to every corner of the organism.

There are certainly many other forms of connectivity among lymphoid cells in the organism, which do not depend directly on the specificity of antibodies. To begin with, it has become clear in recent years that lymphoid cells may release non-antibody “helper” and

¹⁵ It is obvious that all immunological events depend on the specific activities of lymphoid cells. Nevertheless, we still refer to “humoral” as opposed to “cellular” aspects of immunology; the very term “cellular immunology” implies the existence of areas in immunology where the activity of cells is not of major importance, i.e., that there is such a thing as “acellular immunology.” All this has, of course, a clear historical explanation: Immunology was born of the discovery and use of serum antibodies as medical tools, and developed when antibodies were studied as proteins by protein biochemists who were not very concerned about the mechanisms of their formation in the organism.

¹⁶As stated before, it is as yet impossible to describe cellular or molecular details of the operation of these various “factors” produced by lymphoid cells. From the present perspective, however, they are important as mediators of immunologically relevant messages among lymphoid cells, which, instead of utilizing the antigen-binding sites of antibodies, somehow utilize the products of genes at the MHC to convey antigen-specific messages. Some of these factors are only operative on cells carrying the same MHC (or the same I-region) products as the cells from which they were obtained. It is probable, however, that these restrictions may be circumvented by the same maneuvers that make possible the collaboration between histoincompatible T- and B-cells; and that the incompatibility is not absolute, but rather derives from the previous immunological experiences of the donor organism.

¹⁷For example, up to 6% of all the T-cells in Lewis rats were shown to express idiotypes present in alloantibodies formed by Lewis rats against the MHC antigens of the DA (*Eds.* Dark Agouti) strain of rats. These T-cell receptors, which are not immunoglobulin, are shed into the serum and are present in the urine of normal Lewis rats, from which they may be specifically isolated. When normal Lewis rats are immunized with a polymerized preparation of these autologous receptors, they become tolerant to otherwise incompatible transplants of DA cells.

“suppressor factors.” Some of these substances nonspecifically stimulate or inhibit immune events in general: however, some others show the same delicate degree of antigen specificity as antibody reactions, although they seem to be directed at different determinants of the antigen (Munro and Bright, 1976). This latter type of antigen-specific factor is more important in the context we are discussing.

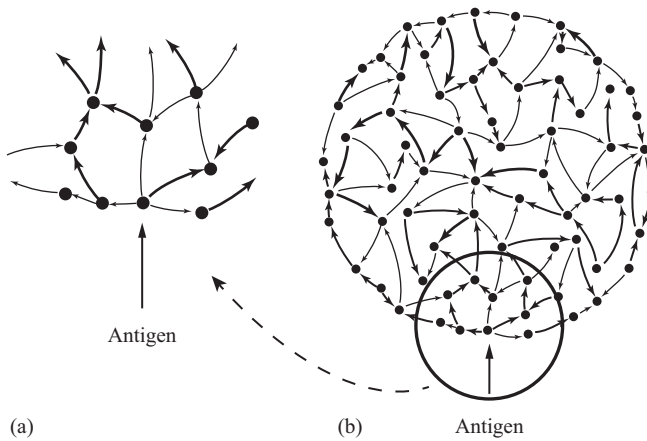
There is solid evidence that these antigen-specific factors are identical to, or contain, products of the I-region of the MHC in the mouse, and that different mouse strains vary not only in their ability to produce the factors, but also in their ability to “accept” the factors—i.e., some strains are known to produce factors that fail to affect the activity of their own lymphocytes, but readily affect those of lymphocytes of other (histocompatible) strains (Munro and Bright, 1976).¹⁶

Besides realizing that there are antigen-specific but antibody-independent interactions among lymphoid cells, it is also important to understand that a great deal of activity seems to be continuously going on between antibody molecules and products of MHC genes. It has been shown, mainly by the work of Binz and Wigzell (1975), that antigen-specific T-cells and B-cells may share membrane molecules with the same idiotypic determinants, although the molecules playing the role of receptors on T-cells are not the conventional immunoglobulin receptors expressed by B-cells. There is a very large proportion of T-cells in normal organisms expressing receptors for alloantigens of the MHC.¹⁷

The precise meaning of these results cannot be seen until we have a better understanding of the roles played by T-cells in immune events. However, these results do indicate that a complex series of interactions, involving on the one hand conventional serum immunoglobulin antibodies, and on the other hand T-cell receptors that somehow involve products of the MHC complex, is permanently going on.

14.4 Network Links and Plasticity

The picture emerging from these considerations of different forms of connectivity in the lymphoid system is, therefore, not of isolated antigen recognition by separate and independent cell clones, but of a series of reciprocal recognition events, of various degrees of specificity and effectiveness, involving the whole of the system.

**Figure 14.1**

Two pictorial representations of the immune network. On the left (a) an antigen is represented interacting with one node (one clone, one set of related clones) of the network, from which a tree of interactions unfolds, idiotypic and otherwise. On the right (b) an expanded view of the same process emphasizes the fact that these interactions constitute a circular network; the effectiveness of the interactions is depicted by the thickness of the arrows.

Different types of lymphocytes, and the various sorts of antibodies they may produce, can be thought of as *links* establishing the connectivity of the network.

A pictorial representation of the network is shown in figure 14.1. The network is depicted as a giant net of interconnected nodes, each representing the actual components of the lymphoid system. Each component enters into interactions with many other components, pictured as incoming and outgoing arrows, by virtue of either recognizing other components or being recognized by them. Such mutual recognitions could be embodied by idiotypic complementarity, through other forms of antibody-mediated connectivity, or directly through cell contacts, as discussed above. The nature of the connecting links is left unspecified in the figure. Whatever these links may be, there will be a *degree* of effectiveness in the recognition events, represented in the figure by different thicknesses in the arrows. The lymphoid system is thus thought of as a giant mobile, with several thousand interconnected nodal points.

In practice, in order to analyze immune events, we generally start at one particular node, representing a particular set of lymphocyte clones. Entering the network from this point yields an unfolding tree, which can potentially branch indefinitely (figure 14.1a). But since the immune system exists in a bounded organism, it is obvious

that such a tree must eventually close onto itself, as if its finest branches were at the same time its finest roots (figure 14.1b). As we know from section 10.4.1, these are complementary views.

In this dynamic mobile the host of interactions occurring at one moment also determine a change in the effectiveness of the links or in the nature of the links themselves. That is, not only does the system undergo large numbers of connecting recognition events, but such events trigger *consequent changes in the states of the lymphoid cells*. Such changes can be either inhibitory or excitatory—that is, they can lead to cell division or blast transformation and antibody production, or they can bring about the halting of these events.

For expositional purposes, we can distinguish two related aspects of the network dynamics: *recognition* and *action*. The changes brought about by the actions triggered by the recognition events makes the lymphoid network a *plastic* one. Which way the system's plasticity will go in adapting to perturbation is the main question in the regulation of the immune system, which we now want to discuss.

14.5 Regulation in the Immune Network

14.5.1

It has often been observed that the immune system and the nervous system share many features. Both systems seem to have evolved for recognition, and they endow organisms with the capacity to respond adaptively to an immense variety of stimuli. Both systems translate their interactions into an internal dynamics of signals. In the lymphoid system the interconnection of elements is made, not through anatomical contact of long axonic branches, but through molecular recognition events that happen during random encounters between freely intermixing lymphoid cells and their soluble products, such as antibodies. In correspondence with this freely distributed connectivity of the lymphoid elements, sensors are not clustered together as in sense organs, but rather are distributed at all points of the network where a receptor is exposed to other macromolecules. The actions resulting from a recognition event are not localized either, and are soon distributed throughout the system.

Both systems exhibit a plasticity in their ontogenetic development. Every recognition event is followed by some form of action that changes the responsiveness of the system. In the nervous system, this is *learning*, and is based on the plasticity of the synaptic contacts, which makes it possible to change, within certain limits, the strength of the connections in the network loops. In the immune

system, changes occur by differential growth of specific clones of lymphocytes and consequent changes in the amounts of their soluble products available for interactions.

What is clearly missing from this analogy, however, is that for the nervous system we have some understanding of how the system can regulate itself (through a hierarchy of levels of command) and the interplay of excitatory and inhibitory signals (cf. chapter 15). We lack a similar understanding of the regulation of the immune system. One important and rather recent development in immunology has been the realization that active inhibitory events do in fact exist.

14.5.2

The specific inhibition of immune responses has been known under the various names: immunologic unresponsiveness, tolerance, paralysis, and suppression. All these names refer to the same basic observation, namely, that under certain conditions, the contact with antigen may reduce rather than enhance subsequent responses to the same antigen, as measured in terms of antibody production or other antigen-induced cellular activities, which may be normally obtained from immunologically naive immune systems of similar genetic and ontogenetic characteristics.

Until very recently, states of immunologic unresponsiveness were seen as negative events, caused by the lack of cells able to respond. More recently, solid evidence was obtained indicating that the unresponsiveness that is measured according to one experimental criterion actually results from the active suppression of the activity of certain clones of lymphocytes by other clones of lymphocytes, denominated *suppressor cells*. This type of suppression can hardly be labeled as unresponsiveness, because it derives from active responses of the system. It has become necessary to reevaluate whether this active type of suppression plays any role in the states of immunological tolerance or paralysis formerly believed to be due to clonal deletion. The present estimate is that clonal deletion may indeed occur in some situations, but more often the "unresponsiveness" of the system has only an operational meaning.¹⁸

Although very little is known about the precise mode of operation of helper or suppressor T-cells, there is a tendency in current immunology to consider their activities as exactly symmetric, as mirror images. That is not necessarily true; suppressor cells may be much more than the mere opposite of helper cells.

The magnitude of an immune event can only be measured as some function of the expansion or contraction of specific

¹⁸Subpopulations of T-cells carrying specific subsets of membrane alloantigens (Ly-2, 3) have been identified as suppressor cells as opposed to "helper" T-cells, which are Ly-1 (Cantor and Boyse, 1975). However, the same population of lymphocytes may act as "suppressors" in one system and as "helpers" in a slightly different system (Eardley et al., 1976). Also, suppressor activities in other systems have been ascribed to Ly-1 positive cells, and not Ly-2, 3 positive cells. Moreover, it is still undecided whether the target of suppressor-cell activity is on other T-cells or on B-cells.

populations of lymphocytes. These events, however, are the net output of webs of inter-connected lymphocytes, and from a certain perspective, all the lymphocytes of the organism are involved in all immune events. Ideally, therefore, the magnitude of an immune event should be assessed not only by the direct consequences of antigen stimulation (such as specific antibody production and cell proliferation) but also by the global repercussions of this stimulation in the network. Current immunological methods are, to a large extent, inapplicable to the assessment of these more global activities of the lymphoid network. Determinations of the total immunoglobulin levels, or "background" levels of cellular activity, may reveal that these global activities exist, but do little to define them. These difficulties have hindered the proper characterization of suppressive immune events.

Suppression is not the consequence of excessive helper activity. Actually, the situation is almost the logical converse of this; lymphocytes that suppress the response of other cells stimulated under optimal conditions may be shown to help these responses when the stimulation is suboptimal. A puzzling detail in this area has been the inability to induce high levels of specific suppressor-cell proliferation in response to the antigen. If suppressor cells were functionally mirror images of helper T-cells, they would be expected to divide similarly to helper cells upon antigen stimulation. This issue is clouded by the fact that we don't know how to quantify suppression directly and may only compare the response of suppressed animals with those of normal controls.

These problems may be perceived differently when it is considered that the activity of any specific clone of lymphocytes is a node in the network, toward which many stimulatory and inhibitory influences may converge. The higher the number of influences converging upon the node, the higher must be its immunological inertia, in the sense that larger portions of the network will be disturbed when the node is disturbed. On the other hand, the lower the number of converging influences, the smaller will be the portion of the network affected by a change in the node; thus, a great deal of clonal expansion, or contraction, may occur with few or weak connections throughout the network.

Most forms of experimental immunization bear little relationship to the types of antigenic stimulation met in normal (ecological) environments. They are induced by artificial routes, frequently utilizing the so-called adjuvant materials, which may boost the responses to abnormally high levels. It might be that these methods induce

major disturbances in circumscribed portions of the network. In contrast, the antigenic stimulation received by the organism in its normal environment—such as the absorption of antigenic materials of dietary and microbial origin from the gut—continuously evokes minor disturbances in major portions of the network. This form of stimulation induces antibody responses that are rather weak compared to the responses induced by artificial immunization, but that correspond more closely to the situation prevailing in the lymphoid system most of the time. Thus, we may be searching for clues to the normal operation of the system by creating rather abnormal conditions of stimulation. This is not uncommon in experimentation, but it must be recognized and acknowledged.

A number of observations are consistent with the notion that the prevalent state of affairs in the lymphoid system has an inertia, which resists attempts to induce sudden and profound deviations in its course of events. Most normal lymphoid-cell activity is suppressive to the performance of the exacerbated type of immune responses that we induce experimentally.¹⁹ Thus suppressor cells, rather than being mirror images of antigen-specific helper cells, may be simply all the cells, antigen-specific and otherwise, that resist displacements in the organization of the network that may be necessary for the production of intense immune responses.

14.5.3

Whatever the molecular mechanism determining which antigenic contacts are stimulatory and which suppressive, it is clear that until recently the role of suppression in the lymphoid system had been grossly underestimated.

The very fact that minute amounts of antigen may elicit the formation of much larger amounts of specific antibodies suggests the existence of mechanisms by which the antigen may be allowed to suppress, rather than to promote, the formation of antibodies. Otherwise, the organism would undergo progressively larger immune responses, which would rapidly reach suicidal proportions, every time it met an antigen that failed to be eliminated shortly after the first contact. The organism must be able to regulate the magnitude of its immune responses by means other than the simple elimination of potentially stimulatory amounts of antigen, simply because it is not always possible to eliminate the antigen very rapidly, or to avoid repeated contacts with the same antigens.

The existence of such a regulatory mechanism is implicit in the fact that the whole of the interactions in the lymphoid system

¹⁹Normal lymphoid cell populations are known to provide a rather more "suppressive" environment to the expansion of adoptively transferred immune cell populations than does irradiated animals, in which the lymphoid system is disrupted. Thus, newborn lymphoid cells are more "suppressive" than adult cells (Köhler et al., 1974). Sublethal irradiation, which destroys large masses of lymphocytes, affects "suppressor" cells more readily than helper cells (Gershon, 1974).

eventually closes onto itself in a network, and that this network must have evolved a balance between excitatory and inhibitory mechanisms. This balance between excitation and inhibition is at the core of the mechanism of immune regulation, and has become a central concern for research. In fact, such balance has proven essential in the self-regulation of other biological systems. Until a balance between excitation and inhibition was described for the nervous system, no detailed analysis of behavioral regulation was forthcoming. In relation to this issue, immunology stands now where neurobiology was 30 years ago.

That the immune system is characterized by the capacity to respond to individual stimulating events while maintaining an overall stability, is strong evidence for active regulatory processes maintaining both humoral and cellular components within bounds. At each moment, the structure of the lymphoid network can be defined by a state specified by a reactive population of cells and its spectrum of interactions or recognition. At each moment, this state determines the form of response that a given perturbation will evoke, that is, which set of actions will follow a set of recognition events. Whatever these actions, they will normally drive the system into a new stable state. We are saying, then, that the immune system functions through closure. In the case at hand the closure's recursion is a highly complex phenomenon, embodying the dynamics of the mutual interdependences of the components of the lymphoid system in sequence. The system's actions are such that they affect what it recognizes. Conversely, what the system recognizes, its links, account for its subsequent actions, by the plasticity that has shaped the network. The formation of anti-idiotypic antibodies is a clear example of this interlocking of recognition and action.

We may now use the term eigenbehavior to designate the stable states of the immune network that arise from such recursive organization. As we said before, Jerne was the first to use the term eigenbehavior to describe the succession of stable states in the immune system in his formulation of the idiotypic network. Thus, the notion of self-determination and the adaptive stability of the immune system are closely associated, and disregarding the system's self-referential constitution amounts to disregarding its mechanism of operation. In other words, the eigenbehavior expresses the coordination of the immune system's tendency to generate a coherent adaptive behavior, while a continuous barrage of perturbations normally induces in the system a constant change of eigenbehaviors.

This capacity to generate a variety or pathway of eigenbehaviors is typical of natural autonomous systems.

It seems obvious that this behavior in the immune network is more ripe for formal modeling than in any other natural, autonomous system. Some attempts have been proposed using differentiable dynamics (Bell, 1970; Hoffmann, 1975; Nicolis and Prigogine, 1977). In my opinion, the algebraic-algorithmic approach of the previous sections should be useful here, since the complexity of the system seems large enough to make differential equations too difficult, but known in enough detail to make algebraic assumptions accessible. It may prove to be a testing ground for how to deal with the question of time change of invariant eigenbehavior—starting, perhaps, with computer simulations rather than exact solutions. We shall not be concerned any more with modeling questions in this chapter, as the intention is to examine the relation of closure to cognitive domains, to which we now turn.²⁰

²⁰ *Eds.* Varela and his colleagues produced several simulation models of immune networks during the 1980s and 1990s, (Calenbuhr et al., 1995; Detours et al., 1994; Stewart and Varela, 1989, 1990, 1991; Varela and Stewart, 1990).

14.6 Cognitive Domain for the Lymphoid System

14.6.1

The structure of an autonomous system specifies a domain of possible interactions with its environment. These permissible interactions are those, and only those, that the components of the system specify, and that are compatible with the maintenance of closure. This domain of interactions we call the cognitive domain.

In the case of the immune system, the specific repertoire of antigen receptors expressed in the system at any moment will determine which antigenic contacts are relevant and which are not. What is relevant for the lymphoid system is given by its organization and its recursive history. Whatever may be recognized as an antigen may be so recognized because it has a degree of resemblance to an already existing set of determinants in the network—an “internal image,” in the language of Jerne. Self-recognition and recognition are one and the same process. This concept runs in the opposite direction to that of self-versus-nonself discrimination, where the specification of what is recognizable is directed outward, and clonal deletion discriminates against the recognition of self.

14.6.2

The altered-self hypothesis (Doherty and Zinkernagel, 1975) is favorable to the present ideas, in that it centers upon the macromolecular characteristics of the organism and the primary source of immune

responsiveness. It proposes the existence of a relationship between the ability of a foreign molecule to “alter” the structure of membrane proteins and its relevance for the activation of T-cells. Early versions of these ideas were offered by Kreth and Williamson (1971). The weakness of the hypothesis lies in the fact that it takes only a half step in the right direction. The hypothesis proposes that the repertoire of responsiveness of T-cells is directed to the alterations of membrane proteins caused by foreign macromolecules, but it fails to state clearly whether the foreign molecule itself is a portion of the determinant recognized by the T-cells, or the “altered” self molecule alone carries the determinant. In either of these alternatives, responsiveness is visualized as directed against something foreign and unpredictable. In the first alternative, the foreign molecule would be a part of the determinant; in the second, a membrane protein, altered by the foreign molecule, would become itself “foreign.”

According to the reasoning developed in the previous sections, all immune events are directed inward, not outward, and the organism perceives the penetration of foreign materials not by recognizing them as foreign, but rather because the foreign materials interfere with ongoing reactions that exist as links in a complex network of interactions.²¹ The organism responds to an “internal image” of the foreign molecule, to its meaning translated in terms of the language previously utilized by the network. Thus, in a way, all immune reactions are “autoimmune” (directed inward) and exogenous antigens are recognized by “cross-reactions.” The “alteration” caused by the antigen has more of a functional meaning, and does not mean that the molecular alterations caused by the direct binding of the foreign molecules to membrane proteins are the antigenic determinants recognized by T-cells. Since these alterations may be expected to be random, there is no fundamental distinction between the mechanism proposed by the altered-self hypothesis and the more orthodox view that the receptors of lymphocytes are directed outward, towards random encounters with antigens of unpredictable structure.

According to the “self-determination” ideas the alteration in self is a concept almost indistinguishable from the concept of “internal image” postulated by Jerne for the idiotypic network, the only fundamental difference being that in the case we are discussing, the “internal images” perceived would be images of membrane proteins that operate as receptors in the network, instead of images of idiotypic determinants on antibody molecules.

²¹ *Eds.* This idea was later treated in a more nuanced manner by Varela and his colleagues. In their work on second-generation immune networks, they came to recognize that naturally autoreactive B- and T-cells were indeed organized as a self-asserting network, but resting lymphocytes (comprising about 85 percent of all lymphocytes) were functionally disconnected from this network and able to respond to external antigens. The production of antibodies by these peripheral systems could perturb the central network and help explain natural tolerance and systemic memory (Varela and Coutinho, 1991; Varela et al., 1993). This proposal had “the major heuristic value of bringing two views of the immune system into compatibility that had formerly seemed irreducible to each other: the clonal selection and the network theories. Within this framework, both theories could be accepted, but each applied to distinct parts of the system” (Coutinho, 2003, 23).

The concept of internal images has the advantage of interpreting immunological reactions as organizational events. The fundamental difference between the contact of antigen molecules with immunized and nonimmunized (immunologically naive) organisms is that the immunized organism will handle the antigen molecule in predictable ways that, to a large extent, are independent of the properties of the antigen. The penetration of foreign macromolecules into vertebrate organisms is, of course, a random event. The nature of the interactions that these molecules will initiate in contact with components of the organism, is unpredictable. However, if these molecules happen to express surface details (antigen determinants) that fit binding sites on antibodies, or other receptor sites on lymphocytes—which were originally reacting with autologous components in the network—then the molecule will be “confused” with these autologous components and be admitted into the network. The potentially chaotic consequences represented by the penetration of foreign materials into the organism are now subjected to the constraints of the organism’s structure and organization, because although these materials produce disturbances in the operation of the lymphoid system, it is the very nature of the system to adapt itself to these disturbances. In other words, the lymphoid system exhibits self-organization; it transforms environmental noise into adaptive functional order. However, it can only do so when this environmental stimulus is perceived as a meaningful internal image; otherwise it will simply constitute environmental noise.

14.7 Genetic and Ontogenetic Determination of the Cognitive Domain

14.7.1

According to the perspective we have developed up to this point, the cognitive domain of an individual’s lymphoid system is determined by the set of receptors expressed in the network, which makes possible the undertaking of a large variety of cellular interactions. All these receptors, including those present on antibody molecules, may be considered as structural components of cell membranes: Although antibody molecules may spend a significant proportion of their existence as free molecules in body fluids, they only play meaningful roles when attached to cell membranes. In addition to antibodies, a large number of membrane proteins, which express a great deal of genetic polymorphism (and thus may function as transplantation alloantigens), such as the products of the major

histocompatibility complex (MHC), play the role of specific receptors governing cellular interactions in the network. Different types of lymphocytes express different types of membrane proteins, which then determine the occurrence of different types of interactions.

Antibodies are clonally expressed on B-lymphocytes from the early periods of the ontogenesis of the lymphoid system, but antibody formation—the differentiation of B-cells into plasma cells for active secretion of antibodies—is only initiated much later and probably require the exposure of the organism to foreign macromolecules. Germ-free animals that are fed (elemental) diets free of macromolecules fail to form immunoglobulins, although they respond slowly to antigen stimulation with normal levels of antibody formation. Other immune functions, which are attributed to T-lymphocytes, are present from early periods in ontogenesis, and very young fetuses are able to reject allogeneic transplants (Sterzl and Silverstein, 1967). Thus, it appears that both T- and B-lymphocytes are able to interact to construct the basis of the network long before birth, and in the absence of extrinsic antigenic stimulation.²²

Whatever their precise mode of specification on cell surfaces may be, antibody molecules, different products of MHC genes, and other membrane proteins (such as the products of Thy-1 and Ly loci) are programmed to initiate cellular interactions that identify the boundaries of the cognitive domain of the lymphoid system on a structural basis, before the organism is exposed to external antigens.²³

At the present, little doubt remains that the products of MHC genes are involved in the structure of the receptors utilized by lymphoid cells during the cellular interactions of the immune response. The fact that these substances may, on the other hand, function as strong transplantation alloantigens has been a most puzzling issue of current immunology. It should be clear, however, that when cells from an organism are transferred to another organism of the same species, which is utilizing a similar (allogenic) set of membrane proteins in the operation of its own lymphoid system, the membrane proteins of the donor organism will readily interfere with the operation of the analogous proteins in the receptor organism. Thus, the strong immunologic responses elicited by the encounter of allogeneic lymphoid cells depend, at the same time, on their similarities and on their differences: Their similarities allow the characteristics of one cell population to fall within the cognitive domain of the other; their differences introduce disturbances in a previously harmonious set of cellular interactions.

²² It is as yet unknown how the clonal expression of diverse antibodies on B-cells is determined. Most probably, the V-regions of immunoglobulins are coded by genes directly inherited from the organism's ancestors, which may or may not be expanded *de novo* during ontogenesis by somatic mutations or recombinations. Similarly, very little is known about the nature of the receptors expressed by T-cells. It is undecided whether T-cell receptors are clonally expressed in the same strict manner as antibodies are expressed on B-cells.

²³ It is probably significant that a complex set of membrane alloantigens, which are expressed solely on embryonic cells, undifferentiated teratomas, and germinal cells of adult organisms and are similar in structure to the products of MHC (H-2) genes, are coded by genes at the T/t locus in the mouse, which is located 14 crossover units to the left of the H-2 complex. Crossovers do not occur between the T/t locus and the H-2 complex, and this whole chromosomal segment is inherited as a unit, a sort of supergene coding for specific membrane proteins that orchestrate many fundamental types of cellular interaction important for differentiation and morphogenesis of vertebrate tissues. Products of the T/t locus and of the H-2 complex may control, respectively, cellular interactions in the embryonic and postembryonic periods, respectively (Artzt and Bennett, 1975).

Another essential dimension of the cognitive domain of the immune system is the ontogenetic transformations it undergoes as a result of the exposure to foreign macromolecules, which starts in the perinatal period. These changes occur as compensations for disturbances introduced into the operation of the network by the random entry of environmental macromolecules, mainly through feeding, microbial contacts, and viral infections.

At the outset, it is important to realize that these transformations are of two kinds. The concept of immunological "memory," through which the organism develops an enhanced capacity to recognize secondary contacts with the antigen, is at the center of immunological thinking. However, it is obviously true that a first contact with the antigen may also result in less antibody formation or lymphocyte blastization upon secondary contact with the same antigen. In other words, the first contact may be tolerogenic rather than immunogenic. It is not generally acknowledged that immunological tolerance is one of the possible consequences of the contact of an adult, fully immunocompetent organism with antigens in its environment. However, there is solid evidence showing that the ingestion of foreign proteins during normal feeding, for example, may be highly tolerogenic, in the sense that it will render the organism virtually unresponsive to a subsequent injection of the same proteins with adjuvant by a parental route. This was demonstrated as early as 1911,²⁴ and has recently been confirmed in a number of different experimental models (e.g., Hanson et al., 1977). Whether similar phenomena govern the unresponsiveness of normal animals to members of the autochthonous bacterial flora of the gut, or whether similar types of tolerance may result from the absorption of antigens through nondigestive mucosal surfaces, remains to be determined. It is, then, imprecise to describe tolerized organisms as unresponsive to the antigen.²⁵ These are simply situations in which the immunological adaptation of the organism results in less overt antibody formation and lymphocyte proliferation.

Through such mechanisms, which we are only beginning to explore, the transformations of the cognitive domain of the immune network in an organism's ontogeny are a combination of its recursivity (its closure) and the fact that it is exposed to random perturbations or fluctuations from the environment (its openness). In other words, it exhibits self-organization and self-regulation, the transformation of environmental noise into adaptive functional order, in a manner similar to all natural systems, such as cells, nervous systems, and animal populations.

²⁴Eds. See Wells and Osborne (1911).

²⁵For example, the tolerance to proteins introduced orally is maintained by active immunological processes, and the responsiveness of tolerized animals cannot be restored by the adoptive transfer of normal or even immunized syngeneic spleen cells (Hanson et al., 1977).

14.7.2

Given the diversity of cellular events in the immune network, the possible diversity of recursive histories (or of pathways taken by self-organization) open to an individual is astronomical. Once an organism has been in contact with a certain variety of macromolecules, its responsiveness will be heavily biased in some direction. It is common knowledge, in fact, that once an animal is immunized against something, it is difficult to make it tolerant, and conversely, once it is tolerant to antigen it is hard to “immunize” it. The strong dependence of the immune system on its recursive history can hardly be overemphasized. To leave it aside would be like studying animal behavior as if the nervous system were not plastic, ignoring its capacity to learn—or like studying the differentiation of a cell with total neglect of its surrounding medium, as if genes could be expressed independent of context. We believe that the possibilities for experimentation and applications of the recursive nature of the immune system are just now beginning to be appreciated. Recent discussions of the concept of “self” in immunology (Grabar, 1974; Tauber, 1976), however, fail to describe the notions of self-determination, recursivity, and closure as we propose herein.²⁶

²⁶ *Eds.* For more recent reflections on the immune self commenting on Varela’s work, see Tauber (1994) and Pradeu (2012).

14.8 A Change in Perspective

I have endeavored to set forth a picture of the immune system that stresses the cooperative nature of events typical of lymphoid cells. Nowhere have specific molecular mechanisms been proposed. We have rather proposed to follow and expand Jerne’s work in viewing the immune system as a network of interactions recursively determining a macromolecular cognitive domain, capable of maintaining and indeed defining an organism’s macromolecular individuality (Vaz and Varela, 1978).

This is, in a way, an extension of the “selective” concepts that replaced “instructive” concepts in the fifties. Selective ideas were introduced in immunology as a result of findings in molecular biology, which required antibodies to be made like other proteins in the organism—i.e., using RNA and not antigen as a template. The ideas now proposed, which require the antigenic stimulus to be somehow relevant to what is already going on in the organism, are more a result of biological and cybernetic considerations. There is no sense in talking about molecular or genetic “information” outside of a context of cooperative interactions. The idea of a collection of lymphocyte receptors directed towards unpredictable outward stimuli is, of course, very appealing, but it is clearly insufficient. Defense

Table 14.1

Prevalent concepts in immunology and alternative concepts proposed as substitutes. From (Vaz and Varela, 1978).

Prevalent	Proposed
Territorial concepts with emphasis on defense against foreign invaders and internal mutant cells.	Integrative, adaptive concepts, with emphasis on the stability of internal signals.
Antigen-centered (input-output).	Organism-centered (autonomy).
Event-centered, with emphasis on specificity.	Network-centered, with emphasis on coordination.
Use of artificial situations to mobilize the system.	Attempt to mimic natural events to mobilize the system.

against infectious disease is a most important aim to be fulfilled, and it is fair to assume that existing organisms inherit the accumulated wisdom of phylogenetic evolution and are equipped with an array of receptors that, regardless of the particular genetic constitution of the individual, is vast enough to cope with the invasion of microorganisms. Defense, however, seems to be more of a side issue to the more important issue of molecular identity, of which the immune system is the essential regulator. Furthermore, epidemiological arguments weigh against the idea that defense against specific infection was the pressure forcing the evolution of genes coding for immune responses against these infections (Black, 1975). In table 14.1 I contrast current immunological concepts with those proposed here.

The picture emerging from this discussion is at variance with many assumptions that have been dear to immunology since the clonal selection theory came to be widely accepted. In many respects, the present view is the logical inverse of that theory. The key concept of Burnet's theory is the notion of self-versus-nonself discrimination. In contrast we started with the assumption of the immune system's autonomy, where the system's identity is none other than the process of cooperative interactions.²⁷ This assumption leads most naturally to the notion of a network and its genetic specification, cognitive domain, self-organization, and recursive history. In this light, established empirical results can be reinterpreted, old problems will disappear, and new ones will emerge.

²⁷ *Eds.* Rather than a rejection of the idea of self-nonself discrimination, this is an attempt to define the self in processual and systemic terms. Subsequent models were more integrative of the clonal selection and the network theories; see the note in section 14.6.2 on second-generation immune networks and discussion in Coutinho (2003) and Pradeu (2012).

Source

(Vaz and Varela, 1978)

15 The Nervous System as a Closed Network

15.1 The System of the Nervous Tissues

15.1.1

Nowhere, it seems, is the philosophy of control more predominant than in the study of the nervous system. In fact, almost every researcher in neuroscience will take as a dogma that (1) the nervous system acts by picking up “information” from the environment and “processing” it, and that (2) this “processing” is adequate because there is “representation” of the outside world in the animal’s (or human being’s) mind.^{1,2} The brain is a machine to produce an accurate picture of the world. Nowhere have the notions developed in the domain of design have taken deeper root than in this field; the brain as a computer is, by now, a commonsense notion.

Vis-à-vis this predominant opinion, it might seem arrogant to propose a viewpoint that is almost the opposite, namely, that the nervous system operates as a closed network with no inputs or outputs, that its cognitive operation reflects only its organization, and that information is imposed on the environment and not picked up from it. The argument, in this case, closely follows the one presented for the immune system. The difference is, however, that the nervous system enjoys a considerably longer history as a field of thinking, and that whatever conclusions are deemed valid for it will have a broader impact on our understanding of what knowledge is. In spite of this difficulty I will maintain that *we should look on the nervous system as an autonomous system*, and that in doing so there are significant consequences for what representation and communication can possibly mean. From this point of view, the approach from the computer gestalt of information processing becomes a restricted approximation. As in the case of the immune system,

¹Consider the following statement: “The brain is an unresting assembly of cells that continually receives information, elaborates and perceives it, and makes decisions” (Kuffler and Nicholls, 1976, 3). And this one: “I have found it necessary to suppose that the perceiver has certain cognitive structures, called *schemata*, that function to pick up the information that the environment offers” (Neisser, 1976, xii).

²*Eds.* In later editions, Kuffler and Nicholls maintain their point, but in reference to the central nervous system, not just the brain. Neisser’s quote deserves some context. As he saw it, the notion of *schemata* was a way of combining information-processing ideas and the ecological approach of his colleague James J. Gibson. The computer metaphor was indeed predominant, but even within this camp, there were interesting differences.

old evidence is reinterpreted, problems disappear, and new ones emerge.

I have learned to look at the nervous system as a closed system from Humberto Maturana, who first proposed this explicitly in 1970 (see also Maturana, 1975, 1978). I consider Maturana's insight a fundamental step. He has provided a basic link between material and systemic processes in the nervous system, with a deep epistemological view of knowledge in man and nature. What emerges is not as yet a well-defined research program, but an open possibility, which I shall interpret and develop further in the context of this book.³

15.1.2

Behavior can be defined succinctly as the structural changes an organism can undergo while maintaining its autopoiesis.⁴ The nervous system allows the organism a varied and rich range of possible behaviors.

In studying an animal's behavior, one is struck by the fact that what we may describe as past interactions determine the conduct in the present as if, embodied in modifications of the nervous system, they acted as causal components that generate behavior. It seems that this basic observation is the fundamental source of the view that the nervous system must have adequate stored representations, whether genetic or ontogenetic (i.e., whether innate or learned). Hence the analogies with computer notions of storage and processing. From the point of view discussed here, this interpretation is unnecessary, for it confuses *two* phenomenal domains: what pertains to the system as a unit, and what pertains to the history of structural coupling. These two domains require different explanations: one operational, one symbolic. By keeping this distinction in mind, the function of the nervous system can be analyzed without contradiction in terms of (1) its *operation*, which is always in the present, as in every deterministic system (and as every neurophysiologist knows), and (2) its *performance*, which we, as observer-community, need to summarize in terms of pathways of structural coupling, e.g., recall, learning, and representation. Let us now rephrase the organization of the nervous system in these terms.

The nervous system is a network of interacting neurons coupled in three ways to the organism of which it is a component:

1. The organism, including the nervous system, provides the physical and biochemical environment for the autopoiesis of the neurons as well as for all other cells, and hence is a possible

³*Eds.* Varela was first and foremost a neurobiologist, and he continued to develop the research program announced here throughout his life, particularly in his work on large-scale neural integration and synchronous neuronal oscillations in the 1990s. See Varela (1995), and Rodríguez et al. (1999); Varela et al. (2001), and Thompson and Varela (2001).

⁴*Eds.* As Varela later recognized, this is an unsatisfactory definition of "behavior" since it does not offer any way of discriminating structural changes in the organism that relate to its agency (e.g., locomotion), from changes simply undergone by the organism (e.g., getting wet in the rain). The reason for this overly broad definition of "behavior" stems from the separation between operational and symbolic discourse, behavior being commonly understood in teleological terms. In later work, Varela sought to bridge this gap and provide a more specific account of behavior, or sense-making, that would also be operational (Varela, 1991b, 1997; Weber and Varela, 2002). To achieve this operational constraint, however, it was necessary to introduce the system-theoretic concept of *adaptivity* (Di Paolo, 2005).

source of physical and biochemical perturbations that may alter the properties of the neurons and thus lead to coupling 2 or 3.

2. There are states of the organism (physical and biochemical) that change the state of activity of the nervous system as a whole by acting upon the receptor surfaces of some of its component neurons, and thus lead to coupling 3.
3. There are states of the nervous system that change the state of the organism (physical or biochemical) and lead recursively to couplings 1 and 2.

Through this coupling the nervous system participates in the generation of the autopoietic relations that define the organism they integrate, and accordingly its structure is subordinated to this participation.

15.1.3

Neurons determine their own boundaries through their autopoiesis, and they are the anatomical units of the nervous system. There are many classes of neurons that can be distinguished by their shapes, but all of them, regardless of the morphological class to which they belong, have branches that put them in direct or indirect operational relations with other otherwise separated neurons.

Functionally (that is, viewed as an allopoietic component of the nervous system), a neuron has a collector surface, a conducting element, and an effector surface, whose relative positions, shapes, and extensions are different in different classes of neurons. The collector surface is that part of the surface of a neuron where it receives afferent influences (synaptic or not) from the effector surfaces of other neurons or its own. The effector surface of a neuron is that part of its surface which either directly (by means of synaptic contacts) or indirectly (through its synaptic or nonsynaptic action on other kinds of cells) affects the collector surface of other neurons or its own. Depending on its kind, a neuron may have its collector and effector surfaces wholly or partly separated by a conducting element (absence or presence of presynaptic inhibition), or it may have both collector and effector surfaces completely interfaced, with no conducting element between them (as, for example, with amacrine cells in the retina).

The interactions between collector and effector surfaces may be excitatory or inhibitory according to the kinds of neuron involved. Excitatory afferent influences cause a change in the state of activity of the collector surface of the receiving neuron, which may lead

to a change in the state of activity of its effector surface, while the inhibitory influences impinging on it “shunt off” the effect of the afferent influences on its receptor surface so that this effect does not reach its effector surface at all, or reaches it with reduced effectiveness.

Operationally, the state of activity of a neuron, characterized by the state of activity of its effector surface, is determined by both its internal structure (membrane properties, relative thickness of branches, and in general all structural relations that determine its possible states) and the afferent influences impinging on its receptor surface. Conversely, the effectiveness of a neuron in changing the state of activity of other neurons depends both on the internal structure of the latter, and on the relative effectiveness of its action on their receptor surfaces with respect to the other afferent influences that these neurons receive. This is so because excitatory and inhibitory influences do not add linearly in the determination of the state of activity of a neuron, but rather have effects that depend on the relative position of their points of action with respect to each other and with respect to the effector surface of the receiving cell. Furthermore, the internal structure of a neuron changes through its life, both as a result of its autonomous genetic determinations and as a result of the circumstances of its operation during the ontogeny of the organism. Thus, neurons are not static entities whose properties remain invariant; on the contrary, they change.

This has three general consequences: (1) There are many configurations of afferent (input) influences on the receptor surface of a neuron that produce the same configuration of efferent (output) activity at its effector surface. (2) Changes in the internal structure of a neuron (whether they arise from the autonomous transformation of the cell, or from its history of interactions in the neuronal network), by changing the domain of states of activity that the neuron can adopt, change its domain of input-output relations, that is, change its transfer function. (3) No single cell or class of cells can alone determine the properties of the neural network that it integrates.

Generally then, the organization of a neuron and its role in the neuronal network that it integrates is not invariant, but changes throughout its ontogeny in a manner subordinated to the ontogeny of the organism, because it is both caused and a cause of the changes that the neuronal network and the organism undergo.

From the descriptive point of view it is possible to say that the properties of the neurons—their internal structure, shape, and

relative position—determine the connectivity of the nervous system and constitute it as a dynamic network of neuronal interactions. This *connectivity* (that is, the anatomical and operational relations that hold between the neurons that constitute the nervous system as a network of lateral, parallel, sequential, and recursive inhibitory and excitatory interactions) determines its domain of possible dynamic states.⁵

Since the properties of the neurons change throughout the ontogeny of the organism, both as a result of their internal determination and as a result of their interactions as components of the nervous system, the connectivity of the nervous system changes along the ontogeny of the organism in a manner recursively subordinated to this ontogeny. Furthermore, since the ontogeny of the organism is the history of its autopoiesis, the connectivity of the nervous system, through the neurons that constitute it, is dynamically subordinated to the autopoiesis of the organism that it integrates.

15.1.4

The foregoing description rephrases standard knowledge about the nervous system—so much so that I shall not substantiate it in any detail (see, e.g., Kuffler and Nicholls, 1976). The question is: What does this all mean operationally? We shall say that, operationally, the nervous system is a *closed network* of interacting neurons such that a change of activity in a neuron always leads to a change of activity in other neurons, either directly through synaptic action, or indirectly through the participation of some physical or chemical intervening element (Maturana, 1970). Therefore, the organization of the nervous system as a finite neuronal network is defined by relations of circularity in the neuronal interactions generated in the network.⁶

Sensory and effector neurons, as they would be described by an observer who beholds an organism in an environment, are not an exception to this, because all sensory activity in an organism leads to activity in its effector surfaces, and all effector activity in its leads to changes in its sensory surfaces.⁷ That at this point an observer should see environmental elements intervening between the effector and the sensory surfaces of the organism is irrelevant, because the nervous system can be defined as a network of neuronal interactions in terms of the interactions of its component neurons, regardless of intervening elements. Therefore, as long as the neural network closes on itself, its phenomenology is the phenomenology of a

⁵Eds. These neurodynamic states would be the focus of Varela's later work. He investigated the hypothesis that transient synchronization in neural cell assemblies (distributed sets of neurons with strong reciprocal connections) underlay the emergence of cognitive acts, such as the moment of perceptual awareness (Varela, 1995). Varela and his team produced empirical verification of these ideas in a series of innovative studies toward the end of his life (Rodriguez et al., 1999; Varela et al., 2001; Le Van Quyen, 2003).

⁶Eds. In later work, Varela provides the following definition of its closure in terms of recursivity in neural activity, which, he insists, must not be understood as lack of interaction with the rest of the body and the environment: "The nervous system is organized by the operational closure of a network of reciprocally related modular subnetworks giving rise to ensembles of coherent activity such that: (i) they continuously mediate invariant patterns of sensory-motor correlation of the sensory and effector surfaces; and (ii) they give rise to a behavior for the total organism as a mobile unit in space" (Varela, 1997, 82).

⁷Eds. On this point, as well as on the relativity of the domain of interactions to the organism, we can see similarities to Jakob von Uexküll's notions of the functional circle and the *Umwelt*, respectively (von Uexküll, 2013). See also Thompson (2007, 153).

⁸*Eds.* The environment is, of course, not irrelevant. What matters here is that there is a dependence between sensory and effector activity and between sensorimotor correlations and the activity of ensembles of interneurons. In fact, a “double” closure is involved—one related to the operational recursivity of the activity of neurons and a sensorimotor closure that modulates these recursive neural dynamics (see figure 3 in Varela, 1997, 82).

⁹*Eds.* The notion of perturbation is equivalent to the one used in dynamical systems theory when two systems are said to be coupled. Randall Beer, an early defender of the use of dynamical ideas in cognitive science, adopted the perturbation perspective in his work on minimally cognitive system, in contrast to the input-output perspective: “It is therefore perhaps most accurate to view an agent and its environment as mutual sources of perturbation, with each system continuously influencing the other’s potential for subsequent interaction” (Beer, 1995, 182).

¹⁰*Eds.* And, implicitly, their interrelations.

¹¹*Eds.* Here and in other parts of the book, the phrase “subordinated to autopoiesis” should be understood in terms of the broad organizational constraints placed by the conditions of self-production and self-distinction of the organism on any of its systems. It does not mean that the nervous system’s activity is determined by this broad constraint. In fact, it has become increasingly clear that the realization of autopoiesis is enabled by neural activity, through either its effects on internal regulation or the regulation of behavior. See e.g., Bechtel and Bich (2023); Bich and Bechtel (2022).

closed system in which neuronal activity always leads to neuronal activity.⁸

This is so even though the environment can *perturb* the nervous system and change its status by coupling to it as an independent agent at any neuronal receptor surface. The changes that the nervous system’s structure can undergo without disintegration (loss of defining relations as a closed neuronal network), as a result of these or any other perturbation, are fully specified by its connectivity, and the perturbing agent only constitutes a historical determinant for the concurrence of these changes.⁹

As a closed neuronal network the nervous system has no input or output, and there is no intrinsic feature in its organization that would allow it to discriminate through the dynamics of its changes of state between possible internal or external causes for these changes of state. This has two fundamental consequences: (1) The phenomenology of the changes of state of the nervous system is exclusively the phenomenology of the changes of state of a closed neuronal network; that is, for the nervous system as a neuronal network there is no inside or outside. (2) The distinction between internal and external causes in the origin of the changes of state of the nervous system can only be made by an observer that beholds the organism (the nervous system) as a unity, and defines its inside and outside by specifying its boundaries.

It follows that it is only with respect to the domain of interactions of the organism as a unity that the changes of state of the nervous system may have an internal or an external origin, and hence that the history of the causes of the changes of state of the nervous system lies in a different phenomenal domain than the changes of state themselves.

15.2 Change and Structural Coupling

15.2.1

Any change in the structure of the nervous system arises from a change in the properties of its component neurons.¹⁰ What change in fact takes place, whether morphological or biochemical or both, is irrelevant for the present discussion. The significant point is that these changes arise in the coupling of the nervous system and the organism through their operation subordinated to the autopoiesis of the latter.¹¹ Some of the changes affect the operation of the nervous system directly because they take place through its working as a closed network; others affect it indirectly because they take place

through the biochemical or genetic coupling of the neurons to the organism and change the properties of the neurons in a manner unrelated to the actual working of the network.

The results are twofold: On the one hand, all changes lead to the same thing, that is, changes in the domain of possible states of the nervous system; on the other hand, the nervous system is coupled to the organism both in its domain of interactions and in its domain of internal transformations.

15.2.2

The connectivity of the nervous system is determined by the shapes of its component neurons. Accordingly, every nervous system has a definite architecture determined by the kinds and the numbers of the different kinds of neurons that compose it; therefore, members of the same species have nervous systems with similar architectures to the extent that they have similar kinds and numbers of neurons. Conversely, members of different species have nervous systems with different architectures according to their specific differences in neuronal composition. Therefore, the closed organization of the nervous system is realized in different species in different manners that have been determined through evolution. In all cases, however, certain general conditions are satisfied.

First, due to its constitution as a network of lateral, parallel, sequential, and recursive interactions, the nervous system closes on itself at all levels, and therefore the mutilations that it may suffer generally leave it a closed neuronal network with a changed structure. Accordingly, the organization of the nervous system is essentially invariant under mutilations, while its domain of possible states, which depends on its structure (and hence on its architecture) is not. Yet, due to its closed organization, whatever is left of the neural network after a partial ablation necessarily operates as a different whole with different properties than the original, but not as a system from which some properties have been selectively subtracted.

Second, there is intrinsically no possibility of operational localization in the nervous system, in the sense that no part of it can be deemed responsible for its operation as a closed network, or for the properties that an observer can detect in its operation as a unity. However, since every nervous system has a definite architecture, every localized lesion in it necessarily produces a specific disconnection (in the sense of Geschwind, 1965) between its parts, and hence a specific change in its domain of possible states.¹²

¹²*Eds.* This paragraph and the second one in section 15.2.3 both reconcile and bypass the classical debate in neuroscience between the theory of localization, according to which discrete areas of the brain have specific cognitive functions, and the theory of equipotentiality, according to which all parts of the brain are equally active and play an equal role in generating behavior. On the one hand, the nervous system always operates as a whole due to its global organizational closure (contrary to localizationism); on the other hand, it has a definite and differentiated architecture, so specific lesions have specific disruptive effects (contrary to the equipotentiality viewpoint).

Third, the architecture of the nervous system is not static, but becomes specified throughout the ontogeny of the organism to which it belongs, and its determination, although under genetic control, is bound to the morphogenesis of the whole organism. This has two implications: The variability in the architecture of the nervous system among the members of a species is determined by individual differences in genetic constitution and ontogeny, and the range of permissible individual variations (compatible with the autopoiesis) is determined by the circumstances in which the autopoiesis of the organism is realized.

Finally, the architecture of the nervous system and the morphology of the organism as a whole define the domain in which the environment can possibly couple to the organism as a source of its deformations.¹³ Thus, as long as the architecture of the nervous system and the morphology of the organism remain invariant, or as long as there are aspects of them that remain unchanged, there is the possibility of recurrent perturbations as recurrent configurations of the environment that couple in the same way to the nervous system and the organism.

¹³ *Eds.* Note that two deeply entangled autonomies, the organism as a whole and the nervous system, play a part of specifying what environmental processes will affect the organism as perturbations, without loss of closure or viability in either case.

15.2.3

Due to its coupling with the organism, the nervous system necessarily participates in the generation of the relations that constitute the organism as an autopoietic unity. Also due to this coupling, the structure of the nervous system is necessarily continuously determined and realized through the generation of neuronal relations internally defined with respect to the nervous system itself. As a consequence, the nervous system necessarily operates as an autonomous system that maintains invariant the relations that define its participation in the autopoiesis of the organism, and does so by generating neuronal relations that are historically determined throughout the ontogeny of the organism through its participation in this ontogeny.

Thus, the changes that the nervous system undergoes as a closed system, while compensating the perturbation that it receives as a result of the interactions of the organism, cannot be localized at any single point in the nervous system, but must be distributed over it in a nonrandom manner, because any localized change is itself a source of additional perturbations that must be compensated with further changes. This process is potentially endless.¹⁴ As a result, the operation of the nervous system as a component of the organism is a continuous generation of significant neuronal relations, and all the

¹⁴ *Eds.* This observation, entailed by the idea of closure, resonates with empirical studies on the brain's so-called default network. In a wakeful state, but at rest without performing an external task, large-scale neural patterns remain active in the human brain, a finding with interesting cognitive and clinical implications (for reviews, see Raichle, 2015; Buckner and DiNicola, 2019).

transformations that it may undergo as a closed neuronal network are subordinated to this. If, as a result of a perturbation, the nervous system fails in the generation of the significant neuronal relations for its participation in the autopoiesis of the organism, the organism disintegrates.

Although organism and nervous system are closed, invariant systems in their organization, the fact that the structure of the nervous system is determined through its participation in the ontogeny of the organism makes this structure a function of the circumstances that determine this ontogeny, that is, of the history of interactions of the organism as well as of its genetic determination. Therefore, the domain of the possible states that the nervous system can adopt as a closed system is at any moment a function of this history of interactions, and implies it. The result is the structural coupling of two constitutively different phenomenologies: the phenomenology of the nervous system (and the organism) as a closed system, and the phenomenology of the environment (including the organism and the nervous system) as a different system. Those two are braided together in a manner such that the domain of the possible states of the nervous system can be seen continuously as commensurate with the domain of the possible states of the environment through constraints that may take many forms. Furthermore, since all states of the nervous system are internal states, and the nervous system cannot make a distinction, in its process of transformation, between its internally and externally generated changes, the nervous system is bound to couple its history of transformations to the history of its internally determined changes of state just as much as to the history of its externally triggered changes of state. Thus, the transformations that the nervous system undergoes during its operation are a constitutive part of its environment.

The historical coupling of the nervous system to the structure of its environment, however, is apparent only in the domain of observation, not in the domain of operation of the nervous system, which remains a closed system in which all states are equivalent to the extent that they all lead to the generation of the relations that define its participation in the autopoiesis of the organism. The observer can see that a given change in the structure of the nervous system arises as a result of a given interaction of the organism, and he can consider this change as a representation of the circumstances of the interaction. This representation as a phenomenon, however, exists only as a symbolic explanation, and has validity only in the domain generated in the observer-community as he maps the environment onto

¹⁵ *Eds.* This paragraph states the critique of representational thinking that was formative for the enactive approach: claiming that cognition depends on the brain containing an internal representation of the outside world confuses the operational domain of brain activity with the domain of the observer. For a similar critique of representational thinking in neuroscience, see Freeman (1999). Freeman's work, particularly Freeman (1975), cited in this chapter, was influential for Varela.

¹⁶ *Eds.* Similar points have been made elsewhere in the book (cf. chapter 9). However, it may seem as if consigning notions of adaptation, learning, memory, and others to the domain of symbolic discourse is a way of declaring the condition of closure as sufficient for explaining how these phenomena arise operationally. This is not so. Adaptation, as the conservation of a relation with the environment whereby closure is not broken, is a (tauto)logical implication of the conservation of closure. But adaptive phenomena have further specificity (goal orientation, learning curves, constraints, interference, etc.), which demand operational explanations beyond the mere fact of closure, such as those provided by notions of sensorimotor equilibration (e.g., Di Paolo et al., 2017) and the regular patterns of neural plasticity.

the behaviors of the organism by treating it as an allopoeitic system. But the referred change in structure of the nervous system constitutes a change in the domain of its possible states under conditions in which the representation of the causative circumstances does *not* enter as a component.¹⁵

15.2.4

If the connectivity structure of the nervous system changes as a result of some interactions of the organism, then the domain of the possible states that it (and the organism) can henceforth adopt also changes; as a consequence, when the same or similar conditions of interaction recur, the dynamic states of the nervous system and, therefore, the way the organisms attains autopoiesis are necessarily different from what they would have otherwise been. Yet, that the conduct of the organism under the recurrent (or new) conditions of interaction should be autopoietic, and hence *appear adaptive* to an observer, is a *necessary* outcome of the continuous closure of both the nervous system and the organism. Since this self-regulatory operation continuously subordinates the nervous system and the organism to the latter's autopoiesis in an internally determined manner, no change of connectivity in the nervous system can participate in the generation of behavior as a representation of the past interactions of the organism: Representations do not belong to the domain of generations of the nervous network. The change in the domain of the possible states that the nervous system can adopt, which takes place throughout the ontogeny of the organism as a result of its interactions, constitutes *learning*. Thus, learning, as a phenomenon of transformation of the nervous system associated with a behavioral change that takes place under maintained autopoiesis, results from the continuous structural coupling of the (determined) phenomenology of the nervous system and the (determined) phenomenology of the environment. The notions of acquisition of representations of the environment or of acquisition of information about the environment in relation to learning do not represent any aspect of the operation of the nervous system. The same applies to notions such as memory and recall, which are descriptions made by the observer of phenomena that take place in her domain of observation, and *not* in the domain of operation of the nervous system; hence they have validity only as symbolic explanations, in which function they are well defined and useful.¹⁶

15.3 Perception and Invariances

I want to delve in more detail into sensory perception, from the perspective of the nervous system as an autonomous system. I find sensory perception a particularly relevant case to look at in more detail because it represents the most obvious source for a common-sense disqualification of the view of nervous performance presented above. It is because of sensory data that we take it for granted that the world presents us with information that must be modeled in the brain. Yet, at the same time, it is also in sensory perception that we find the best examples to show that the way things appear is best understood as a reflection of the structure of the nervous system. Thus sensory perception has accommodated the arguments of the stark realists and naive empiricists of the past two centuries, as well as the outspoken skeptics, from Sextus Empiricus on.

The basic position taken here can be stated thus: Sensory perception cannot be understood as an input process, whereby a stimulus causes an effect in a complex process beyond sense organs. Perception and action cannot be separated, since perception is an expression of the closure of the nervous system.¹⁷ In positive terms, *perception is equivalent to the construction of invariances through a sensory-motor coupling* by means of which the organism becomes viable in its environment. The environmental noise becomes objects through the nervous-system closure.¹⁸

It seems, in fact, amazing that this view of perception has to be stated explicitly, vis-à-vis the dominant view of perception, where “features” can be processed by stages ending with a motor output. Over and over the intrinsic correspondence and interdependence between perception and action has been hinted at—in recent years very explicitly by researchers such as (Held, 1965) and Erich von Holst (1973). In 1950 von Holst wrote a paper (with Horst Mittelstaedt) entitled “Das Reafferenzprinzip”:

The characteristic feature of this new conceptual framework is a rotation of the point of attack through 180°. Rather than asking about the relationship between a given afference and the evoked efference (i.e., about the reflex), we set out in the opposite direction from the efference, asking: What happens in the CNS with the afference (referred to as the “reafference”) which is evoked through the effectors and receptors by the efference? (von Holst, 1973, 141)

von Holst goes on to discuss a variety of cases, from flies to fishes, in which this insight leads to a novel and richer understanding of

¹⁷ Hanson puts it neatly: “People, not their eyes, see. Cameras and eye-balls are blind” (1958, 6).

¹⁸ *Eds.* In Varela’s later enactive work, this conception of perception is reformulated so that perception is equivalent to the bringing forth of invariances (perceptual space, time, objects, object qualities, and affordances) in perceptually guided activity through sensorimotor coupling (Varela et al., 1991). Color perception is used as a study case to illustrate the idea (Thompson et al., 1992). The idea that perception consists in sensorimotor invariances and knowledge of sensorimotor regularities is also central to the sensorimotor theory of perception (O’Regan and Noë, 2001; Noë, 2004).

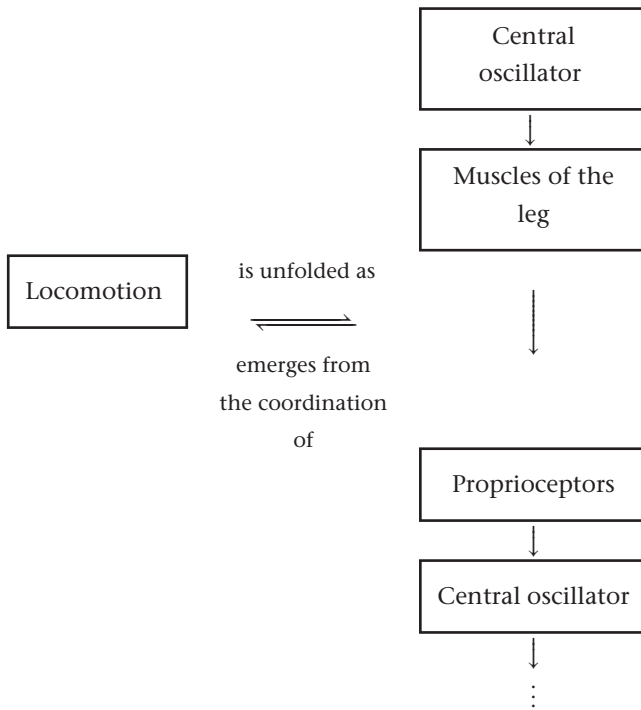
animal behavior. The basic idea that von Holst is introducing quite explicitly (although it has been used implicitly many times) is that of a sensory-motor synapse, a coupling of effector and receptor surfaces as a constitutive feature of the nervous system. This coupling can be seen as occurring at three levels: (1) an internal coupling (as between neurohumoral receptor-effectors), (2) an external direct coupling (as in the spindle-muscle system), and (3) an external indirect coupling via the whole organism (as when a motor action leads to a visual change).¹⁹

¹⁹ *Eds.* Two sources of empirical evidence for the role that these couplings play in stable perception come from experiments in sensorimotor disruptions of the visual field (e.g., Kohler, 1964) and experiments in sensory substitution (e.g., Bach-y-Rita, 2004; Auvray and Harris, 2014). In both cases, coupling (3) is disrupted and stable perception is recovered only through the active formation of new sensorimotor invariants and normally cannot be relearned passively (Bermejo et al., 2020).

Consider, by way of illustration, an animal walking. Let us only be concerned with a lower center that initiates the rhythmic locomotor pattern responsible for the alternate contraction of extensor and flexor muscles. This lower center could be a spinal segment in a mammal, or a thoracic ganglion in an insect (Grillner, 1975). Through such an intrinsic rhythm generator, the animal initiates a step cycle, which is repeated until a change in the animal's state brings the oscillator to a halt. At each phase of the step, the leg movement will activate specific sets of sense organs on the leg—proprioceptors, such as golgi tendon organs or campaniform sensilla—which will produce an afferent traffic of nerve impulses towards the central oscillator. This closes a loop: Efference from the oscillator center causes an afference from the sense organs, which modifies the specific parameters of the oscillator, and so on.

This locomotor behavior can be studied from two different angles. On the one hand we may consider locomotion itself, as a coherent behavior whereby the animal moves from place to place. On the other hand, we can see this global behavior as being produced by the recursive processes outlined above: a sensory-motor loop. These two viewpoints are not, of course, mutually exclusive; in fact, neurophysiologists shift from one to the other in the descriptions (e.g., Pearson, 1976). But it is not clear exactly *how* are they related, so that the global behavior can be explicitly formulated as resulting from the coordination and cooperation of the component operations—in the example at hand, how locomotion is analyzable as a sequence of sensory-motor couplings, and how the specific components of the step cycle actually become coherent. In a diagram:

The situation is, in more general terms, expressed in the description of a network in terms of its unfolding as an infinite tree. We wish to propose now the idea that the behavior exhibited by a network as a whole can be analyzed in terms of the unfolding of its component processes, in such a way that the global behavior can be constructed



from the eigenbehavior of recursive operations. There is no room, in this interpretation, for any opposition between the role of central patterns (G. Brown's proposal in 1911) and sensory afference (C. Sherrington's in 1910).²⁰ Both participate in the coherent generation of an eigenbehavior, which will be continuously modified through perturbations, endogenous and exogenous.

²⁰ *Eds.* See Brown (1911) and Sherrington (1910).

What is interesting here is that the behavior (locomotion), with all its attributes, is clearly distinct from the operation of the nervous system. For the latter, there are only invariances to maintain, and this is accomplished both through fixed circuitry and through new structural changes. For the observer, there are invariances in the structural coupling between nervous system and environment, which he describes as objects for the animal: rocks, walls, and so on. To assume that such objects are an *a priori* for the animal, so that it may "map" them into its nervous system, is unnecessary and in error. However useful such abbreviations of behavior may be ("the animal ran into a wall; it saw its food"), they cannot be taken as operational, but must be clearly seen as symbolic. They don't say much about the structure of the world; they say much about the animal's way of structuring its behavior.

An analogy, due to Maturana, clarifies this well:

"Let us consider what happens in instrumental flight. The pilot is isolated from the outside world; all he can do is manipulate the instruments of the plane according to a certain path of change in their readings. When the pilot comes out of the plane, however, his wife and friends embrace him with Joy and tell him: "What a wonderful landing you made; we were afraid, because of the heavy fog." But the pilot answers in surprise: "Flight? Landing? What do you mean? I did not fly or land; I only manipulated certain internal relations of the plane in order to obtain a particular sequence of readings in a set of instruments." All that took place in the plane was determined by the structure of the plane and the pilot, and was independent of the nature of the medium that produced the perturbations compensated for by the dynamics of states of the plane: flight and landing are irrelevant for the internal dynamics of the plane. However, from the point of view of the observer, the internal dynamics of the plane results in a flight only if in that respect the structure of the plane matches the structure of the medium; otherwise it does not, even if in the nonmatching medium the internal dynamics of states of the plane is indistinguishable from the internal dynamics of states the plane under observed flight" (Maturana, 1978, 42).²¹

²¹Eds. This analogy is problematic. It is meant to show that neural activity is simply the recursive result of changing patterns of neural activity, which is often modulated by the rest of the organism and the environment. It is not *about* such modulations. The problem is that to convey its point, the analogy uses a homuncular metaphor that is suggestive of the very image that it tries to debunk! Not only is neural activity cast in the role of an intentional agent (the pilot), but it is implicitly assumed to be fully in control of the autonomous organism (the flying and landing plane). This homuncular slippage easily invites internalistic and solipsistic interpretations of the approach, as Varela (2011) later recognized.

15.4 The Case of Size Constancy

15.4.1

This analysis may be carried out, of course, in a parallel fashion for human sensory perception. The perception of what we call the three-dimensional world is usually analyzed in terms of certain components (size, distance, depth, and others) that are deemed independent of us and proper to it. What these descriptive notions mean in terms of the functioning of the nervous system, however, is not at all clear.

Consider "size constancy," understood here as a causal disconnection between the perception of an object's size and its retinal image size. This phenomenon can be clearly seen by looking at one's hand at different distances, an observation that reveals that although the linear dimensions of the retinal image of an object change with distance, its perceived size remains relatively constant. Conversely, in the Emmert effect, the size of a postimage, viewed against a screen located closer or farther than its source, appears respectively diminished or enlarged in a proportion that would compensate for the change in size of the retinal image of an object moved between those distances, making its dimensions appear constant.²² Thus, in "size constancy" one has on the one hand a phenomenon that seems analyzable in terms of distance and size of the perceived object, and on the other hand a nervous process that occurs internally in the perceiving organism and that does not seem directly describable in

²²Eds. See Emmert (1881).

these terms. What is the relation between these two aspects of size constancy?

Traditionally its analysis has relied heavily on the first aspect, uncritically using the assumption that notions like distance and size reflect a direct grasping of features of the environment (Thouless, 1931; Anstis et al., 1961; Gregory, 1963, 1966). We may question this assumption and analyze “size constancy” as a neural process of invariance that takes place independently of any “features” of the environment, and show how this approach permits new observations (Maturana et al., 1972); see also (von Holst, 1973).

The most recent elaborate and complete theory of the perceptual effects of “size constancy” is that of Gregory (Anstis et al., 1961; Gregory, 1963, 1966). In discussing illusions such as the Ponzo illusion, he says:

If the constancy scaling tending to compensate for distance were triggered by perspective depth features, then we should expect the observed distortions in the illusion figure ... suggesting that the distortions are produced by constancy scaling when this is misapplied. Since the illusion figures are in fact flat, we can easily see that if the perspective features do set the constancy, it must be inappropriate. (Gregory, 1966, 154).

The basic assumption in Gregory’s theory is that distance is a feature of the environment to be grasped by the perceptual system. This notion provides the linkage between illusions and “size constancy” through the existence of depth features and perspective cues, which, if seen, would constitute a perception of distance and determine the application of the size correction by Emmert’s law. However, we can point to the following evidence for the independent origin of distortions arising in illusions and of size changes arising in the Emmert effect:

1. Consider the Ponzo illusion (figure 15.1). According to Gregory, perspective relations suggest a depth whose perception leads to a misapplied “size constancy” compensation. However, if we transform the perspective context as in figures 15.2 and 15.3, the illusion is maintained, although the depth effect disappears.
2. According to Gregory, in the paper-drawn Necker cube the faces appear equal in size because geometric depth cues are suppressed by distance perception associated with the texture of the paper. This is not so: It suffices to enlarge the size of the drawing (to 20 cm or more) to obtain a noticeable change in the cube’s faces upon inversion, although texture remains equally visible. It is as if the distortion effect increased in a nonlinear fashion with the cube’s dimensions.

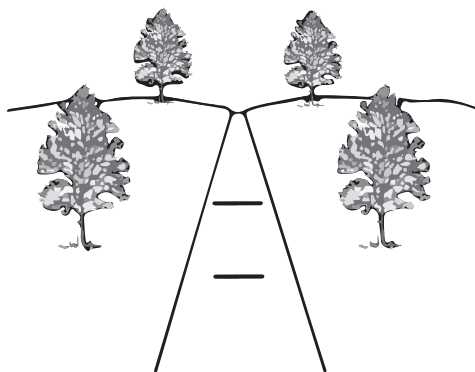


Figure 15.1
Ponzo illusion, depth



Figure 15.2
Ponzo illusion, horizontal

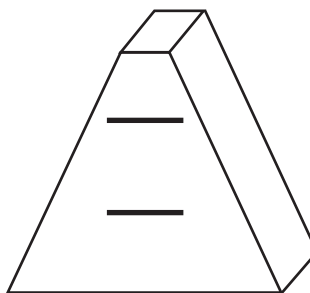


Figure 15.3
Ponzo illusion, vertical

Clearly the notion of perceiving distance is not enough to explain this distortion. We can only say that there are relations in the seen image that bring into play the so-called size distortion.

3. If we obtain a postimage and look through a diverging glass at an object lying at the same distance as the source of the postimage, then although the whole object is reduced in size and thus appears farther away, the size of the postimage is slightly reduced.

Depth features have changed, but "size constancy" would be working in a direction opposite to that expected according to Gregory's theory. There is, then, independence between the effects of what Gregory calls depth perspective cues and "size constancy." Therefore we claim that:

- a. Size illusions such as the Ponzo illusion do not arise as Gregory assumes from the perception of depth and perspective and the application of Emmert's law, but depend on relations present in the visual image that are not contained in the description of depth or perspective.
- b. Apparent changes in size obtained with geometric figures such as the Ponzo illusion are independent of changes in size introduced by "size constancy," insofar as they can be independently produced. In fact, there is no justification in lumping them both in the same category, although it is certainly attractive to believe that these phenomena are related, in view of their apparent similarity. Not only is there no evidence that this should necessarily be so, but, on the contrary, we have shown them to be separable.

I shall not enter into the study of the class of relations that can introduce distortions in the perceived images, and the possible mechanisms through which this effect might be produced. The discussion will be centered on the problem of "size constancy."

What is "size constancy" as a nervous process? To answer this question let us turn to the following evidence.

4. There is a circumstance under which the phenomenon of "size constancy" disappears completely: when the object is viewed through a pinhole. In this situation there is no change in the size of the postimage, neither is there a compensation of the object size with varying distance. Since the effect of a pinhole is to produce infinite depth of focus, this evidence strongly suggests that "size constancy" is related to accommodation. Can we dissociate focusing and visual context to support this suggestion?
5. If after a postimage is obtained the eyes are closed, and while they are closed an effort is made to look at the tip of the nose, then the Emmert effect appears and the postimage shrinks.

Thus with no visual context, Emmert's law can be brought into play. We next look for a dissociation between accommodation and convergence.

6. Using diverging prisms that modify convergence but not accommodation, the size of the postimage remains unchanged.

We are thus led to the conclusion that accommodation is directly related to “size constancy:” The more the accommodation, the more the reduction in size. We can now understand the experiment presented in evidence 3. The accommodation effort goes in the opposite direction of the reduction in size (apparent distance) due to the diverging lens; thus the size of the postimage is decreased.

7. If one has several identical objects at different distances on a visual line, they appear to differ in size, depending on the exact amount of accommodation; while if a single one is moved back and forth, it appears to be constant in size.

Thus the degree of size-change compensation is uniformly applied to the field according to accommodation, and it affects the appearance of all objects in it. The question then arises whether the peripheral effect of accommodation is the significant parameter for size constancy.

8. The effect of the ciliary muscles can be suppressed by atropinization. In this circumstance the Emmert effect remains equally effective.

The correlation that one is led to establish, then, is between the “size constancy” effect and the neural components in accommodation, i.e., the activity of the class of central neurons that control the contraction of the ciliary muscle.

In summary, then, “size constancy” is dependent on accommodation. Furthermore, evidence 8 points to the neural components of accommodation as the only possible correlation with the Emmert effect. This is significant because it reveals that the neural components of a motor event specify a perception.²³ What, then, is “size constancy” as a process? We have shown it to be a correlation between a sensory and a motor phenomenon, such that the state of neural activity that specifies the motor event serves to determine the perceptual effectiveness of the sensory process.

²³ *Eds.* This is even clearer in cases when motor events result in the observer’s self-movement (Combe and Wexler, 2010).

15.4.2

The phenomenon of size constancy is not, according to the preceding results, a function of an independent feature of the *seen* object or scene: it is a function of a given correlation of activity between what takes place in the visual centers and what takes place in the central nuclei that control accommodation. Distance as an independent parameter of the *visible* world does not count: We do

not see distance. The observer cannot obtain (nor expect to obtain) from her description of the changes of an autonomous system a characterization of the independent properties of the source of disturbances. In the case at hand, the reduction in apparent size of an object whose distance from the eye is increased should not, and in fact cannot, be interpreted as arising from grasping distance as a feature of the disturbing source. We must recognize that this effect corresponds to a process that takes place completely within the nervous system, independently of any feature of the environment, although it may be elicited by interactions of the organisms in its environment. How does the notion of distance come about if it is not obtained from the environment? From the previous discussion the answer is obvious: A perception is a process of compensatory changes that the nervous system undergoes in association with an interaction. Correspondingly, a perceptual space is a class of compensatory processes that the organism may undergo. Perception and perceptual spaces, then, do not reflect any feature of the environment, but reflect *the invariances of the anatomical and functional organization of the nervous system in its interactions*.²⁴

Let me say it once more: The question of how the observable behavior of an organism corresponds to environmental constraints cannot be answered by using the traditional notion of perception as a mechanism through which the organism obtains “information” about the environment. A perturbed organism undergoes structural changes that compensate for the perturbations; if the perturbation is repeated, the organism undergoes similar or different changes that compensate for it in the same or in a different manner. The changes that an organism undergoes in compensating for its perturbations may be considered by an observer as descriptive of the perturbing agent, because he establishes a correlation between the conduct that he beholds and the circumstances that he assumes give rise to it. The organism is a system that has its own organization as the fundamental parameter, which it maintains constant through the regulation of many others.²⁵ As an invariant system, the organism compensates deformations and retains its identity as long as it can do so. Thus, that it should display behavior appropriate to the restriction of the environment is to be expected. What requires additional explanation is the way the organism behaves as it does at any moment. Of course, this depends both on its ontogeny and on the evolutionary history of the species to which it belongs. What is significant in the context of the present discussion is that perception and perceptual spaces constitute *operational specifications* of an environment, not apprehensions of features in an

²⁴ *Eds.* This statement prefigures the enactive perspective on perception (Varela et al., 1991; Noë, 2004). However, later work puts more emphasis on the structural and historical aspects of the interaction with the environment, which plays a co-constitutive role in giving rise to perceptual experience, not merely an eliciting role.

²⁵ *Eds.* Interestingly, this statement suggests that Varela may have been thinking of maintaining autonomy as a process involving adaptive regulations, prefiguring the need for a concept of adaptivity (Di Paolo, 2005). This concept, however, is not entailed in the notion of closure as such, since we may conceive of cases of organizational closure in which such regulations are absent.

independent environment. An organism does not extract perceived distance as a characteristic feature of the environment, but generates it as a mode of behavior that is compatible with the environment through a process of invariant compensation of disturbances. Thus, unavoidably, the more plastic the structure of an organism, the more diversified modes of behavior it can generate that generate the environment.

We must realize at this stage of the argument that to view the nervous system as an autonomous system that operationally specifies an environment (a “reality”) is logically equivalent to saying that the system functions inductively: “What happened once will happen again.” That is, once we view the nervous system as autonomous and endowed with structural plasticity, the inevitable consequence is that, whatever the perturbations, these will become organized into a realm of invariances, an environment, held constant unless forced to change under the impact of new perturbations.

15.4.3

Yet another expression of a similar understanding of perception and the nervous system is due to William Powers, using a cybernetic terminology, briefly stated. Powers calls our attention to the fact that a feedback system provided with a given reference signal will compensate for disturbances only relative to the reference point, and not in any way reflect the texture of the disturbance. If we now transpose this homeostat analogy to sensory processes, where the reference signal is given by higher-level signals (such as command interneurons in the case of locomotion), then we immediately get to Powers’s conclusion that behavior is the control of perception. That is, “Behavior is the process by which organisms control their input sensory data” (Powers, 1973, xi). The feedback loop, of course, is only intended in Powers’s treatment as an schematic picture, whereby he builds a functional hierarchy. “The entire hierarchy is organized around a single concept: control by means of adjusting reference signals for first-order systems” (ibid, 78).^{26,27}

15.5 Piaget and Knowledge

If we care to carry this sort of analysis beyond the first stages of sensory perception, considered above, we come upon a view of perception very close to the Piagetian paradigm (see also Richards and von Glasersfeld, 1979). In fact, by iterating eigenfunctions at several levels of central cortical functions, we can envision the emergence of what he calls “permanent objects,” and further on “programs and

²⁶ An interesting collection of articles discussing the impact of this formulation by Powers can be found in *ASC Cybernetics Forum* (vol 8, no.3/4, 1976). In fact, most of the arguments adduced here could be given for the closure theory of the nervous system as well.

²⁷ *Eds.* Powers’s Perceptual Control Theory is in some ways a forerunner of predictive coding models (e.g., Hohwy, 2013) (without the cognitively loaded interpretations in terms of predictions or inferences). Both are often expressed through a similar hierarchical architecture. Powers’s perspective is a useful antidote to the relatively passive view of perception as information gathering, but it presents a view of action as overly subservient to the attaining of perceptual goals. For the enactive approach, rather, action and perception are co-defined moments of the organism-environment coupling (Varela et al., 1991; Di Paolo et al., 2017).

principles" (Piaget, 1937). As von Foerster (1977) had discussed, the Piagetian framework can be rephrased as the emergence of constancy (e.g., objects) as "tokens for [cognitive] eigenbehavior."

I am saying, then, that the Piagetian perspective of biological assimilation can be rephrased very naturally in the context presented here of autonomous systems and structural plasticity.²⁸ In fact, for Piaget's reasoning there are three fundamental concepts: structure, assimilation, and accommodation, three aspects of the fundamental notion of intelligence. Adaptation is, for Piaget (1963, 6), an equilibrium between assimilation and accommodation which the structure undergoes. This adaptation presupposes an underlying coherence, and this is the invariant side of structural change, its organization. "[O]rganization is inseparable from adaptation. They are two complementary processes of a single mechanism, the first being the internal aspect of the cycle of which adaptation constitutes the external aspect" (Piaget, 1963, 7).

²⁸ *Eds.* This has been done for the case of sensorimotor autonomy in chapter 4 of Di Paolo et al. (2017).

It is in this systemic framework that Piaget interprets the emergence of cognitive abilities and the construction of reality in the child (Piaget, 1974). His monumental work is too well known to need discussion here (see, e.g., Furth, 1969). What I want to highlight is the fact that, from the psychological point of view, Piaget was led to an understanding of cognitive operations that can be rephrased, without doing any violence to it, in terms of the *autonomous organization* of the nervous system. In fact, in his *Biology and Knowledge* (1967), Piaget makes it a point to show how cognitive structure and assimilation are functionally isomorphic to general biological phenomena. The view taken here is somewhat in the reverse direction: from the simplest living organization to more complex cognitive systems like the immune and nervous network; and we conclude that these mechanisms for operationally specifying an environment, for shaping a reality, seem to lead without discontinuity into the human realm. I shall have little else to say about the specific forms autonomous mechanisms take in the realm of human cognition; my purpose has been to state the connection. To expand on this area would take an entirely new book, in which we might do justice to the level of thinking initiated by Piaget (1967).

15.6 Interdependence in Neural Networks

15.6.1

There are a number of epistemological consequences of this view of nervous performance, which I will discuss in the next chapter.

There are also a number of consequences for the empirical study of brain function that are worth mentioning at this point. In fact, switching from the study of information pickup to behavior as emerging from invariances not only permits a rephrasing of the recorded empirical facts about the nervous system, but points to new questions.

In particular it seems essential to study the methods through which eigenbehavior of nervous functions can be characterized, beyond the simple level of primary sensory events. There is an evident gap between the level of study of (say) locomotion and Piagetian object invariance. That intermediate range concentrates not so much on single cells or local circuits as on aggregates of neuronal centers in cooperation and competition. For this level of analysis of brain macrostates, there are very few tools and only scanty strategies of representation (Freeman, 1975; John, 1967; Katchalsky et al., 1974).²⁹

²⁹ *Eds.* Varela's later work on large-scale intergration and neuronal synchrony was concerned precisely with developing models and tools for analyzing such brain macrostates (Varela, 1995; Varela et al., 2001). For more on the nonlinear dynamical tools developed by Varela and colleagues, see Le Van Quyen (2003).

The study by Kilmer, McCulloch, and Blum (1969) is a good example of this point of view. They proposed to view the brainstem reticular formation as a mechanism for switching referential states in the central nervous system (such as sleep, wakefulness, sexual behavior). They divide the complex anatomy of the reticular formation into modules in mutual interactions, and assign weight at each module to the distinct referential states. Further rules of module interaction allow the system to resolve the redundancy of potential command.

This model is really a prototype for a whole gamut of possible conceptual models to understand and represent nervous invariances generating interesting behavior (cf. also John, 1967, 1972; Montalvo, 1975; Amari and Arbib, 1977). It is here that the possible overlaps between this view of the nervous system and artificial intelligence can be relevant (e.g., Arbib, 1972). What cooperative computations can possibly mean for the nervous system is, I believe, one of the most important and unthought-of questions highlighted by the present theory.

Yet another concept that must be reconsidered in this light is that of neural representations and representational systems. As conceived of in the fields of engineering and design, a representational system can be defined as a structure whose states or components can be mapped onto, or made to symbolize, the components or states of another structure. Thus a representational system or representational space becomes the structure or manifold in which the correspondence is described. The internal representational systems

in a neural net are memories of “modes” of the world (MacKay, 1969; Szentágothai and Arbib, 1974; Thatcher, 1976).

It seems to me that the understanding of memory and learning in terms of the action and configuration of large populations of neurons with specific spatio-temporal patterns of discharge is a very good and useful one. What seems totally *unnecessary* for this view, and incorrect from the perspective presented here, is that such states correspond or map or model some external world. In fact, there is every indication that there is no uniqueness in the correspondence between a repertoire nervous-system configuration and ambient configurations as seen by an observer. As in the instrumental-flight analogy, all that exists for the nervous system is states or relative activity, and it is obviously necessary to learn how to quantify and specify those well. But this goal of electrophysiology in *no* way requires or presupposes the interpretation of such states as correspondence. They can be as well, and better, interpreted as eigenbehaviors arising out of non-uniquely-specifiable constraints from the structural coupling. Such eigenbehaviors are relevant, not because they correspond or not to ambient features, but because they *can* be produced at all in the dynamics of the nervous system, and they can be changed and modified under structural plasticity. That is, this interpretation sees the nervous system as carving invariances and perceptions out of its structure and organization, shaping an environment. The organization of the nervous system is such as to allow it to specify an order in the random perturbations impinging upon it.³⁰ From this vantage point, representations and representational systems could better be called presentations and presentational systems. Practically, this implies among other things, putting less emphasis on the strategy of looking for correspondences, and more on looking for viability and diversity in the stable patterns generated by a nervous net.

15.6.2

If a car undergoes a structural change because of bumping into a tree (the fender is twisted and the front tire scratched), we do not say that it remembers the accident by storing a memory, nor do we say that it learned from the event by changing its behavior via a representation of trees. These descriptions seem ludicrous because the car is so obviously a human-made artifact. Yet it seems that the same sort of mechanistic and operational description can be applied to highly plastic autonomous systems like the nervous system. Memory requires no record or storage, for it stands only for a

³⁰Eds. Varela later came to appreciate that the influence from the environment impinging on the nervous system is not random; rather, it is richly structured, and that the “carving out” or bringing forth of perceptual invariances happens in and through the nervous system’s coupling with such an environment (Varela et al., 1991). For example, as ecological psychologists emphasize, the ambient light has a rich topological structure, with statistical regularities available to the organism. Thus, in the case of color perception, colors as perceptual invariances emerge through the coupling between the animal’s nervous system and its environment (see Thompson et al., 1992).

history of structural coupling; learning requires no representation, for it stands for structural plasticity. Whatever the observer wants to see or needs to use in symbolic descriptions, such as storage, and representation as mapping, are *not* operational.

It seems astounding to me that the idea of correspondence between brain activity and ambient features has ever been taken seriously. To see a neuron as being responsible for a percept (e.g., Barlow, 1972) is straight vitalism or animism; It attributes to one component of a system all of the properties of a description. I have argued that what pertains to the nervous system is the synthesis of neuronal eigenbehaviors; what pertains to *us* is how we see in terms of *our* own perceptions the performance of a nervous system in its interactions. To assume any sort of mapping between these two distinct phenomenal domains is not only confusing levels of description but setting out in search of a model of a reality that is an always receding mirage.

Sources

(Maturana, 1970), (Maturana, 1978), (Maturana and Varela, 1975), (Maturana et al., 1972).

16 Epistemology Naturalized

16.1 Varieties of Cognitive Processes

16.1.1

The immune and the nervous systems are stupendous examples of the way in which the mechanisms that generate a system's identity, its closure, are commensurate with the mechanisms that generate its cognitive domain, its structural plasticity. In both these cases, invariance is maintained through uninterrupted structural change in such a way that the environment's perturbations can be seen, by an observer, as being *shaped* into a cognitive domain by the system's operation. But the phenomenology of these two systems, although extremely rich, is not unique. In fact, we may generalize from these cases to see that the specification of a cognitive domain is the propensity of *every* autonomous system endowed with structural plasticity.¹

¹*Eds.* See Varela (1991b, 1997), and Di Paolo et al. (2017, 2018).

The main intention of this chapter is to trace some of the implications of this idea more generally, and to combine them with many issues about human knowledge that have formed the scaffolding of the preceding discussion.

16.1.2

The argument is, at this point, straightforward. Let me summarize it briefly. A unity becomes specified through operations of distinction by an observer in a tradition—what we have been calling an observer-community. The distinctions that specify a unit are expressed in terms of necessary relations that hold between the components of a system, its organization. As long as these relations remain invariant, the unit maintains its membership in a particular class of systems. Autonomous systems are specified through their

specific organization (organizational closure). The invariances that correlate with our distinctions and identifications are based on such closure.

This has been one side of a two-part theme. The other side is structure: the specific components (and their properties) that enter into the constitution of a natural system. The fundamental importance of a system's structure is that it, and only it, specifies the texture of the space in which the system operates as a unit, because the component properties will determine which are the possible interactions the system can enter into. This does not characterize the system as a unit, but it does specify which perturbations from the environment enter into its domain of interactions. These interactions can only include those that do not drive the system beyond its organizational invariance. If they do, the system disintegrates or becomes a different one.

We have here another descriptive complementarity. The invariance that characterizes a system can only be studied under perturbations that reveal it, and thus against a background of change. Conversely, change in a system is only in reference to that in it which stays unchanged: its identity as a system. Invariance/change in this context becomes better expressed in the organization/structure duality, which is, once again, a very essential one. We have discussed the use and consequences of these notions for living systems and for autonomous systems in general, as well as ways to characterize organizational invariance. What I wish to do now is explore in more detail the invariance/change complementarity, thus construed as identity/cognition. How a system establishes its identity (its autonomy) correlates with how it generates information; the mechanisms of identity are interwoven with the mechanisms of knowledge.

16.1.3

We have decided to look at change as structural transformation under organizational invariance.² It follows that, although the perturbations from the environment operate through the structure, the consequences of such interactions can only be studied through the way in which the structure embodies an unchanged organization. In particular, autonomous systems will maintain their identity for as long as their closure is maintained. However they will exhibit a history of change as a unit (ontogeny) or a class (evolution) only if their structure can change in a way that does not interrupt closure. As we saw for the immune and nervous systems, the system's

²Eds. The current analysis applies to this particular reading of the invariance/change complementarity, where invariance is neatly consigned to a system's organization and change to its structure. This covers many aspects of the phenomenology of life and mind. But it misses the possibility of understanding *organizational change* (e.g., how processes come together to generate a new, organized system) or what might happen during transitions in organization/identity. Cryogenically preserved bacteria cease to realize their autopoietic organization while their structure remains invariant. Some have reportedly regained autopoiesis after thousands of years (Coghlan, 2009). The usual response is to say that in such cases, we no longer have the same system without querying exactly what happens in these transitions. A broader epistemological framework may be needed to break this hylo-morphic deadlock and look at these questions by capturing the invariants present in *structuring processes and principles of individuation* (e.g., Simondon, 2020).

environment is a constant source of perturbations that act as triggers for structural change.

Notice that the effect of considering a system's environment as a source of perturbations and compensations, rather than inputs and outputs, is far from trivial. In the latter, control-based formulation, interactions from the environment are instructive, constitute part of the definition of the system's organization, and determine the course of transformation. In the autonomy interpretation, the environment is seen as a source of perturbations independent of the definition of the system's organization, and hence intrinsically noninstructive; they can trigger, but not determine, the course of transformation. Accordingly, the outcome of a perturbation—a compensation—reflects the organization and structure of the system; the outcome of an input—an output—reflects a structure attributed both to the environment and to the internal operation of the system. In one case an input (partly) specifies the system's organization and structure; in the other case a perturbation participates in the transformation of an independently specified system. This amounts to saying that in one case we are characterizing a system as organizationally open, and in the other case as organizationally closed. Although complementary, these views are about *different* systems.

The formulation of a system in allonomous terms, where the environment is seen as (instructional) inputs, is peculiar and adequate to the domain of design. For natural systems, a central problem is precisely how ambient perturbations are dealt with by different systems and under change of structure. This is not a problem at all in the domain of design, precisely because an observer specifies, by its use, what the environment should be and how the system ought to perform in it. In natural systems, however, it seems that whatever regularities we see in the interactions of a system with its environment, since they cannot be the reflection of a designer, must be the way in which the identity of the unit can be maintained through structural change. It reflects nothing that the environment contributes to defining the unit as such, but rather the operation of the unit itself. The consequences of this gestalt switch are far from trivial to conceptualize specific systems, as we saw for the case of the cognitive abilities of the immune and nervous system, where the world of antigens and objects "out there" could be completely dismissed in favor of the construction of invariance out of the system's closure.³

³*Eds.* Varela later emphatically rejected the constructivism implied in statements such as this. In an interview, he said: "Classical constructivism does not at all impress me as a convincing mode of thought because it posits one side of the cognitive process as absolute: the organism forces its own logic and its own models on the world. I do not believe that to be the case at all" (Varela and Poerksen, 2006, 40). His enactive epistemology sought to look at bodies as themselves resulting from a process of co-definition with their environment. Retaining the criticism of objectivism derived from the distinction between open and closed systems and from traditions such as constructivism and pragmatism, this is only half the story for the enactive approach because "mind-independent objects are challenged, but object-independent minds never are" (Varela et al., 1991, 233).

The appropriate notion here is that of *structural coupling*. That is, if a system has a plastic structure, and its environment also exhibits a plastic or changing structure, then both environment and system will undergo a process of structural coupling, which the observer-community can describe in terms of trajectories of the system under observation and relate to regularities of the environment's structure. Usually this description of regularities is abbreviated in terms of symbolic descriptions: a peculiar coherence of a trajectory of structural plasticity is symbolized by the recurrent environmental feature that is coupled with it (e.g., a hormone will signal division in a hepatocyte). However, this is in no way an instructional feature of the environment, as this would simply imply a confusion or nondistinction between an operational and a symbolic explanation. Any relation an observer establishes between behavior and specific structural changes of a unit does not reflect a correspondence between system and environment but our capacity to distinguish such regularities. Since one phenomenal domain (behavior) cannot be reduced to another phenomenal domain (structural change), information, however we want to define it, is relative to the observer-community.

16.1.4

The domain of interaction that an autonomous system can enter through structural plasticity without loss of closure constitutes its cognitive domain. Every cognitive domain is adequate for an autonomous system; otherwise it would disintegrate. In fact, adaptation is, from this point of view, a truism (see also Gaines, 1972). Its usefulness as a notion is not operational but symbolic: It points to the kinds of regularities exhibited by structural coupling.⁴ When we say that a species adapts to a niche, we are saying no more or less than that in the sequence of reproductive steps, unities are capable of structural plasticity that results in their viable reproduction. Adaptation is like a frozen accident of the way in which structural plasticity occurred. This seems true in all natural systems we know of, from bacteria on. The uniqueness of the constraints in some simple molecular systems might be the limiting case of the plasticity of adaptation (Eigen, 1971). However, for most of the systems of any interest to us here the resources of structural plasticity are so vastly removed from any direct connection to the perturbation they undergo, that all of their adaptation can be stated in symbolic (or functional) terms. As Jacob (1977) has aptly remarked, adaptation in ontogeny and evolution resembles, not a process of designing, but one of tinkering. In the

⁴Eds. As noted in section 8.3.4, the related idea of *adaptivity*, introduced in Di Paolo (2005), is an operational concept accounting for regulation at the behavioral scale. Adaptivity distinguishes among different kinds of processes during structural coupling. Based on this concept, other ideas can be expressed operationally, such as degrees of breakdown, equilibration, sense-making, behavior, and agency.

end products of tinkering, one sees nothing except the style of tinkering. If this is to be more than a metaphor, it must say that we must relate cognitive actions solely to the identity mechanisms of a unit.

In various stages of evolution, subtle forms of plasticity have emerged, and hence richer cognitive domains, which enhance the unit's viability. We tend to consider the nervous system the upshot of this trend; but from the present point of view, the nervous system does not invent or create cognitive acts. It enlarges the possibilities available to an ongoing autonomy.⁵

16.1.5

I have not dwelt at all on the *necessary conditions* for structural plasticity in autonomous systems, and the host of modeling and representational problems raised by the explicit *relation* between invariance and plasticity. There is little available, in formal terms, to analyze this central problem. Only two main tools are available, namely computer simulation of networks (e.g., Walker and Gelfand, 1978), and, more importantly, the use of differentiable dynamics. A few remarks on the latter are in order.

The idea of self-organization as *order from fluctuations* (von Foerster, 1960), has come to be well known, especially through the perspective of generalized thermodynamics (Nicolis and Prigogine, 1977; Atlan, 1972; Eigen, 1971; Eigen and Schuster, 1978; Morowitz, 1968). More generally, this sort of behavior can be studied from the point of view of abstract dynamical systems, where unlimited complexity in their possible states can arise from endogenous or exogenous perturbations (Smale, 1967; Thom, 1972). I do not intend to expound these well-known ideas here.

As I discussed before (cf. sections 7.2.4, 13.10), I see these tools as one way in which properties of systems, autonomous *or* allonomous, can be expressed. Differentiable dynamics represents, in practice, the most workable framework in which these two points of view can actually coexist and be seen as complementary in an effective way. My argument has been, however, that it would be too limiting to take this framework as the *only* form of formal description, and that for the cases where differentiability and numerical evaluation are irrelevant, we find ourselves almost empty-handed. This is more often than not the case beyond the level of molecular systems and population biology. That is why it is necessary to generalize the classical notion of dynamical stability into a more general framework compatible with and explicitly containing the observer's participation, as partially attempted in part II.

⁵Eds. Niko Tinbergen (1963) proposed that the study of animal behavior breaks down into four kinds of questions: (1) the fitness function of the behavior, (2) its evolutionary history, (3) the mechanisms that cause it, and (4) its development. Varela suggests there is a different question: How does behavior relate to the autonomous organization and the structural processes of an organism's identity?

A very serious shortcoming of the present algebraic-algorithmic framework, as it now stands, is that it offers no way of reconciling invariance and structural change. In contrast, the notion of order from fluctuations is precisely and effectively captured in the differentiable format, and has rightly become very popular. There are two features, however, that are usually missed in these discussions and are worth noting here. First, no clear distinction is made between a system's organization and its environment, although the distinction is used *implicitly* in the way the variables are assumed to be interrelated (cf. section 10.2.2).⁶ As a result, whatever pattern of stability is observed, it is taken to be a reflection of the properties of the components, and not as something proper to the unit's organization that is buried in the apparently harmless interdependencies of the variables. Second, it is all too often forgotten that the pattern observed and the regularities distinguished, the "order," are relative to our observations and not intrinsic to the interacting units or their space. Noise and information are not structural, but relative to the way unit and environment are cut apart. This hardly needs exemplification⁷ any more.

I submit that the interlock between organizational closure and structural plasticity (cf. also Piaget, 1967) can be seen as a general mechanism, of which the order-from-noise principle is a highly *specific* case. I resist the idea of extending this already established nomenclature into cognitive or social events, not only because the use of the word "order" is highly misleading, but because this notion is based on very specific formalism, mostly from physico-chemical insights. It is unclear that we can generalize it to the emergence of invariances in autonomous systems at other levels. It is definitely essential to work towards an understanding of the relations between the unspecified or the random and the new or the emergent, but this needs a sound epistemological basis as to how these notions reflect the observer's perspective. The thermodynamic and Shannonian frameworks should serve as a first step allowing us to see where we are *not*.

16.2 In-formation

One persistent theme in these pages has been the idea that in the cognitive domain specified by the structural coupling of an autonomous system, an observer may distinguish a certain coherence or regular pathways. These regularities constitute *admissible symbolic* descriptions (cf. section 9.5.1). Any component interactions

⁶Eds. An exception to this is the work by Randall Beer on computational modeling of autopoiesis and operational closure, where organization, constitutive processes, and environment are explicitly distinguished (Beer, 2004, 2015, 2020b; Agmon et al., 2016).

⁷In the specific context of Shannon's theory of signals, this has been elegantly pointed out by Atlan (1981). His point is actually obvious once we see it. The transmission between source x (say, the environment) and receiver y (say, the organism) will be seen as increasing in ambiguity or in (uncertainty-reducing) information, depending on whether the observer is looking only at y or at both x and y ; In the latter case, the usual transmitted information $T(x, y) = H(y) - H(x/y)$ shows a reversal of sign: $H(x, y) - H(x) + H(y/x)$ (see Atlan, 1981, for a detailed discussion). He concludes: "The real source of uncertainty, which feeds the source of the channel, is none other than the observer" (Atlan, 1981, 195). This goes hand in hand with our insistence on distinguishing what we see as regularities in the operation of a system and the regularities we see between a unit and its environment. To cross this hierarchical boundary means crossing phenomenal domains.

thus defined as a symbol are generated by the coupling and can only be defined in reference to it. Thus, for an observer the system's functioning shapes its environment, carves a reality for itself out of an undifferentiated background of perturbations, in ways that depend only on the many and varied paths of structural coupling. It is this variety of possible alternatives that makes a natural symbol have its arbitrary quality.

Further, whenever the symbolic components of a cognitive domain can be seen as composable (i.e., have some sort of syntax), this corresponds to an immense evolutionary advantage for the units that generate them. The emergence of second-order autonomous systems is then possible, built on the interdependence of a symbolic domain. Metazoans and the varieties of animal social and ecological life can be thus described. I have emphasized *ad nauseam* that this way of looking at the cognitive, informational processes in natural systems, is quite distinct from the interpretation inherited from the computer gestalt.

This poses a problem. If we want to make apparent a difference in interpretation, it is quite difficult to use the same words and not be misled by the connotations that they have acquired in common and scientific parlance. The word "information" (like the word "order") has been so much associated with representational connotations that it would seem hopelessly lost for any other interpretation. On the other hand, I dislike the introduction of new nomenclature, which tends to cut one's expression from the mainstream of the literature and a possible dialogue. Between the Scylla of misunderstanding and the Charybdis of not talking or talking private talk, I have opted for the first. Thus throughout this book I have used words such as symbol and information, trying to be (perhaps irritatingly) insistent on the perspective from which I am speaking and the consequences that this gestalt switch entails. I may be wrong in this choice, and better ways of talking about the basic intuitions that I have been pursuing may be proposed later. If so, I will be the first to adopt the new language and new words. In the meantime this is the best I can do; let the dialogue on this simply unfold on its own (Maturana and Varela, 1981).

It is for these reasons that I would still like to use the word information, but in its more original etymological sense of *in-formare*, to form within (cf. Bateson, 1972, 420), which corresponds well to the ideas presented here. *We can define in-formation as the admissible symbolic descriptions of the cognitive domains of an autonomous system.* We shall always write it with the hyphen to convey the

⁸*Eds.* A similarly hyphenated sense of *in-information* as *shaping* or *form-taking* can be found in Susan Oyama's critique of the use of information concepts in developmental biology (Oyama, 2000, 14, 159, 214)—an important influence on Varela et al. (1991)—and in Gilbert Simondon's philosophy of individuation (Simondon, 2020, xxii). However, this meaning seems to indicate something more than an admissible symbolic description since it offers a possible way of understanding organizational change (form-taking) through the operative logic of structural processes; see section 16.1.3.

differences of this view from that of information in the computer gestalt.⁸

The differences can be emphasized by putting these two views on the end of a spectrum of shades. On the one end there is information as referential, instructional, representational. On the other end there is in-information as constructed, nonreferential or codependent, conversational. A list of contrasts follows:

<i>Codependent sense</i>	<i>Referential sense</i>
In-information is coherence or regularity, viability.	Information is a mapping or correspondence
Extrinsic, not operational but only relative to an observer who establishes the uses	Intrinsic, operational
Unity is defined autonomously, and relates to it as perturbations	Relation to a unity is through allonomous inputs
Environment or world is defined through invariances relative to the system's operation	Requires a fixed, given world or environment
Observer-community is, explicitly, what detects the regularities	Does not include observer explicitly
In-information is always interpretation	Information is instructive
Generated by structural coupling	Generated by definition

I am not disputing that the computer-based idea of information is necessary and useful in many domains. I am asserting its limited validity for understanding natural systems, discussed in detail for the immune and nervous systems. Furthermore, I think it is important to examine its impact on broader philosophical and ethical grounds, for the way it marks a predominant view of what knowledge and cognition are. Somewhere there is a middle ground. One may see these two extremes not on the *same* level, but rather where the left side is *more encompassing* than the right side.⁹ Referentiality originates by restricting our field of view to fixed interactions and observational viewpoints. The right side is the left in special or limiting cases. The left side is the basic context where in-information or symbols can never be pinned down with substance, with some absolute reference, but they will always be relative to the process of interactions of the domain in which they occur, and to the observer community that describes them. In the language of chapter 10, it is a complementarity in the star sense, not by opposition, but by inclusion and mutual interdependence (cf. section 10.6.1). In this

⁹*Eds.* See section 10.6.2.

sense, every notion of information, symbol, or sign is devoid of substance and is always codependent, and never operational. Not that one cannot make them look very solid in particular situations; the danger is in forgetting that it is we who do so. If we do forget, then information becomes a mythical entity, a vague fluid floating around in nature, the stuff to be found in DNA and languages out there, with nobody specifying them, as a seemingly operational part of the fabric of nature.

16.3 Linguistic Domains and Conversations

16.3.1

We use the term *linguistic domain* to designate the aggregate of symbolic descriptions arising from the structural coupling between two or more autonomous systems. *Communication* is behavior in a linguistic domain.

The idea is clear enough. It amounts to saying that autonomous systems can constitute unities of higher order through their coordination and structural coupling and that these can be described as communicational interactions. The components of the cognitive domain of animal nervous systems that can function in a linguistic way evidently enhance the viability of the participant systems.¹⁰ It is only recently, with the growth of ethology, that we have come to appreciate the true variety of communicative behavior in the living world (Sebeok, 1977; Smith, 1977). Not only can one identify an astounding complexity among the symbols used within one species, as the students of insects have long suspected, but there are cases of linguistic coupling across species beyond the wildest imagination.

My task here is not to recall or expand on the actual texture of different linguistic domains. That is the task of semioticians and ethologists, who must confront their own difficulties (Nauta, 1972; Eco, 1979). Similarly, I am not going to be concerned with the question of when a linguistic domain constitutes a *language*. This is an important discussion (Lenneberg, 1967; Griffin, 1976), with its classical locus of the bee's dance (Gould, 1976). Here, however, I am concerned with the relation between semiosis and autonomy rather than specific forms of semiosis, whether isomorphic to human language or not.

As implied in the definition of a linguistic domain, communication is a generative process. Accordingly, *all* that we have already stated about symbolic explanations for the immune and nervous

¹⁰ *Eds.* Given a linguistic domain, the viability of the participants in that domain is increased compared to if they were not to participate in the domain. This does not mean that species with a linguistic domain are more viable than nonlinguistic ones.

networks applies to communicative behavior as well. Again, communication cannot be understood as instruction or information “transfer” from one organism to another. Whether the semiotic domain is extremely stereotyped (as in tissues interacting through hormones) or highly self-reflexive (as in human language), to put communicative information in a category comparable to energy or matter is misplaced concreteness, and confusing levels of descriptions, as I have said before. I shall not repeat myself again. Animal communication is a network of interactions that has no basis except in its history of coupling and is relative to that history. The signals exchanged during, say, courtship among birds reflect nothing except the possibility of the emergence of those coherences; and in time, such invariants get transformed, rearranged in an ever evolving network of self-dependent elements.¹¹

¹¹ *Eds.* For a compatible perspective on semiosis in relation to a dynamic totality of behaviors, see Rączaszek-Leonardi (2009).

16.3.2

So it is for human language, of course. Everything said is said from a tradition. Every statement reflects a history of interactions from which we cannot escape, for it is what makes human language possible. This constant tension between understanding and breaches of understanding through reinterpretation has been, by and large, a blind spot in western philosophy and scientific attitudes. The analysis of the dynamics and ontology of human understanding is a central theme, I believe, for contemporary thought (Gadamer, 1960, 1976). My intention here is simply to provide some links with our perspective based on the natural world, and see how these two modes of understanding complement and traverse each other.

In fact, a *conversation* has been a basic image used throughout this presentation as a paradigm for interactions among autonomous systems. It is a paradigm as well as a particular instance of an autonomous system, and these two sides of it go together. Its role as an exemplary case of autonomous interaction comes from the fact that a conversation is direct experience, the human experience *par excellence*—we live and breathe in dialogue and language. And from this direct experience we know that one cannot find a firm reference point for the content of a dialogue. There is no methodological escape from dealing with the elusiveness of understanding, and this makes it very evident that whatever is informative in a conversation is intrinsically codependent and interpretational. Whatever is said in order to fix and objectify the nature of a conversation’s content is

said from a perspective, from a tradition, and is always open to question, to revision, to disagreement. This is not failure or weakness, but the heart of the process (Gadamer, 1960).

From another perspective, if we consider a conversation as a totality, there cannot be distinction about what is contributed by whom. Linde and Goguen (1978), for instance, analyzed the discourse structure of a planning session. In their careful descriptions of the structure of the discourse, they found no evidence that the text, as a coherent entity, could be attributed to separate speakers, but it was an alloy of their participation, and exhibited rules and laws that are not reducible to the separate contributions. A similar basic methodological principle is behind Pask's (1975; 1976) approach to teaching machines, where a conversation is a coherent, recursive aggregate.

These ideas are precisely in line with the central theme of this book: that every autonomous structure will exhibit a cognitive domain and behave as a separate, distinct aggregate. Such autonomous units can be constituted by *any* processes capable of engaging in organizational closure, whether molecular interactions, managerial manipulations, or conversational participation (cf. section 7.2.4).¹² I am saying, then, that whenever we engage in social interactions that we label as dialogue or conversation, these constitute autonomous aggregates, which exhibit all the properties of other autonomous units. It is not easy to establish strict criteria for this view of conversations, for their closure is transient and mobile. However, this view is not more laden with difficulties than the predominant way of looking at it in terms of the performance and competence of single speakers.

The difference is that in one case we take language as conveying information and instruction: in the other case we leave aside the individual participant and see the process of conversation and understanding as a distributed, coherent events shared among participants, where meaning and understanding is relative to the recursive process of interpretation within the conversational unit.

I am not a linguist: I offer this as a necessary consequence of my view of natural systems and autonomy. It seems to me to make sense, and to be in harmony with some recent trends—such as discourse analysis (Becker, 1980; Linde, 1980), conversation theory (Pask, 1976), Kristeva's (1969; 1970) work on semiotics as "*pratique signifi-cant*," and most especially the monograph by Flores and Winograd

¹² *Eds.* The idea that social interactions in general—not just human conversations—can be *defined* as an autonomous relational domain between coupled autonomous agents was first proposed by Hanne De Jaegher and colleagues (De Jaegher, 2009; De Jaegher and Di Paolo, 2007; De Jaegher et al., 2010); and further elaborated in the concept of *participatory sense-making* and work on the interactive turn in social cognition research (e.g., Gallagher, 2011). Varela's suggestion that conversations can be considered autonomous units resonates with this work. Varela's later statements about intersubjectivity emphasized affective and ethical aspects such as the intersubjective constitution of concern and the cultivation of a non-egocentric empathy (e.g., Varela and Depraz, 2005; Varela, 1999a). Other enactive work on ethics focuses again on interactive aspects, with emphasis on an ethics of *participation* (Colombetti and Torrance, 2009; Di Paolo and De Jaegher, 2022; Dierckxsens, 2022).

¹³Eds. Following similar evidence and work on participatory sense-making, Di Paolo, Cuffari, and De Jaegher (2018) proposed an enactive approach to *linguistic bodies* that treats language as a situated and distributed system of practices defined by multiple forms of social agency, full embodied engagement, and meaningful participation rather than as a system of exchanges of information.

¹⁴Heidegger has devoted luminous pages to this theme, especially in his articles “Die Zeit des Weltbildes” (in Heidegger, 1950), and “Die Frage nach der Technik” (in Heidegger, 1954).

¹⁵Eds. The reconsideration of the traditional notion of the subject becomes a central motif in Varela’s subsequent enactive research. It is also a central theme of Varela et al. (1991).

¹⁶Eds. The suggestion is that our access to coherences in the social and ecological patterns in which we are embedded is always mediated by such factors as language, research methods, and techniques.

¹⁷In Bateson (1972). I find his terminology in this paper somewhat misleading, but the idea is clear as can be. For further elaboration see also his book *Mind and Nature*, (Bateson, 1979) I benefited greatly from reading drafts of the manuscript and from long conversations with the author.

(1978), which discusses specifically the relation between a biological view of cognition and the understanding of human language.¹³

16.4 Units of Mind

A conversation on the one hand embodies a direct prototype of the way in which autonomous units interact. On the other hand, it is an instance of an autonomous entity in itself, and a very important one, for we are immersed in the ongoing autonomy of our tradition, the ongoing autonomy of a next higher level of interdependence as participants. These two sides of conversations, bring us in fact to the heart of an essential consequence of all that has been said before, namely, the need to *reconsider the traditional notion of subject*.^{14,15}

It is very true that there is a sense in which we must consider individual organisms and their (internal) cognitive processes. However, it is equally true (though somewhat awkward within the tradition we have been raised in) to realize that cognitive processes can also be seen as operating at the *next higher level*, that is, the cognitive processes of the autonomous unit of which we are participants and components. This next higher level can be construed either as purely cultural, or as a mixture of cultural and ecological (as I would prefer). But regardless of this preference, it is obvious that there *is* a next higher level in the coherence of a unit to which we have no direct access, but to which we contribute and in which we exist.¹⁶ To this next higher level belong the characteristics of mind we attribute to ourselves individually; in fact, what we experience as our mind cannot truly be *separated* from this network to which we connect and through which we interdepend. This, I believe, was first stated clearly in Bateson’s essay “Substance, Form and Difference”.¹⁷ His basic insight is that the unit of mind in evolution is not only, and certainly not fundamentally, the skull, but what he calls the “message-in-the-circuit,” and what I would call here the cognitive process of an autonomous unit, at many possible levels.

We can go one step further from here, to notice that *whenever we consider an autonomous unit, it will have two characteristics that make it mindlike*: First, it specifies a distinction between it and not-it, a basic dual split. Secondly, it has a way of dealing with its surroundings in a cognitive (in-formative) fashion, depending on its plasticity. From this point of view, then, mind is an immanent quality, of a class of organizations including individual living systems, but also ecological aggregates, social units of various sorts, brains, conversations, and many others, however spatially distributed or

short-lived. There is mind in every unity engaged in conversation-like interactions.¹⁸

It is somewhat ironic that, in a book devoted to the analysis of individuality and autonomous units, we should come to the conclusion that the traditional notion of subject should be revised. But this interesting flip of the argument is entirely natural if we care to follow the argument *all the way through*, and conclude that there is autonomy beyond what we are used to seeing as individual biological entities, in the collective interactions in a social tradition. The autonomy at the higher level gives a vantage point, from which the individuality of the components in the next lower level is seen in perspective. Both views are, as always, interdependent and complementary—but in this case with a new twist: the fact that we do not have access to the domain of interaction of the unit to which we belong.

Now, I am using the word “mind” here in the sense of cognitive processes, of what is proper to sentient beings, and not of consciousness, awareness, or soul. I am not, in other words, adopting a pantheistic position *à la* Teilhard de Chardin. I am saying that whatever we call mind in human affairs has a similarity to processes that are distributed in nature in its autonomous organizations. That there are differences with the human experience, I doubt not; I am speaking here not of the differences but of the continuity.¹⁹ Such continuity has often been claimed, particularly since Darwin, but always with the subject regarded in the best Cartesian, skull-bound tradition. With Bateson and Maturana, I am suggesting that this continuity goes beyond that, to encompass the cognitive processes of every system that acquires an autonomous structure, *beyond* the single biological individual. This not only gives an entirely different picture of what biosphere and ecology are, but also of what social interactions and traditions are.

This view of mind and nature is perhaps the most interesting, and yet the most immature, of the thoughts that flow from the study of autonomy. I must leave it here in this very rough form.

16.5 Human Knowledge and Cognizing Organisms

16.5.1

I wish to return now to a consideration of human knowledge. This is the last step in closing a circle. We have explicitly assumed a number of epistemological positions in this presentation; I will now complete the argument as to how these positions are in consonance

¹⁸ *Eds.* This key message has shaped the development of the enactive approach. Thompson and Varela (2001) propose different dimensions of embodiment in which this logic plays out. For higher primates, these are cycles of organic, sensorimotor, and social regulation. The logic of autonomy, interactions, and *mindlike* aspects such as normativity has been proposed to be at play in all these dimensions (Di Paolo et al., 2017, 2018).

¹⁹ *Eds.* In enactive terms, this continuity is expressed in the concept of *sense-making* (Varela, 1984b; Thompson, 2007), which describes a relation between an adaptive autonomous system, its history and current situation, and what they entail for the continuation of its form of life in terms of relevant adaptive responses (Di Paolo, 2005, 2009). Sense-making is what is common to all forms of mindedness before other distinctions are introduced, such as between action, perception, thought, and affect, or between “lower” and “higher” forms of mind. When it is valid to speak of a subjective dimension, sense-making entails a relation of interest, concern, or nonindifference.

with the processes and principles derived from the study of natural autonomy. The argument loops around itself, as I suppose it should.

In fact we have insisted over and over that the current and common-sense concepts of control and information are at the heart of an epistemology that needs to be revised. (To these concepts we must add the third in the trio, that of subject, which we just touched upon.) The basic stance taken here is that such revision leads to a naturalized epistemology, weaving together philosophical and empirical insights into a coherent fabric. This follows the tradition represented by Piaget, Bateson, McCulloch, and Maturana in recent years.

What is not generally realized, however, is that these developments force us to give up some of our most inveterate commonsense ideas about the nature of reality and the function of knowledge. Giving up ideas requires something of a wrench. Yet, as Thomas Kuhn has recently said, while the historical study of science shows that the classical view of epistemology is a misfit, no “viable alternate to the traditional epistemological paradigm” has yet been produced (Kuhn, 1970, 121). Let me make my ideas explicit by examining the consequences of a second-order flip, by once again applying the ideas presented for frogs and cells to us, by switching from *observed* system to *observing* systems.

²⁰Eds. There is a strong focus in Varela's work on the status of the organism, which he investigates through the concept of autonomy. This emphasis, which is necessary in the context of approaches that tend to erase the organism from scientific discourse, has often led to the suspicion that the environment plays a lesser role in Varela's account and in the enactive perspective. Theoretically, this is far from being the case, as these passages show. But in concrete investigations, it is true that environmental processes have been under investigated until recently, in comparison with approaches such as ecological psychology. For discussions of the relations between the two approaches in cognitive science and the increasing attention paid to the environment in the enactive approach, see Chemero (2009); Rietveld et al. (2018); McGann et al. (2020); de Haan (2020), and Sepúlveda-Pedro (2023).

16.5.2

Let us try, for a moment, to be naive (in the sense of *inexperienced*, rather than simple-minded) and ask the question: How do we come to have items such as, say, frogs or people, of whom we can say that they perceive other things? Well, in order for a frog to perceive other things, it would seem, we must have a frog and we must have other things. That is, we tacitly assume that the frog must be in an environment. But since we are trying to be naive, we should ask not only how we come to have a frog, but also how we come to have it in an environment. Adding this further question, rather than making it more difficult, makes it easier to answer the first one. If we focused on the frog alone and pondered how it came to be as a thing in its own right, we could not help attributing to it some kind of independent existence; that is, we would have assumed that the frog, as we come to know it, exists independently of the way we distinguish it. In the philosopher's jargon, we would have attributed “ontological reality” to the frog. That is precisely the trap we want to avoid—and we *can* avoid it if only we start out by taking into account *both* the frog *and* its environment with us establishing the link.²⁰

There is no good reason to assume that our experience begins with ready-made objects, animals, and people. It takes a child the better part of two years to assemble such items by coordinating much smaller elements of perceptual and conceptual experience (Piaget, 1937). In any case, all these items that we come to consider more or less permanent must, at some point, have been isolated and “individuated” in the field of our experience. This isolating and individuating necessarily had to be achieved by *us*, for it is we who say that we are aware of them. That is, we must have differentiated them and cut them out from the rest of our experiential field—and by that very act, the rest of our experiential field became their environment.²¹ In terms of the actual operations, performing this act of cutting out may be different from an artist’s drawing the outline of a frog on a sheet of paper, but the two acts are the same in that they simultaneously produce a figure and its ground. Whatever specific item we focus our attention on (or talk about) is experienced within a perceptual (or conceptual) field, which explicitly or implicitly constitutes its environment. The dichotomy of figure and ground, of frog and environment, springs from one and the same set of operations (i.e., focusing attention on and differentiating as repeatable unit a specific part of our experiential field); the two sides are conceptually *connate*—we cannot have the one without the other. Further, besides mechanical interactions, we come to have perceptual or informational ones, involving autonomous entities. In both (the mechanical and the perceptual interactions), it is *we* who observe the event. The leaf, the wind, the frog, and the shadow are all parts of *our* experience, and the events we describe, as well as the differences between them, are the results of the relations we have established between parts of our experience. Now, how do we come to say that an item, such as a frog, perceives things? As we have seen, both the frog and the environmental things it may perceive are parts of *our* experience.

Hand in hand with the establishing of relations goes the effort to explain observed interactions in terms of specific operational relations, in terms of regular processes and functions, and, in some cases, in terms of specific organs that carry out these processes and functions. In the case of the observed organisms’ perceptual interactions with their environment, this effort has been highly successful. In the visual perception of the frog, for instance, an observer may isolate (in the observed frog) eyes that contain a retina with light-sensitive receptors that send electrochemical impulses into a neural network

²¹ *Eds.* The developmental processes that enable human object perception are inherently social (e.g., Yu and Smith, 2016). Distinguishing objects *as* objects in relation to others, with the decentering that this implies, is also a social skill (Di Paolo, 2016).

capable of adding, subtracting, and otherwise being affected by these impulses in such a way that, under certain conditions, they will trigger muscular activity, which in turn will orient and propel the frog in a certain direction.

On the basis of further observation, the observer may then decide that some of the links in the causal chain he has constructed are still too loose, and he may attempt to insert additional steps; or, indeed, he may decide that parts of his analysis are inadequate for one reason or another. It may take the observer a long time to arrive at an at least temporarily satisfactory “explanation” of the frog’s perceptual and behavioral interactions with items in its environment, but there is nothing mysterious about what the observer does: It is no more and no less than establishing relations between parts of his own experience.²²

Hence it is one thing for us, the observers, to say that an organism we are observing perceives, but quite another to say that *we ourselves* perceive. However, the more engrossed we as observers become in establishing operational chains for the perceptual interactions between organisms and things in their environment, the more easily we will begin to think of our own experience as the result of similar or at least analogous interactions with an environment. This seems all the more plausible because we can observe eyes, retinas, and neural networks not only in frogs but also in experiential items which we categorize as organisms of our own kind, which is to say, functionally similar to ourselves.

Indeed, we even can isolate and individuate eyes in that particular experiential item which we have come to “know” as ourselves and as our own bodies. All we have to do is step before a looking glass, and there, as clearly as any other image, we see the part of our experiential field that we call ourselves and that other part which we call our environment. And seeing it in what so plausibly seems an *immediate* fashion makes it almost impossible for us perceivers to realize that what we now categorize as our own environment has a relationship to us, the observers, that is *quite different* from the relationship the frog’s environment had to the observed frog. For what we observers now take to be our own environment is still part of our experience and by no means lies beyond the interface that is supposed to separate the knower from the world he gets to know. That this has to be so becomes clear once we realize that the mirror-self (which, like the frog, is surrounded by an environment) is precisely what we observers experience of ourselves and therefore cannot possibly be we ourselves *as experiencers*.²³

²²Eds. To avoid the misinterpretation that this is a process that occurs in the observer’s head, it is worth reiterating that the processes of detached distinction underpinning an investigation of this kind constitutively entail a community of observers, their traditions, values, technologies, and other elements. See section 16.5.3.

²³Eds. Varela is building toward the fundamental phenomenological insight that experientially, the body is not an object in the world like any external object (Merleau-Ponty, 2012), a theme that is further investigated in Varela et al. (1991) and Thompson (2007). Concerning the difference between seeing a mirror image of oneself and seeing someone’s face, see Harding (1961).

Thus, when observing a frog, one may indeed ask, for instance, how its retinal receptors and neural networks “respond” to a shadow in the environment. Such a question makes sense from the observer’s point of view, because, as observer of the frog, one has independent access to the experiential item one calls “shadow,” and one can, therefore, legitimately invest it with the capability of triggering a perceptual event in another experiential item that one calls “the frog’s visual system.” When observing oneself, however, one is no longer in that privileged position. What we ourselves perceive, whether we call it frog, landscape, or mirror image of ourselves, is simply what *we* perceive; and since we have no way of looking at ourselves and our environment from outside our own experience, we have no possible independent access to whatever it might be that, by analogy to the frog, we would like to hold *operationally* responsible for our perceptions. Strictly speaking, *we do not have access to our cognitive domain*, for we cannot step outside it and see ourselves as a unit in an environment. Herein lies the essential asymmetry between the study of frog’s cognition and the study of our cognition.²⁴

16.5.3

Western rationality and common sense have instead accepted an objectivist scenario almost unquestioningly. Even most thinkers who have pondered problems of epistemology have explicitly or implicitly adopted the view that the activity of “knowing” begins with a cut between the cognizing subject and the object to be known. That is, they *assume* an existing world, an ontological reality of sorts, and once this assumption has been made, it necessarily follows that the knower will have this world as his environment and it will be his task to get to know it as best he can. Knowing, thus, becomes an act of duplicating or replicating what is supposed to be already *there*, outside the knower. The senses become the indispensable mediators that convey “information” on the basis of which the knower can represent to himself a replica of what “exists.”²⁵

I am speaking here, to be sure, about the common sense and epistemology that have dominated the rise of the sciences. I am not saying that there are no alternatives; I am talking about dominant perceptions. And hand in hand with this epistemology goes the rise of the information sciences in the computer gestalt that we have traced in these pages.

Yet, if what I have said so far has any consistency at all, it should now be clear that the first cut, the most elementary distinction we can make, may be the intuitively satisfactory cut between oneself *qua*

²⁴ *Eds.* This crucial observation becomes the point of departure for Varela et al. (1991, p. 3), which describes it in terms of the “fundamental circularity” between science and human experience.

²⁵ *Eds.* Dreyfus and Taylor (2015, p. 12) describe such an epistemology as “mediational” because the separation between the knower and the known must somehow be bridged, often by postulating such constructs as mental representations.

experiencing subject on the one side, and one's experience on the other. But this cut can under no circumstances be a cut between one-self and an independently existing world of objective objects. Our "knowledge," whatever rational meaning we give that term, must begin with experience, and with cuts *within* our experience—such as, for instance, the cut we make between the part of our experience that we come to call "our self" and all the rest of our experience, which we then call our "world." Hence, this world of ours, no matter how we structure it, no matter how well we manage to keep it stable with permanent objects and recurrent interactions, is by definition a world codependent with our experience, and not the ontological reality of which philosophers and scientists alike have dreamed. All of this boils down, actually, to a realization that although the world *does* look solid and regular, when we come to examine it there is no fixed point of reference to which it can be pinned down; it is nowhere substantial or solid. The whole of experience reveals the codependent and relative quality of our knowledge, truly a reflection of our individual and collective actions.²⁶

²⁶ *Eds.* A key development in Varela's thought was to elaborate this notion of a lack of a solid epistemological ground in conversation with Buddhist philosophy, particularly the Madhyamaka tradition and the concept of groundlessness prefigured in these passages (Varela et al., 1991). By the mid-1970s, Varela was already engaged with Buddhism as a practice.

Many shy away at this point from following the logic of this argument, for they see the spectre of solipsism rising, of wild subjectivism, the ghost of anarchism. This, I submit, is pure misinterpretation of both the empirical record and philosophical arguments. Quite simply, to assume that because knowledge is relative (or non-solid) it is therefore arbitrary, would be like saying that because an antigen does not contain information, anything can trigger an immune transformation. The immune system has its regularities, which make it possible for it to have a history, and thus to carve a cognitive domain. We have been raised (and ask these questions) in the midst of a tradition and with a biological structure that we cannot escape or pretend not to have. Thus we do have, by necessity, a world of shared regularities that we cannot alter at whim. In fact, the act of understanding is basically *beyond our will*, precisely because the autonomy of the social and biological system we are in goes *beyond* our skull, because our evolution makes us part of a social aggregate and a natural aggregate, which have an autonomy compatible with, but *not reducible* to, our autonomy as biological individuals. This is precisely why I have insisted so much on talking about an observer-community rather than an observer; the knower is not the biological individual.²⁷ Thus this *epistemology of participation* sees humans in continuity with the natural world, in which knowledge comes into being in autonomous units through an interwoven mesh of frozen histories, like a castle of cards—structured,

²⁷ *Eds.* These comments make it clear that the autonomy perspective and the enactive project that later developed from it decidedly reject internalism, subjective idealism, and solipsism. See, for instance, Varela et al. (1991, 140, 233) and Thompson (2007, 82).

yet bootstrapping its content and solidity from within. Conversely it sees nature as human history, where every factual statement has a hermeneutics from which it derives and which contains its possibilities. The successor to objectivism is not subjectivism, by way of negation, but rather the full appreciation of participation, which is a move beyond either of them, as in a star (cf. section 10.6.1).²⁸

It is by no means easy to adopt this participatory epistemology. Years of efforts directed at demonstrating a correspondence between “knowledge” and an ontological reality are deeply ingrained in our languages and have been foisted on us from the moment we were born. The claim has been “to tell it like it is” rather than to explain how we come to see it the way we do see it. The tradition is strong, overpowering. Even in one’s own thinking, no matter how determined one may be to break away and start afresh, one inadvertently falls back into the conventional track and sees problems where there is no problem. Traditionally we are supposed to play the role of discoverers who, through their cognitive efforts, come to comprehend the structure of the “real” world. Thus we are always prone to revert to some form of realism and to forget that what we are thinking or talking about is under all circumstances *our* experience and that the “knowledge” we acquire is knowledge of invariances and regularities derived from and pertaining to our experience.

16.5.4

If, on the other hand, we do keep in mind that all invariances and regularities are our construction²⁹, this awareness necessarily alters our idea of what is called “empirical investigation” and, indeed, our idea of science itself. We shall come to pay attention to the structure of our concepts and the origin of the categories, rather than assume that any structure and any categories have to be *there* as such.

In traditional terminology, this means that we may see the computer gestalt as an expression of the dominant positivism and its objectivistic understanding of science. My objectivism, I mean here simply the assumption that empirical knowledge, acquired in the scientific format, represents the world independent of the knower and his values, or in a way that only implicates him peripherally. Further, we have been led to assume that objectivism is the *same* as empiricism and sound reasoning.

My claim is that we must differentiate radically between objectivism and empiricism, between an interpretation of what an experience is and the existence of regularities in our collective history at every moment. Natural history has shown us immense potential

²⁸ Eds. Hanne De Jaegher (2021) investigated the existential dimensions of such a *participatory* or *engaged epistemology*. Contact between the knower and the known entails the mutual becoming of both. Knowing demands a paradoxical attitude of *letting be* (i.e., avoiding overdetermining the object of knowledge) without disengaging from it and acknowledging the changes that our acts of knowing induce in it and in us.

²⁹ Eds. Or the result of our joint histories of enaction, as Varela would express later, avoiding internalist or constructivist readings (e.g., Varela, 2011, p. 614).

³⁰*Eds.* This observation, later developed in the enactive literature, is often described as *bringing forth a world* (Varela et al., 1991, 156, 198, 205). Also see Rolla and Figueiredo (2023) and Di Paolo (2023).

for systems to carve their own reality and be rendered viable, without being unique and without representing a stable given world.³⁰ Our knowledge, including science, can be accurately empirical and experimental, without requiring for it the claim of solidity or fixed reference. In fact, in the pursuit of cognitive science, that tradition of objectivity becomes a difficult stumbling block.

This, or course, runs counter to the predominant common sense view of the world. But, in fact, it merely modifies our concept of knowledge in exactly the same way as the theory of evolution has modified our concept of living things. Accordingly, knowledge is true and valid as long as it manages to “survive,” that is, as long as it is not demolished by experience. This is the very solid ground for Popper’s insistence on “falsification” as the actual goal of scientific investigation. But a surviving organism cannot be considered to manifest a description of the environment in which it happens to be *viable*, because an infinite variety of other and different organisms would be just as viable. And similarly, the regularities, rules, and laws that constitute our knowledge at a given time cannot be said to depict or describe an ontological reality, because an infinite variety of other and different regularities, rules, and laws might be just as viable in the “environment” of our experience. (For a somewhat similar view of science, see the work of Paul Feyerabend, 1970.) This, in a sense, is an amplification of the well-known and long-accepted adage that, though we can sometimes disprove a hypothesis, we can never prove one, no matter how often it happens to work out right. The trouble is, when an hypothesis does work out right, we often speak of “confirmation” and begin to believe that we have managed to trap a piece of solid, ontological reality.

16.5.5

According to the biological and systemic views presented in these pages, our models of cognizing systems have come of age. Although we seem to be able to visualize it as an image, we can only *begin* to chart, as an operative process and in a formal fashion, that peculiar elusive relationship between unities and their cognition.

It is a relationship of total interdependence in which relative stability is achieved and maintained through the circularities of interactions. It shows that, in nature and culture alike, every unity that can be called a knower constructs the world she knows and, in doing so, determines her way of knowing. In contrast, there is predominance (and hence power) in the common sense ideas of

control and information-as-representation. This had led philosophy, science, and technology into the attitude that has persistently kept us, philosophers and scientists, out of our own doings, fostering the belief that, in the last analysis, we were not responsible for the world we came to know and manipulate.³¹

Source

(von Glasersfeld and Varela, 1987).

³¹ *Eds.* The fundamental circularity at the heart of the sciences of the mind is the opening and ongoing theme of *The Embodied Mind* (Varela et al., 1991) and the concern of other work in the enactive tradition (e.g., Stewart, 2010; Thompson, 2007). The ethical implications of enactive epistemology hinted at in these passages are elaborated in Varela (1999a) and in recent interventions (Dierckxsens, 2022).

Appendices

A Algorithm for a Tessellation Example of Autopoiesis

This tessellation model, exemplifying autopoiesis in a simple case, was described in chapter 3. We give here the specific algorithm that generates the two-dimensional units illustrated in figures 3.1 and 3.2. The help of various members of the Biological Computer Laboratory at the University of Illinois, especially Heinz von Foerster, Paul Weston, and Kenneth Wilson, is gratefully acknowledged.¹

A.1 Conventions

We shall use the following alphanumeric symbols to designate the elements referred to earlier:

substrate	○	→ S,
catalyst	*	→ K,
link:	◻	→ L,
bonded link	—◻—	→ BL.

The algorithm has two principal phases, concerned, respectively, with the motion of the components over the two-dimensional array of positions, and with production and disintegration of the L-components out of and back into the substrate S's. The rules by which L-components bond to form a boundary complete the algorithm.

The "space" is a rectangular array of points, individually addressable by their row and column positions within the array. In its initial state this space contains one or more catalyst molecules K, with all remaining positions containing substrates S.

In both the motion and production phases, it is necessary to make random selections among certain sets of positions neighboring the particular point in the space at which the algorithm is being applied. The numbering scheme of figure A.1 is then applied, with location

¹*Eds.* The original algorithm was reimplemented with mixed success on different occasions. In the 1990s, Barry McMullin produced a careful examination and commentary of the published algorithm (McMullin, 1997). A frequently encountered problem was that the interior of the membrane tended to be clogged with bonded L particles, which (because they are immobile) are effectively made unavailable for membrane formation and repair. It was necessary to implement an undocumented interaction, chain-based bond inhibition, to reproduce the reported results. See McMullin (2004) for a history of the algorithm and its implementations and a discussion of its wider significance for the practice of conceptual computational modeling.

0 in the figure being identified with the point of application (of course, near the array boundaries, not all of the neighbor locations identified in the figure will actually be found).

Regarding motion, the components are ranked by increasing “mass” as S, L, K. The S’s cannot displace any other species, and thus are only able to move into “holes” or empty spaces in the grid, though they can pass through a *single* thickness of bonded links (BL) to do so. On the other hand the L and K readily displace S’s, pushing them into adjacent holes, if these exist, or else exchanging positions with them, thus passing freely through the substrates S. The most massive, K, can similarly displace free links (L). However, neither of these can pass through a bonded-link segment, and are thus effectively contained by a closed membrane. Concatenated L’s, forming bonded-link segments, are subject to no motions at all.

Regarding production, the initial state contains no bonded links at all; these appear only as the result of formation from substrates (S) in the presence of the catalyst. This occurs whenever two adjacent neighboring positions of a catalyst are occupied by S’s (e.g., 2 and 7, or 5 and 4, in figure A.1). Only one is formed per time step, per catalyst, with multiple possibilities being resolved by random choice. Since two S’s are combined to form one L, each such production leaves a new hole in the space, into which S’s may diffuse.

The disintegration of L’s is applied as a uniform probability of disintegration per time step for each L, whether bonded or free, which results in a proportionality between failure rate and the size of a chain structure. The sharply limited rate of “repair,” which depends upon random motion of S’s through the membrane, random

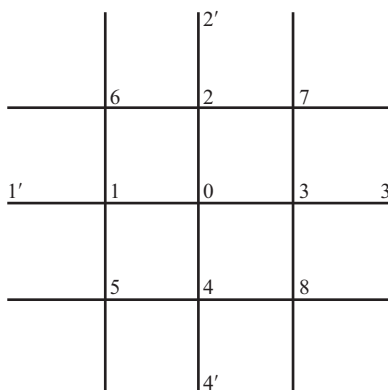


Figure A.1

Designation of coordinates of neighboring spaces with reference to a middle space with designation 0.

production of new L's, and random motion to the repair site, makes the disintegration a very powerful controller of the maximum size for a viable boundary structure. A disintegration probability of less than about 0.01 per time step is required in order to achieve any viable structure at all (these must contain roughly 10 L-units at least, in order to form a closed structure with any space inside).

A.2 Algorithm

1. Motion, first step²

- 1.1 Form a list of the coordinates of all holes h_i .
- 1.2 For each h_i , make a random selection, n_i , in the range 1 through 4, specifying a neighboring location.
- 1.3 For each h_i in turn, where possible, move the occupant of the selected neighboring location in h_i .
 - 1.3.1 If the neighbor is a hole or lies outside the space, take no action.
 - 1.3.2 If the neighbor n_i , contains a bonded L, examine the location n'_i . If n'_i contains an S, move this S to h_i .
 - 1.3.3 Where possible, move the L occupying the location m_i into the specified neighboring location.
- 1.4 Bond any moved L, if possible (rule 6).

2. Motion, second step

- 2.1 Form a list of the coordinates of free L's, m_i .
- 2.2 For each m_i , make a random selection, n_i , in the range 1 through 4, specifying a neighboring location.
- 2.3 Where possible, move the L occupying the location m_i in the specified neighboring location
 - 2.3.1 If location specified by n_i contains another L, or a K, then take no action.
 - 2.3.2 If location specified by n_i contains an S, the S will be displaced.
 - 2.3.2.1 If there is a hole adjacent to the S, it will move into it. If more than one such hole, select randomly.
 - 2.3.2.2 If the S can be moved into a hole by passing through bonded links, as in step 1, then it will do so.
 - 2.3.2.3 If the S cannot be moved into a hole, it will exchange locations with the moving L.

²Eds. For a detailed breakdown and commentary on each step of the original algorithm, see McMullin (1997).

³Eds. K particles may displace free L particles, which in turn may displace S particles. A bonded BL particle cannot be displaced, making them effectively immobile. This contributes to the problem of clogging of prematurely bonded L's.

2.3.3 If the location specified by n_i is a hole, then L simply moves into it.

2.4 Bond each moved L, if possible.

3. Motion, third step³

3.1 Form a list of the coordinates of all K's, c_i

3.2 For each c_i make a random selection n_i , in the range 1 through 4, specifying a neighboring location.

3.3 Where possible, move the K into the selected neighboring location.

3.3.1 If the location specified by n_i contains a BL or another K, take no action.

3.3.2 If the location specified by n_i contains a free L that may be displaced according to the rules of 2.3, then the L will be moved, and the K moved into its place. (Bond the moved L, if possible.)

3.3.3 If the location specified by n_i contains an S, then move the S by the rules of 2.3.2.

3.3.4 If the location specified by n_i contains a free L not movable by the rules of 2.3, then exchange the positions of the K and the L. (Bond L if possible.)

3.3.5 If the location specified by n_i is a hole, the K moves into it.

4. Production⁴

4.1 For each catalyst c_i , form a list of the neighboring positions n_{ij} that are occupied by S's.

4.1.1 Delete from the list of n_{ij} all positions for which neither adjacent neighbor position appears in the list (i.e., a 1 must be deleted from the list of n_{ij} 's if neither 5 nor 6 appears, and a 6 must be deleted 0 if neither 1 nor 2 appears).

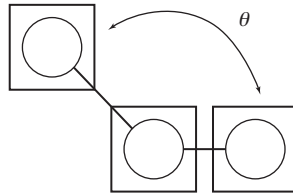
4.2 For each c_i with a non-null list of n_{ij} , choose randomly one of the n_{ij} , let its value be p_i , and at the corresponding location replace the S by a free L.

4.2.1 If the list of n_{ij} contains only one that is adjacent to p_i , then remove the corresponding S.

4.2.2 If the list of n_{ij} includes both locations adjacent to p_i , randomly select the S to be removed.

4.3 Bond each produced L, if possible.

⁴Eds. S particles adjacent to a catalyst always result in a reaction. There is no *reaction probability* parameter.

**Figure A.2**

Definition of the bond angle θ .

5. Disintegration⁵

5.1 For each L, bonded or unbonded, select a random real number d in the range (0, 1).

5.1.1 If $d \leq Pd$ (Pd an adjustable parameter of the algorithm), then remove the corresponding L, and attempt to rebond (rule 7).

5.1.2 Otherwise proceed to next L.

6. Bonding

This step must be given the coordinates of a free L.

6.1 Form a list of the neighboring positions n_i that contain free L's, and the neighboring positions m_i that contain singly bonded L's.

6.2 Drop from the m_i any that would result in a bond angle less than 90° . (Bond angle is determined as in figure A.2.)⁶

6.3 If there are two or more of the m_i , select two, form the corresponding bonds, and exit.

6.4 If there is exactly one m_i , form the corresponding bond.

6.4.1 Remove from the n_i any that would now result in a bond angle of less than 90° .

6.4.2 If there are no n_i , exit.

6.4.3 Select one of the n_i , form the bond, and exit.

6.5 If there are no n_i exit.

6.6 Select one of the n_i , form the corresponding bond, and drop it from the list.

6.6.1 If the n_i list is non-null, execute steps 6.4.1 through 6.4.3.

6.6.2 Exit.

7. Rebond

7.1 Form a list of all neighbor positions m_i occupied by singly bonded L's.

⁵Eds. L particles disintegrate with a fixed probability. If the L particle is bonded, the bonds, oddly, do not disappear. Step 7 attempts to link these bonds to other L particles. This, together with the immobility of BL particles, contributes to the problem of clogging by premature bonding found in independent attempts to reimplement the original algorithm. McMullin and Varela (1997) conclude that an unreported interaction (*chain-based bond inhibition*) was originally in place to inhibit bonding to free L particles in the vicinity of a doubly bonded BL particle.

⁶Eds. This constraint encourages the formation of open chains, but it seems rather arbitrary, as bond angles of 45° are geometrically plausible.

- 7.2 Form a second list, p_{ij} , of pairs of the m_i that can be bonded.
- 7.3 If there are any p_{ij} , choose a maximal subset and form the bonds. Remove the L 's involved from the list m_i .
- 7.4 Add to the bond m_i any neighbor locations occupied by free L 's.
- 7.5 Execute steps 7.1 through 7.3; then exit.

B Some Remarks on Reflexive Domains and Logic

In this book I have used extensively the notion of self-reference as a key to characterize natural autonomy. It is well known that self-referential representations have been the source of much debate and thinking in logic and the foundation of mathematics. It seems proper to say a few words about the connection between these two domains, and to what extent the headaches of one are to be found in the other. At the outset I want to emphasize that these questions on circularities in logic are connected, but are not identical with the substance of the views presented previously.¹ This is so because we have been concerned with recursive, self-referential *processes* in natural systems, rather than with linguistic or logical expressions. This surely makes for a significant difference, as cyberneticians have long known.

In this appendix, I shall first review briefly the standard assumption that reflexivity is impossible in logical and formal systems. Second, I shall discuss in some detail the relation between the calculi of indications and their logical interpretation.

B.1 Type-Free Logical Calculi

It seems very natural to expect *closure* from formalized descriptions, that is, to have the self-referential capacities of natural linguistic descriptions also fully available in their formal embodiment. In particular, it seems almost trivial to extend the classic propositional calculus so that it becomes semantically closed. Yet, as Russell and Tarski made only too clear, such an extension is not easily forthcoming. Indeed, much research on logic and mathematics has been devoted to the possibility of circumventing the encumbrances of types, yet preserving the consistency menaced by paradoxes. These

¹*Eds.* It is true that under some interpretations, the investigations in part II can reveal relevant processual aspects, such as the act of making distinctions, the self-positing status of reentry and wave-forms in sequential time, and the dynamic interpretation of operators. Yet several key processual aspects are not directly addressed in these formalisms, such as, a dynamic temporality including durations, deadlines, rates of change, a more situated spatiality looking at directional flows, spreads, barriers, or any consideration of energetics (though see the discussion in section 13.11).

efforts have had little or no impact. The working mathematician has adopted the safe position offered by the type like axiomatic set theories, which is enough for most of his needs. The logician and philosopher is not so lucky: The liar and related problems do not seem to succumb to the Tarskian doctrine of levels, and the issue is far from settled.

As I read the literature of the past years on this problem, three main categories of approach can be distinguished. First, there are those who follow a Tarskian solution in one form or another and have left self-referential problems as something for the natural linguist to worry about (e.g., Bar-Hillel, 1966). Second, there are those who see some self-reference as being essential if one is not to leave out key parts of mathematics and philosophy, and in general require a distinction between vicious and non-vicious self-reference (e.g., Fitch, 1946; Popper, 1954). Third, there are those who have still maintained as their central interest self-reference as such, and wish to investigate what is necessary for a formal logic to be closed (e.g., Löfgren, 1968; Martin, 1970; Asenjo, 1966). As of now, and as can be guessed by the degree to which self-reference is usually treated as illegitimate, the results of the first approach are plentiful, mature, and in full use; those of the second approach more scanty, still tentative, and enjoying less popularity. The results of the third approach are restricted to a group of specialists and just beginning.²

²*Eds.* For an up-to-date overview of work on paradoxes of self-references, see Bolander (2017).

When no time—or sequence in any sense—is available, circularities can become vicious. They appear as functions that are contained in their own range, as $\in$ -cycles of sets as in Russell's paradox, or finally, as antinomies of the liar type in formal logic. When we contemplate the same circularity as present throughout these levels, we see that to assign a specific value to what has been called vicious is nothing less than introducing a timelike dimension into logic and set theory, not in any form of duration, but in the form of expressions that define a new *domain* not reducible to noncircularity. What at the level of logic appears vicious, at other levels can be seen as very creative indeed.

It is this basic sense of taking self-reference explicitly into account that stands behind a significant amount of recent work, including the use of three-value logic to deal with paradoxes (Chang, 1963; Prior, 1955; Shaw-Kwei, 1954; Skolem, 1960), interpretations of self-reference in natural languages and logic (Herzberger, 1970; Martin, 1967; Parsons, 1974; Post, 1973; Skyrms, 1970; van Fraassen, 1970), and other alternative views on *tertium non datur* (Fitch, 1952; Heyting, 1930; Smullyan, 1957). I shall not discuss all of these authors

here, but instead I want to concentrate on the relevant work of one, Dana Scott, which served to begin the discussion of eigenbehavior.

In a nutshell his position is quite simple: There is nothing intrinsically impossible about a type-free logic. This type-freeness can be expressed in the form of a reflexive domain D such that

$$D \cong [D \rightarrow D],$$

where $[D \rightarrow D]$ is some suitable function space from D to itself (cf. section 13.8).³ Wadsworth (1976) called these isomorphic domain equations. Scott is specifically concerned with the combinatorial version of logic, and with the λ -calculus, which serves as a basis of the investigation, in logic, of a theory of functions (or procedures) in general (Scott, 1971, 1972, 1973). Instead of trying to specify what a λ -expression can mean, the emphasis, since Church, has been mostly on the rules for calculation to reduce one expression to another. But there is more to an expression than what is embodied in the rules. Curry and Feys (1958, 178), for instance, define a paradoxical combinator Y that may be thought of as an application of the famous argument of Gödel (or of Russell's paradox). Yet in their treatment these authors manage to exclude these unwelcome forms because they cannot be reduced to a normal form. It was Scott's basic insight to see that every expression *can* be given a perfectly good meaning, to be discovered *not only* through the reduction rules, but through *other* means as well: those of approximation and limit. These methods were discussed in a sketchy way in section 12.10.2, and more rigorously for continuous algebras of operators in chapter 13. There is a very broad array of issues and possibilities for logic and formal systems to be explored with these order-theoretic notions. Their exploration is just beginning (e.g., Kripke, 1975).

³Eds. This means that there is a mapping between elements in D and operations that map members of D onto other members of D .

This will complete my comments on self-reference and type-free logic. I simply want to point out that there is more to be said about circularities in language than to dismiss them as problematic, and that, in fact, if we do take them at face value there is more richness than meets the eye, as the volume of work cited here indicates. Let us now turn to a more specific topic: the relation between indicational and logical calculi on one hand, and reentry and self-reference on the other.

B.2 Indicational Calculi Interpreted for Logic

The basic way of interpreting an indicational form as a logical proposition was outlined by Spencer-Brown himself in an appendix to his book (1969, 112). It amounts to taking propositional expressions

as a model for the calculus of indications (CI), by establishing the expected correspondence between the expressions of one (CI) with the other (the calculus of propositions) thereby attributing specific properties to indicational forms, which can otherwise be interpreted in a number of other ways. I wish to reconsider here the indicational forms of propositions.

Consider the calculus of indications as a formal language CI. Consider any version of the classical propositional calculus as a formal language PC. Let PC and CI share the same vocabulary of literal variables $A, B, \dots$. Let us define a procedure Π as follows, where A is any expression in PC:⁴

⁴*Eds.* In the following discussion, $\sim B$ means “not- B ”, $B \vee C$ means “ B or C ”, $\vdash A$ means “ A is provable”, and $A \supset B$ means “ A implies B ”.

Definition B.1

Procedure Π :	If A is $\sim B$,	write $\overline{B}$ for A in CI;
	If A is $B \vee C$,	write BC for A in CI;
	If $\vdash A$ in PC,	write $\Pi(A) = \overline{\quad}$ in CI;
	If $\vdash \sim A$ in PC,	write $\Pi(\overline{A}) = \overline{\quad}$ in CI;

Lemma B.2 *To every expression in PC there corresponds an indicational form.*

Proof. Let A be any expression in PC. Then it can be put in its disjunctive normal form A' . We can now apply Π to A' starting from its atomic negations and disjunctions. The result is an expression A'' in CI that corresponds to A . $\square$

Lemma B.3 *Every demonstrable expression in PC is equivalent to the cross $\overline{\quad}$, in CI.*

Proof. Since every demonstrable expression in PC is equivalent to a tautology, the lemma follows immediately from the definition of the procedure and the previous lemma. $\square$

In other words, not only can we transcribe every expression of PC into CI, but its theorems are seen to be those expressions identical to the cross, $\overline{\quad}$. In CI, of course, we not only can calculate this class of expressions, but also those equivalent to the unmarked state, $\quad$, and in general, to any equivalence class we wish to consider. From the semantic point of view, procedure Π amounts to adopting the following interpretation for CI:

$\overline{\quad}$	as “true”;
$\quad$	as “false”;
$\overline{A}$	as “not- A ”;
AB	as “ A or B ”;
$\overline{A}B$	as “ A implies B ”;

and so on.

Note that when we consider the indicational forms of logical propositions, several notions condense into one. Such is the case of the cross, both a value (the marked state) and an operator (the distinctior, the injunction to draw a distinction). When interpreted for logic the double-carry nature of the cross (operator and value) becomes divided into a truth-value (such as true) and the negation (or implication) operator. That these two notions in logic are distinct is obvious, but at the indicational level, and because of the more adequate notation of Spencer-Brown, we can see them condense into just one motion, the cross. Similarly, in logic we must distinguish between the operations “or” and “not,” yet in CI both of these condense into the same property of crosses: containment. Again we have one notion that becomes divided in two when interpreted for logic. It is from this degree of condensation in CI that computational advantages are obtained in considering the underlying indicational form of propositions. For example, the two clearly distinct ideas of

$$A \vee \sim A$$

and

$$A \supset A$$

are seen to have the same form, namely,

$$\overline{A} \mid A.$$

Similarly, general computations are simplified, as for example in the expression E :

$$A \supset C((B \supset C) \supset (A \vee B) \supset C$$

known to be true in PC. By the procedure

$$\Pi(E) = \overline{A} \mid \overline{C} \mid \overline{B} \mid \overline{C} \mid \overline{AB} \mid C$$

we can readily compute

$$\begin{aligned} \Pi(E) &= \overline{A} \mid \overline{B} \mid \overline{C} \mid \overline{AB} \mid \overline{C} \mid C \\ &= \overline{A} \mid \overline{B} \mid \overline{C} \mid \overline{AB} \mid \overline{} \\ &= \overline{}, \end{aligned}$$

so we know it to be true. In general, as readers may convince themselves, calculation in PC becomes almost trivial by considering its underlying indicational form.

In PC expressions are usually detached at the point of implication; in CI they are detached at the point of equivalence of value or identity. This is of little import; Although a consequence need not

be equivalent to its antecedent, this is the case for the class of true expressions, which are the ones derivable in PC. More precisely, we can easily convert the interpretation of PC from CI into an implicational form by the procedure Π , since we have that $A = \top$ when $\vdash A$, and on the other hand $A = B$ when and only when $A \equiv B$, where “ $\equiv$ ” is the biconditional. Thus *modus ponens* follows from the procedure as well, because

$$\begin{array}{c} A, \\ A \supset B, \\ \vdash B \end{array}$$

becomes

$$\begin{array}{c} A = \top, \\ \top B = \top, \end{array}$$

so that

$$\top B = B = \top,$$

by substitution. In other words, in interpreting the CI for PC it becomes only a syntactic question whether we write the calculus in its equivalent or implicational form.

Let us consider now more general conceptual implications of interpreting CI for logic. As we know from Spencer-Brown (1969, 107), the Sheffer postulates can easily be derived from the two simple indicational initials of CI. More generally we can state

Theorem B.4 *Let PC be defined by any list of postulates. Then PC can be derived from the two axioms of CI.*

Proof. We must show that any chosen set of postulates can be derived from the axioms of CI. Let a list of postulates $S_1, S_2, \dots, S_n$ be transcribed according to procedure Π . Since they are all axioms, we must have $\Pi(S_i) = \top$ for all i . But we know that all valid expression of the form $A = B$ can be derived from the two axioms of CI. In particular $\Pi(S_i) = \top$ for all i is derivable. Thus the list of postulates can be derived from the two axioms in CL $\square$

This is interesting: not only does CI provide a framework of great computational capacity for PC, but it also shows that the varieties of algebraic initials are soon to rest on a common ground of utter simplicity. In other words, by interpreting CI for PC we have a calculus for which no initials are necessary. Every algebraic form we may wish to *take* as initials can be derived from its arithmetical

ground, and therefore, in adopting one list of postulates or another for syntactic convenience, we can see the common ground on which we are standing, rather than the diversity of conveniences.

Similarly, it is interesting to note what happens in this light with metalogical results about PC. Clearly, since CI is consistent, so is PC as an interpretation of it. As to completeness, however, again several ideas condense into one at the level of indications. In fact, in the completeness of the CI algebra it is proved that all valid arithmetical expressions are demonstrable, including not only the true ones (thus showing the semantic completeness of PC as first proved by Post⁵), but also any other equivalence class, such as the false ones (thus showing the functional completeness of the two pairs of connectives $(\sim, \vee)$ and $(\sim, \supset)$. Finally, the decidability of PC as derived from CI is simply established by arithmetical computation, and even further, since the completeness proof is a constructive one, we have an effective proof procedure for PC.

That such a well-charted domain as propositional logic can be enriched by considering it as an interpretation of CI is, I submit, a testimony to the true significance of Spencer-Brown's discovery.

Let us now turn to consider self-referential expressions and their underlying indicational forms. In propositional logic we may see self-reference as arising from certain propositions asserting properties or characteristics of themselves. If we denote by Φ any propositional function, we can make this notion more precise as the condition

A is provable if and only if $\Phi(A)$ is provable,

so that A can be taken to assert Φ of itself (Fitch, 1952). If we now transcribe this condition into CI, we obtain

$$A = \top \text{ if and only if } \Pi(\Phi(A)) = \top,$$

or, taking $\phi(A)$ as $\Pi(\Phi(A))$,

$$A = \phi(A),$$

where now ϕ is any indicative function, that is, any algebraic expression in CI. Such an expression *reenters* its own indicative space, and thus is the motivation to define a general form of self-referential indicational forms by an expression of the type

$$A_i = \phi(A_1, \dots, A_i, \dots, A_n), \quad (\text{B.1})$$

i.e., expressions that specify a fixed point. Conversely, any expression in CI of the form (B.1) can be interpreted as a self-referential proposition,

⁵Eds. As part of his doctoral thesis, Emil Post (1921) proved the completeness and consistency of the propositional calculus of the *Principia Mathematica*. He foresaw Kurt Gödel's and Alan Turing's results on the undecidability and incompleteness of general formal systems (Stillwell, 2004).

That we can produce self-reference through expressions like (B.1) is very interesting, for it is a clue as to how CI provides a general device for *designation* for expressions in it, if we choose to interpret it that way. Of course, no closure can be constructed without a global designation mechanism (Smullyan, 1957). In fact, Spencer-Brown uses the identity symbol “=” to indicate value equivalence between expressions. In doing so, he is only following the tradition that defines identity as interchangeability *salva veritate*. In CI, expressions which are of equivalent value are obviously interchangeable without altering the context, whence it is natural to use “=,” and substitutivity as a rule for derivation. When CI is interpreted for logic, as expected, identity is interchangeable with the biconditional $\equiv$, since $A = B$ if and only if $\top = A \equiv B$. Now we may interpret identity also in an extensional context, as interchangeable with designation. If “ p ” designates p , surely they are intentionally different, but it is also the case that we expect them to satisfy substitution *salva veritate*. In languages such as CI and PC, the designation of an expression, since it is equivalent to the expression it designates, can be done through identification. Thus we can interpret some equations of CI in terms of logical designation, as we did for self-referential expressions, and as we may do for other expressions. In this sense “=” is no longer a metalinguistic device, since we can also designate identities of higher orders through the biconditional, as, for example in “ $A = B \equiv C$ ” interpreted as “ A designates $B = C$.”

We have then a designation mechanism in CI (and PC). But, as is easy to suspect, it is not global, in that not all the expressions in CI can be designated. A calculus of propositions where all such propositional functions can be expressed with an unrestricted domain of (propositional) arguments can be called a *closed* propositional calculus. The appropriation of such a name seems adequate insofar as the expressive capacity of the calculus can be maximally used on itself. It is the case, of course, that the classical PC is not closed (Tarski, 1956). For it would require unrestricted self-reference for all propositional functions. In particular, the property “True” is expressible in PC, but does not admit self-reference without contradictions. “True” cannot be an unrestricted propositional function, but only a carefully restricted one. In CI we may write “True” simply by remembering that we have taken $\top$ to be the truth-value true, and thus $\text{Tr}(A)$ can be taken as

$$A \equiv \top$$

or

$$\text{Tr}(A) = A \begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{|c|} \hline \\ \hline \end{array} A.$$

The liar sentence has now the form

$$A = \overline{\text{Tr}(A)} \\ = A \overline{\quad} A \\ = A,$$

which is a reentrant expression with *no* solution in CI, since it yields an antinomic result for either value we assume for *A*. It is, not unexpectedly, an unsolvable self-referential expression. Consequently, in CI, as in PC, we cannot allow unrestricted reentry, since not all expressions have a fixed point. In particular, $A = A$ has none. The designation mechanism in CI (and thus in PC) is a limited one; it cannot be extended without contradictions produced by self-reference.

The paradoxical situations posed by self-reference in CI and PC seem, up to this point, entirely analogous. However, because of the intuitive and notational gains in CL, we can find a constructive solution to this impasse, rather than try to avoid it (chapter 12). The solution is to admit every self-referential situation by taking the antinomic behavior as a *characterization* of self-indication. In a first step this means to admit autonomous values in the arithmetic of the calculus itself, so as to give the variable A in

$$A = \overline{A}$$

the autonomous value

$$A = \square.$$

In the extended calculus of indications (section 12.3.1) I took all instances of self-referential phenomena to bear a “family resemblance,” be it in self-description of languages of any kind, or in self-computation at the level of system organization.

We may, of course, directly interpret the extended calculus of indications (EC) to logic, as we did with the Brownian CI (Varela, 1979a). It would amount to applying the same procedure Π , with the additional interpretation of the autonomous value now as a third truth value. Semantically this third value represents precisely the nature of a self-denying circular statement, also describable, in time, as an oscillation (true-false-true ...). We thus obtain a three-valued logic $L(EC)$, which is syntactically a generalization of a logic first used by Kleene (1938); see also (Schwartz, 1981).

The truth tables of $L(EC)$ are identical to those first described by Kleene and redefined as the variant-standard system S_3 in the study of Dienes (1949). Further, any expression of $L(EC)$ can be immediately transcribed into an expression in Kleene's system (K), simply by rewriting the connectives as defined by II. The real difference between these classical systems and $L(EC)$ lies in the latter's use of the " $=$ " equivalence. In this connection it should be noticed that an expression that is derivable from K 's axioms must have either the value "true" or "undefined," since these are the tautologies of K . Let us transcribe " $\vdash_K p$ " in K for " $p = \Box p$ " in $L(EC)$. Since $L(EC)$ is complete, all identities of the form " $p = \Box p$ " are derivable; therefore all derivable expressions p in K are recovered in $L(EC)$ as exactly those expressions satisfying $p = \Box p$. Conversely, if an expression in $L(EC)$ is such that $p = \Box p$, then we must have $\vdash_K p$, since it has a designated value, and thus it must be derivable in K . This justifies choosing an equivalence form for $L(EC)$; by doing so, not only do we recover all expressions derivable with implication, but we also have access to many other classes of expressions. By taking an implication rule for detachment, we force ourselves to confuse expressions which are " T " or " U " in K . In this sense, $L(EC)$ is richer than the classical systems, with which it coincides in its truth tables. Further, this interpretation provides an axiom system for K , hitherto unknown.

When interpreted for classical logic, self-reference engenders the paradoxes of self-denying sentences. To this problem several authors have addressed themselves, in an attempt to solve it by means of a three-valued logic. Moh Shaw-Kwei (1954) used the three-valued system of Łukasiewicz, interpreting the third value as "paradoxical." He showed that in this case, as in all the family of Łukasiewiczian systems, paradoxes will recur; however, his results do not apply to systems of the K type, where " p implies p " is untrue. Later, Asenjo (1966) proposed a calculus of antinomies, where the third value is taken as "antinomic," with truth tables of the K type; his axioms, however, are incomplete. The present interpretation (Varela, 1979a) extends these authors' attempts by providing a consistent, complete system that can accept self-denying statements as nonparadoxical.

Yet, as might be expected, this simple approach does not really work. The logic obtained is clearly not satisfactorily closed, for in it we lose determinacy of several expressions of normal use (such as $A \vee \sim A$ and $A \supset A$), and although we can now have unrestricted self-reference in the expressions, the range of expressions is greatly reduced. The pervasiveness of three truth values distorts the form of logical propositions to the point of crippling them (cf. section 12.6).

It is necessary to distinguish between infinite and finite self-reference (i.e., vicious and nonvicious self-reference, or in Fitch's (1946) terms, self-reference of the first and second kind). In the latter, a self-referential situation leads to a finite cycle of computations (whether real or conceptual), eventually ending with a definite result. A typical such case in logic is the statement "this statement is true," or in CI the expression $A = \overline{A}$. In infinite self-reference, a self-referring situation leads to an endless loop, where there is no single stable result but an oscillation. Typical, again, in logic is the liar's statement, or in CI the expression $A = \overline{A}$. Here we see why the name infinite self-reference is appropriate, because A can be taken to be the infinite expression

$$A = \overline{\overline{\overline{\dots}}}$$

It is this infinite self-reference, of course, that is currently avoided at all costs in formal systems and scientific discourse. It is this behavior that I proposed to confront constructively by characterizing it in terms of infinite expressions interpretable in time as oscillations of various frequencies and periods (Chapter 12). Interpreted for logic, a Brownian algebra would be a multivalued, infinitary Boolean algebra. These two aspects correspond to one another. Let me explain this. From Scott's work discussed above, we saw that a type-free logic is not forthcoming from a mere juggling of the rules of reduction. This implied that in order to obtain unrestricted reentry we had to extend the algebra of forms to continuous forms (cf. section 12.10.1). When interpreted for values (cf. spatial patterns) this yields an indefinite number of them, rather than just three, as in the case of ECI just discussed. Now, we *do not know* what the details of this logical interpretation of waveform algebras could produce for logic and multivaluedness. According to a brief recent report by Schwartz, it might provide a relation between self-reference and a logic of vagueness (Schwartz, 1981); see also (Gaines, 1976). I do not know; but the grounds for exploration for this and other questions are quite open.

I want to conclude by emphasizing once again, that the calculi of indication are *not* a subtle form of logic. They really intend something quite different as descriptions, and have played, in this book, a very different function. When interpreted for logic, however, these calculi carry with them a clear departure from the usual grounds on which closed logical calculi are considered, particularly in the content and meaning assigned to self-reference. The values introduced in the algebra of indications represent a domain with empirical content, visible in the range of applicability of natural processes,

and only *secondarily* in the meaningfulness of linguistic paradoxes. We are not considering circular expressions as abnormal variants of some standard form of expressions, but as a domain valid in itself, which has its own behavior typified in logic by the atomic circularity "*p* is not *p*."

This epistemological attitude regards circularity as a central notion rather than as a nuisance. Self-reference becomes a necessary tool for scientific description and not an idiosyncrasy of formal speech that can be amputated for formal languages. Indeed, from this point of view, if we are to have a science of living, social, and cognitive systems, and a contemporary epistemology of solid rational foundations, circularity has to be taken at face value, acknowledged as a working and workable notion, and mapped formally.

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